

Strong diffeomorphic approximation in three dimensions for $p > 2$

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Abstract

We resolve the three-dimensional Ball–Evans approximation problem for every finite $p > 2$. Every $W^{1,p}$ homeomorphism between arbitrary bounded domains in $\mathbb{R}^3$ can be approximated strongly in $W^{1,p}$ by smooth diffeomorphisms onto the same target.

1 Introduction and conventions

Let $\Omega, \Lambda \subset \mathbb{R}^3$ be domains, meaning nonempty connected open sets. A homeomorphism $f : \Omega \rightarrow \Lambda$ belongs to $W^{1,p}(\Omega; \mathbb{R}^3)$ when its components and their weak first derivatives are in L^p . Strong $W^{1,p}$ approximation requires convergence of both the maps and these derivatives in L^p . The *fixed-target Ball–Evans approximation problem* asks whether one can choose the approximants to be smooth diffeomorphisms from Ω onto exactly Λ . Here a diffeomorphism and its inverse are smooth on their open domains; this definition makes no assertion about boundary values.

The problem originates in nonlinear elasticity, where injectivity expresses noninterpenetration and the derivative describes local deformation. Ball’s discussions [4–6] connect the approximation question to the variational and computational theory; the original planar approximation paper records his attribution of the question to Evans [28]. Neither ordinary convolution nor sufficiently fine vertex interpolation automatically preserves injectivity. We resolve the fixed-target problem on bounded three-dimensional domains for every finite exponent above two.

Theorem 1.1. *Let $\Omega, \Lambda \subset \mathbb{R}^3$ be bounded domains, let $2 < p < \infty$, and let $f : \Omega \rightarrow \Lambda$ be a homeomorphism in $W^{1,p}(\Omega; \mathbb{R}^3)$. There exist C^∞ diffeomorphisms $f_j : \Omega \rightarrow \Lambda$, each belonging to $W^{1,p}(\Omega; \mathbb{R}^3)$, such that*

$$\int_{\Omega} (|f_j - f|^p + |Df_j - Df|^p) dx \longrightarrow 0. \quad (1.1)$$

Neither boundary regularity, a boundary extension of f , inverse Sobolev regularity, nor a nonvanishing Jacobian is assumed. The theorem concerns forward Sobolev convergence. Smoothness of each inverse does not assert Sobolev convergence of the inverses; strong forward convergence alone also does not imply convergence of an elastic energy with a singular determinant penalty. These are distinct approximation questions; see [5, 24, 37]. The complementary range $1 \leq p \leq 2$ is treated by a different construction in the companion article [36]. No theorem from that range is used here. The full common smoothing argument is included in Appendix A, so the present range theorem has a complete local proof.

1.1 Earlier results and the dimensional obstacle

The planar theory provides both the principal positive precedents and important contrasts. Iwaniec, Kovalev, and Onninen treated the Hilbert-space case $p = 2$ using harmonic replacements in work first circulated in June 2010 [29]. Their extension, first circulated that September, covers every $1 < p < \infty$ on arbitrary planar open sets [28, Theorem 1.1]. Hencl and Pratelli proved the endpoint $p = 1$ by geometric extension and grid constructions, including locally finite piecewise affine approximation [25, Theorem 1.1]. Both results preserve the target and require no inverse Sobolev hypothesis. Thus those features are inherited goals of the problem, not new distinctions of the three-dimensional statement.

Geometric approximation has also been developed under stronger planar input assumptions. Bellido and Mora-Corral control approximation in smaller Hölder exponents on polygonal planar domains when both directions are Hölder [7]. Daneri and Pratelli obtain quantitative bi-Lipschitz approximation with derivative control in both directions for bi-Lipschitz maps [15]; Pratelli treats bi- $W^{1,1}$ maps [37]. These results clarify the distinction between approximating a map by maps with smooth inverses and approximating the inverse in a specified Sobolev norm.

Piecewise affine smoothing is a separate stage. Mora-Corral and Pratelli established a strong planar version with control in both directions [33]. Campbell and Soudský obtained C^1 injective approximation of piecewise affine maps in arbitrary dimension, without asserting smoothness of the inverse [11]. Campbell, D’Onofrio, and Vitek then proved diffeomorphic approximation of locally finite piecewise affine homeomorphisms in dimensions three and four, with uniform and derivative-error control for both the map and its inverse [13, Theorem A]. The remaining task for a general Sobolev input is to obtain the required strong approximation before this smoothing stage. Appendix A includes both the locally bi-Lipschitz-to-PL reduction and a PL smoothing proof with arbitrary positive continuous value tolerances.

The dimension restriction is substantive. Building on the $W^{1,1}$ construction of Hencl and Vejnar [26], and repairing a step in that construction, Campbell, Hencl, and Tengvall obtained counterexamples in dimensions $n \geq 4$ for $1 \leq p < \lfloor n/2 \rfloor$, with Jacobians of both signs on sets of positive measure [12]. Those examples do not decide the three-dimensional problem. Qualitative topology does help in dimension three: the triangulation and PL-approximation theory of Moise and Bing, in Hamilton’s relative formulation, provides the topological flexibility used below [8, 21, 31, 32]. It supplies no strong derivative estimate. In particular, a topological extension chosen after a small-energy exceptional set may have an arbitrarily large Lipschitz constant, so multiplying that earlier energy by its later constant is not a valid approximation argument.

1.2 Proof strategy and organization

We first seek a locally bi-Lipschitz homeomorphism strongly close to f . At points where Df is invertible, suitably chosen small cubes permit exact affine replacements. The principal work concerns the regular rank-one and rank-two data. Their image is volume-null, but their source Sobolev energy need not be small. Discarding them as an exceptional set would therefore lose the approximation theorem. The construction retains them and recovers their gradients through two scalar coordinates.

On each small retained patch, these coordinates have prescribed affine first-order models. Keeping the realized labels close to the original scalar coordinates controls the change in their mean gradients through a boundary integral. Nearly sharp upper bounds for the gradient norms then give strong gradient closeness by uniform convexity. This is why the construction must control both label values and scalar parameter slopes.

1. *Prepare exact regions and retained deficient-rank data.* Section 2 first uses the decomposability

bundle of Alberti–Marchese and the converse to Rademacher’s theorem of De Philippis–Rindler to identify transverse directions in the transported measure [1, 16]. Thin graph bands, related to the null-set constructions of Alberti–Csörnyei–Preiss [2], flatten the retained data onto separated plate patches while preserving their tangential derivatives. These patches determine the exact regions required of a continuous proper surjection $F_b : \Omega \rightarrow \Lambda$, called the carrier. It agrees with affine or PL homeomorphisms over a prescribed open part of the target and has controlled local Sobolev energy outside the marked exceptional interiors that may occur when $2 < p < 3$. Its only non-singleton fibres are specified tame balls, and the proof gives their relative opening to homeomorphisms. The cubical cutoff uses immersions of a two-dimensional skeleton, with the positive-codimension immersion theorem of Hirsch [27] and locally flat Schoenflies [10].

2. *Reduce the variable target to a two-dimensional complex.* Section 3 constructs a Lipschitz map $T : \Lambda \rightarrow \Lambda$ from a strongly convex potential. Its full-dimensional input blocks lie entirely in the carrier’s exact regions; outside these blocks its image is a polyhedral two-complex. This combines classical convex conjugacy and polyhedral diagram geometry [3, 34], with additional protected paired contacts proved here. They place the retained deficient-rank data on rectangular faces with nearly unchanged gradients. On a polygonal face, the two coordinates are a scaled logarithm of the normalized radial fraction and arclength around the perimeter.
3. *Realize these coordinates by source walls.* Section 4 uses annular walls for radial levels and disk walls for perimeter levels. Routing removes unwanted circle intersections without accumulating a derivative factor at each switch or losing the length of a wall’s transverse parameter interval. A family of reference fibre paths, called guards, then pins the realized labels: a misplaced wall would force a fixed positive path cost on too many small spheres. The resulting lower bound grows as the mesh scale tends to zero because $p > 2$. Theorem 4.11 is explicitly conditional on the geometric and energy estimates proved in Sections 5–7.
4. *Supply calibrated meshes and parameter bounds.* Section 5 prepares and normalizes the source walls, preserving the individual edge-count bounds, and constructs their two-colour neighborhoods with simultaneous product coordinates. Normalization and cut-and-paste belong to the normal-surface tradition [20]. Section 6 improves the estimates on selected source patches. In aligned source coordinates, the two scalar gradients are nearly orthogonal in rank two and nearly parallel in rank one. The snapping map in Lemma 6.7 removes the large parameter slopes caused by thin intermediate configurations. Its uniform estimate is only in the direction from new to old coordinates, as illustrated in Figure 4. Broadening then gives the sharp scalar bounds by direct differentiation, not by differentiating an uncontrolled ambient extension.
5. *Recover strong gradients and smooth with a fixed target.* Section 7 chooses local upper bounds for $\int G^p$, where G is the wall-parameter slope density, before choosing guards and the final mesh. These are the capacities used in the wall argument, not Sobolev capacities of sets. Nearly sharp energy and pinned mean gradients yield strong gradient approximation by uniform convexity. A separate compression handles marked exceptional interiors only when $2 < p < 3$; for $p \geq 3$ the volume Lusin property eliminates them. The proof ends by correcting the reserved full-rank cubes and applying the complete smoothing theorem of Appendix A. The strict target maps are self-homeomorphisms, and the final local replacements preserve their old images; the resulting homeomorphic approximants are onto the fixed target.

The central quantitative distinction is between derivatives of the two retained parameters and derivatives of a homeomorphism filling the complementary chambers. Only the former enter the principal energy estimates. The constant and error choices are recorded in Remark 7.5; they ensure that every uncontrolled constant is fixed before the error required to absorb it.

1.3 Conventions

Unless stated otherwise, all domains, closures, local finiteness, and compactness are taken relative to the indicated open manifold. A compact set in an open domain has positive distance from its Euclidean complement. A PL map or triangulation is locally finite. A locally bi-Lipschitz homeomorphism has finite Lipschitz constants in both directions on sufficiently small neighborhoods; no global bound is implicit.

The symbol $|A|$ denotes the Frobenius norm of a matrix, and $\|A\|_{\text{op}}$ its operator norm. We use the Frobenius norm in sharp gradient estimates. Equivalent norms give the same strong convergence in Theorem 1.1. The symbols $\mathcal{L}^m$ and $\mathcal{H}^m$ denote Lebesgue and Hausdorff measure. Integrals on parameterized traces include their stated multiplicities. The differential along an m -dimensional parameter space is denoted D_m when it must be distinguished from the full differential.

A *fine value tolerance* on a domain U is a positive continuous function on U . To approximate finely means to meet a prescribed such tolerance pointwise. Whenever inverse control is needed on a compact localization, it is included in the prescribed tests. Locally finite constructions permit tolerances tending to zero toward the ends of the open domains; no positive global lower bound is assumed.

Constants called *absolute* depend only on dimension and on fixed universal reference shapes. A constant C_p may also depend on the fixed exponent. Other dependencies are specified when the constant is introduced. A finite constant that depends on a chosen compact collection of jets or charts is fixed before an error is required to be small against it.

A homeomorphism between connected oriented domains has a constant topological orientation. If necessary, reflect the target, perform the construction for the orientation-preserving map, and undo the reflection. This convention imposes no extra hypothesis on the Sobolev differential.

Lemma 1.2 (Local tolerances). *Let $U \subset \mathbb{R}^n$ be open and let $a : U \rightarrow (0, \infty)$ be continuous. For every $L > 0$ there is a positive L -Lipschitz function d on U such that*

$$d(x) \leq \min\{a(x), 1, L \text{dist}(x, \mathbb{R}^n \setminus U)\}.$$

There are also positive smooth minorants satisfying prescribed positive continuous upper bounds on their values and first derivatives. One may additionally impose positive constant bounds on a locally finite family of compact regional tests. Finite collections of such requirements can be combined, and error budgets on a countable compact exhaustion can be made summable.

Proof. Extend $e(x) = \min\{a(x), 1, L \text{dist}(x, \mathbb{R}^n \setminus U)\}$ by zero outside U , and put

$$d(x) = \inf_{y \in \mathbb{R}^n} \{e(y) + L|x - y|\}.$$

This function is L -Lipschitz and is bounded above by e . It is positive at an interior point x : on a small closed ball about x , e has a positive minimum, and outside that ball the distance term has a positive lower bound.

For a smooth minorant take a smooth locally finite partition of unity $\{\chi_i\}$ with compact supports in U and put $b = \sum_i c_i \chi_i$, with positive constants c_i . Given positive continuous bounds v, w , choose

$$c_i \leq 2^{-i-2} \min \left\{ \inf_{\text{supp } \chi_i} v, \frac{\inf_{\text{supp } \chi_i} w}{1 + \|D\chi_i\|_{\infty}} \right\}.$$

Then $b > 0$, $b \leq v$, and $|Db| \leq w$. Replace a finite list of requirements by their minimum. For countably many regional tests use a locally finite compact cover with compact enlargements, refine the bounds on each intersecting support, and allocate total errors by a convergent positive series. Only finitely many regional requirements meet a given compact support. $\square$

We will also use the following elementary global observation. If $H : U \rightarrow V$ is a homeomorphism and a continuous map $G : U \rightarrow V$ satisfies

$$|G(x) - H(x)| < \frac{1}{2} \text{dist}(H(x), \mathbb{R}^n \setminus V),$$

then the straight homotopy stays in V . When V is bounded, it is a proper homotopy: the preimage of a compact target set is contained, uniformly in the homotopy parameter, in the H -preimage of a compact set a positive distance from ∂V . This observation will be used together with local injectivity and degree, not as a substitute for local injectivity.

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2 Retained rank data and an exact carrier

Our aim is to retain the source energy carried by the regular rank-one and rank-two points while preparing most of the target for exact local replacements. The retained data will first be flattened onto separated plates, with only a small change to their derivatives. On opposite sides of each plate we select paired target rectangles. The carrier of Proposition 2.13 will agree with affine or PL homeomorphisms over neighborhoods of these rectangles and over an open set with arbitrarily small residual target volume, while agreeing with the chosen map near the retained source data. The order of these two requirements explains its two rounds: endpoint neighborhoods are fixed first; the remaining target volume is prescribed afterwards.

The carrier may collapse isolated tame balls. Such fibres permit the energy-controlled local replacements below and can later be opened topologically. For $2 < p < 3$, the replacements can also leave marked source interiors whose energy is treated by a separate compression. The final two subsections prepare that compression and the eventual full-rank correction used in Section 7.

2.1 Regular points and fine tolerances

Throughout this section, $p > 2$ and the topological orientation of f is positive. A reflection reduces the other orientation to this one. All regularity assertions below are local on the open sets; no boundary regularity is involved. A map has the volume Lusin property N if it maps sets of three-dimensional Lebesgue measure zero to sets of measure zero.

We use the weighted area formula in the form recorded by Simon [38, Chapter 2, Section 3, Equations (3.4)–(3.5)]. The general Lipschitz-decomposition background for Sobolev maps is provided by Bojarski–Hajlasz [9, Theorem 3(1)]. The stronger ordinary differentiability and image-null assertions needed here are proved next, using openness and two-dimensional Sobolev estimates on spheres.

Lemma 2.1 (Regular points of a Sobolev homeomorphism). *Let $u : G \rightarrow G'$ be a homeomorphism between open subsets of $\mathbb{R}^3$, with $u \in W_{\text{loc}}^{1,p}(G; \mathbb{R}^3)$ and $p > 2$. There is a Borel set $R_u \subset G$ of full measure such that, for $x \in R_u$, the weak differential $A = Du(x)$ is also the Fréchet differential and*

$$\lim_{r \downarrow 0} \sup_{0 < |z-x| \leq r} \frac{|u(z) - u(x) - A(z-x)|}{|z-x|} = 0, \quad \lim_{r \downarrow 0} \frac{1}{|B_r|} \int_{B_r(x)} |Du - A|^p = 0. \quad (2.1)$$

If u has positive topological orientation, then $\det A > 0$ at all points of R_u where A is invertible. Moreover,

$$\mathcal{L}^3(u(\{x \in R_u : \text{rank } Du(x) < 3\})) = 0. \quad (2.2)$$

If $p \geq 3$, u has the volume Lusin property N locally, and hence on G . Finally, for every fixed coordinate direction, almost every parallel coordinate-plane section of G has image under u of three-dimensional measure zero. This last assertion holds for every continuous $W_{\text{loc}}^{1,p}$ map, without injectivity.

Proof. We first pass from Sobolev differentiation in mean to ordinary Fréchet differentiation. At almost every x , with $A = Du(x)$,

$$w_r(\xi) := \frac{u(x + r\xi) - u(x)}{r} - A\xi \longrightarrow 0 \quad \text{in } W^{1,p}(B_2; \mathbb{R}^3). \quad (2.3)$$

For completeness, the gradient assertion is the Lebesgue differentiation Theorem applied to Du . To obtain the value assertion, subtract the affine function with gradient A and apply Poincaré's inequality on concentric balls. The difference between its mean on a ball of radius t and on the concentric ball of radius $t/2$ is bounded by

$$Ct \left(\frac{1}{|B_t|} \int_{B_t(x)} |Du - A|^p \right)^{1/p}.$$

Summing on dyadic radii and using the precise value $u(x)$ gives the value assertion in (2.3).

Fix $0 < d < 1/8$. For every $z \in B_1$, slicing the annulus $B_{2d}(z) \setminus B_d(z)$ gives a radius $\rho \in (d, 2d)$ for which

$$\int_{\partial B_\rho(z)} (|w_r|^p + |D_\tau w_r|^p) \leq d^{-1} \|w_r\|_{W^{1,p}(B_2)}^p.$$

The trace agrees with the continuous restriction. Sobolev embedding on the two-dimensional sphere, using $p > 2$, therefore gives

$$\sup_{\partial B_\rho(z)} |w_r| \leq C_{p,d} \|w_r\|_{W^{1,p}(B_2)}.$$

The constant is independent of z . Each coordinate of the homeomorphism $u_r(\xi) = (u(x+r\xi) - u(x))/r$ attains its maximum and minimum on the boundary of $\overline{B}_\rho(z)$: an extremum in the interior would contradict openness of u_r . Comparing the affine function $A\xi$ at z and on this sphere yields

$$\sup_{z \in B_1} |w_r(z)| \leq C\|A\|_{\text{op}}d + C_{p,d}\|w_r\|_{W^{1,p}(B_2)}.$$

First let $r \downarrow 0$, then $d \downarrow 0$. We obtain uniform affine approximation on $B_r(x)$, which is equivalent to the first assertion of (2.1). Intersecting the full measure sets used here gives a Borel choice of R_u .

If A is invertible, uniform approximation on a sufficiently small sphere makes $u(x+r\xi) - u(x)$ homotopic, through maps avoiding zero, to $rA\xi$. The local degree of the homeomorphism is therefore $\text{sgn det } A$, proving the orientation assertion.

The set of Fréchet differentiability points admits a countable Lipschitz decomposition. Indeed, restrict first to points where $|Du| \leq m$ and

$$|u(z) - u(x) - Du(x)(z - x)| \leq |z - x| \quad (0 < |z - x| < 1/m),$$

and then partition into sets of diameter less than $1/m$ in an interior exhaustion. On each resulting set the restriction of u is Lipschitz. Such sets cover every differentiability point after varying m . On a Lipschitz extension of each restriction, its approximate differential agrees almost everywhere on that set with Du . The area formula, applied to the subset where $\text{rank } Du < 3$, now proves (2.2). Source null subsets of these Lipschitz pieces have null images as well, so the conclusion concerns the entire regular deficient-rank set, not merely a source full-measure subset of it.

For the volume property N , it suffices to use $p = 3$ locally. For an interior cube Q , choose a sphere of radius comparable to its side length, enclosing Q and contained in a fixed enlargement $C_0Q \Subset G$. The sphere oscillation inequality and slicing give

$$(\text{osc}_Q u)^3 \leq C \int_{C_0Q} |Du|^3. \quad (*)$$

Here openness again bounds coordinate oscillations inside the sphere by those on its boundary. If $E \Subset G$ is null, choose an open neighborhood $O \Subset G$ of E with arbitrarily small $\int_O |Du|^3$. A Whitney decomposition of O may be chosen with $C_0Q \subset O$ and bounded overlap of these enlargements. Summing $(*)$ over the cubes that cover E bounds the outer volume of $u(E)$ by $C \int_O |Du|^3$. This proves the claim by exhaustion. When $p > 3$, local L^3 integrability follows from Hölder's inequality.

For the last assertion, Fubini gives a continuous $W^{1,p}$ restriction to almost every coordinate-plane section. On a compact square in such a section, divide into squares Q of side length a , allowing uniformly bounded enlargements inside a slightly larger compact square. Planar Morrey's inequality and Hölder's inequality give

$$\begin{aligned} \sum_Q \text{diam}(u(Q))^2 &\leq C \sum_Q a^{2-4/p} \left(\int_{2Q} |D_\tau u|^p \right)^{2/p} \\ &\leq C \left(\sum_Q a^2 \right)^{1-2/p} \left(\sum_Q \int_{2Q} |D_\tau u|^p \right)^{2/p}. \end{aligned}$$

This bound is independent of a . Uniform continuity gives

$$\max_Q \text{diam}(u(Q)) \longrightarrow 0, \quad \sum_Q \text{diam}(u(Q))^3 \longrightarrow 0.$$

A countable exhaustion proves the three-dimensional null-image assertion. $\square$

Remark 2.2 (Fine value tolerances). Lemma 1.2 supplies, for every positive continuous function a on an open set G and every $L > 0$, a positive L -Lipschitz minorant $a_0 \leq a$ with $a_0(x) \leq L \operatorname{dist}(x, \mathbb{R}^3 \setminus G)$. We use these minorants to combine the fine value requirements below, including distance to the target boundary and previously fixed compact-set accuracies.

2.2 Flattening the deficient positive ranks in the target

The regular deficient-rank image has zero target volume, but the integral of $|Df|^p$ over its source preimage need not be small. We therefore organize this part of the map rather than discard it. The next construction moves almost all of its weighted data onto separated flat plates and preserves their tangential derivatives.

Let R_f be the regular set from Lemma 2.1, and set

$$D = \{x \in R_f : 1 \leq \operatorname{rank} Df(x) \leq 2\}, \quad \mu(E) = \int_{E \cap D} (1 + |Df|^p) dx, \quad \nu = f_#(\mathbf{1}_\Omega \mathcal{L}^3).$$

The finite measure μ is the measure used for all losses in this preparation. Boundedness of Ω and $f \in W^{1,p}$ ensure its finiteness.

The decomposability bundle $V(\nu, y)$ records the tangent directions of rectifiable curves whose averaged measures are absolutely continuous with respect to ν . We use the precise locality and differentiability properties cited in the proof below to find a direction transverse to this bundle on the retained data.

Lemma 2.3 (A forbidden cone for the transported measure). *Let $V(\nu, y)$ be the decomposability bundle of ν . Then*

$$\operatorname{im} Df(x) \subset V(\nu, f(x)) \quad \text{for almost every } x \in \Omega. \quad (2.4)$$

Moreover, $V(\nu, y) \neq \mathbb{R}^3$ for ν -almost every $y \in f(D)$. Consequently, for each $0 < \lambda < 1$, $f(D)$ can be partitioned, up to a ν -null set, into finitely many measurable pieces with associated unit vectors e such that

$$|\operatorname{proj}_{V(\nu, y)} e| < \lambda/10 \quad \text{on the corresponding piece.} \quad (2.5)$$

In coordinates $(h, v) \in e^\perp \times \mathbb{R}$, the bundle on that piece contains no nonzero vector in the double cone

$$\{(a, b) : |a| \leq |b|/\lambda\}.$$

Proof. We use the decomposability bundle and its strong locality property from Alberti–Marchese [1, Section 2.6 and Proposition 2.9(i)]. In particular, tangent directions of measurable families of rectifiable curve measures belong to $V(\nu, \cdot)$ whenever the aggregate curve measure is absolutely continuous with respect to ν . Weighted restrictions are allowed: a density can be absorbed by restriction and the layer-cake formula.

Fix a coordinate direction e_i and an interior rectangular box. For almost every line in that direction, the restriction of f is absolutely continuous. It is also injective, so the one-dimensional area formula identifies the pushforward of $|\partial_i f| dt$ with length measure on the image curve. Integration over the transverse coordinates gives the aggregate measure

$$f_#(\mathbf{1}_{\operatorname{box}} |\partial_i f| dx) \ll \nu.$$

The images are rectifiable for almost every line; if necessary they may be split into bounded-length pieces or parametrized by arclength. At almost every parameter where $\partial_i f \neq 0$, their tangent line is $\operatorname{span}\{\partial_i f\}$. The defining property of the bundle therefore gives $\partial_i f(x) \in V(\nu, f(x))$ for almost every

x in the box where that derivative is nonzero. A countable box exhaustion and the three coordinate directions prove (2.4).

By Lemma 2.1, $f(D)$ has zero volume. Suppose that $A \subset f(D)$ has positive ν measure and $V(\nu, y) = \mathbb{R}^3$ on A . The differentiability Theorem of Alberti–Marchese [1, Theorem 1.1(i)] says that every Lipschitz function on $\mathbb{R}^3$ is ordinarily differentiable $\nu|_A$ -almost everywhere. The converse to Rademacher’s Theorem of De Philippis–Rindler [16, Theorem 1.14] then implies $\nu|_A \ll \mathcal{L}^3$, which is impossible since $A \subset f(D)$ is null. This proves properness. This invocation uses the measure-theoretic converse to Rademacher, rather than an uncited inference from the existence of individual curve families.

Choose a finite sufficiently fine net of directions on the unit sphere. Every proper linear subspace has a unit normal, so one of these directions satisfies (2.5). Assign the first such direction measurably. If (a, b) belongs to the displayed cone and is nonzero, then

$$\frac{|b|}{\sqrt{|a|^2 + b^2}} \geq \frac{\lambda}{\sqrt{1 + \lambda^2}} > \lambda/2,$$

whereas any vector $w \in V(\nu, y)$ obeys $|e \cdot w| < (\lambda/10)|w|$. The cone is therefore forbidden. $\square$

The simultaneous null-set conclusion below is a graph-parametrized form of Alberti–Marchese’s Lemma 7.5. Its compact-convex selection mechanism is Rainwater’s lemma, as presented in [1, Lemma 7.1 and Corollary 7.2]. We give the argument because one common exceptional set for every allowed graph is essential to the subsequent flattening.

Lemma 2.4 (A simultaneous graph-null set). *Fix orthogonal coordinates $(h, v) \in \mathbb{R}^2 \times \mathbb{R}$ and $0 < \lambda < 1$. Suppose a finite measure σ is concentrated on a Borel set A and is absolutely continuous with respect to ν , and that $V(\nu, y)$ contains no nonzero vector in $\{(a, b) : |a| \leq |b|/\lambda\}$ for σ -almost every y . There is a Borel set $A_0 \subset A$ of full σ measure such that*

$$|\{v : (h(v), v) \in A_0\}| = 0 \tag{2.6}$$

for every $1/\lambda$ -Lipschitz graph h on any interval.

Proof. We first work in a compact rectangle $X = H \times I$, where H is a closed horizontal box and I a compact vertical interval. Let Γ be the compact family of all $1/\lambda$ -Lipschitz maps $h : I \rightarrow H$, equipped with the uniform topology. Write

$$\rho_h = (v \mapsto (h(v), v))_{\#}(\mathcal{L}^1|_I), \quad \mathcal{C} = \left\{ \int_{\Gamma} \rho_h dP(h) : P \in \mathcal{P}(\Gamma) \right\}.$$

The map $h \mapsto \rho_h$ is weakly continuous. Thus $\mathcal{C}$ is convex and weakly compact in the finite measures on X .

Every $\rho \in \mathcal{C}$ is singular with respect to $\sigma|_X$. Indeed, otherwise there would be a nonzero measure $\sigma_0 \leq \rho$ with $\sigma_0 \ll \sigma|_X$. Writing $a = d\sigma_0/d\rho$ and $\rho = \int \rho_h dP(h)$ yields

$$\sigma_0 = \int_{\Gamma} a \rho_h dP(h).$$

Each ρ_h is equivalent to length measure on its graph, with bounded positive density. Density restriction and layer-cake therefore express σ_0 as a family of rectifiable curve measures. Its aggregate is absolutely continuous with respect to ν . The tangents of the graphs must belong to $V(\nu, \cdot)$ at almost every point charged by this family, while their directions $(h'(v), 1)$ lie in the forbidden cone. This contradicts $\sigma_0 \neq 0$.

For $0 \leq \phi \leq 1$ continuous on X , put

$$L(\phi, \rho) = \int_X \phi d\rho + \int_X (1 - \phi) d\sigma.$$

Mutual singularity and regularity of finite measures imply $\inf_{\phi} L(\phi, \rho) = 0$ for every $\rho \in \mathcal{C}$. The compact-convex minimax Theorem, with $\mathcal{C}$ as its compact side [39, Theorem 4.2], applies to this continuous affine functional and gives

$$\inf_{\substack{\phi \in C(X) \\ 0 \leq \phi \leq 1}} \sup_{\rho \in \mathcal{C}} L(\phi, \rho) = 0.$$

Choose ϕ_n with

$$\int_X (1 - \phi_n) d\sigma + \sup_{h \in \Gamma} \int_I \phi_n(h(v), v) dv \leq 2^{-n}.$$

Tonelli's Theorem shows that $\phi_n \rightarrow 1$ at σ -almost every point, while, for each individual graph, $\phi_n(h(v), v) \rightarrow 0$ for almost every v . Hence the single Borel set $\{y : \phi_n(y) \rightarrow 1\}$ has full $\sigma|_X$ measure and is null on every graph in Γ . No intersection of graph-dependent full-measure sets is being taken.

To pass to arbitrary graphs, use countably many rectangles with strict interior margins and exhaust the measure by their interiors. On the vertical interval of such a rectangle, restrict the graph and project its horizontal component to the closed box H ; this preserves the Lipschitz constant and does not change graph points inside the smaller rectangle. A graph given on a shorter interval extends with the same Lipschitz constant by constant extension at its endpoints. Taking a countable union of the resulting full-measure, locally graph-null sets proves (2.6). Intersect this set with A at the end. $\square$

We next convert graph-nullness into finitely many thin bands. The cone order, antichain decomposition and Lipschitz slabs have their counterparts in the null-set constructions of Albritton–Csörnyei–Preiss [2, Sections 2.2 and 3]. The uniform near-hit estimate is the compactness mechanism used in [1, Lemma 4.12, Step 1]; we retain its proof here.

Lemma 2.5 (Finite thin bands). *Let K be a compact subset of the interior of a rectangular chart $H \times I \subset \mathbb{R}^2 \times \mathbb{R}$, and assume that K satisfies (2.6). For every $\delta > 0$, there are finitely many piecewise affine functions $g_1, \dots, g_N$ on H , each with Lipschitz constant at most $C\lambda$, and a common length $L > 0$ such that*

$$NL < \delta, \quad K \subseteq \bigcup_{j=1}^N \{(h, v) : |v - g_j(h)| \leq L/2\}.$$

The values of the functions, and their bands, can be kept in a fixed compact subinterval of the interior of I . Here the containment $\subseteq$ means that the union contains an open neighborhood of K .

Proof. Take a cubical grid of spacing w and let P_w be the centers of the grid boxes that meet K . For distinct centers define

$$(h, v) \prec (h', v') \iff v' - v \geq \lambda|h' - h|.$$

This is a strict partial order: transitivity follows from the triangle inequality, and distinct comparable centers have $v' > v$. Because vertical grid coordinates differ by integer multiples of w , successive centers in a chain have vertical separation at least w .

The horizontal coordinates of a chain, interpolated linearly as a function of v and extended constantly beyond its first and last nodes, form an allowed $1/\lambda$ -Lipschitz graph. All such graphs lie in a fixed compact graph family, after slightly enlarging H and I while keeping the chart margins. For this family set

$$\omega(r) = \sup_{h \in \Gamma} |\{v \in I : \text{dist}((h(v), v), K) \leq r\}|.$$

Then $\omega(r) \rightarrow 0$ as $r \downarrow 0$. Otherwise compactness would give uniformly convergent graphs along a sequence $r \downarrow 0$; the upper limit of the near-hit indicators is bounded by the indicator that the limiting graph meets K . Reverse Fatou and (2.6) give a contradiction.

Around each node of a chain take a vertical interval of radius $w/4$. These intervals are disjoint and, for small w , lie inside the fixed interval with margins. On each such interval the graph is within $C_\lambda w$ of K . If n_w is the longest chain length, it follows that

$$\frac{1}{2}n_w w \leq \omega(C_\lambda w), \quad \text{hence} \quad n_w w \rightarrow 0.$$

Give every center its longest-chain rank. Centers of the same rank form an antichain, and two of them satisfy $|v' - v| < \lambda|h' - h|$. In particular, their horizontal coordinates are distinct and their heights extend to a λ -Lipschitz function on H by the scalar Lipschitz extension formula. Clip the extension to an interior vertical interval containing all the center heights. Piecewise affine interpolation on a sufficiently fine regular triangulation then approximates it to $o(w)$ with Lipschitz bound $C\lambda$.

There are at most n_w such functions. A band of length $L = C_2 w$ about each function contains the boxes assigned to it with a strict margin, provided C_2 is fixed sufficiently large. Thus the bands contain a neighborhood of K , and $NL \leq C_2 n_w w \rightarrow 0$. The initial margins and a sufficiently small w give the final assertion. $\square$

Proposition 2.6 (Target flattening). *Given $\varepsilon > 0$, $0 < \lambda < 1/100$, and a positive continuous fine tolerance a on Λ , one can retain a compact set $K \subset D$ with $\mu(D \setminus K) < \varepsilon$ and construct a Lipschitz map $S_0 : \Lambda \rightarrow \Lambda$ with the following properties.*

1. *The map is the identity outside finitely many disjoint rectangular boxes compactly contained in Λ , $\text{Lip}(S_0) \leq C$, and $|S_0(y) - y| < a(y)$. The constant C is absolute. For $0 < \kappa \leq 1$,*

$$S_\kappa = (1 - \kappa)S_0 + \kappa \text{id}$$

is a bi-Lipschitz homeomorphism of Λ onto itself.

2. *With $F_\kappa = S_\kappa \circ f$ for $0 \leq \kappa \leq 1$,*

$$|DF_\kappa| \leq C|Df| \quad \text{a.e. on } \Omega, \quad |DF_\kappa - Df| \leq C\lambda|Df| \quad \text{a.e. on } K. \quad (2.7)$$

Moreover, $F_\kappa \rightarrow F_0$ strongly in $W^{1,p}(\Omega; \mathbb{R}^3)$, and $\text{rank } DF_0 = \text{rank } Df$ almost everywhere on K .

3. *The compact set $F_0(K)$ is contained in finitely many compact flat plate patches with pairwise disjoint ambient neighborhoods. Each patch lies in a plane of slope at most $C\lambda$ in one of the chosen coordinates. Each patch has an assigned rank, either one or two; the rank-one patches can in addition retain any prescribed small directional grouping of $\text{im } Df$.*
4. *There is a compact zero-volume set $Z \Subset \Lambda$, contained in finitely many piecewise affine graph sheets, such that*

$$S_0 : \Lambda \setminus S_0^{-1}(Z) \rightarrow \Lambda \setminus Z \quad (2.8)$$

is a locally bi-Lipschitz homeomorphism. Its only non-singleton fibres are the collapsed vertical intervals over points of Z . Consequently F_0 is a homeomorphism from $\Omega \setminus F_0^{-1}(Z)$ onto $\Lambda \setminus Z$.

For almost every target point outside Z , its unique preimage under this last homeomorphism either is a regular point of F_0 with positive invertible differential and the power-mean test in (2.1), or belongs to a fixed source null set. When $p \geq 3$, the second alternative occurs only on a target null set.

Proof. We describe the measure selections before the geometric construction. Since μ is finite, first discard an arbitrarily small μ -mass so that the remaining jets and the reciprocals of their nonzero singular values are bounded. Partition by rank and, at rank one, by a finite angular partition of the range lines. Lemma 2.3 further partitions the data among finitely many directions e . The measure $f_{\#}\mu$ is absolutely continuous with respect to ν , so every ν -full choice in these pieces is also full for the weight being retained. Apply Lemma 2.4 to obtain simultaneous graph-null data in each direction.

By inner regularity choose compact subsets of the finitely many pieces with small total loss. Their images are disjoint compact sets, hence have positive separation. In each assigned coordinate frame take a sufficiently fine box grid inside that separation margin and inside Λ . Its cut planes can be chosen to have zero $f_{\#}\mu$ measure: for each coordinate there are only countably many levels with positive mass. Discard small neighborhoods of the finitely many cuts and take compact subsets again. We have reduced the problem, with arbitrarily small loss, to finitely many disjoint boxes, with the compact data strictly inside each box and a fixed rank and direction assignment there. In what follows K^* denotes the compact target data in one such box.

Apply Lemma 2.5 to K^* . Sort the starts of the length- L bands pointwise, writing them as $a_1(h) \leq \dots \leq a_N(h)$. Order statistics of finitely many piecewise affine functions are piecewise affine after finite subdivision, and retain their common Lipschitz bound. Define

$$s_1 = a_1, \quad s_j = \max\{a_j, s_{j-1} + L\} \quad (2 \leq j \leq N). \quad (2.9)$$

These starts remain $C\lambda$ -Lipschitz, since equivalently $s_j = \max_{i \leq j}(a_i + (j-i)L)$. Their bands have disjoint interiors in each vertical column.

No point of an original band is lost. To verify this, group the successive new intervals into maximal contiguous clusters. A cluster starts at an unchanged start $s_i = a_i$, and every advanced interval is attached to the top of its preceding cluster. Since the original starts are sorted, the portion below an advanced start belongs to that preceding cluster; the remaining portion is covered by the advanced interval itself. This proves coverage by induction. The total upward displacement is at most NL , so choosing NL small preserves all vertical margins.

Choose a horizontal cutoff $0 \leq \chi \leq 1$, compactly supported in the horizontal interior of the box and equal to one near the horizontal projection of K^* . Choose a nondecreasing Lipschitz function $\eta(v)$ that is zero below a fixed level above every band and is one near the top of the box. Its transition is contained strictly above all bands. The term involving η compensates above the bands for the vertical length removed below it, so each column keeps its endpoints. For $[t]_{[0,L]} = \max\{0, \min\{t, L\}\}$, define

$$q_0(h, v) = v + \chi(h) \left(- \sum_{j=1}^N [v - s_j(h)]_{[0,L]} + NL\eta(v) \right), \quad S_0(h, v) = (h, q_0(h, v)). \quad (2.10)$$

Choose NL after the cutoffs so small that

$$NL \operatorname{Lip}(\chi) \leq \lambda, \quad NL \operatorname{Lip}(\eta) \leq 1, \quad (2.11)$$

and so that NL is below the prescribed fine motion tolerance and all unused box margins.

The correction in (2.10) vanishes below all bands, above the transition of η , and near the horizontal boundary. Thus S_0 equals the identity near the boundary of the box. For fixed h , the vertical map is nondecreasing, with slope $1 - \chi(h)$ on each band and slope $1 + \chi(h)NL\eta'(v)$ on the upper compensation interval. Elsewhere its slope is one. These statements hold almost everywhere and imply the corresponding monotonicity for the Lipschitz function. In particular the slope lies in $[0, 2]$.

The horizontal bound is independent of the number of bands. At any point of a piecewise affine region at most one truncated summand in (2.10) is unsaturated. Every fully saturated summand is the constant L . The horizontal gradient of the sum therefore has norm at most $C\lambda$. Since the whole correction has size at most NL , the cutoff term is bounded by $NL\text{Lip}(\chi)$. Continuity across the finitely many pieces yields

$$|R(h, v) - R(\tilde{h}, \tilde{v})| \leq C\lambda|h - \tilde{h}| + C|v - \tilde{v}|, \quad R(h, v) = q_0(h, v) - v. \quad (2.12)$$

Thus $\text{Lip}(S_0) \leq C$ and $|S_0 - \text{id}| \leq NL$. Each vertical column is mapped onto itself, so the whole box is mapped onto itself. Use this formula in each of the finitely many disjoint boxes and the identity elsewhere. Extension by the identity to $\mathbb{R}^3$ is globally Lipschitz, as is seen by integrating along line segments across the finitely many boxes.

For $\kappa > 0$, the vertical slope of $q_\kappa = (1 - \kappa)q_0 + \kappa v$ is at least κ and at most two. The horizontal Lipschitz bound is still uniform. Each column map is therefore a bi-Lipschitz increasing homeomorphism fixing its ends; its inverse depends Lipschitz-continuously on the horizontal coordinate, with a bound depending on κ . Patching with the identity proves the asserted bi-Lipschitz property of S_κ . No inverse bound uniform as $\kappa \downarrow 0$ is asserted.

Lipschitz composition and the ACL characterization of Sobolev maps give the first bound of (2.7). For the second, work in the assigned coordinates near a retained source point and use (2.12) along almost every source coordinate line. At almost every such point,

$$|\partial_i(R \circ f)| \leq C\lambda|\text{proj}_{e^\perp} \partial_i f| + C|e \cdot \partial_i f| \leq C\lambda|\partial_i f|,$$

where Lemma 2.3 supplies the last inequality. Summing the three coordinate estimates proves the claim, also on band junctions: it does not require differentiability of S_0 at the target datum. The horizontal component of DF_0 is exactly $\text{proj}_{e^\perp} Df$. The projection is injective on $V(\nu, f(x))$, so this component has rank $\text{rank } Df$ on the retained data. Conversely, Lipschitz composition cannot increase rank here. To see this without differentiating S_0 at the target point, combine Fréchet differentiation of f with approximate differentiation of F_0 : the inequality $|F_0(x+z) - F_0(x)| \leq C|Df(x)z| + o(|z|)$ implies $|DF_0(x)v| \leq C|Df(x)v|$ for every v , by taking approximate limits in narrow cones about v . Thus $\ker Df(x) \subset \ker DF_0(x)$, proving equality of ranks. Finally the exact identity

$$F_\kappa = (1 - \kappa)F_0 + \kappa f$$

and $F_0 \in W^{1,p}$ prove strong convergence as $\kappa \downarrow 0$.

We next identify the exceptional set of values of S_0 exactly. On a band, when $\chi(h) = 1$, its image height is

$$b_j(h) = s_j(h) - (j - 1)L.$$

Let

$$Z_{\text{box}} = \bigcup_{j=1}^N \{(h, b_j(h)) : \chi(h) = 1\}, \quad Z = \bigcup_{\text{boxes}} Z_{\text{box}}. \quad (2.13)$$

These are compact sets inside their boxes. Each b_j is piecewise affine with slope at most $C\lambda$, so Z has zero volume. When adjacent bands touch, their displayed image values coincide; the resulting fibre is the entire contiguous closed interval. These and only these intervals are the non-singleton fibres. Indeed, if $\chi(h) < 1$, the whole column is strictly increasing. If $\chi(h) = 1$, it is strictly increasing outside the closed union of bands.

For a value outside Z there is exactly one preimage. Near that preimage the vertical slope has a positive lower bound: either $1 - \chi$ is bounded below locally, or the point is at positive distance from all closed bands. Together with (2.12), this gives a local Lipschitz inverse. This proves the precise homeomorphic-complement statement (2.8). The selected data lie where $\chi = 1$ and in the union of bands, and hence their images lie in Z .

It remains to arrange genuine separated plate patches, rather than merely a finite union of sheets. Refine each piecewise affine graph into its finitely many affine pieces, including their lower-dimensional strata. Assign every output datum to one containing affine plane; if sheets coincide or meet, make one measurable choice. Use the finite pushforward of the retained source weight under F_0 for this assignment. Inner regularity gives disjoint compact subsets of the assignments with an arbitrarily small further loss. These compacts have disjoint ambient neighborhoods. In each plane, take a sufficiently fine rectangular grid, choose its cut lines to have zero assigned measure, and remove thin neighborhoods of the cuts. The remaining compact pieces lie strictly inside flat rectangles with margins, and the rectangles and ambient neighborhoods can be chosen disjoint using the preceding separation. Pull back the retained compact subsets through F_0 within the already compact source data. This gives the final compact K and all the claimed patch properties. The initial chart separation prevents saturation from mixing different rank or range-line assignments. In particular, no claim that sheet intersections or projected singular measures have zero assigned mass is needed. On the source subset assigned to a given plate, the normal component of F_0 is constant. Locality of weak derivatives therefore puts the range of DF_0 in that plate's tangent plane almost everywhere on the subset. Together with rank preservation and (2.7), this also preserves the prescribed approximate line direction in the rank-one case, with an additional $C\lambda$ angular error. Distribute the losses in this and the preceding selections so that their sum is less than ε .

Lastly, put $U = \Omega \setminus F_0^{-1}(Z)$. On U , F_0 is a Sobolev homeomorphism with positive orientation onto $\Lambda \setminus Z$. Apply Lemma 2.1 on a countable interior exhaustion. Its nonregular source points form a fixed null set $E \subset U$, and the image of its regular rank-deficient points is null. This proves the asserted dichotomy for almost every target point. For $p \geq 3$, the volume property N makes $F_0(E)$ null as well. For $2 < p < 3$, that image is not assumed null and is retained as the stated exceptional alternative. $\square$

2.3 Protected endpoint pairs

The next step selects a pair of target rectangles on opposite sides of each retained plate. Their geometric purpose is to give exactly two contact points for each slope in a central rectangle. In the convex quantizer of Section 3, these paired contacts produce a planar output face containing the eventual quantized images of the retained data. The carrier must first make neighborhoods of the endpoint rectangles exact affine or PL regions; the central data themselves will remain untouched by those implants.

We continue to measure losses by the finite measure introduced before Lemma 2.3:

$$d\mu(x) = \mathbf{1}_{\{1 \leq \text{rank } Df \leq 2\}}(x)(1 + |Df(x)|^p) dx.$$

Only regular points, as in Lemma 2.1, enter this measure. Every compact restriction and every finite measurable partition below may discard a prescribed arbitrarily small amount of this finite

measure. In particular, a finite sequence of such restrictions can be made with a prescribed total loss. We apply Proposition 2.6, and write $F_0 = S_0 f$ and $F_\kappa = S_\kappa f$ for its maps.

Lemma 2.7 (Protected wells). *Let finitely many separated compact pieces of the retained F_0 -image lie on flat plates, with relatively compact disjoint ambient chart neighborhoods. Let a positive continuous motion scale δ on Λ be prescribed. After an arbitrarily small μ -loss, one can choose, on each plate, an origin z , a unit vector n as close as desired to its normal, a number $d > 0$, and nested protected horizontal rectangles in $n^\perp$ with the following properties. There is a nonnegative smooth compactly supported function b_1 on Λ , extended by zero to $\mathbb{R}^3$, such that*

- (i) $\Delta b_1 \leq C$ with an absolute constant, and b_1 is as small as prescribed relative to δ^2 . In particular $b_1 = o(\text{dist}(\cdot, \mathbb{R}^3 \setminus \Lambda)^2)$ at the boundary.
- (ii) For every h in a protected horizontal rectangle, put $q = z + h$. The affine function

$$L_h(x) = q \cdot x - \frac{1}{2}|q|^2 + d^2$$

supports $\psi_1(x) = |x|^2/2 + b_1(x)$ globally, and its contact set consists exactly of $q + dn$ and $q - dn$. In particular,

$$\sup_x \{q \cdot x - \psi_1(x)\} = \frac{1}{2}|q|^2 - d^2. \quad (2.14)$$

For a fixed compact rectangle and any fixed neighborhoods of these two endpoint sheets, the supporting gap is uniformly positive outside the neighborhoods.

- (iii) The assigned central data $X = F_0(x)$ have $|\langle X - z, n \rangle| < d/4$. For almost every assigned x with respect to μ , both points

$$Y_\pm(X) = X + n(\pm d - \langle X - z, n \rangle) \quad (2.15)$$

belong to genuine homeomorphic target neighborhoods of F_0 and have either a regular full-rank inverse point or an inverse point in a fixed source null set. The latter alternative is needed only when $2 < p < 3$.

All assertions persist after shrinking the protected horizontal rectangles. The endpoint assignments can be restricted to compact sets on which their inverse points and their regular or exceptional types have fixed separated compact ranges.

Figure 1 shows the geometry of these pairs. The endpoint rectangles will become exact target regions in the first round of Proposition 2.13; the present lemma selects their positions and inverse types.

Proof. First work on one plate. Choose a horizontal cutoff $0 \leq \theta \leq 1$ which is one on a neighborhood of a compact rectangle strictly larger than the rectangle to be protected. Its support is compact in the horizontal chart. Choose a smooth nonnegative function β , supported in $[-2, 2]$, such that

$$u \mapsto u^2/2 + \beta(u)$$

has minimum 1 exactly at $u = \pm 1$. For example, first prescribe this function to be $1 + (u^2 - 1)^2$ for $|u| \leq 3/2$; in the remaining transition region the unperturbed function $u^2/2$ is already larger than 1, so the nonnegative bump can be smoothly cut off there without creating another minimum. In the coordinates $x = z + h + vn$, set

$$b_z(x) = \theta(h)d^2\beta(v/d). \quad (2.16)$$

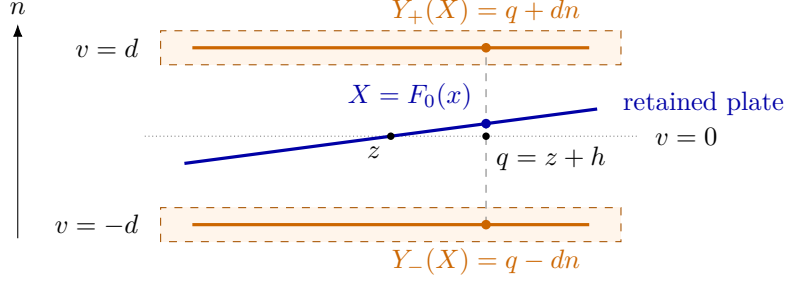


Figure 1: A section through a retained plate and its paired endpoint rectangles; horizontal rectangles appear as intervals. The chosen normal n may be slightly tilted from the plate normal, so the datum $X = F_0(x)$ need not lie in the central plane $v = 0$. Both endpoints have the same horizontal coordinate h . The dashed boxes indicate neighborhoods that the carrier will later make exact. The diagram shows F_0 data, before the positive parameter κ is chosen.

Choose d much smaller than the horizontal cutoff margin and the ambient chart margin. All supports can then be made disjoint. Since

$$\Delta b_z = d^2 \beta(v/d) \Delta_h \theta + \theta(h) \beta''(v/d),$$

choosing $d^2 \|\Delta_h \theta\|_\infty \leq 1$ gives an absolute upper bound for Δb_z . Disjointness gives the same bound for their sum b_1 . Its size is $O(d^2)$ on each chart, so the finitely many d 's may also enforce every prescribed δ^2 bound. The support is compact in Λ , proving the stated boundary decay.

For a protected h_0 , subtract L_{h_0} and write $q = z + h_0$. Within the chart the difference is

$$\psi_1(x) - L_{h_0}(x) = \frac{1}{2} |h - h_0|^2 + \frac{1}{2} v^2 + b_1(x) - d^2.$$

Where $\theta = 1$, the one-dimensional choice of β makes this nonnegative, with equality precisely when $h = h_0$ and $v = \pm d$. Where $\theta \neq 1$, the horizontal margin has been chosen so that $|h - h_0|^2/2 > d^2$. The other bumps are nonnegative and have disjoint supports. The same distance estimate applies outside the chart. This proves the global contact statement and (2.14). On compact h_0 -sets the gap is uniform away from the endpoint neighborhoods: otherwise a sequence of almost contacts would, by quadratic growth and compactness, limit to a third contact.

Initially the central data lie on the original flat plate. Choose d in a small positive interval $[d_-, d_+]$, and then restrict n to a sufficiently small open cap about the original normal that $|\langle X - z, n \rangle| < d_-/4$ for all central data in the compact patch. All preceding supports, cutoffs and margins may be chosen uniformly over this parameter range. Give n and d smooth probability densities in the cap and interval. For fixed X , the map in (2.15) has the form $X + rn$, where $r = \pm d - \langle X - z, n \rangle$. In sphere coordinates its volume Jacobian has absolute value r^2 : differentiation in d supplies $\pm n$, and each angular derivative has tangential part $r dn$. Here $|r| \geq 3d_-/4$. Thus each endpoint law is absolutely continuous for three-dimensional volume.

By Proposition 2.6, almost every target point off the closed sheet-value set has exactly the stated regular or exceptional type. Fubini applied to the finite assignment measure therefore gives a parameter choice for which both endpoints have an allowed type for μ -almost every assigned point. This is an averaging argument in three target dimensions; it uses no absolute continuity of the horizontal assignment measure. Push that measure to the horizontal patch. Split according to the two endpoint types simultaneously and restrict each part to compact subsets of its genuine inverse charts. The inverse endpoint maps are continuous there. Further compact restriction gives

separated compact source ranges for the finitely many classifications, with arbitrarily small loss. The central compact data remain separated from all endpoint inverse data because their F_0 -values lie on the central sheet, whereas the endpoints lie in its homeomorphic complement. A finite rectangular refinement with smaller margins supplies the rectangles in the statement. $\square$

2.4 Implanting exact cores by cubical collapse

The endpoint selection has located two compact sets of target values over each retained plate. Their inverse data are either regular full rank or belong to a source null set, but the maps near these values are not yet required to be affine or PL. We now construct such exact neighborhoods. The regular data will permit an energy estimate with a universal constant. For exceptional inverse data, which occur only when $2 < p < 3$, the construction will confine the uncontrolled cost to marked source cubes. The data are selected using F_0 , while the initial replacements will be made in the homeomorphism F_κ for a sufficiently small $\kappa > 0$.

The local replacement has a simple normalized goal. Given a homeomorphism close to the identity near the boundary of a cube, make it exactly the identity on the inner cube, leave its outer boundary values and its image unchanged, and control the energy in the intervening shell. The output is allowed to collapse isolated tame balls. Those fibres can subsequently be opened while preserving previously exact target regions; the opening itself supplies no energy estimate.

The energy estimate comes from sampling the input near a two-dimensional cubical skeleton. Small source charts of size ℓ are immersed at one common scale s ; the factors in the derivative of the lift cancel, and the weighted overlaps of the samples are summable. We first construct that immersed skeleton, then carry out the replacement and identify its fibres.

Write

$$B = [-1, 1]^3, \quad t(x) = \max_{1 \leq j \leq 3} |x_j| - 1, \quad B_r = \{x : t(x) < r\}.$$

Thus the notation B denotes the closed inner cube, whereas B_r is open. Boundaries have zero volume, so this convention does not affect any integral below. The constants in the next two Lemmas depend on the fixed cubical construction and, when indicated, on p ; they do not depend on the input homeomorphism or on the sufficiently small dyadic number s .

Lemma 2.8 (An immersed cubical skeleton). *There are positive constants b, C_0, C_1 and, for every sufficiently small dyadic s , a locally finite conforming triangulation $\mathcal{T}_s$ of $\{0 < t < 2s\}$ with the following properties. If $D \in \mathcal{T}_s$ has scale ℓ_D , then*

$$\text{diam } D \asymp \ell_D, \quad \ell_D \asymp \min\{s, t(x)\} \quad (x \in D),$$

where the second assertion allows the uniform constants implicit in $\asymp$. The simplices have uniformly controlled shapes and finitely many local configurations up to translation and rescaling. There is a smooth orientation-preserving local diffeomorphism i on an open neighborhood $\mathcal{U}$ of their two-skeleton, including a neighborhood of the outer boundary $\{t = 2s\}$, such that

$$i = \text{id} \text{ near } \{t = 2s\}, \quad |i(x) - x| \leq C_0 s.$$

The neighborhood contains, at every simplex, a neighborhood of its skeleton of width $b\ell_D$. On a slightly smaller such neighborhood,

$$|Di| \leq C_1 \frac{s}{\ell_D}, \quad |(Di)^{-1}| \leq C_1 \frac{\ell_D}{s}, \quad |D^2 i| \leq C_1 \frac{s}{\ell_D^2}. \quad (2.17)$$

There are also uniform local inverse charts: a chart centered at a point of the smaller neighborhood contains a source ball of radius $b_1\ell_D$, and its image contains the corresponding target ball of radius b_2s , for fixed $b_1, b_2 > 0$.

More precisely, after splitting into a bounded number of charts per simplex, the formulas for i have the form

$$i(x_* + \ell\xi) = y_* + sI(\xi), \quad (2.18)$$

where I belongs to a fixed finite list of smooth formulas on fixed compact chart neighborhoods, and y_* belongs to a fixed rational subdivision of the s -grid. For fixed s there are only finitely many possible y_* in the bounded region used by this construction. The map i is permitted to have overlaps between different local charts.

Proof. Set $\lambda_k = 2^{-k}s$, $k \geq 0$. Tile

$$\{\lambda_k < t < 2\lambda_k\}$$

by axis-aligned cubes with side comparable to λ_k ; a fixed dyadic subdivision can be used throughout. This is possible with exact alignment because 1, s , and all λ_k are dyadic. Across consecutive layers, a face of a coarser cube is subdivided to match its finitely many finer neighbors. Triangulate each resulting face by coning its subdivided boundary to its center, and cone these triangles to the center of the cube. Shared faces receive the same triangulation. The neighbor-size ratios are bounded, and all vertex positions in a normalized cube belong to a fixed finite set. This proves the shape and local-configuration assertions. The harmless finite modifications needed near $t = 2s$ can be made on fixed dyadic levels. In particular a thin outer band can be reserved for the identity map.

We construct the immersion successively near vertices, edges, and faces. All neighborhoods in this construction are in the source; there is no requirement that their images be disjoint. Choose at each vertex v a representative incident scale ℓ_v . The ratios between incident scales belong to a finite list. Away from the reserved outer band, choose a nearest point y_v of the s -grid and prescribe

$$i(x) = y_v + \frac{s}{\ell_v}(x - v)$$

on a small vertex neighborhood. These source neighborhoods are disjoint when their relative radii are small enough. On the outer band instead prescribe the identity germ. Its normalized scale ratio is bounded above and below, since $\ell_v \asymp s$ there. All target anchors, including those in the identity band, lie on a fixed rational subdivision of the s -grid. Adjacent anchors differ by at most Cs .

Consider an edge core outside the vertex neighborhoods. Its two end germs are already prescribed. Join these germs by a regular immersed arc in a ball of radius Cs containing them. This can be done relative to the end intervals by the one-dimensional relative immersion Theorem. Choose two transverse vector fields along the arc, equal to the prescribed transverse derivatives on the end intervals. They give a positive three-frame. Relative to its positive scalar end frames, the homotopy class of this frame path can be made trivial: one full rotation of the two transverse vectors changes the class in $\pi_1(GL^+(3)) \cong \mathbb{Z}/2$. Thus choose the rotation, if needed, so that the entire frame path is homotopic to the path of positive scalar frames, with its ends fixed. Thickening the framed arc gives an orientation-preserving local diffeomorphism on a sufficiently thin edge neighborhood. It agrees exactly with the old affine formulas on the end collars. Distinct truncated edge cores have disjoint source neighborhoods.

For a triangular face, remove the already treated vertex and edge neighborhoods. Choose a rounded disk in the remaining face whose boundary collar lies in those neighborhoods. The prescribed local diffeomorphism on that collar provides both a map of the collar and a positive three-frame. Its frame loop is null-homotopic. Indeed, move the loop within the treated neighborhood onto the

triangle perimeter: each edge frame path has the trivial relative class just prescribed, and the vertex frames are positive scalars. Concatenating these paths therefore gives the trivial loop in $GL^+(3)$.

The target ball is contractible, so the collar map extends as a smooth base map of the disk into a slightly larger ball of radius Cs . The null-homotopy of the three-frame extends its restriction to the two fixed tangent directions of the face. We have consequently obtained a formal immersion of the disk that is the differential of the prescribed map near its boundary. The relative Smale–Hirsch Theorem, in source dimension two and target dimension three, gives an actual immersed disk agreeing with the prescribed map on a smaller boundary collar; see [27, Theorem 5.7]. We use the target open ball as the target manifold in that Theorem. Thus the extension stays in a ball of radius Cs . In particular this is not an application of a relative immersion extension theorem in equal dimensions.

To thicken the disk, choose a positive transverse column ν along it, equal to the old normal derivative on the boundary collar. This choice extends because the set of positive transverse columns over an oriented tangent two-frame is a contractible half-space. In flat source coordinates (u, r) use the first-order formula

$$\phi(u) + r\nu(u).$$

It is a local diffeomorphism for sufficiently small $|r|$. On the boundary collar, its difference from the old exact formula is $O(r^2)$, with derivative difference $O(|r|)$. A fixed smooth cutoff in u blends it with that old formula. After reducing the normal width, the differential remains positive and nonsingular, and the formulas agree exactly on the region retained from the old neighborhood. Distinct face cores have disjoint source neighborhoods; their overlaps with the old neighborhood are precisely the regions where this matching has been imposed. This constructs a smooth positive local diffeomorphism on a joint neighborhood of the entire two-skeleton.

We now justify all uniformity assertions, which are not consequences of the qualitative immersion Theorem alone. Normalize an incident source configuration by its scale ℓ and normalize target lengths by s . There are finitely many source configurations and incident scale ratios. Differences between the target anchors are bounded integer vectors on the fixed rational grid, so they also have only finitely many possibilities. The identity prescriptions occur only in the outer layers, where the normalized ratios and positions have finitely many possibilities. First choose one edge extension for each of these finitely many edge data, including its collar formula and framing. Next choose one face extension for each of the finitely many face data determined by those choices. Fix these choices once and for all. This gives a finite list of normalized smooth formulas on compact subneighborhoods.

On each such compact subneighborhood the least singular value is positive and all first and second derivatives are bounded. Take the minimum of the finitely many positive margins and the maximum of the finitely many derivative bounds. Shrinking once more gives a common relative neighborhood width. The inverse function Theorem, with these uniform bounds and a common reduction of the chart radius, gives source inverse-chart radii comparable to ℓ and image radii comparable to s . Scaling back gives (2.17) and (2.18). Finally every formula lies within Cs of its anchor, and every anchor is within Cs of its source configuration. Hence $|i(x) - x| \leq C_0s$. $\square$

Lemma 2.9 (Cubical cutoff). *There are constants $C > 2$ and $c > 0$ with the following property. Let $1 \leq p < \infty$, let s be sufficiently small and dyadic, and let h be a homeomorphism on an open neighborhood of $\overline{B_{Cs}}$, onto its image, with $h \in W_{\text{loc}}^{1,p}$. Suppose that*

$$|h(x) - x| \leq cs \quad \text{whenever } |t(x)| \leq Cs. \quad (2.19)$$

One can replace h inside B_{2s} by a continuous Sobolev map F , keeping a neighborhood of its boundary

unchanged, so that

$$F = \text{id} \quad \text{on } B, \quad F(B_{2s}) = h(B_{2s}), \quad (2.20)$$

$$\int_{0 < t < 2s} |DF|^p \leq C_p \int_{|t| \leq Cs} |Dh|^p. \quad (2.21)$$

The only nonsingleton fibers of F are tame closed balls over a countable set of target points. Their diameters tend to zero at every accumulation, and those target points can accumulate only on ∂B . These fibers can be opened simultaneously to give a homeomorphism, altering F only over arbitrarily small mutually disjoint target neighborhoods of their values. This last assertion is topological; it does not assert a Sobolev energy bound for the opened maps. If h has volume Lusin property N , then so does F .

Proof. The lift and its energy. Apply Lemma 2.8. Increase C so that all auxiliary image neighborhoods and inverse-chart neighborhoods are contained in $\{|t| < Cs\}$. This is possible because their centers have $0 \leq t \leq 2s$ and their radii are at most a fixed multiple of s . Put $\rho(x) = \min\{s, t(x)\}$ in the open shell. This is a Lipschitz scale function and is comparable to ℓ_D on every simplex D .

On a smaller skeleton neighborhood define $H(x)$ by the near- x inverse branch of

$$i(H(x)) = h(i(x)). \quad (2.22)$$

The right side differs from $i(x)$ by at most cs . Uniform inverse charts in Lemma 2.8 therefore make this definition possible when c is sufficiently small and give

$$|H(x) - x| \leq Ac\rho(x). \quad (2.23)$$

The branches agree on overlaps. To see this, reduce c so that any two candidate values near a given source point lie in the same injectivity chart; continuity of the local inverse then gives agreement on overlapping neighborhoods. In particular H is continuous and is locally a composition of homeomorphisms.

It is also globally injective on the smaller skeleton neighborhood. Suppose $H(x) = H(x')$ and write $\delta = Ac$. Then

$$|x - x'| \leq \delta(\rho(x) + \rho(x')).$$

Since ρ is Lipschitz, choosing $\delta < 1/10$ makes $\rho(x)$ and $\rho(x')$ comparable and gives $|x - x'| \leq 3\delta\rho(x)$. With a further fixed reduction of c , the points x, x' and their common lifted image lie in one injectivity chart for i . Equation (2.22) and injectivity of h first give $i(x) = i(x')$, and injectivity of that chart then gives $x = x'$. Thus H is an embedding. On the reserved outer band $i = \text{id}$; the same is true at the nearby lifted points when c is small. Consequently $H = h$ on a neighborhood of $\{t = 2s\}$.

The weak chain rule in a local chart gives

$$DH(x) = Di(H(x))^{-1} Dh(i(x)) Di(x) \quad \text{for almost every } x.$$

The two scale factors in (2.17) cancel, so $|DH(x)| \leq A|Dh(i(x))|$. Each used part of a simplex is covered by a uniformly bounded number of injectivity charts of i . Change variables separately in those charts. Their Jacobians satisfy $|\det Di| \geq a(s/\ell_D)^3$, and their images lie in $B(x_D, As)$ for any fixed $x_D \in D$, after increasing A . We obtain

$$\int_{D \cap \text{dom } H} |DH|^p \leq A_p \left(\frac{\ell_D}{s}\right)^3 \int_{B(x_D, As) \cap \{|t| \leq Cs\}} |Dh(y)|^p dy. \quad (2.24)$$

No global injectivity of i has been used in this change of variables.

For completeness, fix a dyadic depth $\lambda \leq s$ and a target point y . Simplices of scale comparable to λ whose balls $B(x_D, As)$ contain y lie in a cubical layer of thickness $O(\lambda)$ within an $O(s)$ -ball. This set has volume at most $As^2\lambda$, including near edges and corners of the cube. Since the simplex interiors are disjoint and have volume at least $a\lambda^3$, there are at most $A(s/\lambda)^2$ such simplices. Summing (2.24), and then interchanging the nonnegative sum and integral, gives

$$\begin{aligned} \sum_D \int_{D \cap \text{dom } H} |DH|^p &\leq A_p \sum_{k \geq 0} \left(\frac{2^{-k}s}{s} \right)^3 \left(\frac{s}{2^{-k}s} \right)^2 \int_{|t| \leq Cs} |Dh|^p \\ &\leq A_p \int_{|t| \leq Cs} |Dh|^p. \end{aligned} \quad (2.25)$$

This summation is the quantitative reason for using the immersion: the local sampling volume contributes a third power, whereas the number of overlapping samples in one layer contributes only a second power.

Topological hole filling. We next fill the parts of the simplices where the lift was not constructed. These fillings will supply a homeomorphism of the shell; we will then collapse their images to remove their uncontrolled energy. In each normalized simplex choose nested smooth convex balls, obtained by smoothing and slightly shrinking that simplex,

$$V_D \Subset V_D^+ \Subset U_D \Subset D.$$

They can be chosen homothetic about a common interior point after fixing one smooth convex model for each simplex shape. Choose V_D large enough that $D \setminus \overline{V_D}$ and a two-sided collar of ∂V_D lie in the lift neighborhood. The three successive margins are fixed positive multiples of ℓ_D . Taking c small relative to those margins gives

$$H(\partial V_D) \Subset V_D^+.$$

The sphere $H(\partial V_D)$ is locally flat, since H is an embedding of a neighborhood of ∂V_D . By the generalized Schoenflies Theorem, it bounds a tame closed ball K_D ; see [10, Theorem 5]. Its bounded side lies in V_D^+ : the complement of the convex ball V_D^+ is connected to infinity and misses the sphere. Extend the boundary homeomorphism $H|_{\partial V_D}$ to a homeomorphism $\overline{V_D} \rightarrow K_D$.

These extensions fit on the correct sides of the lifted boundaries. Indeed the open shell with all closed hole interiors removed is connected through the skeleton to the outer band. Its lifted image misses every $H(\partial V_D)$ by injectivity of H . For a fixed D , an outer-band point can be chosen outside V_D^+ , so this connected image lies on the exterior side of that sphere. Different balls K_D are disjoint, since they lie inside different simplex interiors. Therefore filling all holes and using H elsewhere gives an injective map on the shell.

Extend it by the identity on B and call the resulting map G . This extension is continuous. On the lift region its displacement is $O(\rho)$ by (2.23); within a filled hole, both the source and image lie in the same simplex of diameter $O(\ell_D) = O(t)$. Thus $G(x) - x \rightarrow 0$ as $x \rightarrow \partial B$ from the shell. Lifted points stay outside B because their displacement is less than $t(x)/2$, and filled points stay outside B because $K_D \Subset D$. Hence there is no new failure of injectivity at the inner boundary.

The map G is consequently a continuous injection on $\overline{B_{2s}}$, with boundary values h . It is a homeomorphism onto its compact image. Invariance of domain and separation by the boundary sphere show that

$$G(B_{2s}) = h(B_{2s}) :$$

both are the bounded component enclosed by the same embedded boundary $h(\partial B_{2s})$. Thus G glues to h outside this cube. At this point G need not have a finite Sobolev energy in the holes.

Collapsing the uncontrolled fillings. We eliminate that uncharged energy by a map in the output coordinates. Choose the balls U_D so that the ones in the outermost layer leave a fixed positive multiple of s to ∂B_{2s} . All other layers are farther from that boundary. By (2.19) and a boundary-degree argument, $B_{2s-cs} \subset h(B_{2s})$. Reducing c therefore ensures $U_D \Subset h(B_{2s})$ for every D . The same choice leaves a whole outer band free of these supports.

On U_D construct a Lipschitz radial squeeze Q_D that is constant on $\overline{V_D^+}$, equals the identity near ∂U_D , and is a homeomorphism from $U_D \setminus \overline{V_D^+}$ onto U_D with the collapsed point removed. Explicitly, in the uniformly controlled polar coordinates of the smooth convex model, replace the radial variable by a continuous function that is zero up to the radius of V_D^+ , increases linearly on the next radial interval, and equals the original radius near ∂U_D . The fixed relative buffer widths give a Lipschitz constant independent of D and s . Extend by the identity outside U_D . The supports are disjoint. Their diameters tend to zero toward ∂B , so the resulting map Q is continuous there and equals the identity on B . The uniform Lipschitz bound holds across the seams as well, by decomposing any line segment into the portions inside the supports and their complement. The map Q maps $h(B_{2s})$ onto itself.

Set $F = Q \circ G$. On a neighborhood of every filled hole this map is constant, because $K_D \Subset V_D^+$. Elsewhere it is locally the Lipschitz composition $Q \circ H$, and

$$|DF| \leq \text{Lip}(Q)|DH| \quad \text{almost everywhere there.}$$

The local Sobolev statements glue across hole boundaries because the map is constant on two-sided collars of those boundaries. They glue across simplex faces because the lift formulas already agree there. Equation (2.25) proves (2.21) on the open shell.

There is also Sobolev gluing at its limiting inner boundary. One direct verification uses absolute continuity on lines. For almost every line parallel to a coordinate axis, the restrictions in each open component of the shell are locally absolutely continuous and their derivatives are integrable, by Fubini and the energy bound. An absolutely continuous function on compact subintervals with integrable derivative and a continuous endpoint limit extends absolutely continuously to that endpoint. The line intersects the cube boundary only finitely many times, except for a null family of lines contained in boundary planes. Since F is continuous and is the identity inside the cube, these restrictions glue to an absolutely continuous restriction on the whole line. This proves the asserted Sobolev regularity and identifies the weak derivatives. The exterior seam is unchanged on a neighborhood, so no additional gluing issue occurs there.

Fibres and relative opening. The collapsed fibers have an exact description:

$$F^{-1}(a_D) = G^{-1}(\overline{V_D^+}),$$

where a_D is the center to which Q_D collapses. They are tame closed balls, since G is a homeomorphism and the output buffers are tame balls. Every other fiber is a singleton. In any compact subset away from ∂B there are finitely many simplices; at the remaining accumulation points their scales tend to zero. Moreover $\text{diam } G^{-1}(\overline{V_D^+}) \leq A\ell_D$. To verify this last assertion outside the filled hole, use (2.23): a lifted point mapping into V_D^+ lies within $O(\ell_D)$ of D and has comparable Whitney scale. Inside the hole the assertion is immediate. This proves the stated null-size and accumulation properties.

Here is the claimed exact opening procedure. Around each a_D take an arbitrarily small target ball W_D with closure in U_D , mutually disjoint from the other chosen balls. Their radii can be reduced further at will. On $Q_D^{-1}(W_D)$ replace the radial collapse by a strictly increasing radial

homeomorphism onto W_D , agreeing with Q_D near its boundary; one can first take a small centered ball in the polar coordinates if necessary. Leave Q_D unchanged outside $Q_D^{-1}(W_D)$. Composing all these changes with G gives a homeomorphism and changes F only on $F^{-1}(\bigcup_D W_D)$. The source diameters at accumulating holes tend to zero, so both the map and its inverse remain continuous at ∂B . Equivalently, one may use a boundary collar and Schoenflies to fill each of these smaller neighborhoods. No regularity of these topological fillings has been used in the energy estimate.

Finally assume that h has property N . On the lift region, locally $H = i^{-1} \circ h \circ i$, with smooth diffeomorphic charts on both sides. Such compositions inherit property N , and so does their Lipschitz postcomposition by Q . The filled-hole neighborhoods map to countably many points. On B the map is the identity, and the common inner boundary has zero volume and maps to itself. These countably many local descriptions prove property N for F . $\square$

2.5 Opening isolated ball fibres

We record the precise relative form of the topological opening that is used both during and after the carrier construction. A null family below means that only finitely many members meeting a fixed compact localization have diameter greater than any prescribed positive number.

Lemma 2.10 (Relative opening). *Let $F : \Omega \rightarrow \Lambda$ be a continuous proper onto map obtained by locally finite image-preserving applications of Lemma 2.9 in genuine homeomorphic neighborhoods. Assume that subsequent applications avoid earlier fibre values and their accumulation sets. Let P be the countable set of non-singleton fibre values. Then its fibres are tame closed balls, form a null family on compact localizations, and their value accumulations have singleton fibres. If $C \subset \Lambda$ is relatively closed and $C \cap P = \emptyset$, there is a homeomorphism $H : \Omega \rightarrow \Lambda$, as finely close in values to F as prescribed, such that*

$$H = F \quad \text{on } F^{-1}(C). \quad (2.26)$$

One can require agreement on a neighborhood of every compact target test already separated from P .

Proof. For a single cutoff the assertion about fibres is part of Lemma 2.9: its nontrivial fibres lie inside separate Whitney tetrahedra, and their diameters tend to zero at the exact inner seam. The corresponding target points accumulate only at that seam, where the fibre is a singleton. A later operation in a genuine homeomorphic neighborhood changes none of those fibres. Local finiteness of the operations therefore preserves this description and the null-family property on every compact localization. In particular every $v \in P$ is isolated from the other values in P and from their accumulation set.

Choose mutually disjoint small target balls B_v about these values. For each fixed v their radii can be made smaller than its distance from C , from the other exceptional values and accumulation seams, and from all fixed compact tests to be preserved. These are positive individual clearances; a common lower bound is neither asserted nor needed. Enumerate the values and impose in addition radii tending to zero, with any prescribed fine value-error bound. Properness gives upper semicontinuity of compact fibres: if V is a neighborhood of $F^{-1}(v)$, a sufficiently small B_v satisfies $F^{-1}(\overline{B}_v) \subset V$. Otherwise points outside V mapping to a sequence converging to v would have a subsequence in a compact preimage, contradicting the definition of the fibre.

The inverse of ∂B_v is an embedded locally flat sphere: F is a homeomorphism on a neighborhood of this sphere. It bounds a tame ball containing the fibre, either directly by the radial squeeze description in Lemma 2.9, or by the locally flat Schoenflies Theorem [10, Theorem 5]. The preceding upper semicontinuity places this ball inside a tame neighborhood compactly contained in Ω . Replace F on the ball by a ball homeomorphism extending its boundary parametrization. To keep an outer

collar fixed, first use two nested target balls and do this only on the inner one; use the old map on the annular collar.

The replacements have disjoint target images, so the resulting map is one-to-one and onto. At an accumulation seam its difference from F tends to zero, since a change inside B_v has size at most $\text{diam } B_v$. It is thus continuous everywhere. It is a homeomorphism by invariance of domain, and its image is precisely Λ . Alternatively, properness and the null-family property also give continuity of the inverse at the singleton seams. The balls avoid C , which proves (2.26). Enlarging a compact test to a small closed neighborhood before choosing the balls gives the last assertion. $\square$

2.6 Exceptional boxes and regular affine cores

We apply the cutoff in two ways, according to the endpoint inverse classification. For exceptional data the assignment measure may be singular, so the first lemma selects source cubes with little lost assignment mass and small volume and facet energy. The second lemma uses regular affine jets to obtain an energy bound with a universal constant; the accuracy needed to apply it may depend on the fixed jet bounds.

Lemma 2.11 (Exceptional inverse data). *Let E be a compact source null set contained in finitely many genuine homeomorphic neighborhoods of a continuous $W_{\text{loc}}^{1,p}$ map F , where $p > 2$. Let σ be a finite measure supported on E , and let $g \in L_{\text{loc}}^1$ be nonnegative. For any $a, b > 0$, any prescribed source and target size bounds, and any closed sets separated from E , one can choose finitely many cubes U with disjoint closures, avoiding those closed sets and compact in the given neighborhoods, such that, after a σ -loss less than a , the retained data lie with positive prescribed relative margins in their interiors and*

$$\int_{\bigcup U^+} g < b, \quad \sum_U \ell_U \int_{\partial U}^{\text{ext}} g d\mathcal{H}^2 < b. \quad (2.27)$$

Here U^+ are sufficiently small enlarged neighborhoods, and the superscript ext means that each facet is allowed a fixed small tangential extension. The facet positions can be chosen to be L^1 Lebesgue points in their normal variable, and their F -images have zero three-dimensional volume. The relative insets and the further margins may be prescribed arbitrarily small before the final grid phase is selected.

Proof. Take an open neighborhood V of E , compact in the stated charts and disjoint from the forbidden sets, on which $\int_V g$ is arbitrarily small. Such a neighborhood exists by absolute continuity of the integral. Choose a cubical grid with small pitch r ; all cubes and the small facet extensions that meet E then lie in V . Replace each grid cube by a concentric inset of side $(1 - \tau)r$, with $\tau > 0$ small. Their closures are disjoint. Average the phase over $[0, r)^3$. For any fixed point, the fraction of phases for which it lies in a gap or an additional relative margin is $O(\tau)$, with the additional margin included in this bound. Integrating against σ controls the expected lost assignment mass, without any assumption on σ .

For the facet costs, Fubini in the normal coordinate gives

$$\mathbb{E} \sum_U \ell_U \int_{\partial U}^{\text{ext}} g d\mathcal{H}^2 \leq C \int_V g.$$

The same estimate holds for the finitely many tangentially extended facets, with a fixed overlap constant. Choose τ so that the expected lost mass is much smaller than a , and choose V so that the expected facet cost is much smaller than b . Markov's inequality then leaves a set of phases of positive measure on which both required bounds hold. Almost every phase additionally has all facet

coordinates as L^1 Lebesgue points of the normal slices of g . Lemma 2.1 gives zero image volume for these generic facets, including their finite tangential extensions. Intersecting with these full-measure phase conditions costs nothing. Only finitely many cubes meet the compact set E . A final compact restriction of the retained assignment set gives strict positive margins. Decreasing r beforehand enforces the source sizes and, by continuity of F on the compact localization, the target sizes. $\square$

Lemma 2.12 (Regular affine implantation). *Let F be a positively oriented $W_{\text{loc}}^{1,p}$ homeomorphism on a neighborhood of a compact set of regular full-rank source points. Suppose its jets and inverse jets are bounded there and its Fréchet and power-mean differentiation tests hold uniformly below a fixed radius. In sufficiently small target grid boxes one can implant exact positive affine cores, with arbitrarily small relative grid margins and with disjoint enlarged source comparison boxes. For a box of target side R using the affine jet b at one of its assigned inverse points, the new energy in its source support is at most*

$$C_p \frac{|Db|^p R^3}{\det Db} \leq C_p \int_{V_b} |DF|^p, \quad (2.28)$$

where V_b is the source comparison box obtained as the affine preimage of a buffered target box. The constant depends only on p and the fixed cubical cutoff construction, not on the condition number of Db . The choices may be made relative to previously fixed disjoint compact sets and exact cores.

Proof. Choose a representative x_b whose image belongs to the pertinent grid box, and put $b(x) = F(x_b) + DF(x_b)(x - x_b)$. The source preimage under b of an enlarged target box lies in a ball of radius $C_b R$ about x_b . The Fréchet test, followed by the change of variables $y = b(x)$, makes

$$\sup |Fb^{-1}(y) - y|/R \quad \text{and} \quad R^{-3} \int |D(Fb^{-1}) - I|^p$$

as small as required. The accuracy is chosen after the finite jet and inverse-jet bounds; this is where those bounds affect the construction. One may also require

$$\int_{V_b} |DF - Db|^p \leq 2^{-p-1} |Db|^p |V_b|.$$

Consequently the normalized derivative integral is $O_p(R^3)$, whereas $\int_{V_b} |DF|^p$ is bounded below by a fixed multiple of $|Db|^p |V_b|$. The volume $|V_b|$ is a fixed shape multiple of $R^3 / \det Db$.

Apply Lemma 2.9 to Fb^{-1} in the target-normalized coordinates and then precompose its result with b . It is the identity on the inner normalized core, hence the resulting source map equals b on the affine-source core. Multiplication by Db and the source Jacobian give the first bound in (2.28); the preceding lower bound gives the second. An inverse condition number is not used in this final comparison.

For disjointness, inset not only the core but every used enlarged normalized box from its outer grid box. Make the value-test error smaller than the resulting target margin. Then F maps the affine-source comparison box inside the associated outer grid box. Since F is one-to-one on these neighborhoods, source comparison boxes corresponding to disjoint target boxes are disjoint. Compact separation handles different inverse charts and all previously reserved data. Uniform tests are obtained whenever needed by compact restriction in the finite assignment measure before the mesh is chosen. This argument does not require the grid centers themselves to be regular image points. $\square$

2.7 An exact carrier with prescribed residual volume

We can now build the carrier. The first round uses the endpoint assignments to produce exact neighborhoods of both rectangles in each pair. Those neighborhoods and their buffers must be fixed before the quantizer can specify how small the residual target volume should be. After these volume thresholds are given, the second round adds exact cores throughout the remaining target, keeping the first-round regions and the retained source data fixed.

Proposition 2.13 (Two-round exact carrier). *Prescribe a positive continuous source value tolerance $a : \Omega \rightarrow (0, \infty)$ before applying Proposition 2.6. Use there a target tolerance a_Λ satisfying*

$$a_\Lambda(y) \leq \frac{1}{4}a(f^{-1}(y)),$$

and then fix the resulting rank data and endpoint protections of Lemma 2.7. Fix also an enlargement factor $A > 1$ for the marked collars, before assigning their facet budgets, and let $\varepsilon > 0$. After a total μ -loss smaller than ε , a first round of choices gives $\kappa > 0$, a compact retained source set K , a continuous proper onto carrier $F_b^1 : \Omega \rightarrow \Lambda$, an exact open target set O_1 , and compact endpoint rectangles strictly inside O_1 . These choices precede any prescribed residual-volume thresholds. Fix a locally finite target Whitney tiling $\mathcal{Q}$, and choose arbitrary numbers $a_Q > 0$ for $Q \in \mathcal{Q}$, after O_1 and the endpoint buffers have been determined. A second round gives $F_b : \Omega \rightarrow \Lambda$ and $O \supset O_1$ such that:

- (i) *F_b is continuous, proper and onto, belongs to $W_{\text{loc}}^{1,p}$, and has only the countable tame ball fibres of Lemma 2.10. It is a genuine homeomorphism on target neighborhoods away from their values and accumulation seams. For $p \geq 3$ it satisfies volume Lusin N .*
- (ii) *O is a locally finite union of exact affine or PL core images. Every compact subset of such a core has a neighborhood on which F_b and its inverse are exactly those of its affine or PL chart. In particular no point of O has a nontrivial fibre, and*

$$\mathcal{L}^3(Q \setminus O) < a_Q \quad (Q \in \mathcal{Q}). \quad (2.29)$$

The compact endpoint rectangles retain their fixed buffers in O_1 .

- (iii) *$F_b = F_\kappa$ on a neighborhood of K . It meets the prescribed fine value tolerance relative to f . The choices of κ and of the retained central margins ensure that $F_\kappa(K)$ stays in the protected central plate neighborhoods.*
- (iv) *When $2 < p < 3$, there is a locally finite family of marked source cubes U with disjoint closures, away from K , such that*

$$\int_{\bigcup U} (1 + |Df|^p) < \varepsilon. \quad (2.30)$$

For $p \geq 3$ this family is empty. Off the marked cubes the energy has the universal buffered regional estimate described below.

- (v) *Write $C_U(w)$ for a cube-radius collar of physical width w about ∂U . For the prescribed enlargement factor A , the widths $w_U > 0$ can be chosen after both rounds, with disjoint enlarged marked cubes, so that*

$$\sum_U \frac{\ell_U}{w_U} \int_{C_U(Aw_U)} |DF_b|^p < \varepsilon. \quad (2.31)$$

The same bound, multiplied by the p th power of the Lipschitz constant, holds after a uniformly Lipschitz output composition. The enlarged cubes may still have original $(1 + |Df|^p)$ -mass less than 2ε .

The regional estimate in (iv) has the following meaning. Suppose a locally finite family of regional tests A_j and two successive small open buffered enlargements $A_j^{[1]}$ and $A_j^{[2]}$ are prescribed before the implants are made. All implants may be chosen so fine that

$$\int_{A_j \setminus \bigcup U} |DF_b|^p \leq C_p \int_{A_j^{[2]}} |Df|^p \quad \text{for every } j. \quad (2.32)$$

Here each enlargement has positive local buffer, the supports and comparison boxes meeting A_j fit inside its first enlargement at the corresponding round, and C_p is independent of all selected jet condition numbers, residual thresholds, and PL reference-map constants. In particular the tests may be reserved full-rank cut shells or ordinary regions outside prescribed interiors. The same construction works for summably weighted such tests toward the ends of the open domains.

Proof. We give the choices in order. The initial flattening tolerance gives

$$|F_\kappa(x) - f(x)| = (1 - \kappa)|S_0(f(x)) - f(x)| < \frac{1}{4}a(x) \quad (0 \leq \kappa \leq 1).$$

All subsequent replacements keep their old boundary values and their old local images. Choose their compact supports so small that the oscillation of the map being replaced gives

$$|F_b^1 - F_\kappa| < a/4, \quad |F_b - F_b^1| < a/4.$$

The positive minorants in Lemma 1.2 permit these choices simultaneously on each locally finite family. Thus the final carrier satisfies $|F_b - f| < a$. These choices control the additional implantation motion; the first displayed bound has already reserved the flattening motion before the rank and endpoint data were fixed.

First round: fixing the exceptional boxes. Classify both endpoint signs simultaneously as in Lemma 2.7. In the range $2 < p < 3$, the exceptional inverse assignments have compact ranges in a fixed source null set. Apply Lemma 2.11, with $g = 1 + |Df|^p$, to enclose them in finitely many marked cubes U , discarding only a prescribed small assignment weight. Make all these cubes and their enlarged neighborhoods disjoint from K and from the retained regular inverse assignments. Give their source integrals and extended-facet budgets arbitrarily small shares of ε . Retain the exceptional inverse points with strict inner margins. In the grid phase selection impose the generic-facet condition for both F_0 and the original map f . Lemma 2.1 supplies full-measure sets of phases for both conditions, so their intersection retains the same budget choices. The inset fractions can be as small as desired and are fixed at this stage.

Inside each U , fix a slightly smaller cubical core U^- containing its retained inverse data with room, and leave enough space for the cutoff support to remain compact in U . The associated compact endpoint set L_U lies in $F_0(\text{int } U^-)$. Choose a compact target neighborhood W_U of L_U inside this open image and inside the genuine homeomorphic target neighborhood of F_0 . These margins are chosen using F_0 , before κ or a PL reference map is fixed.

First round: fixing the regular grids and all margins. First compact-restrict the regular endpoint assignments so that their jets, inverse jets and differentiation radii for F_0 are uniform. Choose small relative grid margins. On each plate we will use the same shifted horizontal square grid for both endpoint signs. Choose its pitch R sufficiently small that every pertinent square has a fixed classification for each sign, using compact separation of the classification sets. For an exceptional sign, require that the whole inset endpoint rectangle lies in the interior of its previously chosen W_U . For regular signs, the uniform differentiation tests in Lemma 2.12 permit the required normalized value and gradient accuracy on every buffered box. These upper bounds on R are independent of the grid phase.

Now select that phase. For every fixed datum, the fraction of phases lost to the grid margins is bounded by the relative margin fraction. Fubini therefore gives a phase losing arbitrarily little of the finite horizontal assignment measure. Trim compactly inside its surviving squares, and protect smaller rectangles once more for the actual central data.

For each square with a regular sign, choose a representative endpoint inverse and its affine jet b . Form the target three-box of horizontal and normal side R , centered at the grid center at height $\pm d$. The uniform F_0 tests make $F_0 b^{-1}$ close to the identity on every buffered box, including in normalized gradient power mean. Fix these finitely many tests and their affine-source comparison boxes. The source boxes are disjoint because their F_0 -images stay strictly inside disjoint outer grid boxes. Prior compact separation also makes them disjoint from the marked cubes and from K . All geometric tests and central margins are now fixed.

First round: choosing κ and making the implants. Take $\kappa > 0$ small enough that all the regular value and gradient tests also hold for $F_\kappa b^{-1}$. This follows from $F_\kappa \rightarrow F_0$ in values and in $W^{1,p}$ on each fixed compact. Require also that $F_\kappa(K)$ remains within the nested central margins and that $F_\kappa^{-1}(W_U) \Subset \text{int } U^-$ for each exceptional core.

The latter inverse requirement follows from uniform inverse convergence on compact target sets in the homeomorphic complement of F_0 . Indeed, if inverse points for F_κ failed to converge uniformly to those for F_0 , properness and the fine motion bound would give a convergent subsequence limiting to a different preimage under F_0 . Uniqueness in the genuine inverse neighborhood excludes this. The same compact inverse argument applies to sufficiently fine homeomorphic approximations of the now fixed F_κ .

Apply Lemma A.2 to choose a PL homeomorphism h_{ref} finely approximating F_κ . Choose it so fine that $h_{\text{ref}}^{-1}F_\kappa$ satisfies the normalized identity tolerance of Lemma 2.9 on each exceptional cube, and so that $h_{\text{ref}}^{-1}(W_U) \Subset \text{int } U^-$. Inverse continuity on the compact localizations supplies the first test, and the preceding inverse-margin argument supplies the second. Apply the cutoff with core U^- and support compact in U , and postcompose by h_{ref} . The result equals h_{ref} on U^- , so its exact core image contains the whole protected endpoint rectangle with room. Since h_{ref} and its inverse are locally bi-Lipschitz, this replacement has finite local $W^{1,p}$ energy; its possibly large constant is used only inside marked cubes.

For each regular sign, use the cutoff construction of Lemma 2.12 with the fixed jet and the F_κ tests. The resulting map is exactly b on its core, whose image contains the inset endpoint rectangle. These supports are disjoint from the exceptional changes, and all supports avoid a neighborhood of K . The regular cost is

$$C_p |Db|^p R^3 / \det Db \leq C_p \int_{V_b} |DF_0|^p \leq C_p \int_{V_b} |Df|^p,$$

where the final absolute factor is supplied by Proposition 2.6. The comparison uses the fixed F_0 jet tests; the F_κ tests were required to be equally accurate before applying the cutoff. This normalization is why no condition number enters the final energy charge.

Call the first-round map F_b^1 , and let O_1 be the union of slightly smaller exact core images. Each is a polyhedral affine or PL image and has an exact neighborhood with buffer. The compact endpoint rectangles lie strictly inside this open set. The map is proper, onto and continuous because the finite changes preserve local images and agree with the old map near their outer boundaries. Lemma 2.9 supplies its local Sobolev regularity and its ball-fibre description. It also preserves volume N when $p \geq 3$. The fibre values and their accumulation sets have zero target volume: the former are countable, and the latter lie on finitely many exact affine or PL core boundaries.

Second round: choosing the sets to fill. The first-round choices, including O_1 and its endpoint buffers, are now fixed. Prescribe all the a_Q . In the interior of each Whitney cube Q , remove O_1

from the work region and avoid its boundary, the first-round fibre values and their accumulations, the images of marked-cube boundaries, and $F_b^1(K) = F_\kappa(K)$. Except for O_1 , all these exclusions have zero volume. The last one is null because the retained rank-deficient regular image under f is null and S_κ is Lipschitz. The first marked-cube boundaries were selected to have null images under f ; near them $F_b^1 = F_\kappa = S_\kappa f$, and the Lipschitz map S_κ preserves that nullity.

On the remaining genuine homeomorphic portions, Lemma 2.1 gives the following exhaustive alternative at almost every target point: its preimage is regular full rank, or belongs to a fixed source null set. Regular deficient images have zero volume. When $p \geq 3$, volume N eliminates the second alternative. Compactly restrict these measurable target sets, separating their classifications, their inverse charts, and the inside and outside of the first marked cubes. The target volume lost in Q can be made smaller than any fixed fraction of a_Q . All selected sets are compact in the remaining work region. In particular their source preimages are compact and all necessary separations are positive.

Second round: implants. For exceptional assignments outside the old marked cubes use Lemma 2.11 with $g = 1 + |Df|^p + |DF_b^1|^p$. This is locally integrable, and the exceptional source set is null. Choose new marked cubes with summably small source and facet budgets, disjoint from all earlier marked cubes, from the regular assignments, and from K . Their supports have images compact inside the current Whitney interior. For exceptional assignments already inside an old marked cube, use smaller reference-map implants there; no new marked cube is required. A suitable PL reference is available on the genuine homeomorphic neighborhoods in use. One way to obtain it globally is to apply Lemma 2.10 to F_b^1 with a closed neighborhood of all chosen compact tests fixed, and then use Lemma A.2. The resulting reference approximates F_b^1 arbitrarily well on these tests. The same inverse-margin argument as in the first round makes the chosen endpoint or volume assignments fall within its exact core images.

For regular assignments use ordinary three-dimensional target grids and Lemma 2.12, now with the jets of F_b^1 . Compact restriction gives the uniform tests required for a sufficiently fine grid. Predetermine arbitrarily small insets, then make the value and gradient tests correspondingly fine. Each comparison box meeting an old marked cube is chosen wholly inside it; exterior comparison boxes miss every marked cube. The previously compactly separated exceptional assignments and new marked cubes are avoided. The regular source comparison boxes in this round are disjoint, and their costs are bounded by $C_p \int |DF_b^1|^p$ on those boxes.

For exceptional assignments, grid phase averaging against ordinary target volume pulled back to the null source set controls the loss to source insets. For regular assignments, the target grid margins have arbitrarily small volume. Therefore in each Q the retained assignments can be covered by exact cores with total additional loss less than the unused share of a_Q . Only finitely many compact assignments and implants are needed in each Whitney cube. Whitney local finiteness and properness then make all second-round supports locally finite in target and source. These arguments prove (2.29). They also show that second-round supports miss a neighborhood of every old marked-cube boundary and of the compact retained set K .

Let F_b be the resulting map and O the union of the old and new buffered exact core images. The first-round endpoint cores have been kept fixed. Image preservation and local finiteness give properness and surjectivity. The local Sobolev, Lusin and fibre properties follow as in the first round. The additional accumulation seams are only the new exact core boundaries, a locally finite family of zero-volume polyhedra. Thus Lemma 2.10 applies to the combined carrier.

Universal regional charging. Outside the marked cubes no reference-map constant enters the construction. At the first round disjoint regular comparison boxes give the energy bound by $C_p \int |Df|^p$; elsewhere $|DF_\kappa| \leq C|Df|$. At the second round disjoint regular comparison boxes give the bound by $C_p \int |DF_b^1|^p$. If a support or comparison box meets A_j , choose it small enough to

lie in $A_j^{[1]}$ at that round. At the preceding round use $A_j^{[2]}$ for this first enlargement. Combining the two disjoint sum estimates proves (2.32). There are only two rounds, so the number of buffer enlargements and the multiplication of universal constants are fixed. Local finiteness of the tests allows all required support sizes to be chosen simultaneously from positive local minorants. For countably many prescribed weighted tests, take the usual compact exhaustion and summable budgets. This uses no positive lower bound for a buffer width at the ends of the domains.

Marked-cube collars. The enlargement factor A was fixed before the marked cubes were selected. Choose all initial and added marked-cube budgets so small that (2.30) holds and their extended-facet budgets, including the factor C_A in the slice estimate below, sum to an arbitrarily small number. Near the boundary of a first marked cube the carrier still equals F_κ , whose derivative is bounded by $C|Df|$. Near a newly marked cube it equals the unchanged F_b^1 . The modifications have positive distance from each fixed boundary: later supports are locally finite over its compact image and avoid it. Thus a possibly very thin unchanged neighborhood of each ∂U remains.

Choose w_U within that neighborhood. The collar is contained in six normal slabs of thickness $O(Aw_U)$ over the extended facets. The L^1 Lebesgue property of the selected normal slices yields

$$\limsup_{w \downarrow 0} \frac{\ell_U}{w} \int_{C_U(Aw)} |DF_b|^p \leq C_A \ell_U \int_{\partial U}^{\text{ext}} g_U d\mathcal{H}^2,$$

where $g_U = C|Df|^p$ for old cubes and $g_U = |DF_b^1|^p$ for added cubes. Give each cube a further summable error share and decrease its width to realize this estimate. The earlier facet budgets imply (2.31). Widths can simultaneously make enlarged cubes disjoint and their additional original energy and volume arbitrarily small. Finally, the Sobolev chain inequality $|D(L \circ F_b)| \leq \text{Lip}(L)|DF_b|$ proves the assertion about Lipschitz output compositions. This completes all parts of the Proposition. $\square$

2.8 Full-rank regions reserved for the final correction

The carrier construction is now established. The next three results prepare a separate step in the eventual completion: reserving finitely many full-rank cubes, then making a preliminary locally bi-Lipschitz approximant affine on those cubes. In Section 7, the cubes and their energy tests will be reserved before applying the carrier construction; the actual correction will be performed only at the end. Here the replacement must remain a homeomorphism. We therefore return to the normalized cube $B = [-1, 1]^3$ and the shell coordinate $t(x) = \max_j |x_j| - 1$ of Lemma 2.9, and first prove its smooth-input version.

Lemma 2.14 (Smooth cubical cutoff). *Under the hypotheses of Lemma 2.9, suppose in addition that h and its inverse are smooth on the neighborhoods in use. For every $\varepsilon_0 > 0$ there is instead a locally bi-Lipschitz homeomorphism F onto the same image, equal to the identity on B and equal to h near and outside ∂B_{2s} , such that*

$$\int_{0 < t < 2s} |DF|^p \leq C_p \int_{|t| \leq C_s} |Dh|^p + \varepsilon_0. \quad (2.33)$$

The local bi-Lipschitz constants may depend on h, s , and ε_0 . In particular no uniform inverse Lipschitz bound is asserted as $\varepsilon_0 \downarrow 0$.

Proof. Use the construction in Lemma 2.9 up to the hole-filling stage. The lift is now smooth on its neighborhood, and the lifted hole boundaries are smooth embedded spheres. The smooth three-dimensional Schoenflies Theorem identifies their bounded sides with smooth balls; see [23, Section 1.1, Theorem 1.1]. A prescribed smooth orientation-preserving sphere parametrization

extends over a ball: after a ball identification, isotope the resulting sphere diffeomorphism to a rotation, and insert that isotopy in an annulus before using the rotation in the central ball. Smale's theorem supplies a jointly smooth isotopy for the fixed sphere diffeomorphism [40, Theorem 6]; for the stronger smooth dependence on the sphere diffeomorphism, see also Li–Watts [30, Theorem 1.5]. Using collar coordinates first makes the extension agree with the already prescribed lift on a smaller two-sided boundary collar. Hence each hole has a smooth, and in particular bi-Lipschitz, filling. It remains essential to prove that their constants can be chosen uniformly over the infinitely many Whitney layers.

Fix h and s . By (2.18), the normalized lift in a local chart is

$$\xi \mapsto I^{-1} \left(\frac{h(y_* + sI(\xi)) - y_*}{s} \right), \quad (2.34)$$

where the inverse is the specified nearby branch. Overlaps with neighboring charts are part of the same finite pattern data. There are finitely many choices of I and finitely many possible target anchors y_* , since s is fixed and the sampling region is bounded. The normalized hole and buffer shapes also have finitely many choices. Thus there are only finitely many normalized smooth collar maps on hole boundaries, not merely a bounded family of arbitrary boundary maps. Choose a smooth filling once for each of these finitely many data. On their compact closures take the maximum of the finitely many forward and inverse Lipschitz constants. Call it $L(h, s)$. The physical source and output coordinates of a hole are both rescaled by its scale ℓ_D , so this constant does not change with the depth of the Whitney layer.

The resulting pre-squeeze homeomorphism G therefore has a uniform finite bound for $|DG|$ in all filled holes, and the analogous inverse bounds there. Its lift pieces have such bounds as well: there are only the finitely many normalized maps (2.34). The agreement on collars provides the same local bounds across all finite interfaces.

Replace the nonstrict radial squeezes by strictly increasing ones. For a common $0 < \alpha < 1$, let $Q_{D,\alpha}$ be the homothety of ratio α about a_D on V_D^+ , interpolate monotonically to the identity in the surrounding fixed buffer, and keep the identity near ∂U_D . Their forward Lipschitz constants are uniformly bounded independently of α , and their inverse constants are finite for every fixed $\alpha > 0$. On each filled hole,

$$D(Q_{D,\alpha} \circ G) = \alpha DG.$$

Consequently

$$\sum_D \int_{V_D} |D(Q_{D,\alpha} \circ G)|^p \leq \alpha^p L(h, s)^p \sum_D |V_D| \leq \alpha^p L(h, s)^p |B_{2s} \setminus B|.$$

Choose α so that this is below ε_0 . On the remaining regions the proof of (2.25) applies without change, with the same uniform upper bound for the Lipschitz constant of the squeezes. This proves (2.33).

For clarity, local bi-Lipschitz regularity includes the inner seam. For these fixed data the maps and their inverses have uniform finite local Lipschitz bounds off ∂B , by the finite normalized descriptions and the positive squeeze parameter. They extend continuously as the identity on ∂B . On a small line segment crossing a face, edge, or vertex of that cube, decompose the segment into its open subintervals off the cube boundary. Integrate the common Lipschitz bound on those intervals and use continuity at their endpoints; on portions inside the cube the map is the identity. This gives the same finite Lipschitz bound across the seam. Apply the identical argument in the image to the inverse, which is also the identity on B . Thus the final map is locally bi-Lipschitz throughout the open replacement domain and glues to h as claimed. $\square$

Remark 2.15. The cutoff Lemmas apply in affine output coordinates. More explicitly, let

$$b(x) = y_0 + A(x - x_0), \quad A \in GL^+(3), \quad r > 0,$$

and normalize a map k by

$$\widehat{k}(\xi) = \frac{1}{r} A^{-1}(k(x_0 + r\xi) - y_0).$$

If $\widehat{k}$ satisfies the hypotheses of either cutoff Lemma, transforming the resulting map back makes it equal to b on $x_0 + rB$ and preserves its old image and its outer boundary collar. Writing

$$S = \{x_0 + r\xi : 0 < t(\xi) < 2s\}, \quad A^{\text{band}} = \{x_0 + r\xi : |t(\xi)| \leq Cs\},$$

the smooth estimate becomes

$$\int_S |Dk_{\text{new}}|^p \leq C_p |A|^p |A^{-1}|^p \int_{A^{\text{band}}} |Dk|^p + |A|^p r^3 \varepsilon_0. \quad (2.35)$$

This follows from $D_\xi \widehat{k} = A^{-1} D_x k$ and the source Jacobian r^3 . The last additive error can therefore be prescribed arbitrarily in the original coordinates as well.

Proposition 2.16 (Postponed full-rank correction). *Let $p > 2$ and let $f : \Omega \rightarrow \Lambda$ be an orientation-preserving Sobolev homeomorphism between bounded domains. Let R be the set of its regular differentiation points at which Df has rank three. Let $P \Subset \Omega$ be any previously protected compact set disjoint from R . Given $\varepsilon > 0$, one can choose finitely many pairwise disjoint buffered source cubes, disjoint from P , with centers $x_j \in R$, radii r_j , and positive affine jets*

$$A_j = Df(x_j), \quad b_j(x) = f(x_j) + A_j(x - x_j),$$

such that the inner cubes $K_j = x_j + r_j B$ satisfy

$$\int_{R \setminus \bigcup_j K_j} |Df|^p < \varepsilon, \quad \sum_j \int_{K_j} |Df - A_j|^p < \varepsilon. \quad (2.36)$$

Their buffered cutoff bands

$$E_j = \{x_j + r_j \xi : |t(\xi)| \leq Cs\}$$

can have arbitrarily small total f -energy, even after multiplication by all the finitely bounded affine factors $C_p |A_j|^p |A_j^{-1}|^p$. On those bands the normalized f is within $cs/4$ of the identity.

Write $Q_j = x_j + r_j B_{(C+1)s}$ for the buffered outer cubes. After these choices there are numbers $v_j > 0$ and $\beta > 0$ such that every smooth diffeomorphism $k : \Omega \rightarrow \Lambda$ satisfying

$$\sup_{Q_j} |k - f| < v_j \quad \text{for each } j, \quad \sum_j \int_{E_j} |Dk|^p < \beta$$

can be changed only inside the associated outer cubes $x_j + r_j B_{2s}$ to a locally bi-Lipschitz homeomorphism $\tilde{k} : \Omega \rightarrow \Lambda$ with

$$\tilde{k} = b_j \quad \text{on } K_j.$$

The sum of the derivative-error integrals on the inner cubes and their modified shells is less than ε , after starting the selection with smaller error prescriptions if necessary. The cubes and tests may also be chosen to retain any prescribed positive continuous value tolerance for the final modification.

A locally bi-Lipschitz preliminary map in $W^{1,p}(\Omega; \mathbb{R}^3)$ can be used in place of k by first applying Theorem A.1, preserving the finite value and band-energy tests with room to spare.

Proof. We spell out the order of choices. Since $|Df|^p$ is integrable, choose a compact set $K \subset R \setminus P$, with arbitrarily small discarded full-rank energy, on which

$$|Df(x)| + |Df(x)^{-1}| \leq M$$

for some finite M . This is possible by inner regularity and truncation of the finite values of the two matrices. All points of K may be taken to be both Fréchet differentiation points and L^p differentiation points of Df . The positive determinant of their derivatives follows from the orientation and the local-degree test at an invertible Fréchet differential. Fix M before making any shell-energy choices.

Take the common dyadic $s > 0$ small. At every $x \in K$ there are arbitrarily small radii r such that the buffered outer cube

$$Q = x + rB_{(C+1)s}$$

has closure in $\Omega \setminus P$, and, with $A = Df(x)$ and $b(z) = f(x) + A(z - x)$,

$$\sup_{z \in Q} |A^{-1}(f(z) - b(z))| < \frac{cs}{4} r, \quad (2.37)$$

$$\int_Q |Df - A|^p < \eta |Q|. \quad (2.38)$$

Here $\eta > 0$ can be prescribed after M and s . The first inequality follows from Fréchet differentiability, using $|A^{-1}| \leq M$; the second follows from L^p differentiation. Both persist at all sufficiently small radii, so these outer cubes form a fine Vitali family centered on K . Their radii can additionally be bounded by any prescribed local upper bounds needed for value accuracy and clearance.

Use Vitali selection and then a finite truncation to obtain disjoint outer cubes Q_j missing only an arbitrarily small amount of the weighted measure $\mathbf{1}_K |Df|^p dx$. The closures may be made disjoint as well: after the finite selection, shrink the cubes by arbitrarily small amounts, retaining the strict differentiation tests and allowing an arbitrarily small additional weighted loss. Start those tests with strict margins before this shrinking. The fixed ratio between an outer cube and its inner cube is preserved by rescaling its r_j . Set $A_j = Df(x_j)$ and use the notation in the statement.

Each buffered band E_j and each outer-minus-inner region $Q_j \setminus K_j$ has volume at most $As|Q_j|$, with A depending only on the fixed constant C . Equation (2.38) and $|X + Y|^p \leq 2^{p-1}(|X|^p + |Y|^p)$ give

$$\sum_j \int_{E_j \cup (Q_j \setminus K_j)} |Df|^p \leq A_p(M^p s + \eta) \sum_j |Q_j| \leq A_p(M^p s + \eta) |\Omega|. \quad (2.39)$$

The same differentiation test gives

$$\sum_j \int_{K_j} |Df - A_j|^p \leq \eta |\Omega|. \quad (2.40)$$

Thus the discarded full-rank energy consists of the initial compact truncation, the finite Vitali truncation, and a part bounded by (2.39). This proves (2.36) after decreasing all the error prescriptions. It also proves the stronger band statement: every affine cutoff multiplier is at most $C_p M^{2p}$, so choose s and then η to make

$$C_p M^{2p} A_p(M^p s + \eta) |\Omega|$$

as small as required. Notice that M was fixed before s and η ; there is no bound being chosen after the bands to which it must apply. Equation (2.37) is exactly the stated normalized value test for f .

Now fix this finite family. Require the preliminary smooth map k to satisfy

$$\sup_{Q_j} |k - f| < \frac{csr_j}{4M} \quad \text{for each } j. \quad (2.41)$$

Combining this with (2.37) shows that

$$\widehat{k}_j(\xi) = r_j^{-1} A_j^{-1}(k(x_j + r_j \xi) - f(x_j))$$

is within $cs/2$ of ξ on $|t(\xi)| \leq Cs$. In particular it meets the closeness hypothesis of Lemma 2.14 with a fixed positive margin. Apply that Lemma separately in these disjoint cubes, and return to the original coordinates as in Remark 2.15. Choose the finitely many additive errors to have arbitrarily small sum. The new maps are exactly b_j on K_j , agree with k on outer collars, and have the same images as k in their respective replacement cubes. Therefore they glue to a locally bi-Lipschitz homeomorphism onto the same fixed target Λ .

Let $S_j = \{x_j + r_j \xi : 0 < t(\xi) < 2s\}$ denote the modified shells. Equation (2.35) yields

$$\sum_j \int_{S_j} |D\tilde{k}|^p \leq C_p M^{2p} \sum_j \int_{E_j} |Dk|^p + \delta,$$

where $\delta > 0$ is freely prescribed. Hence

$$\sum_j \int_{S_j} |D\tilde{k} - Df|^p \leq 2^{p-1} \left(C_p M^{2p} \sum_j \int_{E_j} |Dk|^p + \delta + \sum_j \int_{S_j} |Df|^p \right). \quad (2.42)$$

The last term was made small in (2.39). Choose a positive upper bound for $\sum_j \int_{E_j} |Dk|^p$ after the finite affine bounds are known, and take δ small. Equations (2.40) and (2.42) then give the asserted total derivative accuracy on the corrected regions.

To retain a prescribed positive continuous value tolerance, first make the cubes small enough that the oscillation of f on each buffered cube is much smaller than that tolerance there. This is allowed by the fineness of the Vitali family and continuity of f . Also impose correspondingly small value errors on $k - f$. Every modified value lies in the old image of its replacement cube. Thus its distance from f at the source point is bounded by that cube's oscillation of f plus the chosen value error of k . These quantities can be made smaller than the requested tolerance.

Finally suppose the available preliminary map is only locally bi-Lipschitz. Apply Theorem A.1 with sufficiently small strong $W^{1,p}$ error and its fine value control on the finitely many compact buffered cubes. Strong convergence preserves their energy upper tests after allowing a strict margin, since

$$\int_{E_j} |Dk|^p \leq 2^{p-1} \int_{E_j} |Dg|^p + 2^{p-1} \|Dk - Dg\|_{L^p(\Omega)}^p$$

for a preliminary map g . The fine value control preserves (2.41). The preceding smooth correction therefore applies. Any change outside the selected cubes at this preliminary smoothing stage has the arbitrarily small strong error prescribed in Theorem A.1. $\square$

2.9 Compressing the marked interiors

Lemma 2.17 (Subcritical marked-cube compression). *Suppose $2 < p < 3$, and let U and w_U be the marked cubes and collars of Proposition 2.13. Fix any additional error $\varepsilon_0 > 0$. Let $k : \Omega \rightarrow \Lambda$ be a*

locally bi-Lipschitz homeomorphism. Suppose on thinner collars $C'_U(w_U)$, containing a fixed relative share of the outer half-collar, one has

$$\sum_U \frac{\ell_U}{w_U} \int_{C'_U(w_U)} |Dk|^p \leq b. \quad (2.43)$$

There are disjoint slightly enlarged cubes $\tilde{U} \supset U$, whose boundaries lie in these outer collars, and source self-homeomorphisms supported in their closures, such that the resulting locally bi-Lipschitz map $\tilde{k}$ agrees with k outside $\bigcup \tilde{U}$ and satisfies

$$\sum_U \int_{\tilde{U}} |D\tilde{k}|^p \leq C_p b + \varepsilon_0. \quad (2.44)$$

Its value change on each $\tilde{U}$ takes place inside $k(\tilde{U})$. Thus previously required fine compact value accuracy is preserved by choosing the marked cubes sufficiently fine. In particular, if the costs in (2.43) are bounded by a fixed multiple of (2.31) plus errors summable after multiplication by ℓ_U/w_U , all marked-interior costs can be made arbitrarily small. No convergence to a fixed boundary trace is assumed.

Proof. Slice the outer collar of each U by concentric cube boundaries. The coarea formula for the cube radius and (2.43) give a slightly larger cube $\tilde{U}$ such that

$$\sum_U \ell_U \int_{\partial\tilde{U}} |Dk|^p d\mathcal{H}^2 \leq Cb. \quad (2.45)$$

Choose its radius also to be a Lebesgue point of the surface integral as a function of cube radius. Almost every radius has this property, so it is compatible with the averaging inequality. The boundaries are chosen on the outer side to ensure that all of the previously exempt interior U is included. The available disjoint collar enlargements make the cubes $\tilde{U}$ disjoint.

Write a fixed such cube as $x_0 + a[-1, 1]^3$, and use coordinates $x = x_0 + at\theta$, where $0 \leq t \leq 1$ and $\theta \in \partial[-1, 1]^3$. The edges between facets have zero volume and may be omitted in integration. For $0 < \alpha < 1/2$ and $0 < \rho < 1/2$, let Φ be the radial self-homeomorphism with radial function

$$\varphi(t) = \begin{cases} (1 - \rho)t/\alpha, & 0 \leq t \leq \alpha, \\ 1 - \rho + \rho(t - \alpha)/(1 - \alpha), & \alpha \leq t \leq 1. \end{cases}$$

It fixes the boundary and is bi-Lipschitz for these fixed positive parameters. On the inner cube, k has a finite Lipschitz constant L on the compact cube, so

$$\int_{\{t < \alpha\}} |D(k \circ \Phi)|^p \leq Ca^3 L^p \alpha^{3-p}.$$

Choose α so small that this is below an assigned summable error. This is exactly where $p < 3$ is used.

On the outer region, $|D\Phi| \leq C(\varphi(t)/t + \varphi'(t)) \leq C/t$. Cube polar integration therefore gives

$$\int_{\{\alpha < t < 1\}} |D(k \circ \Phi)|^p \leq C_p a^3 \int_{\alpha}^1 t^{2-p} \int_{\partial[-1, 1]^3} |Dk(x_0 + a\varphi(t)\theta)|^p d\mathcal{H}^2(\theta) dt.$$

For fixed α , let $\rho \downarrow 0$. The inner surface integral is sampled from radii in $[1 - \rho, 1]$. The chosen outer radius is an L^1 Lebesgue point. After the linear substitution from t to $\varphi(t)$, the weight t^{2-p} is

bounded by a constant depending on the fixed α ; hence Lebesgue differentiation applies and the limsup of the last expression is at most

$$C_p a \int_{\partial \tilde{U}} |Dk|^p d\mathcal{H}^2 \int_{\alpha}^1 t^{2-p} dt \leq C_p \ell_U \int_{\partial \tilde{U}} |Dk|^p d\mathcal{H}^2.$$

The last constant may depend on p , since $\int_0^1 t^{2-p} dt = (3-p)^{-1}$. Take ρ small enough to add only another assigned summable error. Applying this procedure on all cubes and using (2.45) proves (2.44).

Each source modification fixes its boundary. Local finiteness and positive fixed radial slopes therefore preserve local bi-Lipschitzness and the global homeomorphism onto the same target. All new values on a cube belong to its old image under k . If k is finely close to f , uniform continuity of f on compact localizations and the original fineness of the cubes make the resulting oscillation as small as the required compact accuracy. The proof sampled the actual ambient derivative of k on a newly chosen radius, and nowhere used convergence of derivatives on a previously fixed boundary. $\square$

3 Convex product quantizers and the exterior carrier

Throughout this section $p > 2$. We use the carrier F_b , its exact target set O , the smaller exact set O_1 , and the protected endpoint boxes supplied by Lemma 2.7 and Proposition 2.13. We construct a Lipschitz target map T whose three-dimensional input blocks lie in O , where the carrier is already exact. Outside those blocks, T takes values in a polyhedral two-complex. The retained deficient-rank gradients must survive on its faces. Small central prisms will then be removed from the input space to produce an exterior with standard annuli and disks. A final local replacement makes the carrier exact near this new boundary, ready for the source-wall construction.

The carrier F_b is a proper continuous surjection in $W_{\text{loc}}^{1,p}$ and has the relative opening property of Lemma 2.10. On the inverse of O it agrees locally with specified affine or PL homeomorphisms. The compact retained rank data are denoted by K ; near K the carrier equals F_κ . All smallness prescriptions may vary continuously toward the boundary. A locally finite prescription therefore imposes only finitely many accuracy requirements on each compact set.

The construction combines classical convex conjugacy and proximity [34] with the polyhedral geometry of power diagrams [3]. The contact exclusion, protected paired arrays, orthogonal input products and properness estimates required here are established below; they are not consequences of those general constructions alone.

3.1 Convex contacts and product cells

At a smooth contact point, a function minus its affine support has a local minimum, so its Laplacian is nonnegative. We therefore force contacts into O by making the Laplacian negative outside O . The small residual volume allows this without losing the protected paired supports or appreciably changing the potential.

Lemma 3.1 (Excluding contacts by a Poisson perturbation). *Let b_1 be the protected-well perturbation of Lemma 2.7. Fix a positive 1-Lipschitz function δ on Λ , extended by zero outside, with $\delta \leq \text{dist}(\cdot, \mathbb{R}^3 \setminus \Lambda)$. The well sizes can first be chosen sufficiently small, and the residual volume prescriptions for $\Lambda \setminus O$ in Proposition 2.13 can subsequently be chosen sufficiently small, so that there is $b_2 \in C^\infty(\Lambda)$, zero outside Λ , with the following properties. For an absolute C ,*

$$\psi(x) = \frac{1}{2}|x|^2 + b_1(x) + b_2(x), \quad \Delta\psi \leq C \text{ in } \Lambda, \quad \Delta\psi < 0 \text{ in } \Lambda \setminus O. \quad (3.1)$$

The function b_2 vanishes on neighborhoods of all protected endpoint boxes, every prescribed affine support indexed by a protected horizontal slope remains a global support of ψ , and, with arbitrarily small $\epsilon > 0$,

$$|b_1 + b_2| \leq \epsilon \delta^2, \quad b_1 + b_2 = o(\text{dist}(x, \mathbb{R}^3 \setminus \Lambda)^2) \quad (x \rightarrow \partial \Lambda). \quad (3.2)$$

Proof. Choose closed Whitney cubes Q with interior enlargements Q^+ and bounded overlap of the enlargements. For each cube the disallowed set $E_Q = Q \setminus O$ is compact. Its measure is smaller than a threshold that has not yet been specified. Smooth a neighborhood of E_Q inside Q^+ to obtain $0 \leq \rho_Q \leq 1$, equal to one near E_Q and with arbitrarily small support measure in units of Q . This is possible by outer regularity, provided the residual threshold is small enough. In normalized coordinates put this density in a fixed three-dimensional torus $\mathbb{T}$ containing Q^+ , and solve

$$\Delta u_Q = -A\rho_Q + \frac{A}{|\mathbb{T}|} \int_{\mathbb{T}} \rho_Q, \quad \frac{1}{|\mathbb{T}|} \int_{\mathbb{T}} u_Q = 0.$$

Here A is fixed, larger than $3 + \sup \Delta b_1 + 2$. For completeness, the required smallness is already a consequence of the periodic Green kernel: its singularities are $O(|x|^{-1})$ and those of its first derivative are $O(|x|^{-2})$. Both kernels belong to $L^{q/(q-1)}$ for fixed $q > 3$. Convolution and Hölder's inequality therefore give

$$\|u_Q\|_{\infty} + \|Du_Q\|_{\infty} \leq C_q A \|\rho_Q\|_{L^q} \leq C_q A |\text{supp } \rho_Q|^{1/q}.$$

The mean-zero Green solution is smooth because the density is smooth. Multiply it by a cutoff equal to one on Q and supported in Q^+ , and return to physical coordinates with the factor $\ell(Q)^2$. The resulting v_Q satisfies $\Delta v_Q \leq \varepsilon_Q$ everywhere and $\Delta v_Q \leq -A + \varepsilon_Q$ on E_Q ; its value and gradient are as small as desired in the corresponding scaled units. The errors here include the mean compensation and both cutoff terms, which involve only u_Q, Du_Q .

There is a smooth cutoff χ , zero on neighborhoods of the compact protected endpoint boxes and one outside O_1 , whose transition is contained in O_1 . Its derivatives on each Q^+ are fixed before the residual thresholds. Set $b_2 = \chi \sum_Q v_Q$. The sum is locally finite. In

$$\Delta(\chi v_Q) = \chi \Delta v_Q + 2D\chi \cdot Dv_Q + v_Q \Delta \chi$$

the possibly large negative term is harmless for an upper bound, and all other terms can be made arbitrarily small per cube. Bounded overlap, with the errors chosen below 1 in total at every point, proves the first inequality in (3.1). At a point outside O the cutoff is identically one locally and at least one E_Q contributes $-A$; this proves the strict negative inequality. Make the value bounds on each enlargement smaller also than the prescribed local multiple of δ^2 , and let that multiple tend to zero with boundary distance. This gives (3.2), after the initial analogous choice for b_1 .

It remains to check the global supports. For a fixed compact horizontal endpoint box, subtract one of its prescribed affine supports from $|x|^2/2 + b_1(x)$. Its only zeros are the two prescribed endpoint points for that slope, by the two-well test. Uniformly over the compact set of slopes, it has a strictly positive minimum on every compact set outside the endpoint neighborhoods. At infinity the gap grows quadratically. The added perturbation vanishes in the endpoint neighborhoods. Choose its absolute value elsewhere below half the preceding gap, using the minimum over the compact slope set. There are only finitely many protected patches, by Lemma 2.7, and their supports have been separated. The minimum of their positive gap requirements is positive away from the endpoint neighborhoods. Taking the smaller of these requirements preserves all supports simultaneously. All these requirements determine only smaller residual thresholds, after the endpoint neighborhoods have been fixed, as allowed by Proposition 2.13. $\square$

Lemma 3.2 (Contact curvature and dual separation). *Let ϕ be the lower convex envelope of ψ from Lemma 3.1, and let*

$$F(q) = \phi^*(q) = \sup_x \{q \cdot x - \psi(x)\}.$$

There is an absolute C_0 such that ϕ is $C^{1,1}$ and $0 \leq D^2\phi \leq C_0 I$ in the distributional sense. With $m = 1/C_0$ reduced if necessary, every actual contact maximizer x_i at slope q_i satisfies

$$F(q) \geq F(q_i) + x_i \cdot (q - q_i) + \frac{m}{2} |q - q_i|^2. \quad (3.3)$$

For any requested fine motion tolerance the perturbations can be chosen so that all contact maximizers at q lie within that tolerance of q . For $q \notin \Lambda$ the unique maximizer is q and $F(q) = |q|^2/2$. For $q \in \Lambda$ every maximizer lies in O ; over a compact subset of Λ their union is compactly contained in O .

Proof. The function ψ has quadratic growth at infinity. A supporting plane of its convex envelope consequently has a compact, nonempty set of contacts, and each point of the envelope on that plane is a convex combination of at most four contacts. One way to see attainment is to minimize the average of ψ over probability measures of fixed barycenter. Quadratic growth gives tightness and a bounded second moment; passage to the limit gives a minimizer. Separation by a supporting plane forces its support onto the contact set, and Carathéodory's Theorem reduces it to four points.

At a contact in Λ , the smooth function ψ minus its supporting plane has a minimum. Thus $D^2\psi$ is nonnegative there; (3.1) bounds each eigenvalue above by C . At a contact on or outside $\partial\Lambda$, the zero extension and (3.2) give the quadratic Taylor expansion of $|x|^2/2$ with an $o(|v|^2)$ remainder. If $x = \sum \alpha_j x_j$ is a contact combination, use $x_j + v$ and $x_j - v$ as competing combinations to obtain

$$\limsup_{v \rightarrow 0} \frac{\phi(x+v) + \phi(x-v) - 2\phi(x)}{|v|^2} \leq C_0.$$

Here C_0 is absolute and independent of the contact combination. On any line this implies semiconcavity with constant C_0 : if, after subtracting $(C_0 + \varepsilon)t^2/2$, the function dipped below one of its chords, its chord-subtracted minimum would be interior. The symmetric second difference at that minimum is nonnegative, contrary to the displayed strict negative upper bound. Let $\varepsilon \downarrow 0$. Convexity and semiconcavity imply differentiability and the global bound

$$\phi(x+v) \leq \phi(x) + D\phi(x) \cdot v + \frac{C_0}{2} |v|^2.$$

At a contact x_i , $D\phi(x_i) = q_i$. Substituting $v = (q - q_i)/C_0$ into the definition of the conjugate proves (3.3).

Write $b = b_1 + b_2$ and $d = |x - q|$ for a maximizing contact. Comparison with $x = q$ gives

$$\frac{1}{2}d^2 \leq |b(x)| + |b(q)| \leq \epsilon\{(\delta(q) + d)^2 + \delta(q)^2\}.$$

For sufficiently small ϵ this yields $d \leq C\sqrt{\epsilon}\delta(q)$. When $q \notin \Lambda$ it gives $d = 0$. Choosing the positive Lipschitz minorant δ and then ϵ proves the fine motion assertion and keeps all contacts inside Λ for interior q . A contact in $\Lambda \setminus O$ would have nonnegative Hessian and therefore nonnegative Laplacian, contrary to (3.1). Finally, maximizing contacts form a closed graph and remain in a bounded set over bounded slopes. For slopes in a compact subset of Λ their union is therefore compact; since it is contained in O , its clearance from O^c is positive. $\square$

Proposition 3.3 (A proper product quantizer). *Fix $0 < c < m$ with $c < 1$. One can choose a locally finite family of interior sites in Λ , with one contact at each ordinary site and the two protected*

contacts at each paired site. The family can include the rectangular paired arrays constructed in Lemma 3.4. Index these site and contact pairs by i , writing them as (q_i, x_i) , and define

$$G(q) = \sup_i \{F(q_i) + x_i \cdot (q - q_i) + \frac{c}{2}|q - q_i|^2\}, \quad T = (\partial G)^{-1}, \quad (3.4)$$

where the supremum also includes the support based at every $q_i \notin \Lambda$. Here q is an output of T , and $y \in \partial G(q)$ is an input. Then $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is single-valued and c^{-1} -Lipschitz, equals the identity outside Λ , and restricts to a proper onto map of Λ to itself, as finely close to the identity as prescribed.

Inside Λ , $G - c|q|^2/2$ is a locally finite maximum of affine functions. Its diagram has compact convex polyhedral cells. Write $p_i = x_i - cq_i$. Over the relative interior of a face P , whose active indices are $I(P)$, its exact input product is

$$\partial G(q) = cq + \text{conv}\{p_i : i \in I(P)\}, \quad T(cq + v) = q \quad (q \in \text{relint } P). \quad (3.5)$$

The two factors have orthogonal direction spaces of complementary dimensions. Every retained support owns a genuine three-dimensional output cell C_i . Its closed input block

$$X_i = cC_i + p_i \quad (3.6)$$

is compactly contained in O . These closed blocks are pairwise disjoint and have locally finite, mutually disjoint protective neighborhoods within the exact regions of the carrier. Outside their union the output of T lies in the two-skeleton of the diagram.

Proof. Localizing the active supports. For a support L_i appearing in (3.4), (3.3) gives

$$L_i(q) \leq F(q) - \frac{m-c}{2}|q - q_i|^2. \quad (3.7)$$

A sufficiently fine locally finite net gives $G \geq F - e$ for any prescribed positive continuous error e on Λ : for each compact set, contacts and slopes are bounded, so the support at a nearby site converges uniformly to F there. A Whitney refinement with compact buffers gives these estimates simultaneously. Also $G \leq F$. Outside Λ its own support is present, so $G = F = |q|^2/2$ there. Choose $e(q)$ so small that

$$\tau(q) = \sqrt{2e(q)/(m-c)} < \frac{1}{4} \text{dist}(q, \Lambda^c),$$

and impose any additional fine upper bounds on τ needed below. Every active site is within $\tau(q)$ of q by (3.7). Exterior supports and sites outside that compact interior neighborhood are uniformly below the supremum. The remaining interior sites are finite locally, so the supremum is attained and gives the asserted polyhedral diagram. This argument also applies on all closed cell boundaries.

The proper Lipschitz map. The function G is c -strongly convex and grows quadratically. Consequently $G(q) - y \cdot q$ has a unique minimum for every y . Strong monotonicity gives

$$(y - y') \cdot (T(y) - T(y')) \geq c|T(y) - T(y')|^2,$$

proving the Lipschitz bound. At a point outside Λ , equality $G = F$ and differentiability of F imply $\partial G(q) = \{q\}$: every supporting affine function for G there also supports F . Thus T is the identity outside, and no interior point is sent outside. For interior q , (3.5) expresses each input as a convex combination of

$$cq + p_i = x_i + c(q - q_i).$$

The contact motion bound and the bound $|q - q_i| \leq \tau(q)$ make every such point arbitrarily finely close to q ; in particular it lies in Λ . This proves surjectivity onto Λ . Arrange $|T(y) - y| < \frac{1}{4} \text{dist}(y, \Lambda^c)$,

with analogous inverse motion control in terms of q . A sequence of inputs approaching $\partial\Lambda$ then has outputs approaching that boundary. Since Λ is bounded, inverse images of compact subsets are compact, which proves properness.

Orthogonal products and exact input blocks. Subtracting the common quadratic in (3.4) gives slopes p_i . Equalities between its active affine functions have normals $p_i - p_j$. The tangent space of a relatively open face is the common kernel of these equalities, so their span is its entire normal space. This proves the complementary orthogonal product assertion and the subgradient formula. The closure of a product over P is meant here when a face is closed; a boundary point can have further incident products in its complete fibre.

An ordinary support at its own site is strictly above every support from another site, by (3.7). The gap is uniformly positive locally: nearby distinct sites are separated, and the remaining sites, including the exterior family, have positive distance from that site. It therefore wins on an open set. At a paired site the two contacts have different normal components; each wins on one open side of their common tie plane while the same other-site gap persists. Thus all retained supports own full cells.

Two offsets from different sites cannot coincide. Indeed strong monotonicity of ∂F , obtained by adding the two inequalities (3.3), yields

$$(x_i - x_j) \cdot (q_i - q_j) \geq m|q_i - q_j|^2.$$

If $p_i = p_j$, its left side is $c|q_i - q_j|^2$, a contradiction. At one site discard identical contacts. If two closed blocks met, uniqueness of T would give the same output q and then the same offset, another contradiction.

Choose τ smaller also than the local contact clearance in O divided by $2c$, using compact neighborhoods of the slopes. Then every $x_i + c(q - q_i)$ for $q \in C_i$ lies in O . A fixed cell cannot approach $\partial\Lambda$, since $|q - q_i| < \text{dist}(q, \Lambda^c)/4$ throughout it. Boundedness makes C_i and X_i compact. Over a compact input set, motion control confines outputs and active sites to compact interior sets, so only finitely many blocks occur. Pairwise disjoint compact blocks in this locally finite family admit pairwise disjoint neighborhoods in O , reduced further to the prescribed exact carrier neighborhoods. Finally, over the interior of a full output cell the product is the singleton graph $y = cq + p_i$, which is precisely its full input block. This proves the last assertion. $\square$

3.2 Calibration and strict approximation

For the rest of the construction fix c universally, for example $c = \min\{m, 1\}/2$. It is independent of every later accuracy parameter; constants depending only on this choice are absolute.

Lemma 3.4 (Paired arrays, redistribution, and phase calibration). *The choices in Proposition 3.3 can be made together with a locally bi-Lipschitz homeomorphism $J : \Lambda \rightarrow \Lambda$, equal to the identity near every closed X_i , such that TJ has an absolute Lipschitz bound. Put*

$$\hat{F} = JF_b, \quad q_0 = T\hat{F}.$$

Apart from arbitrarily small prescribed source weight, q_0 maps the retained rank-one and rank-two data into the interiors of horizontal rectangular true two-faces. The carrier $\hat{F}$ retains the exact data over the blocks and its relative opening property, and $|Dq_0| \leq C|DF_b|$ almost everywhere. On the retained compact rank- k data, $k = 1, 2$,

$$Dq_0 = a_*^{-1} \text{proj}_{n^\perp} DF_\kappa, \quad |Dq_0 - Df| \leq C(\lambda + 1 - a_*)|Df|, \quad (3.8)$$

where $c < a_* < 1$ may be as close to one as required and λ is the preceding flattening accuracy. The rank is exactly k and $\ker Df \subset \ker Dq_0$. For rank one the rectangle can have any previously prescribed small short-to-long aspect, with its first axis arbitrarily close to the range line of Dq_0 . Further compact margins, sector avoidance, and uniform jet tests may be imposed by trimming.

Proof. On a protected patch write $q = z + h + q_v n$. At the sites $q_i = z + u_i$ of a Cartesian lattice in $n^\perp$, with periods d_1, d_2 , retain both contacts $x_i^\pm = z + u_i \pm d_\perp n$. The protected support identity is

$$F(z + u) = \frac{1}{2}|z + u|^2 - d_\perp^2.$$

Exclude nonmatching sites from a narrow tube about a compact interior part of the plane $q_v = 0$. The tube width is chosen after the error and active-site tolerances in the preceding proof, and the periods are chosen much smaller still. A point in the tube has a paired site at distance at most its width plus $\sqrt{d_1^2 + d_2^2}$; local continuity and the support formula give the required dual approximation there. Outside use a sufficiently fine allowed net. Thus exclusion does not invalidate the previous approximation.

More explicitly, on the central plane the deficit of the nearest pair is $O(d_1^2 + d_2^2)$, whereas every nonmatching site has, on a smaller protected interior, a fixed positive distance and hence a fixed positive deficit by (3.7). Decrease the periods until the former is smaller. Dominance persists in a vertical neighborhood. The affine parts of two matching pairs differ there by the nearest-node quadratic comparison with coefficient $1 - c > 0$. Their bisectors are therefore exactly the Cartesian bisectors, and the two signs tie exactly on $q_v = 0$.

The subgradient formula now gives, for protected inputs $y = z + h + vn$ with $|v| < d_\perp$,

$$T(y) = z + (Q_{c,1}(h_1), Q_{c,2}(h_2))_{\text{horiz}}. \quad (3.9)$$

In a period centered at node u of pitch d , the scalar map is $u + (h - u)/c$ for $|h - u| \leq cd/2$, and is constant, equal to the appropriate output period endpoint, on each remaining half-gap. At a tie all tied nodes have both signs, so their convex hull includes the entire interval $|v| \leq d_\perp$; thus the formula also holds on the seams. The pertinent full output rectangles in $q_v = 0$ are true faces of the diagram, not arbitrary two-dimensional subsets of cells.

The scalar map Q_c has slope $1/c$ on only a fraction c of each input period and collapses the complementary gaps. We enlarge this fraction to a close to one, so that most retained data see the nearly unit slope $1/a$. For this purpose choose a Lipschitz cutoff $a(h, v) \in [c, a_*]$, equal to a_* on a neighborhood of the protected values and equal to c off a compact subset of the separability region $|v| < d_\perp$. In a period define R_a to map the interval of fraction a linearly onto the interval of fraction c , and the two complementary intervals linearly onto their counterparts, fixing the endpoints. Thus $Q_c R_a = Q_a$, where Q_a is the same scalar map with c replaced by a . We must also check that the two simultaneous redistributions remain invertible when a varies with the input. In centered coordinates the inverse of R_a is

$$B_a(w) = \begin{cases} (a/c)w, & |w| \leq cd/2, \\ ad/2 + \frac{1-a}{1-c}(w - cd/2), & cd/2 \leq w \leq d/2, \end{cases}$$

with the odd extension on the negative half-period. Hence $|\partial_a B_a| \leq d/2$, including across the fixed inverse breakpoints. Define simultaneously

$$J(h, v) = (R_{a(h,v),1}(h_1), R_{a(h,v),2}(h_2), v).$$

For given output (w, v) , its inverse solves

$$h_j = B_{a(h,v),j}(w_j), \quad j = 1, 2.$$

Choosing the periods so that $\sqrt{d_1^2 + d_2^2} \text{Lip}(a) < 1/2$ makes this a contraction on $\mathbb{R}^2$, after extending the periodic formulas and putting $a = c$ outside the support. It has a unique solution, Lipschitz in w, v . The forward formulas are Lipschitz for the fixed $a_* < 1$; consequently J is bi-Lipschitz on this patch. It is the identity off the patch and maps it onto itself. The disjoint patches assemble locally finitely.

The composite $Q_c R_a = Q_a$ has ordinary input slopes bounded by $1/c$ and parameter derivative bounded by $C_c d$; the latter follows either from its formula $u + (h - u)/a$ on the middle interval or from continuity at the moving breakpoints. The chosen period bound makes TJ uniformly Lipschitz, independently of how close a_* is to one. The support is strictly between the two endpoint heights; a full block there would require an extreme sign, at height $\pm d_\perp$. Thus the support avoids all closed full blocks, with a neighborhood. The same is true of its image. The carrier opening property transfers under the target homeomorphism J , and the derivative bound follows from the Lipschitz chain rule.

All these choices can be uniform over lattice phases. Fix nested horizontal boxes $H_{\text{inner}} \subseteq H_{\text{array}} \subseteq H_{\text{endpoint}}$, the excluded tube, and the cutoff support first. For every phase retain all lattice nodes in H_{array} , taking both periods below one tenth of the nesting margins. The nearest-node deficit is bounded by $C(d_1^2 + d_2^2)$ for every phase, whereas the nonmatching-site gap on the inner box is fixed. Thus a common separability slab $|v| < d_\perp - \sigma$ contains the fixed cutoff support for all phases. Its separation from full blocks and the inverse contraction estimate are phase-independent.

Choose the phases after the compact data, cutoffs, and periods have been fixed. For any fixed datum, uniform independent translations of the two lattice coordinates place it in their middle intervals with probability a_*^2 . Fubini's Theorem for the finite source weight therefore gives a phase losing at most the prescribed multiple of $1 - a_*^2$, with room for further losses. Conditional on membership in these intervals, its two normalized output coordinates are uniform on the output rectangle: this follows directly from the linear formula and translation-invariance of phase measure. Consequently any prescribed zero-area subset of a normalized rectangle, such as its center, perimeter, or finitely many sector rays, is avoided for almost every retained datum for almost every phase. Neighborhoods of those sets have losses tending to zero by the same averaging. This argument uses no absolute continuity of the projected data measure. Independently shifted finer rectangular subdivisions give the same conclusion after discarding fitting margins.

Where the retained data have $a = a_*$ and both middle intervals are strict, differentiation of (3.9) gives (3.8); the preceding flattening bound and $|n - e| \leq C\lambda$ give its error estimate. The Lipschitz composition annihilates every direction in $\ker Df$ at a source point where the compared jets exist. After the preliminary positive singular-value bounds have been fixed, choose the errors smaller than those bounds, so the rank cannot decrease. It cannot increase because of the kernel inclusion. The first lattice axis follows the selected rank-one range direction; its angular errors were chosen smaller than the assigned aspect. Finally continuity, regularity of the finite source weight, and differentiation allow compact trimming to all claimed interior and uniform jet margins. $\square$

Lemma 3.5 (Strict quantizers). *The map T has proper locally bi-Lipschitz approximations $T_\eta : \Lambda \rightarrow \Lambda$ with the same target. Given any positive continuous local tolerance $\epsilon_0(y)$, they can be chosen with*

$$|T_\eta(y) - T(y)| < \epsilon_0(y), \quad |DT_\eta(y) - DT(y)| < \epsilon_0(y) \quad \text{for a.e. } y.$$

On every product piece the derivatives of T are its affine tangential derivatives, agreeing on directions tangent to common strata. If h is locally bi-Lipschitz, the pushforward by Th of volume restricted to

each product interior is absolutely continuous with respect to the dimensional measure on its output stratum.

Proof. For $q = T(y)$ set

$$T_\eta(y) = q + \eta(q)(y - q),$$

where $\eta > 0$ is smooth, bounded above by a small constant, and has a small global Lipschitz constant. Such functions with arbitrary local upper bounds on their value and first derivative are obtained from a locally finite smooth partition of unity: choose the coefficient of each term smaller than all required bounds on its support, divided by the local overlap count and the first derivative bound of that term. Put $M = \sup_\Lambda |y - T(y)| < \infty$. For $q_j = T(y_j)$, monotonicity and direct subtraction give

$$(T_\eta(y_1) - T_\eta(y_2)) \cdot (q_1 - q_2) \geq (1 - \|\eta\|_\infty - M \text{Lip } \eta) |q_1 - q_2|^2 \geq \frac{1}{2} |q_1 - q_2|^2.$$

Thus differences of outputs control differences of q . Subtracting again in the defining formula, and using a positive lower bound for η on each compact set, controls $|y_1 - y_2|$ locally by the output difference. This proves injectivity and local inverse Lipschitz bounds, including the case $q_1 = q_2$, when the output difference is exactly $\eta(q_1)(y_1 - y_2)$. Forward Lipschitz bounds follow from those of T and η . Fine motion keeps the segment from y to q inside Λ and makes T_η proper, as in Proposition 3.3. Invariance of domain gives an open image; properness gives a relatively closed image. Connectedness of Λ proves surjectivity.

At almost every input,

$$D(T_\eta - T) = \eta(T)(I - DT) + (y - T) \otimes (D\eta(T)DT).$$

Properness first transfers any local input tolerance to a local output tolerance; choose $\eta, D\eta$ below it in the preceding construction. This proves both requested estimates. On a product piece (3.5) is orthogonal projection to its face, with factor c^{-1} in tangential directions. Its restriction to a common stratum is the same map from either side, proving tangential agreement. Fubini on the product, followed by the change-of-variables inequality for a locally bi-Lipschitz h , proves the dimensional absolute continuity assertion. At a zero-dimensional output stratum the map Th is constant and its weak derivative vanishes almost everywhere on that level set. $\square$

3.3 Face coordinates and the graph exterior

Lemma 3.6 (Subdividing the true faces). *The true two-faces of the quantizer diagram have locally finite, conforming subdivisions into convex polygonal subcells D with chosen centers d_D . Protected rectangular faces may be kept as rectangles or subdivided by finer rectangular grids. Elsewhere the ratio of diameter to center-boundary clearance has an absolute bound away from arbitrarily small prescribed neighborhoods of the original vertices. Inside those neighborhoods it has a finite bound depending only on the original face. The latter constants can be confined to regions of arbitrarily small weighted $\int |Dq_0|^p$ cost. Additional rays from a protected center to inserted perimeter vertices can be chosen to avoid almost every retained datum.*

Proof. In a compact convex face P , choose a maximal separated net of spacing d and intersect its planar Voronoi cells with P . Maximality bounds each cell's diameter by Cd , while separation gives the intersection of a disk of radius cd about its generator with P . Fix an interior disk of P once. Interpolating the generator toward its center by a distance proportional to d gives a disk of radius cpd in that intersection: convexity carries the fixed interior disk into P , and its interpolated center and radius both remain within the small disk about the generator. Thus the clearance ratio is

bounded by a finite constant depending on P . Outside a prescribed neighborhood of the vertices and for sufficiently small d , at most one supporting edge line cuts this small disk. A disk cut by one half-plane through or outside its center contains a disk of one fixed fraction of its radius. The clearance constant there is absolute. Choose the center d_D of each cell in the indicated inscribed disk.

Match the subdivisions of a shared original edge by overlaying the finitely many incident subdivisions. This inserts only collinear perimeter vertices and does not change any cell or its clearance. Choose the finite nets generically first, so no Voronoi vertex lies on an original edge and each incident Voronoi segment meets that edge transversely. The finitely many nonzero angles and strict convexity margins have a positive minimum. Changes of division positions on original edges and the corresponding changes in adjacent segments are taken below these margins and preserve cyclic order. They therefore preserve convexity and the clearance bounds. Collinear overlay nodes may slide along a side without changing its polygon. For a fixed interior point of a protected rectangle, the ray from its center through that point intersects the perimeter at one location. A continuously distributed perturbation of an extra perimeter vertex equals that location with probability zero. Fubini for the finite retained measure gives simultaneous avoidance, even for singular data. The original center, corner, and sector-ray exclusions have already been arranged by Lemma 3.4. All choices are finite per face and locally finite in Λ .

A Sobolev map has zero weak derivative almost everywhere on any one of its level sets: apply the scalar statement to each coordinate, which follows by truncating that coordinate to a shrinking interval about the level and using the Sobolev chain rule. In particular $Dq_0 = 0$ almost everywhere on $\{q_0 = v\}$ for an original vertex v . On a compact localization, dominated convergence gives

$$\lim_{\rho \downarrow 0} \int_{\{|q_0 - v| < \rho\}} |Dq_0|^p = 0.$$

The map q_0 is proper and locally Sobolev, so the localizations needed for a compact target neighborhood are compact source localizations. Fix the original-face clearance factors first, and then choose the vertex neighborhoods to make these integrals smaller than any given error divided by those factors. This also permits any previously fixed collar weights. Use a summable family of errors for the locally finite vertices. Only after these neighborhoods have been fixed take d sufficiently small. The arbitrary original-face constants thus multiply only the separately paid vertex contributions. $\square$

Lemma 3.7 (Polar coordinates and their sharp rectangular bound). *For each subcell D , put*

$$R_{D,e}(t, s) = d_D + r(\xi_e(s) - d_D), \quad r = \exp(t/L_D),$$

where ξ_e is arclength on a perimeter subedge and the same arclength label is used in all incident sectors. Choose L_D comparable to the maximum radius of D ; in a protected rank-one rectangle choose it as the long-axis half-length and choose d_D to be the geometric midpoint of the rectangle. On an open sector,

$$|D_q t| + |D_q s| \leq C_D/r, \quad |\partial_t R_{D,e}| + |\partial_s R_{D,e}| \leq Cr. \quad (3.10)$$

The first constant depends only on the clearance ratio, and the second is absolute. For a rectangle of half-length L and half-width γL , and its unit long-axis vector v_1 ,

$$|D_q t(v_1)| + |D_q s(v_1)| \leq (1 + \gamma)/r, \quad \max\{|\partial_t R|, |\partial_s R|\} \leq \sqrt{1 + \gamma^2} r. \quad (3.11)$$

Proof. Translate d_D to zero. If the supporting line of e is $\nu \cdot \xi = b > 0$, then $r = (\nu \cdot q)/b$, $Dr = \nu/b$, and $Dt = L_D \nu/(br)$. Also

$$\xi'(s)Ds = \frac{1}{r} \left(I - \frac{q \otimes \nu}{br} \right),$$

when restricted to the sector plane. Since b is bounded below by the center clearance and $|q|/r$ by the maximum radius, these are the first estimates. Directly, $\partial_t R = r\xi/L_D$ and $\partial_s R = r\xi'$, proving the second. For the rectangle, on a short end $q_1 = \pm rL$, so $|Dt(v_1)| = 1/r$ and $|Ds(v_1)| \leq \gamma/r$. On a long side r depends only on the short coordinate, so the two values are 0 and $1/r$. Finally $|\xi| \leq L\sqrt{1+\gamma^2}$ and $|\xi'| = 1$. Splitting a straight side only changes the additive origin of s , so none of the estimates changes. $\square$

Proposition 3.8 (The graph exterior and its two colours). *Choose a small $r_{u,D} > 0$ in every subcell. In input target coordinates remove the full closed blocks X_i and the open cores of the interval prisms over $d_D + r_{u,D}(D - d_D)$, with their ends attached to the full blocks. The closure M_y of the remainder is a locally finite PL three-manifold with boundary. The radial projection in each subcell, identity on subedges, defines a continuous map*

$$\pi T : M_y \longrightarrow \Gamma,$$

where Γ is the subedge graph. Each component of M_y is a possibly noncompact graph handlebody. In particular every PL embedded compact interior two-sphere bounds a PL ball and

$$H_2(M_y; \mathbb{Z}) = 0. \tag{3.12}$$

For intermediate levels there are two standard families of proper walls: $A_D(t)$ is the perimeter at common radial label t , times the face-offset interval; $B_e(s)$ is the entire inverse of the intermediate subedge point under πT . Walls within one colour are disjoint. Each A is an incompressible annulus and each B is a disk. An incident pair has two boundary intersection points, one on each annular end; a nonincident pair has none. Consequently transverse individually standard walls with this boundary incidence have one spanning intersection arc for each incident pair, and otherwise only circles; every such circle is contractible in both walls.

Figure 2 shows one face product and the two wall directions. The proof also accounts for the offset polygons and polytopes over true edges and vertices.

Proof. Over an open true face the product formula is a polygon times an interval perpendicular to it, whose two ends are faces of the two adjacent full blocks. Over an open true edge it is an interval times a convex polygon; over a true vertex it is a convex three-dimensional offset polytope. These products glue by their common faces because they are subdifferentials of the same piecewise-quadratic function. The offset dimensions are exactly complementary by Proposition 3.3. Removing the central subpolygons and the full blocks therefore leaves the usual half-space or corner neighborhood at each boundary point; subdivisions turn all such corners into PL half-ball charts. This proves the manifold assertion. The prisms are disjoint and attach along disjoint disks after the $r_{u,D}$ have been chosen small. Their boundaries and the full-block boundaries give its locally finite boundary. The radial projection agrees on every shared subedge. On a true edge the map T is the projection along its polygonal offset fibre, so it gives the same value for all incident face products. This proves continuity of πT through the true edges and vertices.

Cut M_y at one standard meridian B over an interior point of each subedge. We verify that every resulting chamber is a ball. Fix a vertex w of the subdivided graph and a true face P incident there. Inside P , the annular portion projecting to the truncated star at w is a disk $A_{P,w}$: for each incident subcell take the radial strip from the first cut on one side of w , around w , to the adjacent cut.

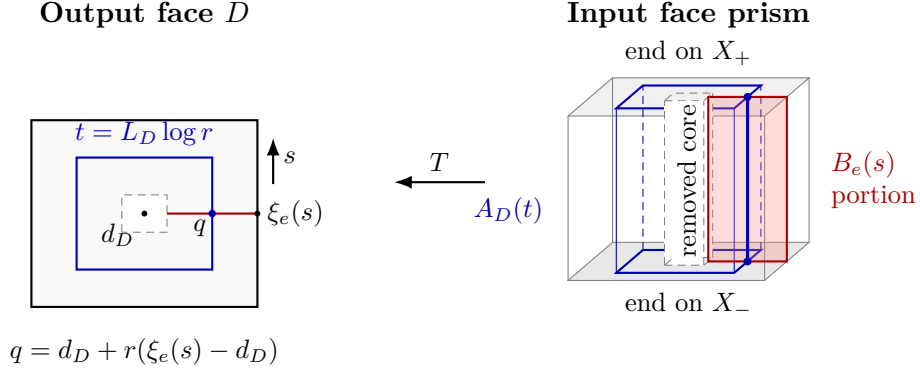


Figure 2: A rectangular face and its orthogonal input product, schematically and at independent scales; $X_{\pm}$ denote the adjacent full blocks. The quantizer T collapses the normal interval and rescales tangential directions by c^{-1} . After removing the central core, a radial perimeter times the interval is the blue annulus $A_D(t)$. The red rectangle is only the portion of $B_e(s)$ in this prism. Over a true edge, the complete meridian disk also contains its offset polygon and the spoke rectangles of all incident faces. The blue intersection interval joins the two ends of the annulus on the adjacent full blocks.

Successive strips meet along one radial side. Their boundary is a single polygonal curve alternating cut spokes and inner arcs; if $w \in \partial P$, add the one outer boundary arc through w . This is a disk fan, with empty intersection with ∂P in the interior case and one boundary arc in the boundary case. Its product with the face-offset interval is a ball.

If w is interior to a true face this product already is the chamber. If it is interior to a true edge, start with the polygon-offset prism over the truncated edge segment. Each incident face product attaches along one side disk, with disjoint relative interiors of the attaching disks. Attaching a ball to a ball along a boundary disk yields a ball: identify the disk with the flat disk in a half-ball chart and flatten the two collars, then extend the resulting boundary identification radially. This argument is PL after subdivision. Thus the edge chamber is a ball.

At a true vertex start with the full convex offset polytope. The incident edge products attach on its facets. The incident fan is the normal fan of that polytope: the equalities between active affine supports identify exactly the facet-edge incidences in question. After the edge attachments, each $A_{P,w}$ product attaches along the disk over its two-sided outer boundary arc and the face-offset interval. The two sides meet along the corresponding polytope edge. These are boundary disks whose relative interiors are disjoint from the other attaching disks. The same ball-gluing argument completes the vertex chamber. It also shows that its frontier meets adjacent chambers exactly in the chosen meridian disks. This local verification covers degeneracies of the original diagram; it uses the full active offset polytope rather than a simplicity assumption on the vertex.

Restoring the meridian collars therefore attaches ordinary one-handles to disjoint zero-handles. This is a graph-handlebody presentation, locally finite on compact sets. It retracts onto a one-dimensional graph, proving (3.12). For irreducibility, a compact PL embedded sphere meets only finitely many handles; include them and their endpoints in a finite graph chunk, cutting its remaining handles farther away from the sphere. Each component of this finite chunk is an ordinary compact handlebody. To recall the standard elementary argument, cut such a handlebody along its meridian disks. Put the PL sphere transverse to them and use an innermost intersection circle and the adjacent disk in the cut ball to remove intersection circles by disk surgery and isotopy. Induction reduces to a sphere in a ball, which bounds a ball by the PL Schoenflies Theorem; restoring each

removed disk collar restores a ball on the same side of the sphere. Equivalently this is the induction that attaching a boundary one-handle to a union of balls preserves irreducibility. The bounded ball lies in the finite chunk, hence in M_y .

The A product is visibly an annulus. Its core maps under πT to the reduced perimeter loop of D in Γ . That loop traverses successive distinct perimeter edges without cancellation, so every nonzero power is nontrivial in the free fundamental group of the graph. The induced map on the annular fundamental group is therefore injective; in particular the annulus is incompressible. At an artificial edge, B is the union of two spoke rectangles along their common interval, a disk. At a true edge it consists of the convex offset polygon and one spoke rectangle on each corresponding boundary interval. Gluing the rectangles to this polygon successively along boundary intervals again gives a disk. Their descriptions prove disjointness in each colour and the stated boundary incidence.

For transverse individually standard walls the intersection is a compact one-manifold, with boundary precisely the prescribed wall boundary intersections. An incident pair therefore has exactly one arc, joining those two endpoints, and any remaining components are circles. The arc joins the two annular ends and is spanning. Every circle bounds a disk in the B wall, hence is nullhomotopic in M_y ; injectivity of the annular fundamental group makes it nullhomotopic in A as well. On a disk or annulus such a simple nullhomotopic circle bounds a disk. This proves the last assertions. $\square$

3.4 Exact boundary data and edge matching

The target exterior M_y now has standard annuli and disks, but its new central-prism boundaries need not lie in the exact region of $\widehat{F}$. To pull these walls back with fixed PL boundary data, we first make the carrier exact near those boundaries. The changes are confined below a prescribed radial level a_D , leaving the calibrated data at higher radii unchanged. Their new local energy need only be finite: the radial margin lets the later target map be constant on the changed region. The construction must also retain enough control of fibres to preserve sampled edge intersections when it is opened.

Lemma 3.9 (A carrier exact on the central tubes). *Let $p > 2$. Let $T : \Lambda \rightarrow \Lambda$ have the properties of Proposition 3.3, and let $\widehat{F} : \Omega \rightarrow \Lambda$ have the carrier properties retained in Lemma 3.4. In particular, $\widehat{F}$ is a proper continuous $W_{\text{loc}}^{1,p}$ carrier with connected fibres and a countable set E_0 of nonsingleton values, it has the relative opening property of Lemma 2.10, and its prescribed maps over neighborhoods of the closed full blocks X_i are PL homeomorphisms. Use the locally finite face subdivision and central prisms of Proposition 3.8. Fix, separately from the central radii, a positive low activation radius a_D in every subcell D .*

The radii $0 < r_{u,D} \ll a_D$ can be chosen so that the resulting closed exterior M_y has the following additional properties. There exist a PL homeomorphism $h_ : \Omega \rightarrow \Lambda$, a proper continuous $W_{\text{loc}}^{1,p}$ map $\overline{K} : \Lambda \rightarrow \Lambda$, and*

$$F_* = \overline{K}h_*, \quad M_x = h_*^{-1}(M_y), \quad q_* = TF_*,$$

such that:

1. $\overline{K}$ is the identity on an open neighborhood of the full blocks and of all the removed central prisms. Consequently $F_*^{-1}(M_y) = M_x$, and $F_* = h_*$ on an open neighborhood of ∂M_x . These boundary data are PL and have the originally prescribed values near the full blocks.
2. $\overline{K}$, and hence F_* , is a fine limit of actual homeomorphisms with these exact data. Its fibres are nonempty and connected, and only countably many of its target values have nonsingleton fibres.

3. All changes from $\hat{F}$ occur in compactly localized, locally finite face neighborhoods whose old and new labels lie in $r < a_D$. The changes meet any prescribed fine value tolerance. Thus $F_* = \hat{F}$ in the complementary high region. On every compact portion of M_x the labels q_* , and the radial and perimeter coordinates used there, have finite ambient local Sobolev budgets. No uniform bound for these new low-region budgets is asserted.

There is a locally finite closed family S of possible limiting inner cube seams in the changed regions. Away from S , the construction has the local fibre formula proved below. The portions of S at which the construction is not already exact lie away from ∂M_x .

Proof. Fixing the target geometry and the reference homeomorphism. We first fix the target geometry before choosing the final PL approximation. Choose slightly enlarged, pairwise disjoint polyhedral balls around the X_i , contained with room in the exact PL regions of $\hat{F}$, and fix smaller open protection neighborhoods there. These protections, the central prisms, and all boxes and sampling margins specified next depend only on the prescribed target geometry and on the geometric constants of Lemma 2.9; they do not depend on a map h_* or K_0 .

Include the properness requirements at this stage. Fix a compact exhaustion $C_m \subset \text{int } C_{m+1}$ of Λ . Choose a positive 1-Lipschitz function τ so small that

$$\tau(z) < \frac{1}{4} \text{dist}(z, C_m) \quad \text{whenever } z \notin \text{int } C_{m+1}.$$

Such a minorant exists: only finitely many of these conditions apply near any fixed point, and the compact gaps make each applicable bound positive. All geometry and approximation choices below include the requirement that the resulting target-coordinate maps g , and also K_0 , move each y by less than $\tau(y)/10$.

Over a true face, the quantizer has an orthogonal product description with an interval as normal fibre. At each of its two ends the attachment disk of a sufficiently small central prism lies in one of the fixed exact neighborhoods. Choose an elongated affine box in the open face prism: its transverse inner rectangle contains the central cross-section with positive margin; its longitudinal inner interval contains the portion outside those two exact neighborhoods; and both longitudinal end strips of its larger box lie strictly inside the exact neighborhoods. The larger box remains strictly between the two endpoints of the full normal interval. It therefore avoids the full blocks themselves. First decrease $r_{u,D}$, relative also to the aspect of D , to put this whole transverse construction in $r \ll a_D$; then choose the shell thickness in normalized box coordinates. Decrease these transverse and shell sizes also to meet the fixed $\tau/10$ motion bound: in an affine box the lift displacement and every hole-buffer diameter are bounded by a fixed box constant times the normalized shell size, while the inner block will be the identity. This bound applies to all the later strict fills as their images lie in those same buffers. The two end margins absorb both that thickness and the sampling enlargements required below. Distinct boxes and their sampling neighborhoods are disjoint when they belong to distinct central prisms, and the neighborhoods form a locally finite family. This follows by choosing them inside the open face products and using their compact attachment margins. The tube portions beyond the inner longitudinal interval are already exact.

Here and below a sampling neighborhood includes a little more than the immersion image itself. The enlargement is needed for the degree tests in the fibre argument. All these neighborhoods are chosen with compact closure in the indicated face and end regions. Now choose a positive 1-Lipschitz minorant $\rho \leq \tau/10$ of all the resulting target value-closeness requirements, including their affine normalization factors. Open $\hat{F}$ by Lemma 2.10 and apply Lemma A.2, relative to the fixed enlarged protection balls, to obtain a PL homeomorphism h_* satisfying

$$|h_*(x) - \hat{F}(x)| < \rho(\hat{F}(x))/10.$$

Define $K_0 = \widehat{F}h_*^{-1}$. At $y = h_*(x)$, the Lipschitz property of ρ gives $|K_0(y) - y| < \rho(y)/9$. Thus all previously fixed sampling and properness tests hold. The map K_0 is proper, locally Sobolev, exactly the identity on the protected neighborhoods, and inherits the old relative opening property. Its local Sobolev constants may depend on h_* ; only their finiteness is used below.

The motion bounds already chosen imply, for every resulting map g and its sufficiently close homeomorphic approximants,

$$g^{-1}(C_m) \subset K_0^{-1}(C_{m+1}). \quad (3.13)$$

Indeed, $|g(y) - K_0(y)| < \tau(y)/5$, whereas $\tau(K_0(y)) > 9\tau(y)/10$. If $K_0(y) \notin C_{m+1}$, the defining bound for τ therefore prevents $g(y)$ from lying in C_m . All these requirements preceded the choice of h_* .

The lifted replacement. We describe a single normalized box. Let $B = [-1, 1]^3$ be its inner block, and use the shell, mesh, omitted balls V_σ , buffer balls V_σ^+ , and skeleton immersion i of Lemma 2.9. Here σ indexes shell tetrahedra; D continues to denote a face subcell. At depth scale l , the immersion has chart radii comparable to l in its domain and to s in its image. On each injective chart U ,

$$|Di| \leq Cs/l, \quad |D(i|_U)^{-1}| \leq Cl/s.$$

Prescribe $i = \text{id}$ on a whole outer mesh band, leave that band free of cap supports, and retain the old map there. On all tetrahedra whose fills are needed in the protected longitudinal end strips, put the immersion samplings inside the exact region of K_0 . Their lifts and fills are then the identity on an open neighborhood, including the necessary collars. This is a relative use of the skeleton construction in Lemma 2.9.

On the remaining usable lift region define the near-point branch

$$H(x) = (i|_{U_x})^{-1}(K_0(i(x))), \quad i(H(x)) = K_0(i(x)). \quad (3.14)$$

The sampling requirements give $|K_0 - \text{id}| \leq \varepsilon_{\text{lift}}s$, where $\varepsilon_{\text{lift}} > 0$ is a fixed closeness constant below all chart and buffer margins. Hence $|H(x) - x| \leq C\varepsilon_{\text{lift}}l$. The near-point branches agree on overlaps; their existence and agreement do not require injectivity of K_0 . Neither does the chain-rule estimate in the cutoff proof. On each tetrahedron it bounds the lift energy by

$$C_p(l/s)^3 \int_{\text{enlarged image sampling}} |DK_0(y)|^p dy.$$

Summing over depth scales gives the finite shell bound of that proof. Affine normalization introduces finite constants depending on this fixed elongated box, which is harmless here.

For every nonidentity omitted ball, use the elementary forward squeeze J_σ which collapses V_σ^+ to its center v_σ , is a homeomorphism off that closed ball onto the punctured output region, and is the identity near the tetrahedron boundary. The lifted boundary of the omitted hole lies strictly inside V_σ^+ . Define the new map as $J_\sigma H$ where the lift is used and as the constant v_σ throughout the omitted hole. These definitions agree on a whole collar. Identity-filled holes use neither a cap nor a nonidentity fill. Write J_{cap} for the disjoint union of the squeezes. Set the map equal to the identity on B and to K_0 beyond the replacement. This defines $\overline{K}$.

Continuity at the inner seam follows since all displacements on a depth- l tetrahedron are $O(l)$. More precisely, the near-branch lift and every cap stay on the exterior side of B . Indeed the Whitney scale comparability gives $\text{dist}(\sigma, B) \geq \alpha l$ for one fixed $\alpha > 0$. Our choice of $\varepsilon_{\text{lift}}$ includes $C\varepsilon_{\text{lift}}l < \alpha l/2$ in the lift bound. Every lifted point is then strictly exterior to B . Each cap is supported in an exterior tetrahedron and maps its support into itself, so cap postcomposition and the omitted-hole constants are strictly exterior as well, at every depth scale. The summed lift

estimates, followed by the Sobolev gluing argument in Lemma 2.9, prove local $W^{1,p}$ regularity. Constant holes cost no energy. At the outer band the immersion is the identity and no cap acts, so the seam is unchanged on a neighborhood. The reserved outer band and sufficiently small closeness bounds also keep cap supports and their inverse action inside the replacement image; hence no source outside the box is changed. Local finiteness now proves these assertions simultaneously for all boxes. Their low-region margins ensure that both old and new actual labels stay below a_D .

Homeomorphic openings and connected fibres. Apply the old opening property to obtain homeomorphisms $k_n \rightarrow K_0$ finely, preserving the exact regions and end-strip samplings. Lift k_n by the same near-point branches. The homeomorphic cutoff argument of Lemma 2.9 fills the omitted balls topologically, with images inside V_σ^+ and the required boundary values. Replace each cap by a strictly increasing radial squeeze tending to J_σ . The resulting maps $\bar{k}_n$ are homeomorphisms, agree with the exact data, and preserve the outer replacement image. On every omitted hole their outputs converge to the declared constant, irrespective of the choice of topological fill. Elsewhere the lift formula gives convergence; at the inner seams the $O(l)$ bound gives uniform convergence by first discarding sufficiently small depth scales. Local finiteness gives convergence on compacts, and diagonal choices give any prescribed fine accuracy.

By (3.13), the inverse images under all sufficiently close $\bar{k}_n$ of a fixed compact lie in one compact. In particular $\bar{K}$ is proper and onto: take a convergent subsequence of $\bar{k}_n^{-1}(y)$ for a given y . Its fibres are connected as well. For two points a, b in a fibre over y , choose closed balls $\bar{B}(y, \rho_n) \subseteq \Lambda$, with $\rho_n \downarrow 0$, containing both $\bar{k}_n(a)$ and $\bar{k}_n(b)$. Their inverse images are compact connected sets containing a, b in a common compact. A Hausdorff-convergent subsequence has connected limit in $\bar{K}^{-1}(y)$ containing a, b . This proves the claim.

Local fibre identification and exact boundary data. The following formula gives countability of the new exceptional values and transfers the wall tests in the next lemma to old relative-opening tests. Suppose y is not a cap center and set $x' = J_{\text{cap}}^{-1}(y)$. In a nonidentity lift region this inverse is unique, and x' lies outside the closed collapsed buffers V_σ^+ . Their fixed clearance from the omitted holes V_σ gives a ball $B_s = B(x', \alpha l)$ in the usable lift region, with an injective immersion chart on a larger neighborhood U . The joint immersion neighborhood of Lemma 2.8 supplies these charts also across ordinary skeleton seams; the limiting inner seams S are treated separately below. Choose $B_t = B(i(x'), \beta s)$ inside the enlarged image sampling neighborhood, reducing the fixed $\beta > 0$ so that its chart pullback lies in $B(x', \alpha l/2)$. This is possible by the inverse derivative bound Cl/s .

The closeness constant was chosen with $\varepsilon_{\text{lift}} < \beta/2$ and $C\varepsilon_{\text{lift}} < \alpha/2$. On ∂B_t , closeness to the identity gives degree one for K_0 at $i(x')$ and excludes that value from the boundary image. Hence the old fibre meets B_t and misses its boundary. Connectedness puts the entire old fibre inside B_t . In fact every point of this fibre lies within $\varepsilon_{\text{lift}}s$ of $i(x')$, by the same closeness test there.

Its chart pullback is therefore well inside B_s . Equation (3.14) and the near-branch choice identify this pullback with the local new fibre. The relative displacement bound excludes preimages on ∂B_s . Connectedness of the new fibre, applied once more, excludes any other preimage. Consequently

$$\bar{K}^{-1}(y) = (i|_U)^{-1} \left(K_0^{-1}(i(J_{\text{cap}}^{-1}(y))) \right). \quad (3.15)$$

All margins can be kept uniformly positive after shrinking to a small target neighborhood W of y . Thus this formula holds for *every* value in W , with one injective chart U , and not merely for its center. The same argument crosses skeleton seams; the outer band uses the identity immersion.

Take a countable cover by these charts. Outside cap centers, (3.15) shows that a nonsingleton value must arise from a chart inverse of an old exceptional value. There are countably many such values. Equivalently, a local diffeomorphism has discrete, hence countable, point preimages in this second-countable domain. Identity-filled regions have open identity behavior and thus singleton

fibres by connectedness. A point of an inner seam is also isolated in its fibre: on the inner side the map is the identity, and nearby exterior outputs stay strictly exterior even after the cap operation. Its entire fibre is therefore singleton. Finally, in an unchanged open region a singleton old fibre gives an isolated point in the new fibre, so a nonsingleton new fibre meeting that region must have an old exceptional value. These cases exhaust the construction and prove countability.

The identity neighborhood of the removed thick cores has singleton fibres, again because each such point is isolated in its connected fibre. It follows that $\overline{K}^{-1}(M_y) = M_y$, including the boundary, and gives the asserted exact boundary data after composing with h_* . The pullbacks of the affine inner cube boundaries form the stated locally finite family S ; the end strips make its nonexact portions disjoint from a neighborhood of ∂M_x . Finally, T is locally Lipschitz, and on M_y every central radius is positive. On compact portions only finitely many product and sector charts occur. Their radial and perimeter functions have finite Lipschitz constants there, and near the boundary F_* is PL. Sobolev composition, with local Lipschitz extensions of these coordinate functions, supplies the claimed finite ambient budgets. $\square$

The later wall estimates count intersections along the carrier's mesh edge paths. Opening the carrier must preserve these very hits, or the counts would no longer describe the pulled-back walls. The next lemma does this for the sampled wall families, using their whole-wall avoidance of nonsingleton values.

Lemma 3.10 (Exact edge matching when opening the central carrier). *Use the carrier and exterior of Lemma 3.9. Let $\mathcal{E}$ be a locally finite family of mesh edges in M_x , and let $\mathcal{W}$ be a locally finite family of sampled standard walls of Proposition 3.8. Assume:*

1. *the entire union $W = \bigcup \mathcal{W}$ is relatively closed in M_y and avoids all nonsingleton values of $\overline{K}$;*
2. *the actual hit set $H = (\bigcup \mathcal{E}) \cap F_*^{-1}(W)$ is finite on compact localizations, and no hit outside the exact regions lies on S .*

Then there are homeomorphisms $f_n : \Omega \rightarrow \Lambda$ converging finely to F_ , equal to F_* on a neighborhood of ∂M_x and on the other prescribed exact regions, for which*

$$f_n^{-1}(W_0) \cap e = F_*^{-1}(W_0) \cap e \quad \text{for every } W_0 \in \mathcal{W}, e \in \mathcal{E}.$$

In particular each restriction maps M_x onto M_y with the exact prescribed boundary map. The hypotheses concern sampled edge tests; they make no relative-opening assertion for arbitrary new closed tests accumulating through the inner lift scales.

Proof. Work first in the target-domain coordinates given by h_* . Consider a hit x outside an exact region, and write $y = \overline{K}(h_*(x))$. Because W avoids all new nonsingleton values, y is not a cap center. The seam-avoidance assumption permits a target neighborhood V in which the chart formula (3.15) holds with fixed margins. For each wall meeting y , take a compact patch P of that wall in V , containing a relative neighborhood of y . The formula shows that

$$C_P = i(J_{\text{cap}}^{-1}(P)) \quad \text{is compact and satisfies} \quad C_P \cap E_0 = \emptyset. \quad (3.16)$$

Indeed, a nonsingleton old fibre at any value of C_P would give a nonsingleton new fibre at the corresponding point of P , which is excluded by the hypothesis on the *whole wall*.

If the hit lies in an unchanged open region, the same conclusion uses a direct patch P . Its old fibre at y is singleton: otherwise the old connected fibre would have an isolated point where the new fibre is singleton. Properness then places the old inverse image of a sufficiently small neighborhood of y entirely inside the unchanged region. To see this, a contrary sequence of values tending to y and

preimages outside that region has a convergent preimage subsequence, contradicting the singleton fibre. For every value on the smaller wall patch the old fibre consequently agrees with the new one and is singleton. Hits in exact open regions require no additional tests.

There are only finitely many hits on each compact localization. Thus their neighborhoods can be chosen locally finitely, avoiding S wherever required. Only finitely many depth scales occur in each compact collection of these charts. Choose the old patch images in the compactly localized sampling neighborhoods used in the preceding Lemma. The resulting family of compact sets C_P is locally finite in Λ ; in particular its union is relatively closed. Adjoin the closed protected exact data, slightly inside their exact neighborhoods. All these old target tests avoid E_0 . Apply Lemma 2.10 to obtain arbitrarily close old homeomorphisms k_n agreeing with K_0 over every one of these tests and retaining the exact strips. Here agreement over C_P means the exact old evaluations on $K_0^{-1}(C_P)$; since these fibres are singleton and k_n is bijective, it also means equality of the inverse evaluations.

Lift and fill these homeomorphisms as before. For $y \in P$ the fixed branch and (3.16) give exactly the same inverse image of $J_{\text{cap}}^{-1}(y)$ under the new lift as under the carrier lift. Choose the strict radial cap openings so that their inverse maps agree with J_{cap}^{-1} on all tested patches. This is possible by changing a collapse only over an arbitrarily small ball about its center: each center has positive distance from the relatively closed union of the tests relevant to it, although no uniform distance is required. The ordinary piecewise radial interpolation is strictly increasing inside that ball and unchanged outside it. Null mesh sizes at the limiting seams and the reserved outer bands retain the convergence and support conclusions of the preceding proof. The resulting homeomorphism therefore has exactly the carrier inverse over every tested wall patch; global uniqueness follows from its bijectivity.

Shrink the source hit neighborhoods so their old images, and then their sufficiently close new images, meet walls only in the chosen patches. On the rest of the edges, fine closeness prevents new hits. Explicitly, after removing these neighborhoods, the edge remainder on each compact localization has compact image disjoint from the relatively closed wall union, hence a positive gap. An exhaustion and a positive minorant impose all these local gap conditions simultaneously. This proves the claimed equality for every wall and edge, and composition with h_* returns to source coordinates. The exact boundary neighborhood is retained throughout, so the homeomorphisms preserve M_x and its prescribed boundary map as asserted. $\square$

Remark 3.11. The edge hypotheses can be obtained by the later ACL and coarea sampling step: choose edges for which the relevant label paths are absolutely continuous, arrange that their intersections with the nonexact inner seams have parameter measure zero, and avoid the resulting null sets of scalar labels when sampling levels. Finite variation gives finitely many hits at almost every sampled level. Countably many exceptional values exclude only countably many radial or intermediate meridian levels; a graph vertex is not an intermediate meridian value. These sampling facts are hypotheses of the preceding Lemma, rather than an assertion that arbitrary edges or arbitrary sampled levels satisfy them.

4 Wall routing, guard barriers, and realization

Fix $2 < p < \infty$. We use the quantizer and carrier of Proposition 3.3 and Lemma 3.9, the exterior of Proposition 3.8, and the notation

$$M_y \subset \Lambda, \quad M_x = h_*^{-1}(M_y), \quad F_* = \overline{K}h_*, \quad q_* = TF_*, \quad q_0 = T\widehat{F}.$$

In particular h_* is fixed, $q_* = Th_*$ near ∂M_x , and the prescribed map on every full block is fixed. Both exteriors are irreducible and have vanishing absolute second homology. The two standard

wall families are the annuli $A(D, t)$ and meridian disks $B(e, s)$ of Proposition 3.8. Their radial and perimeter coordinates are

$$t = L_D \log r, \quad q = d_D + r(\xi_e(s) - d_D), \quad r \geq r_{u,D} > 0.$$

All families and coordinate decompositions below are locally finite. A compact wall is understood as a proper wall in the manifold with boundary M_x or M_y .

This section separates the topological conclusion from its quantitative inputs. The pre-stack geometry is supplied in Proposition 6.10; the capacities used to choose guards are fixed in Lemma 7.3, and Proposition 7.4 constructs the initial walls with those bounds. In particular, a large Lipschitz constant for a subsequently chosen extension is never used as an input to the guard construction.

4.1 Sampling transverse labels

Fix $0 < \zeta < 1$. Let j index a sample in one radial or perimeter feature. Write $[0, w_j]$ for its root-parameter interval and

$$p_j : [0, w_j] \longrightarrow I_j$$

for its increasing affine identification with a label interval about the sample. Within one feature the intervals I_j are ordered and disjoint, and $|I_j| \asymp c_* w_j$, where $c_* > 0$ is fixed and small. The number w_j measures transverse parameter length, not physical collar thickness. Routing will retain a finite union $E_j \subset (0, w_j)$ of closed parameter intervals with

$$|E_j| \geq (1 - \zeta)w_j. \tag{4.1}$$

The same affine p_j is used on every component of E_j . We first construct stairs for arbitrary such sets. Before the retained intervals are chosen, we use the whole-interval case $E_j = [0, w_j]$ to test labels. After realization, the stairs will be constant off $p_j(E_j)$, so their nonzero derivatives occur only where a wall parameter has been prescribed.

Lemma 4.1 (Stairs and sampling). *On a feature interval of length ℓ , one can choose N equal weights $w_j = w$ with $Nw/\ell = 1 + O(\zeta)$, and construct an endpoint-fixing nondecreasing stair S , constant outside the selected $p_j(E_j)$, such that its total increase over each I_j is ℓ/N . Its slope with respect to the occupied root parameter is at most $1 + O(\zeta)$, and its Lipschitz constant in the target label is $O(c_*^{-1})$. The stair can be made arbitrarily close to the identity on each compact feature.*

The choices allow finitely many forbidden intervals of arbitrarily small prescribed total length. For a fixed finite list of integrable edge-count tests, the weighted sample counts have expected density at most $1 + O(\zeta + c_)$ against label measure. They satisfy the corresponding upper bounds with arbitrarily small additive errors and probability tending to one as the slots are refined.*

Proof. Remove the forbidden intervals and divide each remaining component into slots. Choose the numbers of slots approximately proportional to component lengths. Taking w sufficiently small makes both rounding errors and the difference of each slot length from w as small as required. This also works for finitely many features using one common w ; no exact divisibility is required. Sample in each slot outside a small endpoint margin wide enough to contain I_j . The total omitted relative length and the rounding loss can be included in ζ .

Give S increase ℓ/N at constant speed over the occupied root length $|E_j|$, and no increase on the complementary pieces. Its speed is

$$\frac{\ell}{N|E_j|} \leq \frac{\ell}{Nw(1 - \zeta)} = 1 + O(\zeta).$$

Since $p'_j = |I_j|/w \asymp c_*$, the target-label slope is $O(c_*^{-1})$. Start the first flat portion at the left endpoint and use all N increments, so the final value is the right endpoint. The ordered slots show that the displacement is bounded by their largest length, the rounding error, and the total forbidden length. These quantities can all be prescribed small. The same conclusion holds when $E_j = [0, w_j]$, a version used before selecting stable occupied intervals.

For a nonnegative integrable count function n , sampling uniformly in the allowed middle of a slot J_j gives

$$\mathbb{E}[w_j n(\tau_j)] = \frac{w_j}{|J_j^{\text{allow}}|} \int_{J_j^{\text{allow}}} n(t) dt.$$

The density is at most $1 + O(\zeta + c_*)$. For bounded n , independence and $\max_j w_j \rightarrow 0$ make the variance of the weighted sum tend to zero. For general $n \in L^1$, truncate it, apply this argument to the bounded part, and use the first moment of the integrable tail. A finite list of tests can be imposed simultaneously. On an exhaustion, feature weights and additive errors can be chosen locally, with summable failure allowances. $\square$

4.2 The two-colour pre-stack input

Before routing, a pre-stack is a product neighborhood of a sampled source wall, equipped with its transverse scalar parameter. Opposite colours may still meet in circles as well as in the prescribed spanning arcs. The following definition records the simultaneous product structure and the pointwise slope control used in routing and in the guard estimates.

Definition 4.2 (Pre-stack contract). Let M_x, M_y be the source and target exteriors and let $h_* : M_x \rightarrow M_y$ be the fixed reference homeomorphism. A pre-stack system consists of a locally finite, countable, two-colour family of compact collars

$$\mathcal{E}_j : \Sigma_j \times [0, w_j] \longrightarrow C_j \subset M_x, \quad u_j(\mathcal{E}_j(z, \lambda)) = \lambda, \quad w_j > 0,$$

whose central sampled wall is the level $u_j = w_j/2$, subject to the following conditions.

1. Each $\mathcal{E}_j$ is a bi-Lipschitz homeomorphism onto its image. For colour A , Σ_j is an annulus whose core is essential in M_x ; for colour B , Σ_j is a disk. Each level is properly embedded, and $C_j \cap \partial M_x = \mathcal{E}_j(\partial \Sigma_j \times [0, w_j])$. The collars of a single colour are disjoint, including their boundaries. The family of closed collars is locally finite.
2. There is a specified affine increasing correspondence from $[0, w_j]$ to a true target label interval I_j . On ∂M_x every level has exactly the boundary trace of its specified standard target wall, pulled back by h_* . The increasing- u_j coorientation agrees with that target label coorientation at every boundary component; for an annulus this includes both ends. The target incidence says that an incident opposite-colour pair has two boundary intersection points, one on each annulus end, and that a nonincident pair has none.
3. For each opposite-colour pair j, k , the intersection $C_j \cap C_k$ is a finite disjoint union of bi-Lipschitz product boxes

$$Q \times [0, w_j] \times [0, w_k], \quad u_j = \lambda, \quad u_k = \mu,$$

where Q is a circle or a compact interval. The interval endpoints are exactly the parts of that box on ∂M_x . These are full parameter rectangles: no box terminates at an interior value of either parameter. The level sheets meet transversely in these product coordinates. There are no further intersection pieces. In particular, for an incident pair every level pair has

the specified single spanning arc, together with possible circles; a nonincident pair has only possible circles.

4. In each collar the subdivision by all its opposite-colour boxes is jointly bi-Lipschitz trivial over u_j , preserving every concurrent opposite parameter. Precisely, for some reference level there is a bi-Lipschitz product parametrization that carries every intersection box, every complementary piece, and every boundary piece at the reference level to its counterpart at each level; on an opposite box it preserves u_k . Its restriction to a boundary piece remains in ∂M_x . No uniform quantitative bound for this trivialization is required.
5. The ambient PL structure and the physical metric are compatible in the following explicit sense. Every locally finite arrangement of pieces obtained by finitely many constant or affine parameter cuts on each compact set admits common ambient coordinate changes that are locally bi-Lipschitz in the physical metric and make the pieces PL simultaneously. These changes preserve the manifold boundary and the specified incidences. Separate unrelated PL charts for individual pieces do not suffice for this requirement.
6. Every u_j is Lipschitz on its closed collar in the physical metric. Any subsequent estimate using a density G assumes, as an additional quantitative input, a nonnegative locally measurable G dominating the point-to-base upper local slopes

$$\text{lip}_{C_j} u_j(x) = \limsup_{\substack{y \rightarrow x \\ y \in C_j, y \neq x}} \frac{|u_j(y) - u_j(x)|}{|y - x|}.$$

Use the adjacent maxima at internal interfaces and include both parameter bounds at an essential intersection box. On exceptional collar-boundary or mesh-interface strata one may enlarge G to the actual local Lipschitz bounds of the fixed compact collars. These strata, in the common working coordinates or their bi-Lipschitz images, have zero volume; the enlargement therefore changes no volume capacity. This convention does not require a two-point Lipschitz bound across gaps of a nonconvex union of pieces. The contract alone asserts neither an integral estimate for G nor independence of that estimate from later choices of guards.

The full parameter rectangles and the simultaneous trivializations let the pieces cut out by one colour be tracked as the other root parameter varies, while preserving the concurrent opposite parameter. The common ambient coordinates serve a different purpose: they permit the eventual endpoint walls to be cut and filled in one PL structure. Constructions of these data, including the sufficient test in Lemma 5.8, are given in Sections 5 and 6.

4.3 Diagonal routing and conservation of periods

The switching construction is related to regular exchange in normal-surface theory [20]. Here the additional requirement is to preserve transverse parameter length and exact boundary labels, including along a possibly infinite rooted assembly. We work in the full source exterior M_x , including any boundary-collar continuations used in preparing the input walls. The constructions in this subsection assume the pre-stack contract of Definition 4.2. They are qualitative until explicit parameter-slope bounds are imposed. We also use the previously established irreducibility and vanishing $H_2(M_x; \mathbb{Z}) = 0$ of the exterior.

Fix one level in each of two intersecting opposite-colour collars. A circle of intersection bounds a disk on the disk wall. It cannot be essential on the annular wall, since an annular core is noncontractible in M_x . Consequently every intersection circle is trivial on both walls. A pair with no

prescribed boundary incidence has no intersection arc. An incident pair has exactly two boundary intersection points, one on each end of the annulus, and hence exactly one proper intersection arc. That arc spans the annulus. Any additional intersections are circles.

In a circle box write its coordinates as

$$(\theta, u, v) \in S^1 \times [0, a] \times [0, b], \quad a = w_j, \quad b = w_k.$$

Reverse u or v in this box if necessary. We arrange that the rim $v = 0$ of the strip on a j -layer faces the exterior of its bounded inner disk, and that the rim $u = 0$ on a k -layer has the same property. Call these two sides OUT and the sides $v = b$ and $u = a$ IN. The pattern trivializations ensure that these choices hold throughout the box. Introduce

$$d = a - u + v. \quad (4.2)$$

For $d \in (0, a + b) \setminus \{a, b\}$ the segment of this level in the parameter rectangle has one endpoint on OUT and the other on IN. Its product with S^1 is an annulus. On OUT the two sides give the d -intervals $[0, a]$ and $[a, a + b]$, while on IN they give $[0, b]$ and $[b, a + b]$. Thus these annuli match the combined OUT parameter length to the combined IN parameter length bijectively, by partial affine maps of slope 1 or -1 . The corner values $0, a, b, a + b$ are excluded. No diagonal replacement is made in an arc box: both original parameters are retained there. Figure 3 shows this parameter matching.

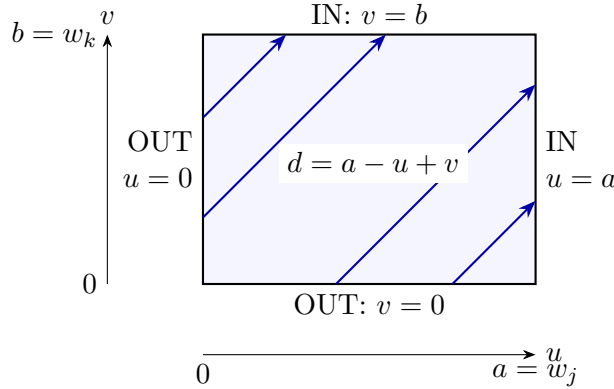


Figure 3: The circle-box parameter rectangle. Each regular d -segment joins one OUT side to one IN side, with partial parameter transport of absolute slope one. Its product with the circle is the replacement annulus. Corner values are discarded; essential arc boxes retain both parameters and are not routed.

Proposition 4.3 (Routing and good parameters). *For every pre-stack root j , outside a null subset of $(0, w_j)$, the diagonal rule constructs a connected, oriented, properly embedded surface $L_{j,\lambda}$ whose boundary, with coorientation, is the prescribed boundary of the original j -level at parameter λ . Different constructed surfaces are disjoint except for the prescribed essential arc intersections between opposite-colour roots. Each such surface is obtained by attaching punctured disks and annuli along a rooted tree; it may have ends at infinity. The following stronger local finiteness holds for each constructed surface: for every compact $K \subset M_x$, only finitely many of its reached states have an original parent collar meeting K .*

Proof. Cut every original level along the open strips belonging to circle boxes. The circles are trivial and disjoint on that level. Their bounded disks are therefore nested or disjoint. There is one

root piece containing the actual boundary of the wall, and, for an annulus, both actual boundary components belong to this same piece. The root is an annulus or a disk with finitely many open disks removed. Every other piece is a punctured disk. Each nonroot piece has exactly one IN rim, namely its rim toward its parent in the nesting, and any remaining rims are OUT. The root has no IN rim. All essential arc boxes lie on root pieces: a spanning arc or a proper arc with endpoints on the actual boundary cannot enter a bounded disk through one of its disjoint circle boundaries. There are finitely many pieces in any one collar, and they are tracked over its whole parameter interval by the pattern trivialization.

A state is a pair (P, t) consisting of such a tracked piece and its local parameter. Starting at the root state (R_j, λ) , follow every OUT rim through its diagonal annulus to the matched IN rim, and continue at that state. Every nonroot state has exactly one incoming predecessor wherever the matching is defined: its unique IN rim has precisely one OUT partner. Root states have none. This proves that no state reached from a root can repeat. Indeed a repetition along an ancestral path would give a cycle of predecessors, which could never terminate at the root. If two distinct ancestral paths reached the same state, following its unique predecessors backwards would force their paths, root indices, and root parameters to coincide. This also rules out duplication between different root-parameter starts. The states reached from one start consequently form a tree, with finite branching at each state. Cycles or infinite backward orbits that are not reachable from any root play no role.

There are countably many tracked pieces and countably many finite transition itineraries. On its domain each itinerary maps the starting parameter to a local parameter by $t = \lambda + c$ or $t = -\lambda + c$. A corner or breakpoint condition therefore excludes at most one starting value for that itinerary. Discard the union of these countable sets. All transitions then occur along nondegenerate segments and are local partial isometries of parameter intervals.

Here is the estimate that controls possibly infinite trees. For a tracked piece P belonging to collar C_k , let $N_P(j, \lambda)$ count the states of the tree started at (j, λ) that use P . The image parameter sets in $(0, w_k)$ of all valid itineraries from all roots to P are pairwise disjoint. Otherwise one state would have two backward ancestries. Each itinerary preserves one-dimensional Lebesgue measure. Decomposing its valid domain into its measurable partial intervals and using countable additivity gives

$$\sum_j \int_0^{w_j} N_P(j, \lambda) d\lambda \leq w_k. \quad (4.3)$$

The valid domains are measurable: finite itineraries impose affine interval conditions, and the excluded parameter set is countable.

For a compact K , let $\mathcal{P}(K)$ be all tracked pieces whose original parent collars meet K . This set is finite, by local finiteness of the compact collars and the finite number of pieces in each collar. Summing (4.3) yields

$$\sum_j \int_0^{w_j} \sum_{P \in \mathcal{P}(K)} N_P(j, \lambda) d\lambda \leq \sum_{P \in \mathcal{P}(K)} w_{k(P)} < \infty.$$

It follows that for almost every parameter of every root, only finitely many reached states have parent collars meeting K . Intersect these full-measure sets over a countable compact exhaustion of M_x . Call the remaining parameters good, also requiring avoidance of all corner values.

For a good parameter, glue all reached pieces to the associated diagonal annuli. The gluing is locally finite in the ambient manifold: a band meeting a compact set lies in a circle box, hence in the original collar of its parent state, and those states are finite in number by goodness. Each piece or band is compact and includes its attaching rims. Thus the assembly is locally closed and

properly embedded. Each rim has exactly two local half-surface pieces, giving an ordinary surface chart there. The pre-stack charts make the same assertion at corners and at the actual manifold boundary. Connectedness follows from the tree construction. Only the root piece meets the actual boundary, so the boundary of the assembly is exactly the original root boundary.

Orient the root according to its prescribed coorientation in the oriented three-manifold. At each tree edge orient the attached band and then its child piece so that the common boundary orientations cancel. There is no orientation consistency obstruction because no state is attached by two ancestral paths. All child pieces are planar and all bands are annuli. A finite tree assembly, capped at its frontier rims by disks, is therefore a disk or annulus of the original root type: removing an inner disk and attaching a punctured disk followed by disks at its remaining holes simply replaces that inner disk by a disk.

Finally, two assemblies cannot meet in one original piece at the same parameter or in one diagonal annulus, by uniqueness of backwards ancestry and the bijective diagonal matching. Their only possible remaining intersections are in arc boxes. Those boxes belong to root pieces on both sides and are exactly the retained essential intersections. A single assembly contains only one root, so these intersections cause no self-intersection. $\square$

Remark 4.4. The argument uses ordinary Lebesgue length in each original parameter interval and partial isometries between those intervals. The widths w_j need not be equal or commensurable. Goodness controls states whose *entire original collars* meet a compact set; controlling only the routed fragments would not suffice for the cap argument below.

The routed surface has the right boundary, but it need not have been obtained by a compact isotopy. To compare its crossings with those of the standard wall, we truncate the routing tree and cap its frontier. This reduces every compact path test to a compact homology calculation.

Proposition 4.5 (Conservation of oriented periods). *Let $L = L_{j,\lambda}$ be a good routed surface. Let $W_{j,\lambda} \subset M_x$ be the pullback by h_* of the standard cut at the exact affine label corresponding to $\lambda \in (0, w_j)$. Give both surfaces the coorientation of increasing standard parameter. For every compact path γ with endpoints on $\partial M_x \setminus \partial L$, their relative oriented intersection numbers agree:*

$$I(\gamma, L) = I(\gamma, W_{j,\lambda}). \quad (4.4)$$

Here intersection numbers may be defined after general-position perturbation relative to the endpoints. If the original path has finitely many hits on L , its signed local crossings give the same number and

$$|I(\gamma, L)| \leq \#(\gamma \cap L),$$

where hits are counted with their occurrences along the path.

Proof. Truncate the routing tree at a finite depth n . At every frontier OUT rim, cap with the original inside disk bounded by that rim in its original wall at the corresponding local parameter. Orient each cap to cancel the frontier boundary. The result is a finite compact oriented two-chain A_n . The caps need not be disjoint from one another or from the truncated assembly; only their boundary identities are used. Every cap is contained in the original parent collar of its frontier state.

Let K be a compact neighborhood of the path, or of any prescribed compact homotopy of paths relative to their endpoints. Goodness says that only finitely many reached states have original collars meeting K . Their depths have a finite maximum. If n exceeds that maximum, every omitted piece and every new cap lies outside K . Taking a slightly larger compact neighborhood first shows that A_n agrees with L in a neighborhood of the path or homotopy. This is the reason for using whole parent collars in the definition of goodness.

The actual boundary of A_n is the root boundary. The pre-stack boundary contract, including agreement of coorientations at both annular ends, gives

$$\partial A_n = \partial W_{j,\lambda}$$

as oriented one-chains. A common subdivision and boundary parametrization makes these chains literally identical. Hence $Z_n = A_n - W_{j,\lambda}$ is an absolute compact two-cycle. Since $H_2(M_x; \mathbb{Z}) = 0$, it bounds a finite compact three-chain B_n .

We spell out why the endpoints on the manifold boundary cause no extra intersection term. First take the path in the interior except at its endpoints, using a boundary collar and keeping the endpoints fixed. Neither endpoint lies on the support of Z_n at the boundary, since the two surface boundaries agree and the endpoints avoid that common boundary. Push the cycle and its bounding chain into the interior by a sufficiently small collar push. During this push the cycle avoids neighborhoods of the two endpoints, so its intersection with the path is unchanged. The pushed three-chain is disjoint from the path endpoints. The boundary formula for oriented intersections, after general position, therefore gives

$$I(\gamma, Z_n) = I(\gamma, \partial B_n) = 0.$$

The same formula applies to the pushed chains; the notation suppresses that harmless push. Since $A_n = L$ near the path, this proves (4.4). For a homotopy relative to the endpoints, choose one n for its entire compact image. The same finite-chain argument proves homotopy invariance. Equivalently, one can push that homotopy into the interior away from the fixed endpoint neighborhoods before applying the intersection formula.

If the unperturbed path has finitely many hits, choose disjoint small parameter intervals about those hits, each mapping into a two-sided surface chart. Before and after its isolated hit the path lies on a definite local side. Its local algebraic crossing is 1, -1 , or 0, according to these two sides and the coorientation. A sufficiently small general-position perturbation has exactly that total algebraic crossing in the chart. The same reasoning works in a proper half-space chart at a boundary hit; push its path segment slightly inward and keep its ends fixed. Outside these intervals there are positive gaps on compact remainders, so no new crossings are needed. Summing proves the asserted agreement and the bound by the number of hits. No isotopy relative to any later guard system has been used. $\square$

4.4 Graph potentials and guards

On M_y put $\beta = \pi T \in \Gamma$, with subedge lengths inherited from Euclidean arclength. Define α in a metric star having one branch for each subcell D , with branch coordinate $b = -t \geq 0$ and common node $b = 0$. Thus all $r = 1$ points have the same α . The coordinates are continuous and locally Lipschitz in the product charts. If $q = T(y)$, then

$$|q - \beta| \leq Cb. \tag{4.5}$$

At a fixed compatible β on one branch, the physical point varies with b at speed at most C ; at fixed b in one subcell it varies by at most the Euclidean displacement of β . These follow directly from the polar formula and $L_D \asymp \max |\xi_e - d_D|$.

For a layer rooted at $A(D, t_L)$ write α_L for its standard star label. For a layer rooted at $B(e, s_L)$ write β_L for its standard perimeter label. The following potentials, each 1-Lipschitz in the

appropriate graph metric, detect distance from the entire physical support of the root layer:

root	star potential	perimeter potential
$A(D, t_L)$	$\text{dist}(\alpha, \alpha_L)$	$\text{dist}_{\mathbb{R}^3}(\beta, \partial D)$
$B(e, s_L)$	$\text{dist}(\alpha, \mathcal{S}_e)$	$ \beta - \beta_L $

(4.6)

Here $\mathcal{S}_e$ is the subtree consisting of the common node and all full branches corresponding to subcells incident to e .

To verify detection, suppose both displayed distances are small. In the A case, if the guard is on the same branch, compare at a nearby perimeter point of ∂D and then at the same b . On another branch, small star distance implies that both branch heights are small, so (4.5) applies. In the B case, on a compatible branch compare using the same b and β_L ; on an incompatible branch small distance to $\mathcal{S}_e$ makes the guard height small. Thus the physical distance to the standard support is at most a fixed multiple of the sum of the two potential values. Both potentials vanish on that support. Moreover, when the potential tests the colour opposite to the root, it vanishes identically along every feature having an essential intersection with the root.

Definition 4.6 (Guards and prior capacities). Choose a locally finite compact cover of M_x by smaller buffered coordinate regions K_a , with larger regions U_a carrying fixed bi-Lipschitz ball or half-ball charts. All smaller regions have positive margins in the larger charts. Fix finite numbers C_a as upper capacities for

$$\int_{U_a} G^p \leq C_a. \quad (4.7)$$

These numbers, the coordinate bounds and the prescribed continuous label accuracies are fixed before guards, sample weights and the final resolution are chosen.

A guard system is obtained from a sufficiently fine homeomorphic approximation h_g to F_* , agreeing with h_* on the exact boundary data and respecting the exteriors. Choose a locally finite source net of sufficiently small local spacing. For every net point g , choose two paths in the constant Th_g fibre from g to boundary anchors whose h_* -images are extremal points of the target fibre on distinct full blocks, and let κ_g be their union. Thus κ_g is compact and connected, and each of its points has a path within κ_g to a boundary anchor. Write $q_g = Th_g(g)$ and require Th_g to be sufficiently close to q_* . The pre-stacks must be disjoint from all κ_g .

The guards have a locally uniform positive reach, independent of the net spacing and the approximating h_g . Indeed a target fibre in the exterior has paths to at least two extremal endpoints on distinct full blocks, also over true edges and vertices. Those endpoints have positive mutual separation on compact label tests. Their source positions use the fixed boundary map h_*^{-1} , so a guard through any one point has uniformly positive diameter. Properness of Th_g makes the family of guards locally finite: if a guard meets a fixed compact set, its label belongs to a compact label set, and its net point belongs to the compact inverse image of that set.

For later use, the order in Definition 4.6 can be implemented without a relative isotopy fixing guards. Reserve tiny forbidden parameter neighborhoods of the guard labels in each feature before sampling. Their total length can be arbitrarily small. Properness bounds the number of relevant guards for each compact feature using the prescribed net; hence the accuracy of h_g can be chosen before the actual gaps are placed. Carrier levels outside slightly enlarged gaps have positive buffers from the guards. Make the mesh and boundary motions smaller than these buffers. The preparation in Proposition 5.6 ends with wall pieces lying only in tetrahedra incident to original possible edge hits. Controlled placement, thickening and cell-sized snapping then stay within the buffers. This controls the final wall supports; no normalization isotopy relative to the guards is required. This argument is

useful only when (4.7) remains valid at all such smaller resolutions; that is the independent capacity assertion of Lemma 7.3, with the geometry of Proposition 6.10. Proposition 7.4 carries out the preparation with these choices.

Lemma 4.7 (Crossing lower bound). *Assume the pre-stack hypotheses of Definition 4.2 and its slope majorant G , and let the pre-stacks avoid the guards. Suppose a guard label has physical distance at least $\delta > 0$ from the standard support of a good routed layer L . On a fixed compact separation test, sufficiently accurate feature stairs give a constant $c_\delta > 0$ such that every compact rectifiable path σ joining L to that guard satisfies*

$$\int_\sigma G d\mathcal{H}^1 \geq c_\delta. \quad (4.8)$$

The constant depends on the fixed geometry and separation test, not on the sample weights or the number of switches in an itinerary.

Proof. We may assume $\int_\sigma G d\mathcal{H}^1 < \infty$; otherwise the conclusion is immediate. The coarea bound below will make the weighted crossing sums absolutely integrable.

Use the whole-interval stairs of Lemma 4.1, before restricting to stable occupied intervals. They act on the star and on Γ , fixing vertices and each edge setwise; on the star use $b = -t$ with its orientation reversed. Choose one of the potentials U in (4.6) giving a separation comparable to δ . For each root j of the tested colour set

$$c_j(\lambda) = \frac{d}{d\lambda} U(S(p_j(\lambda))). \quad (4.9)$$

These coefficients are uniformly bounded. In the opposite-colour case they vanish almost everywhere on all essentially incident features. The potential at the boundary anchor of L is as close to zero as required, whereas the potential at the guard anchor stays quantitatively positive.

Extend σ along L and along the guard to a path γ from a boundary anchor a_L of L to a boundary anchor a_g of the guard. For almost every tested root parameter with nonzero coefficient, the added portions avoid the corresponding routed layer: the guard avoids all pre-stacks, distinct routed layers are disjoint except at essential root boxes, and coefficients on those excluded essential features vanish. The layer containing the chosen starting point itself is a single null parameter value. Let ℓ denote the tested graph coordinate, either α or β . The weighted crossing identity is

$$\begin{aligned} \sum_j \int_0^{w_j} c_j(\lambda) I(\gamma, L_{j,\lambda}) d\lambda &= \sum_j \int_0^{w_j} c_j(\lambda) I(\gamma, W_{j,\lambda}) d\lambda \\ &= U(S(\ell(h_*(a_g)))) - U(S(\ell(h_*(a_L)))). \end{aligned} \quad (4.10)$$

Here the sum is over roots of the tested colour. The first equality is Proposition 4.5. For the second, choose a fixed piecewise regular representative of γ , homotopic relative to its endpoints. It meets only finitely many standard features; its graph-coordinate paths are piecewise Lipschitz. Oriented one-dimensional level counting and the Fundamental Theorem of Calculus on every graph edge give the displayed difference of $U \circ S$ at the two anchors. There is no error accumulated along a cycle or a long path.

On the other hand, one-dimensional coarea bounds the integrated number of hits on σ . Within an unchanged pre-stack piece it uses u_j ; within a transition rectangle it uses $d = w_j - u + v$. The upper local slope of d is at most the sum of the two original slopes. Partial transition itineraries have derivative $+1$ or -1 , and Proposition 4.3 gives a unique reachable ancestry for every piece-value and every diagonal-segment value. Thus coarea pays for all reached root layers there without an

itinerary multiplicity. Same-colour disjointness bounds overlap; at an essential box count both parameters. Apply coarea on the closed subsets of path portions and partition the affine itinerary rules countably if necessary. Interface slopes use the convention in Definition 4.2. More explicitly, the fixed parameter function is Lipschitz on each closed collar. On the closed subset of times when a rectifiable path lies in that collar, its scalar trace has a Lipschitz extension; at almost every density time its absolute derivative is bounded by the point-to-base upper slope times the path speed. The one-dimensional coarea inequality on that subset therefore uses exactly this G . Hence

$$\sum_j \int_0^{w_j} \#(\sigma \cap L_{j,\lambda}) d\lambda \leq C \int_\sigma G d\mathcal{H}^1.$$

The right side is finite by assumption. Hence almost every count is finite and the weighted signed sum in (4.10) is absolutely integrable. The appended portions contribute zero for almost every parameter with nonzero coefficient, so the displayed coarea bound controls its absolute value. The bounded coefficients and the positive potential difference prove (4.8).

There is no infinite list of conflicting stair accuracies hidden in this argument. The perimeter projection is proper. Near a compact guard perimeter-label test only finitely many subcells and subedges occur, and they use finite lists of accuracy requirements. For a remote A feature use distance to its whole ∂D ; for a remote B feature use distance to its whole closed subedge. These supports are invariant under their stairs, and only stair accuracy near the guard label is needed. The proper map πq_* sends a locally finite compact source cover to locally finite compact perimeter-label sets. Choose locally finite, relatively compact enlargements of these sets and accuracies smaller than their margins. This supplies consistent local prescriptions on all features. $\square$

Lemma 4.8 (Spherical path estimate). *Let $p > 2$, and let Σ_R be a sphere of radius R , either whole or intersected with a closed half-space containing its center. There is a constant C_p such that, for any two points $z_1, z_2 \in \Sigma_R$ and nonnegative measurable g with finite surface L^p norm, some rectifiable path in Σ_R joining them has cost at most*

$$C_p R^{1-2/p} \left(\int_{\Sigma_R} g^p d\mathcal{H}^2 \right)^{1/p}. \quad (4.11)$$

Fixed bi-Lipschitz chart changes alter only the constant.

Proof. Scale first to $R = 1$. Choose a fixed cap strictly above the equator parallel to the cutting plane. From an endpoint on or above the equator, shortest arcs to cap points stay in the upper hemisphere. For an endpoint below the equator, restrict cap points to those with nonnegative horizontal scalar product with its radius direction. This retains a fixed fraction of the cap. The initial vertical velocity of the shortest arc is then nonnegative. While its height is negative, the geodesic equation gives nonnegative second vertical derivative; once it becomes positive it cannot return negative before its positive endpoint on an arc shorter than π . The arc therefore stays above the height of its starting point and inside the prescribed half-space.

Average the cost of these arcs over the permitted cap fraction. Spherical polar integration from the endpoint gives a kernel bounded by $C/\sin\theta$ against surface measure. Its conjugate power is integrable precisely because $p' = p/(p-1) < 2$; the endpoint and antipodal singularities both satisfy $\int_0^\epsilon \theta^{1-p'} d\theta < \infty$. Hölder's inequality bounds the averaged cost by $C_p \|g\|_{L^p}$. Do this at both endpoints. Join the resulting two cap points by a shortest arc in the cap's containing upper hemisphere; averaging also this middle cost has the same bound. Some choice of the two cap points has total cost bounded by the sum of the averages. Scaling surface measure and length restores the factor $R^{1-2/p}$. The same proof bounds the costs after a fixed chart change, using its length and surface-measure constants. $\square$

Proposition 4.9 (Guard barrier). *Fix the data of Definition 4.6, the capacities C_a and a positive continuous local layer-support accuracy. There are sufficiently fine net spacings and sufficiently small approximation and stair errors, depending only on these prior data, with the following property. If a pre-stack system satisfies (4.7) and avoids the resulting guards, then every point of every good routed layer is within the prescribed local accuracy, under q_* , of that layer's standard physical support.*

Proof. It suffices to impose countably many locally finite compact tests with positive accuracy constants. Suppose one fails: at a point x of a good layer, its physical-support discrepancy exceeds δ . The layer is connected to its actual boundary, where its label is exact. Uniform continuity of q_* on a buffered chart therefore gives a path stretch of the layer of diameter at least $a > 0$ on which the discrepancy exceeds $\delta/2$. The number a can be chosen from the fixed smaller-chart margins, the modulus of continuity and δ , independently of the layer and sampling. This also applies near the manifold boundary, using half-ball charts; a discrepant point cannot end its path in the exact boundary without leaving this discrepant neighborhood.

For small h , pack at least ca/h disjoint chart balls of radius $2h$ centered along this stretch. A maximal separated selection on a connected set of diameter a gives this elementary packing bound. Place a guard net point within $h/2$ of each center. Its label is still separated by a fixed fraction of δ from the layer support, by uniform continuity and the chosen approximation accuracy. The layer stretch and the guard each meet every chart sphere of radius between h and $2h$ about the center: each has a point inside that radius and a connected continuation outside it. For the guard this uses the prior uniform reach; for the layer it uses the diameter of the selected stretch and smaller-chart margins.

Every spherical path between such two hits has G -cost at least a fixed $c_\delta > 0$ by Lemma 4.7. Apply Lemma 4.8 to the whole or truncated sphere. For almost every radius $R \in [h, 2h]$,

$$\int_{\Sigma_R} G^p d\mathcal{H}^2 \geq cR^{2-p}.$$

Integrating in radius gives at least ch^{3-p} in each chart ball. The disjoint balls therefore force

$$\int_{U_a} G^p \geq cah^{2-p}. \quad (4.12)$$

Since $p > 2$, choose h using the previously fixed C_a to contradict (4.7). All constants used before choosing h come from the fixed test, not the pre-stack resolution or weights. Spheres whose centers are near the boundary are exactly the half-space intersections in Lemma 4.8. Paths are allowed in the whole manifold chart, including through guards; no estimate on a punctured chart is used. A locally finite choice of spacings, with positive continuous minorants for the requested errors, proves all the compact tests simultaneously. $\square$

4.5 Stable occupied stacks and their realization

Choose the layer-support errors of Proposition 4.9 also small enough toward infinity to have two consequences. A layer with a fixed compact standard support is confined to a compact source set; and on any fixed compact source set only finitely many root indices can occur. To make these choices explicitly, take a compact exhaustion of the locally finite label complex, enlarge each member by a compact neighborhood, and use properness of q_* to pull these neighborhoods back. Require the local error outside the corresponding inverse image to be less than its positive separation from the smaller label compact. Conversely, on a compact source test choose the error small enough to keep any meeting standard support in a fixed compact enlargement of its q_* image. Only finitely many compact features, and finitely many samples on each feature, occur there.

Lemma 4.10 (Stable intervals). *Under these confinement choices, a good routed layer is a compact disk or annulus of its root type. For every root j and every $\zeta > 0$ one can choose finitely many separated closed intervals $E_j \subset (0, w_j)$ satisfying (4.1), such that their routed layers form compact, locally finite, bi-Lipschitz product stacks. These products preserve all concurrent essential parameters and the exact boundary data. Different stacks are disjoint except for the genuine essential-pair products.*

Proof. A good tree visits only finitely many states whose entire initial collars meet a given compact set. Confinement therefore makes the whole reached tree finite. It has no frontier, and attaching its finite tree of punctured disk pieces merely fills the root's inner disk holes by disks. Thus its completed surface has the original disk or annulus topology and is compact.

All corner and breakpoint values have been discarded. Finitely many strict affine itinerary inequalities then remain valid on an open interval about the starting parameter. The same finite tree and switching sides persist throughout this interval. The good parameters have full measure, so a finite collection of separated closed intervals inside these stability neighborhoods captures at least $(1 - \zeta)w_j$ of the root interval. Locally finite confinement persists by continuity from the dense good parameters.

For product triviality use the pre-stack trivializations under the affine parameter transports on each reached piece. In a circle box, parametrize each nonvanishing diagonal segment between its fixed pair of sides by fractional affine position. Its product with the circle gives a band bundle. Match the boundary-circle parametrizations of this band to the adjacent piece trivializations: relative to a reference parameter, lift the orientation-preserving circle reparametrizations continuously to monotone maps of $\mathbb{R}$ commuting with integral translation, and interpolate their lifts across the band. On a compact stability interval both the maps and their inverses have finite Lipschitz bounds, as do the segment fractions. Gluing the finitely many bundles gives a bi-Lipschitz product. Essential boxes lie on the root piece and their other parameter is preserved by the original trivialization, so the modifications do not alter them. The compatible changed-coordinate PL hypothesis in Definition 4.2 applies to all endpoint arrangements. $\square$

Theorem 4.11 (Conditional wall realization). *Fix the quantizer, carrier, exterior and boundary data above. Fix $0 < \zeta < 1$, $c_* > 0$, the requested locally positive label and value accuracies, and locally finite buffered charts with finite capacities C_a . Choose guards and feature-stair accuracies in the order of Definition 4.6 and Proposition 4.9. Suppose the resulting sampled pre-stacks satisfy Definition 4.2, avoid these guards, and admit a slope majorant G obeying the previously fixed capacities (4.7). The sample intervals are ordered, have the affine parametrizations p_j , and satisfy the slot, gap and accuracy choices of Lemma 4.1.*

Then there are finite unions of occupied closed intervals E_j satisfying (4.1) and a locally bi-Lipschitz homeomorphism

$$h : M_x \longrightarrow M_y, \quad h|_{\partial M_x} = h_*|_{\partial M_x},$$

which sends every occupied routed layer at parameter λ to the standard layer at label $p_j(\lambda)$, with the same affine parametrization on all occupied components. It extends by the prescribed h_ on the removed cores and tubes. Moreover $q_h = Th$ is as finely close to q_* on M_x as prescribed.*

The Theorem uses only the preassigned capacities for its choice of guards. Existence of a system meeting its hypotheses at that resolution is the separate content of Proposition 6.10, Lemma 7.3, and Proposition 7.4.

Proof. Lemma 4.10 supplies the occupied stacks. We first construct h with their exact parameter correspondence and then prove the asserted label accuracy.

Cut at both endpoints of every occupied B interval, thus at multiple gates on every subedge. The corresponding target components are balls by the graph-handlebody description in Proposition 3.8; this includes the intermediate chambers along a subedge. A target ball boundary consists of a punctured-sphere patch of ∂M_y and its distinct gate disks. At the source the corresponding fixed boundary patch, capped with the new gates, is an embedded sphere. Trim a small product collar of the manifold boundary and carry this sphere into the trimmed copy. It bounds a ball by irreducibility. The ball lies on the inward side: the discarded collar escapes to the original manifold boundary without meeting the sphere, so it cannot be contained in the ball. Pulling back from the trimmed copy gives the required ball bounded by the original patch and gates.

No additional gate lies inside this ball, since its boundary lies outside the prescribed patch and it is disjoint from the ball's gates. Along every gate, the two prescribed patch-balls occupy its two local sides, as already seen near its rim.

The patch-ball interiors are disjoint. Perform the following comparison in the preceding trimmed construction. For any two patch-balls, choose a point just inside each of their distinct boundary patches, lying outside the other ball. If the capped spheres share gates, their opposite local sides permit arbitrarily small compatible collar pushes there that make the spheres disjoint while preserving these two points. If their interiors overlapped, the pushes could also retain a point of that overlap. The resulting disjoint spheres would then bound nested balls, contradicting one of the two distinguished points. Hence the original interiors cannot overlap.

There is no leftover component. Indeed each patch-ball minus its gates is a relatively open and closed connected region off the gate system. A compact transverse path from an unassigned component to an outer boundary patch would have finitely many gate crossings by local finiteness; it can neither have a first entry through a gate, whose two sides are already assigned, nor reach the endpoint without such an entry. Thus the source and target B -ball decompositions correspond exactly.

On each gate, the endpoint A incidences are disjoint proper arcs with prescribed endpoint pairs. They correspond relative to the circle boundary by ordinary cutting of a disk along proper arcs. Cutting an endpoint annulus along all its spanning gate arcs leaves disks. Their boundary slots on the two annulus ends identify the pieces, even if the spanning arcs twist. There are at least two distinct cuts on every sampled subedge: each occupied interval has two endpoints and each feature is sampled. Thus successive arcs leave actual disk pieces with slots on both ends; with more than two arcs the slot not containing another endpoint uniquely identifies adjacent pieces. Along a gate interior there is one adjacent piece on each local side, in the two distinct gate balls.

These A disks lie in the matching B balls, as their end segments already do. Their boundary loops correspond on the ball spheres. A disjoint family of proper tame disks in a ball cuts it into balls, indexed by the sphere patches cut along those loops. Each such patch reaches an open patch of the manifold boundary: on a gate, every complementary region of disjoint proper arcs reaches an open rim interval. Consequently the endpoint chamber incidence agrees on the source and target. The collar structure makes membership in the interior of an occupied interval constant on an endpoint cell interior, and its boundary patch determines the corresponding active or inactive side.

All of these disk and ball recognitions can be made in the common changed polyhedral coordinates required by Definition 4.2. Their finite PL models are bi-Lipschitz equivalent to ordinary disks and balls. Tame Schoenflies and the relative PL structure are the qualitative topological inputs; see [10, Theorem 5] and [21, Theorem 2.2]. No uniform constant for these models is claimed or required.

It remains to map the occupied interiors, not just their endpoint walls. A region where both colours are active is an arc times the two occupied parameter intervals. Its two boundary end patches already carry the prescribed correspondence h_* . Extend this as a product, interpolating the interval coordinate along the arc between its two increasing endpoint correspondences, and preserve

the two root parameters exactly. In a B stack, removing the interiors of these opposite-colour strips from each disk fibre leaves disk fibres. In an A stack the analogous removal of spanning strips leaves disk fibres as well. Their incidence was identified above, and Lemma 4.10 gives products over the remaining single active parameter that preserve all concurrent box parameters.

The boundaries of each such disk fibre are already assigned, either on a both-active product or on ∂M_x . In disk-product coordinates extend them by coning. More explicitly, for a jointly bi-Lipschitz family of circle maps g_λ , the map

$$(r\theta, \lambda) \mapsto (rg_\lambda(\theta), \lambda)$$

is jointly bi-Lipschitz, as is its inverse; the factor r controls the parameter variation at the disk center. The boundary circle maps have these bounds because their finitely many pieces and the product charts do. This defines the single-active portions compatibly with the both-active ones.

These active products occupy the appropriate endpoint cells completely: their fibres reach the true boundary, their side walls are precisely the prescribed cuts, and the collar/product structure makes them open in the relevant cell interiors and closed up to their indicated boundary. Every remaining open cell is therefore a neither-active ball. Its boundary map is already assigned and extends by coning in bi-Lipschitz ball coordinates. The resulting finite-on-compacts maps glue to a locally bi-Lipschitz map in both directions; use the common polyhedral coordinates at their interfaces. Source and target local finiteness gives a global homeomorphism. On the cores and tubes use h_* , matching the boundary values.

Finally remake the stairs with their increments allocated only on E_j ; Lemma 4.1 still bounds their root-parameter speeds. We now pin the point label $q_h(x)$, rather than the support of an individual routed layer. Fix a value $q_x = q_h(x)$ and a guard κ_g . Take any point z on the constant- q_x fibre and any point $y \in \kappa_g$. The target product-fibre description gives a piecewise linear path from $h(z)$ to a boundary anchor; pulling it back by h gives a rectifiable path ν_z from z to a boundary anchor a_z . For almost every occupied parameter, ν_z misses its routed layer: by exact realization, a hit would require q_x to have that single standard label. Choose a path $\nu_y \subset \kappa_g$ from y to a boundary anchor a_g . This path misses every pre-stack, hence every routed layer. For any rectifiable connector σ from z to y , form the boundary-to-boundary path $\gamma = \nu_z^{-1} * \sigma * \nu_y$.

Use the point-distance potentials for the star and perimeter coordinates of q_x . The comparison (4.5) shows that, if the guard label is separated from q_x , one of these potentials separates the two labels. Its value at q_x is zero before applying the stairs. Sufficiently small stair displacement keeps its value at the first transformed anchor small and its value at the guard anchor bounded away from zero. Apply the period identity to γ and weight its crossings by the occupied-stair coefficients as in Lemma 4.7. The two appended fibre paths contribute zero for almost every parameter with nonzero coefficient. Thus the potential difference at a_z, a_g is bounded by the weighted unsigned crossings on σ , and hence by $C \int_\sigma G$. The coefficient bound changes only by $1 + O(\zeta)$. The same invariant-support tests for remote subedges and vertices make these prescriptions locally consistent.

If $q_h(x)$ differs too far from $q_*(x)$, follow a path in its constant- q_x fibre toward an exact boundary anchor. Since q_h is constant along this path and q_* agrees with it at the anchor, continuity of q_* produces a fixed-diameter stretch with a definite discrepancy. Pack small balls along this stretch and use nearby guards. The connector bound just proved applies to every spherical path between the fibre and a guard. The sphere argument of Proposition 4.9 therefore gives (4.12), contradicting the same previously scheduled capacity. The initial guard spacings accommodate both the earlier layer-support tests and these point-label tests. This proves the requested accuracy of q_h ; the argument estimates labels after T , not Dh . $\square$

4.6 Target stairs, parameter costs, and strict approximation

Fix the homeomorphism h and occupied intervals supplied by Theorem 4.11. Thus h sends every occupied root parameter $\lambda \in E_j$ to its standard label $p_j(\lambda)$, agrees with the prescribed map on full blocks, and respects the central tubes. We now collapse the unused target label intervals. This makes the derivative of the resulting singular composition depend on the prescribed scalar parameters.

The activation radii and full-cell layer widths are fixed target data; the occupied stair intervals are supplied by h . Denote the radial and perimeter stairs by S_D and S_e . For a fixed activation radius a with $r_{u,D} \ll a \ll 1$, use a nondecreasing Lipschitz cutoff H_a with an absolute Lipschitz bound,

$$H_a(u) = 0 \ (u \leq a), \quad H_a(u) = u \ (u \geq 3a), \quad 0 \leq H_a(u) \leq u,$$

with the choices made locally feature by feature. On the output two-complex define

$$P_0(d_D + r(\xi_e(s) - d_D)) = d_D + \rho_D(r)(\xi_e(S_e(s)) - d_D), \quad (4.13)$$

$$\rho_D(r) = H_a(\exp(S_D(L_D \log r)/L_D)) \quad (r \geq r_{u,D}), \quad (4.14)$$

with $\rho_D = 0$ farther inward. The shared subedge stairs make these formulas agree across sectors and subcells. By making the stair accuracy small relative to L_D , one has

$$\rho_D(r) \leq Cr, \quad |\rho'_D(r)| \leq C/c_*. \quad (4.15)$$

The derivative of the output with respect to one occupied root parameter is $O(r)$. Above the central transition, its output log-radius or perimeter parameter has speed at most $1 + O(\zeta)$. The first bound cancels the polar $1/r$ appearing in weak coarea estimates when the true and final labels are pinned. No ratio of subedge count to subedge length enters these speeds.

Extend P_0 to a full convex output cell using a fixed interior point z_i . Write a point as $z_i + b(y - z_i)$ with y on the cell boundary. Let $\lambda(b)$ be zero away from a thin boundary layer and rise to one at $b = 1$. On the boundary, let $\mathcal{P}_\lambda$ interpolate the radial factor between r and $\rho_D(r)$ and the perimeter factor between s and $S_e(s)$, separately and linearly. Set

$$P_0(z_i + b(y - z_i)) = z_i + b(\mathcal{P}_{\lambda(b)}(y) - z_i). \quad (4.16)$$

Here P_0 denotes the final singular map; $\mathcal{P}_0$ is the identity boundary map. This notation avoids identifying those two uses of zero. The interpolation preserves each boundary sector and subcell.

Tangential Lipschitz bounds in (4.16) depend only on the fixed cell geometry, the subcell aspect bounds and c_* . The speed in λ tends uniformly to zero with the radial cutoff scale and stair displacement. In fixing these target data, first choose the boundary-layer thickness using the fixed tangential bounds and absolute continuity of the relevant integrals; then make the factor displacements small compared with that thickness. Thus the extra full-cell energy of P_0Th can be made summably small. This uses that $h = h_*$ on the full input blocks and that T is the prescribed homothety there. On full blocks outside these thin boundary layers the composition remains q_0 .

Proposition 4.12 (Parameter cost after realization). *On occupied pre-stack pieces, a final active root parameter has local differential $\pm du_j$. On a routed circle box it has local differential*

$$\pm d(w_j - u_j + u_k) = \pm(-du_j + du_k), \quad (4.17)$$

with the local coordinate reversals understood. At such a switched point only one root scalar is active. At an essential box both original root scalars persist independently. Consequently the exterior derivative costs of P_0Th depend on these parameter differentials and the speeds in (4.15), with no factor involving Dh or the number of switches.

Proof. Every partial itinerary is a composition of translations and reflections of the root parameter. Thus its derivative is $+1$ or -1 , irrespective of its length. A diagonal segment has one reachable ancestry, so there is only one active scalar on the corresponding circle annulus. Essential boxes were never switched and retain both root parameters. Under the realization, each occupied interval corresponds by the original affine map p_j to its target label. The subsequent stair has bounded forward speed against that root parameter by Lemma 4.1. In inactive label directions the stair is constant. The chain rule in the piecewise product coordinates therefore yields the asserted local formulas and the $O(r)$ output factors; interfaces have measure zero. This accounts for the derivative entirely in terms of the original parameter functions. The qualitative disk and ball extension supplies no additional energy constant. $\square$

Proposition 4.13 (Strict target compositions). *For a fixed realization h as above, the singular composition P_0Th admits locally bi-Lipschitz homeomorphic approximations $P^\varepsilon T_\eta h$ with arbitrarily small prescribed local $W^{1,p}$ differences and value errors. Prescriptions may be summable on a compact exhaustion. Only local finiteness of the uncontrolled strict derivatives is required in regions whose energy is subsequently removed by a source collapse.*

Proof. Replace the radial and perimeter factors by

$$(1 - \varepsilon_D)\rho_D(r) + \varepsilon_D r, \quad (1 - \varepsilon_e)S_e(s) + \varepsilon_e s,$$

with positive, sufficiently small constants on each feature. Each factor is strictly increasing with a positive lower slope on a compact feature. The interpolating boundary maps in (4.16) are then homeomorphisms at every radius. Polar coordinates, including their centers, and the conical cell coordinates give both local Lipschitz directions. Hence P^ε is a locally bi-Lipschitz self-homeomorphism of Λ . Piecewise Lipschitz derivative bounds for the differences of the factors imply

$$P^\varepsilon Th \longrightarrow P_0Th \quad \text{in } W_{\text{loc}}^{1,p}$$

as the feature parameters decrease, with arbitrary compact-local error prescriptions.

Now fix P^ε and h , and use the strict quantizer T_η from Lemma 3.5. The composition convergence requires checking the tangential derivative at a collapsed stratum, not merely invoking uniform convergence. Use stairs and cutoffs that are piecewise smooth, with finitely many pieces on each compact. On the relative interior of a true face or edge, the pushforward of source volume by Th is absolutely continuous with respect to the corresponding dimensional measure: this follows from the product fibres of T and local bi-Lipschitzness of h . Sector cuts and stair knots therefore have null inverse image within these strata.

At a generic face point, the derivatives of the full-cell conical extension on face-tangent vectors converge to the boundary transformation's derivatives. At a generic edge point approached through an incident face sector, an edge-tangent vector keeps r fixed and the perimeter derivative tends to the common subedge derivative at $r = 1$. In any incident full-cell cone the conical radial coordinate has derivative zero on that edge-tangent vector; thus the same compatibility holds on approach through the full cell. On a true-vertex level set $D(Th) = 0$ almost everywhere. Consequently the tangential limit supplied by DT_η gives

$$D(P^\varepsilon T_\eta h) \longrightarrow D(P^\varepsilon Th) \quad \text{almost everywhere locally.}$$

The fixed piecewise bounds give an integrable local majorant after η and its gradient are restricted as in Lemma 3.5. Bounded convergence proves the local power estimates. Properness and local finiteness allow a diagonal choice meeting summable compact prescriptions. Before passage to the limit, $T_\eta h$ is itself locally bi-Lipschitz, so the ordinary piecewise chain rule applies. $\square$

5 Meshes and wall neighborhoods

This section constructs wall neighborhoods satisfying the geometric hypotheses of Definition 4.2, with fixed-factor bounds for their scalar parameter slopes. Section 6 will improve those bounds on the calibrated source patches. The analytic selection of the edge tests and the guard-independent integral capacities is made in Lemma 7.1 and Lemma 7.3; Proposition 7.4 transfers their counts to the actual starting walls. We separate those selections from the geometric estimates: all statements below concerning an edge count are conditional on that count, and their constants do not depend on the final sample weights or on the mesh resolution.

Two different estimates are needed. Away from the calibrated source patches, a fixed multiple of the weighted edge counts suffices to bound the scalar parameter slopes. On those finitely many patches, the later gradient argument requires nearly sharp bounds: a fixed multiplicative loss would remain in the approximation error. We first construct the ordinary collars. Section 6 then arranges the calibrated collars around nearly planar disks. That construction controls thin intermediate collars by a change of coordinates and broadens selected unchanged portions to obtain the sharp scalar bounds. All these estimates concern the collar parameters; the derivative of the eventual ambient wall realization is not estimated here.

We work in the polyhedral exterior pulled back by the fixed map h_* . Its working coordinates are locally bi-Lipschitz and piecewise smooth in physical coordinates. A compact set meets finitely many working charts and finitely many smooth pieces. A *fixed local constant* may depend on these charts, their incidence numbers, and the already chosen feature lengths, but not on subsequent collar thinning, guard gaps, sample weights, or mesh refinement. In calibrated islands it may also depend on a fixed compact set of basis matrices and on the slot margin specified below. A constant called universal has none of these additional dependencies.

5.1 Compatible fine meshes and lattice insertions

The boundary square complex has the following form. Squares on exposed full blocks use normalized (t, s) coordinates, and squares on the vertical sides of the inner tubes use normalized interval-height and s coordinates. Across a seam the varying coordinate agrees affinely, possibly after reversal. If that coordinate carries an intermediate label, both sides carry the same feature with the same normalized correspondence; otherwise it is unused on both sides. Thus the tested boundary levels are axis lines before the modifications below.

Lemma 5.1 (Fine meshes and prescribed lattice cores). *Fix the working polyhedral structure and a boundary product collar. There are arbitrarily fine, locally finite triangulations with the following properties.*

1. *Their boundary meshes have bounded shapes and bounded incidence inside each normalized square. Their volume shapes and incidences have fixed local bounds.*
2. *Any prescribed positive local upper bound for mesh size can be met. In finitely many inset coarse-simplex cores one may require a common constant background step H and exact Kuhn meshes of fixed bases A on prescribed smaller patches.*
3. *Transition shape constants depend on the fixed bases and coarse charts, but are independent of H and of the final guard requirements. Most of each smooth coarse-chart interior can instead be meshed by tetrahedra with universally bounded physical shapes. All remaining transition and chart-error regions may be confined to predetermined sets with arbitrarily small integrals of any fixed locally integrable weights.*

These statements remain valid after pushing the boundary to a cut at depth $d_0(x) > 0$, with chart constants independent of the smallness of d_0 , provided its absolute slope in square coordinates is bounded.

Proof. Give the vertices of the coarse triangulation a global order. On each coarse tetrahedron use the corresponding ordered Kuhn-simplex coordinates. The dyadic subdivisions restrict to the same subdivisions on common faces; on a right-triangle face they are the subdivisions by side parallels. Choose a positive mesh-size function δ with arbitrarily small absolute Lipschitz constant, subordinate to all required upper bounds. Refine until the simplex scale is at most a fixed small multiple of δ throughout that simplex. Positivity and the small slope imply that neighboring stopping levels, including levels throughout a vertex star, differ by a bounded amount.

Make the subdivision conforming by overlaying the nested subdivisions on a common face, splitting their edges, triangulating the resulting face pieces, and coning the interior pieces. There are only finitely many normalized configurations when the level difference is bounded. Thus this operation has a finite shape list and bounded local incidence. It need not alter a uniform Kuhn region whose star has no hanging data. To reserve a constant core, choose its preferred dyadic size much smaller than the allowed slope times its distance from the coarse facets. The infimum of these preferred values plus the allowed slope times path distance is a Lipschitz minorant. The unit-complex metric separates each fixed star interior from remote simplices, so this construction is positive locally and permits the stated constant cores. Ordinary positive piecewise-linear minorants give the same construction on a locally finite complex.

The lattice insertion uses the classical empty-sphere construction of Delaunay [17]. Its buffer, jitter and determinant estimates are supplied here. We describe the insertion of a second lattice, since the absence of a cut-dependent sliver constant is useful later. Suppose first that the two lattice bases have orthogonal columns in one fixed positive metric. In mesh units, retain pure portions of the two lattices separated by a buffer much wider than the covering radius. Fill the intermediate region with a separated finite-density net of free sites, allowing a small fixed jitter of each site. The net has a bounded covering radius and bounded local candidate counts. Choose the buffer width so that a Delaunay cell can contain fixed sites from at most one pure lattice.

Place the free sites successively, avoiding small neighborhoods of the affine spans of at most three previously placed or fixed nearby sites. There are boundedly many conditions at every step. Their excluded volumes can be made smaller than the allowed jitter volume. The affinely independent prefixes of fixed lattice points have fixed positive discrete determinant gaps. Induction therefore gives a positive mesh-unit lower bound for the volume of every nondegenerate candidate tetrahedron. Affinely dependent prefixes cannot occur in a nondegenerate tetrahedron. Resolve nonsimplicial Delaunay cells by a pulling order that agrees, on each deep pure lattice portion, with the lattice-coordinate order producing the Kuhn tetrahedra. The covering radius bounds Delaunay locality. Far enough inside a pure portion the cells are precisely its metric-orthogonal boxes, so the construction attaches to the unchanged Kuhn meshes. It can be performed with the outer lattice extended indefinitely, avoiding an artificial outer boundary issue.

For an arbitrary inserted basis $A = U\Sigma V^T$, first couple the identity lattice with $U\Sigma$ in the Euclidean metric. Then couple $U\Sigma$ with A in the metric $(U\Sigma)^{-T}(U\Sigma)^{-1}$. Both pairs have orthogonal columns in their respective metrics. A pure intermediate band, with the same Kuhn order on both sides, joins the two constructions. All constants depend on the fixed bases and buffer ratios, not on H .

On a smooth coarse-chart piece, take finitely many inset patches on which the chart derivative is close to an invertible matrix B . Insert the basis B^{-1} there. Physical tetrahedra on the smaller patches then have universal shapes. The matrices and their inverses have fixed bounds on each coarse compact.

Having fixed these bounds, make the patch margins, uncovered sets, and neighborhoods of coarse facets and nonsmooth chart loci small in the chosen integrable weights. Absolute continuity allows this, since the latter loci are locally finite null sets and the transition constants just constructed do not involve the shrinking physical margin. Take H smaller afterward. Summable local prescriptions handle the noncompact exterior.

Finally, in collar coordinates (x, d) send $d = 0$ to $d = d_0(x)$ by a monotone piecewise-linear depth change which is the identity for $d \geq Cd_0(x)$. Its depth slopes can be bounded above and below by fixed positive constants, and its cross slopes by a fixed multiple of $|\nabla d_0|$. It preserves x . Transporting the coarse collar coordinates by this map proves the last assertion. $\square$

5.2 Boundary warping, straight traces, and joint collars

Choose the cut depth d_0 within the region of exact h_* behavior, with several multiples of d_0 still in that region. It can be arbitrarily small and have a small bounded absolute slope. Throughout the boundary construction impose $\delta \ll d_0$.

Lemma 5.2 (Boundary trace preparation). *The boundary mesh and the central wall traces can be arranged so that:*

1. *each central trace in a boundary triangle is a straight normal arc with at most one hit on an individual edge;*
2. *near each selected sample, a closed label interval has a piecewise-linear ribbon continuation matching the normal-disk prism construction, and distinct intervals of one colour are disjoint;*
3. *the two-colour pattern throughout the outer collar is jointly locally bi-Lipschitz equivalent to the cut pattern times depth, preserving both parameters;*
4. *the inverse label slopes and the edge-coarea bounds on the modified collar have fixed local bounds independent of d_0 , of the excluded belt widths, and of the final mesh and sample scales.*

In normalized square coordinates, if a feature has physical label length ℓ , an occupied interval of label length comparable to $c_ w_j$ gives on each crossed boundary edge a fractional width at least*

$$\frac{c\varepsilon c_* w_j}{\ell H_0}. \quad (5.1)$$

Here H_0 is the local dyadic square scale, $\varepsilon > 0$ is fixed before final sampling, and $c > 0$ is independent of the subsequent choices.

Proof. Compression and inverse response. Write $z = (z_1, z_2)$ for the original normalized square coordinates; their axis levels carry the prescribed labels. The warped spatial position will be denoted by x . Thus all exclusions of label belts below are made in the z coordinates, before the warp. When $H_0 = 2^{-k}$ and $H_0/2 \leq \delta(x) \leq H_0$, set $\chi(x) = 2(1 - \delta(x)/H_0)$. Let ϕ_k be an increasing piecewise-linear map of the unit interval, symmetric under reversal and fixing its endpoints. On the interior of each dyadic bin of length H_0 , except for narrow endpoint belts, it has slope ε and image in an $O(\varepsilon H_0)$ window about the bin midpoint. In the belts it expands to join the fixed bin endpoints. Its minimum slope is ε , regardless of belt width. Choose the relative belt lengths summably small over k . After a compact feature's finitely many relevant scales have been specified, exclude all their belts from sample points and their closed label intervals.

For z in a square define x by

$$x_i = (1 - \chi(x))\phi_k(z_i) + \chi(x)\phi_{k+1}(z_i), \quad i = 1, 2. \quad (5.2)$$

Since $|\phi_k - \phi_{k+1}| \leq CH_0$ and $\text{Lip } \chi \leq 2\text{Lip } \delta/H_0$, the right side has Lipschitz constant at most $C\text{Lip } \delta$ in x . Choose this less than $1/4$. The Contraction Theorem gives a unique x for each z . Conversely, at a fixed x , each coordinate equation is inverted by a strictly increasing scalar map of minimum slope ε . This proves that the map is onto and one-to-one and that its inverse has Lipschitz constant at most C/ε , independently of belt widths. At dyadic scale changes the formulas agree. Reversal symmetry and the common seam coordinate make them compatible on the square complex.

The same construction with the right side interpolated with z_i turns the warp on. At every stage the minimum scalar response is at least ε . A level $z_i = \text{constant}$ is a graph in the other coordinate with slope $O(\text{Lip } \delta)$. In an allowed bin configuration its normal response to z_i is $\varepsilon(1 + O(\text{Lip } \delta))$: differentiate the scalar contraction with the running coordinate held fixed. This assertion also follows by difference quotients at the finitely many piecewise-linear breaks. For partial turn-on the response has the same positive lower bound. The needed other inverse ordinate exists by the scalar monotonicity already proved.

Use normalized depth $d/d_0(x)$ as an independent product coordinate for these operations. Turning on the warp moves points by $O(\delta)$. Converting back to physical depth adds at most $C(\delta/d_0)(1 + |\nabla d_0|)$ to the relevant inverse response bounds. It therefore causes no inverse power of the collar thickness in the final estimate.

Mesh placement and straight traces. The possible positions of allowed traces near one triangle star lie within $C(\varepsilon + \text{Lip } \delta)H_0$ of a bounded list of references: blend the two relevant bin midpoints using χ at a fixed nearby point. There are boundedly many such references per star, and they stay a fixed multiple of H_0 from the square's axis ends. Perturb each mesh vertex by a small shape-preserving fraction of its local size, freely in a square or along a macro seam, and keep actual square corners fixed. We may require, with a fixed $c > 0$, that vertices stay cH_0 from relevant axis references, that non-seam edges have both coordinate components of magnitude at least cH_0 , and that all edges avoid the cH_0 balls about relevant pair-reference points. Here lists from both endpoint stars and all incident triangles are included.

For completeness these requirements can be imposed simultaneously. Give the allowed vertex perturbations independent uniform distributions on fixed small balls or seam intervals. Each bad event involves only boundedly many such variables. Its probability tends uniformly to zero as $c \downarrow 0$, as follows. Normalize the local mesh scale to one; the perturbation balls and intervals then have fixed positive sizes. With a free endpoint, condition on the other endpoint. Except when that endpoint lies within $\sqrt{c}$ of the forbidden pair-reference point, the edge-point distance test excludes a strip of width $O(\sqrt{c})$ in the free endpoint's ball. The exception itself has probability $O(\sqrt{c})$ if the other endpoint is movable; a fixed corner already has a fixed positive clearance. For endpoints on two different macro sides near a corner, write them as $(a, 0)$ and $(0, b)$ and the pair-reference point as (r, s) . The line determinant is $ab - as - br$. Outside $|b - s| < \sqrt{c}$, a distance at most c excludes only $O(\sqrt{c})$ of the allowed a interval; the omitted b interval has probability $O(\sqrt{c})$ as well. The axis-reference and edge-component tests give ordinary interval exclusions of vanishing length. These bounds are uniform in the normalized configurations. Fixed corner coordinates are already separated from the references; a seam edge is automatically separated from pair references. Take the boundary mesh fine enough that no edge joins two actual square corners. There is then no exception involving two immovable endpoints. Each variable belongs to boundedly many tests, even at a high-valence actual corner, because that corner is fixed and every other variable belongs to bounded stars in at most two squares. The classical Lovász local lemma of Erdős and Lovász [19], in the formulation recalled in [35, Theorem 1.1], therefore applies after making the single-event probability smaller than $1/(e(D + 1))$, where D bounds the dependency degree. Apply it on finite exhaustions and pass to a convergent subnet in the compact product of closed perturbation sets;

impose twice the desired slack before passing to the limit. This gives all requirements on the locally finite complex. Now fix c and take ε and $\text{Lip } \delta$ much smaller than its associated tolerances.

The jitter extends to the collar: interpolate the vertex movement piecewise linearly on each square and taper it over depth $O(\delta)$, keeping it constant on an initial product band. Its slopes are bounded, its size is a small fraction of the mesh, and it respects seams. Working triangulation coordinates are pulled back by this change.

In a fully warped triangle every allowed trace is consequently a single graph arc. Its endpoints are uniformly away from triangle vertices, and its running-coordinate span is at least $c'H_0$. Opposite-colour crossings are transverse and stay away from edges. At a moving endpoint the running speed is at most $C(c)$ times the normal speed, by edge obliqueness; a macro edge parallel to the arc cannot be an endpoint edge. The same pair of endpoint edges persists through each allowed bin interval. Replace the arc by its chord at fixed running coordinate. If its endpoints are $(a_i(z), b_i(z))$, the normal response of the chord at a fixed running coordinate is a convex combination of

$$b'_i(z) - m(z)a'_i(z),$$

where $m(z)$ is the chord slope. The endpoint cone and the smallness of m give the lower bound $c'\varepsilon$. Chords of disjoint full cuts of a convex triangle have the same order. Convex interpolation from the original arc to its chord therefore preserves order and the response bound, and agrees across triangle edges.

Ribbons and continuation through the cut. We must also match the closed parameter intervals to affine disk prisms. On either side of a sample, take the ordered stair triangulation of the chord times a half-interval, moving one endpoint per stair triangle between the specified end placements. At fixed abstract arc fraction, its parameter velocity is the relevant endpoint secant velocity. It lies in the same response cone. Choose the label interval much shorter than the local mesh scale; then the arc-fraction tangential derivative has positive running component at least $c'H_0$, while its normal slope remains small. Interpolate between the varying chord and this stair ribbon in both coordinates at fixed oriented arc fraction. The same determinant calculation, namely positive running speed times positive normal response with a smaller cross term, gives a family of ordered graph arcs. The edge trajectories are monotone between the same endpoints, so each half-ribbon fills exactly the shell between its end arcs. Equivalently its positive Jacobians and embedded boundary give degree one. Hence intervals stay disjoint, opposite crossings stay transverse, and the central sample chord is unchanged. Equal subdivision of the arcs, used later, causes no loss: tangential corrections involve differences of endpoint displacements.

Perform these stages successively across the outer cut collar, starting with the identity at the true boundary $d = 0$ and reaching the prepared chord-and-ribbon pattern at the cut $d = d_0(x)$. Thus the true boundary traces remain the prescribed h_* traces at every label. Inside the cut mirror the central homotopy back to the unwarped central curves, keeping a straight-chord product germ at the cut. Stretch the remaining depth coordinate back to the full exterior, with fixed two-sided slope bounds and identity past several multiples of d_0 , and pull back the starting carrier walls there. This changes only the exact h_* region and preserves every individual wall's topology. The normal boundary arcs have the required per-wall per-edge count. On the modified collar, inverse level maps on allowed strips have the fixed bounds just proved, so one-dimensional coarea on edges has fixed local constants. The stretched exact data have the same property. These constants may involve feature lengths, $1/\varepsilon$, and $1/c_*$, but not d_0 , belt widths, guards, or resolution. Formula (5.1) follows by converting label speed $c'\varepsilon/\ell$ to a fractional edge displacement on an edge of scale H_0 .

The simultaneous product collar. At this point each family has a continuation through the collar. We now construct a common product identification that preserves both parameters. At

each normalized depth every used trace in a whole square is a graph $x_i = F_{z_i}^i(x_{\bar{i}})$ with small cross-slope, strict order, and locally two-sided Lipschitz dependence on each occupied label interval. Adjacent triangle pieces have strictly increasing running-coordinate spans. Fill gaps between the occupied intervals by linear interpolation at fixed running coordinate, fixing the axis endpoints. The gaps have positive local response for the fixed data by strict separation and compactness. All these definitions agree at macro seams. Solve the two graph equations jointly by contraction. The resulting depth-dependent homeomorphisms of the square complex, compared with the map at the cut, identify both families with the cut pattern times depth and preserve both parameters. Their local bi-Lipschitz constants need not be uniform, which is sufficient for this assertion. $\square$

5.3 Generic volume edges and count estimates

Lemma 5.3 (Generic edges and conditional count charging). *Away from the constrained boundary collar, the meshes above admit shape-preserving ambient jitter for which their edges are Sobolev ACL edges with the usual chain rule. If W_e is the sum of sample weights over central wall hits of both colours on an edge, samples can then be chosen so that, on every unmodified edge,*

$$W_e \leq (1 + C(\zeta + c_*)) \text{Var}_e(q_*) + \epsilon_e, \quad (5.3)$$

with separate-colour versions and arbitrarily small prescribed positive errors ϵ_e . Here variation means the appropriate star or perimeter-graph variation; in a smooth sector it can instead be tested by the individual scalar directional derivatives. On the modified boundary collar the corresponding coarea bounds use the fixed local inverse-label constants of Lemma 5.2. The expected ambient charges for powers of the edge-variation means have bounded overlap and fixed local density factors independent of mesh resolution.

Proof. Interpolate small random physical vertex vectors by the working simplex barycentric coordinates and multiply by a cutoff vanishing at the constrained collar. Choose the local amplitudes to make the ambient displacement have a small local Lipschitz constant. In poorly shaped charts use smaller fixed local fractions; in a regular physical bulk use a universal fraction; in a calibrated patch use a prescribed arbitrarily small fraction. Include chart derivatives and incidence constants when fixing these fractions. Taper only in the exact region, where fixed label Lipschitz bounds are available, and take the mesh much finer than taper depth. The resulting ambient maps are locally bi-Lipschitz; fine proper displacement and degree give global homeomorphisms.

At a fixed edge parameter outside the cutoff region, at least one endpoint's barycentric coefficient is bounded below. The perturbed point therefore has a density at most Ch^{-3} in a neighborhood of scale h , with C depending only on the fixed jitter fraction and local charts. Its speed in normalized edge time is at most Ch . Fubini applied first to smooth approximations of the Sobolev labels and then to their gradients gives ACL, the chain rule, and the expected integral estimates. The neighborhoods lie in bounded mesh stars, so multiplying by adjacent tetrahedron volumes and summing gives bounded overlap. At inner limiting lift seams the edge-time set is almost surely null; ACL makes its label length zero. This suffices for generic sample avoidance without requiring finitely many seam hits.

One-dimensional coarea gives integrable hit-count functions on the finite feature tests associated with each compact edge. The stratified slot sampling, performed after the edges are fixed, gives (5.3), its partial-colour versions, and any finite collection of directional tests with the stated errors. Generic samples avoid vertices, nongenuine chart breaks, tangencies, and corners, and have finitely many hits on each compact edge. Near the normalization cut use the PL general-position counts and the straight-chord germs from Lemma 5.2; elsewhere isolated crossings can be cleaned off a smaller cut collar.

Away from modifications, the graph speeds are those of q_* ; on a high-radius sector they are bounded by the appropriate aspect factor times $|Dq_0|/r$. On collar modifications the fixed inverse level bounds from Lemma 5.2 replace them. Local irregular-chart factors charge only the preselected error regions and slightly enlarged buffers. For simultaneous finite aggregate tests use Markov's inequality with room; compact-local upper tests can be assigned summable failure probabilities. This Lemma is a count estimate, not yet an estimate for the final derivative. $\square$

5.4 Preparing and normalizing the source walls

The edge estimates are used on actual locally flat source walls. Proposition 7.4 supplies those systems by a common opening of the carrier, with the prescribed boundary collars and finite isolated edge hits. Counts alone do not supply such surfaces. The two results below prepare one colour at a time: first make its walls PL without adding hits, then replace them by normal disks while retaining every individual wall-edge upper bound. Neither replacement is required to preserve the other colour.

All PL assertions below concern the abstract triangulation. They remain applicable when its realization in the physical exterior is given by locally bi-Lipschitz charts. For the standard exterior these charts are obtained from sector boxes with coordinates $(t, s, \text{interval height})$, from edge products with the edge coordinate and the offset-polygon coordinates, and from the vertex offset polytopes. The logarithmic radial coordinate is used only outside the removed inner polygon. Common boundary cells have affine identifications after normalizing the interval coordinates. Subdivision therefore gives a compatible locally finite triangulation.

Lemma 5.4 (Relative PL preparation with edge counts). *Let M be a three-dimensional PL manifold, possibly with boundary, with a locally finite triangulation. Physical coordinates on M may be obtained from its PL coordinates by locally bi-Lipschitz homeomorphisms. Let (S_j) be a locally finite disjoint family of compact, properly embedded, locally flat surfaces. Suppose that each S_j avoids the mesh vertices and has only finitely many, isolated intersections with mesh edges. On a neighborhood of ∂M suppose the surfaces already have prescribed PL product collars, with transverse edge crossings and straight traces in boundary triangles. The protected collars may be made smaller before the construction.*

Then the family can be replaced by a family of PL surfaces of the same individual topological types, retaining the prescribed smaller boundary collars, such that the number of intersections of each individual surface with each individual edge does not increase. Every retained interior edge intersection can be required to have a flat transverse disk germ. The changes can be confined to any prescribed neighborhood of the original family and can be arbitrarily small in a prescribed positive continuous position tolerance. No bound on the Lipschitz constants of these preparatory changes is asserted.

Proof. We first explain the preparation at an isolated interior edge hit x . Choose a surface chart in which the relevant part of S_j is a proper unknotted disk, with no other sheet present, and choose the support to avoid all other mesh edges and the prescribed boundary collar. In the original PL coordinates the edge is straight near x . Take a short polygonal cylinder about it, with both caps disjoint from the surface. The cap positions are chosen first, and then the radius is decreased. Isolation of the hit and continuity of the inverse surface parametrization ensure that every intersection with the cylinder side lies in a small subdisk of the surface chart. Make the surface PL and transverse on a neighborhood of this side, leaving a neighborhood of the axis unchanged. This local preparation follows by approximating the flattening chart on the compact annular region

by a PL chart and extending the small embedding change across its collar. Its support has positive distance from the axis and from the fixed part of the surface.

There are now finitely many intersection loops on the cylinder side. An innermost loop that is null-homotopic in the side annulus bounds a disk on that side whose interior misses the surface. Replace the corresponding surface subdisk by a small pushoff of that side disk, matching the unchanged surface along a collar of the loop. The surface remains a locally flat disk, the operation creates no axis intersections, and it removes that loop without adding side intersections. Continue until all remaining loops are essential in the side annulus. Every remaining loop has linking number 1 or -1 with the extended straight axis, so its bounded surface subdisk contains the unique possible axis hit. These subdisks are nested. Replace the outermost one by an unknotted disk in the cylinder with the same rim, meeting the axis once transversely and flat near that crossing. Such a disk is obtained by isotoping its essential rim to a cross-section in an outer annular collar and adjoining the flat cross-section. If no side loop remains, the surface misses the cylinder: its caps are disjoint from the surface and the boundary of the surface chart is outside the cylinder. Thus the final number of hits in this support is zero or one. The old and new proper locally flat disks agree near their outer rim and split the enclosing chart ball into balls; the relative disk-extension consequence of generalized Schoenflies [10, Theorem 5, p. 76] therefore extends the disk replacement to an ambient homeomorphism fixed at the enclosing ball boundary. All these supports may be chosen disjoint and locally finite. In particular the construction preserves surface type and same-colour disjointness.

Retain smaller PL balls at the surviving crossings, together with a smaller prescribed boundary collar. Outside these protected sets, every surface has a positive gap from the mesh edges on each compact set. It remains to approximate the surfaces while preserving these gaps and the protected PL germs. Here are the relative triangulation details. In a compact neighborhood of a finite subsystem, cut along the collared locally flat surfaces. Triangulate the two copies of each cut surface compatibly, extending the triangulations already prescribed at their rims and in the crossing balls. Hamilton's existence construction, started from the prescribed boundary collars [21, Theorem 2.1 and Section 3, p. 69], extends these triangulations over the cut manifolds; gluing the copies back gives a PL structure in which all the surfaces are PL. This structure agrees with the original one on smaller protected neighborhoods. Equivalently, to arrange the boundary agreement before extending, the uniqueness of the PL structure on a surface supplies an isotopy between the two boundary triangulations, and a collar extends that isotopy to the interior.

The relative three-dimensional PL approximation Theorem [21, Section 3, Theorem 2.2] now gives a PL homeomorphism from this auxiliary PL structure to the original structure, arbitrarily close to the identity and equal to the identity on the smaller protected neighborhoods. Choose its position tolerance below the gaps to all unprotected edge segments. Its images of the surfaces are PL; they retain exactly the retained edge hits and create no others. PL general position relative to the protected germs is then imposed with the same gap restriction. Only the surface sets, and not a pre-existing parametrizing homeomorphism, have been prescribed to be PL.

For the locally finite family, choose pairwise disjoint small regular neighborhoods of its members, with locally finite closures. The preceding compact construction is performed in these neighborhoods, retaining their frontiers and the protected boundary data. The constructions agree with the identity near the frontiers and therefore glue. Local finiteness supplies continuity in both directions and preserves all the asserted local restrictions. The same argument applies in the abstract PL coordinates when the physical charts are bi-Lipschitz. $\square$

Remark 5.5. The edge-count conclusion uses the flat crossing germs and the positive gaps on the remaining edge segments. Uniform approximation by itself does not control the number of intersections with an edge. The preparation is applied to one colour at a time; it imposes no relative

condition on the other colour. The preceding proof uses the classical relative triangulation, PL approximation, and locally flat Schoenflies Theorems in dimension three only in their qualitative forms.

The following argument follows Haken's normalization method [20, Chapter I, §5], using disk and bigon moves that reduce edge weight. Its needed conclusion keeps a separate bound for every wall and every mesh edge, so we give the relative argument rather than substitute a total-weight normalization theorem.

Proposition 5.6 (Normalization with individual edge bounds). *Let M be an orientable irreducible PL three-manifold with a locally finite triangulation. Let (S_j) be a locally finite family of pairwise disjoint, compact, properly embedded PL surfaces in general position. Each S_j is a disk or an annulus; in the annulus case its core is noncontractible in M . Assume that the prescribed boundary traces are straight normal arcs in boundary triangles and that*

$$\#(S_j \cap e) \leq 1 \quad \text{for every boundary edge } e \text{ and every } j.$$

Then there is a locally finite disjoint family (S'_j) with the same individual topologies, coorientations, and exact boundary traces, consisting of normal triangles and quadrilaterals, such that

$$\#(S'_j \cap e) \leq \#(S_j \cap e) \quad \text{for every } j \text{ and every edge } e.$$

For a finite family the replacements can be obtained by an ambient isotopy relative to the boundary. Consequently any initially prescribed upper bounds on the individual edge counts are retained. This assertion concerns one colour only; it imposes no requirement to preserve the other colour.

Proof. Every surface under consideration is incompressible in the following form: a simple closed curve in its interior that is null-homotopic in M bounds a disk in that surface. This is immediate for a disk. On an annulus a simple closed curve not bounding a disk represents a core, contradicting the core hypothesis if it is null-homotopic in M .

First suppose that the family is finite. Among its positions obtained by ambient isotopies relative to the boundary and satisfying the initial individual edge bounds, choose one minimizing the total number of edge hits; subject to this, minimize the total number of circular components of intersection with faces. These are nonnegative integers and the class of admissible positions is nonempty. All moves used below have final positions within every individual edge bound.

Suppose that a face contains an intersection circle. Choose an innermost circle in the entire family. Its face disk D has interior disjoint from all the surfaces. Incompressibility supplies a disk E on the surface containing the circle. The union $D \cup E$ is an embedded sphere in the interior of M , after rounding its corner, and hence bounds a ball. No other surface is in this ball: it is disjoint from the sphere and has boundary on ∂M . The part of the same surface outside E is also outside, since it is connected and reaches its actual boundary. Thus E can be moved across the ball to a small pushoff of D , without moving any other surface. The new disk has no edge hits. Face-interior collars make its seam transverse without creating new face circles. The move either decreases edge weight or, with that weight unchanged, decreases the number of face circles, a contradiction.

If an intersection arc in a face returns to the same edge, choose a returning arc cutting off an innermost face bigon. Its interior is disjoint from the surface family. The edge side of the bigon has no further hit: any such hit would continue into the face on the bigon side and yield an arc there. The edge is not a boundary edge, since the returning arc belongs to one wall and would give two hits of that wall on the edge. The usual bigon isotopy removes the two hits and is supported away from every other edge and from ∂M . One precise description is to move the edge segment across

the empty bigon and slightly past its surface side, and then apply the inverse ambient isotopy to the surfaces. This again decreases edge weight. Hence all face intersections are normal arcs.

Consider now the pieces of the surfaces in a tetrahedron. They have nonempty polygonal boundary; a closed piece would be a whole connected component of an original surface, whereas every such component has actual boundary. If a piece is not a disk, the collection of pieces has a compression disk whose interior misses the entire collection. To justify this assertion, cut the tetrahedron along bicollars of all pieces. If some inclusion of a surface copy into an adjacent cut region fails to be injective on fundamental groups, the Loop Theorem produces the asserted disk. Theorem 2.A.5 of [41, p. 13] allows a surface contained in the manifold boundary; apply it to the interior of that surface copy. Round the polygonal corners and push the disk interior off the boundary of the region. Extend its boundary back to the original piece through that piece's product collar; the collars are disjoint, so this does not meet any other piece. If all those inclusions were injective, reconstruct the tetrahedron as a graph of spaces. Its vertex spaces are the connected cut regions, and its edge spaces are the bicollars of the connected surface pieces; an edge may have both ends at the same cut region. The two attaching maps for every edge space induce the assumed injections. The graph-of-spaces normal form therefore injects each piece's fundamental group into that of the tetrahedron. A connected compact surface with boundary other than a disk has nontrivial fundamental group, whereas the tetrahedron has trivial fundamental group. This is impossible.

Let D be the resulting compression disk and let $\gamma = \partial D$ lie in the interior of a piece P . Incompressibility of the whole wall supplies a disk E in that wall with boundary γ . Since γ is essential in P , the disk E is not contained in P . Choose a path in E from γ to a point outside P . Its first exit from P crosses a component β of ∂P . The connected curve β is disjoint from $\gamma = \partial E$ and meets $\text{int } E$, so it lies entirely in $\text{int } E$. Thus E contains a whole boundary polygon of P . This polygon is in the interior of the wall and thus meets only interior edges of M . In particular E has at least one edge hit. The sphere $D \cup E$ bounds a ball. As above, all the other walls and the part of this wall outside E lie outside that ball because they reach ∂M . Replacing E by a pushoff of D removes its edge hits and introduces none, contrary to minimality. All tetrahedron pieces are consequently disks.

Disjoint proper disks cut a tetrahedron into balls, with a dual tree. Traversing an edge of the tetrahedron gives a walk on this tree. If one disk is met twice, that walk contains an immediate backtrack. There is therefore an edge segment with no intervening surface hit whose endpoints lie on the same disk. Join its endpoints by an arc in that disk, interior except at its ends. In the intervening complementary ball the two arcs lie on the boundary sphere and form a simple closed curve. A boundary disk pushed into the ball gives an empty bigon, with interior disjoint from all surfaces and from the other tetrahedron faces and edges. Its edge side cannot be a boundary edge by the per-wall boundary-edge hypothesis. The bigon move again decreases weight without increasing any individual edge count. Thus every disk meets each tetrahedron edge at most once.

A simple normal loop on the tetrahedron boundary meeting each edge at most once separates the four vertices into two nonempty sets. It crosses exactly the edges joining the two sets. A $1 + 3$ partition gives a triangle and a $2 + 2$ partition gives a quadrilateral. Straightening the face arcs and replacing the filling disks by compatible standard normal disks preserves the surface family: proper tame disks in a ball with the same boundary pattern are carried to one another by a boundary-relative ball homeomorphism. The construction is an ambient isotopy relative to the boundary, and establishes the finite assertion.

For a locally infinite family, exhaust its index set by finite subfamilies and perform the finite construction. A fixed wall initially meets only finitely many edges, because it is compact and the triangulation is locally finite. Its normalized disks can occur only in tetrahedra incident to

these edges, since no new edge hit is permitted. There are finitely many such tetrahedra, and only finitely many normal combinatorial possibilities under the fixed hit bounds. Include the wall labels, coorientations, matching data across faces, and order of hits along edges among these finite data. A successive subsequence extraction stabilizes all these data wall by wall. For a compact subset of M , only finitely many walls can appear in this extraction: the finitely many relevant tetrahedron stars are compact, and any candidate wall originally met an edge in those stars. Initial local finiteness applies. Thus the stabilized data realize a locally finite, simultaneously disjoint normal family. Each whole wall has stabilized on its finite support, so its topology and prescribed boundary placements are retained. Ordered placements on interior edges and standard fillings realize the data; boundary-edge placements are the originally prescribed ones. No convergence of an infinite ambient isotopy is needed. $\square$

5.5 Normal disks and affine shell prisms

Lemma 5.7 (Normal templates). *In a normalized tetrahedron fix an endpoint margin $\gamma > 0$. For every normal disk type use its triangle, or the two triangles of a quadrilateral on one chosen combinatorial diagonal. We call these pieces the main triangles and, in the quadrilateral case, call their shared diagonal the main crease. Require all crossing-edge fractions to lie in $[\gamma, 1 - \gamma]$. Compatible within-colour edge orders give disjoint templates. Their separations are at least $c(\gamma)$ times the minimum edge gap. Between two ordered placements of the same disk type the ordered affine stair prisms give a PL homeomorphism onto the entire shell. If its scalar parameter has range of length at most w , then*

$$|\nabla u| \leq C(\gamma) \max_{e \text{ crossed}} \frac{w}{\Delta_e}, \quad (5.4)$$

where Δ_e is the fractional width on the crossed edge. Physical and other working coordinates add their usual fixed shape and scale factors.

Proof. Write the vertex partition as $I \mid J$. For a quadrilateral both groups have two vertices. In barycentric probability coordinates write

$$\lambda = ((1 - u)a, ub), \quad a_0 + a_1 = b_0 + b_1 = 1, \quad r_{ij} = \frac{t_{ij}}{1 - t_{ij}}.$$

Choose diagonal 00, 11. On the triangle 00, 01, 11 the disk is the graph

$$\frac{u}{1 - u} = \frac{r_{01}a_0 + r_{11}a_1}{(r_{01}/r_{00})b_0 + b_1}, \quad r_{11}a_1b_0 \leq r_{00}a_0b_1, \quad (5.5)$$

with the symmetric formula on the other half. The two agree on the diagonal. The displayed expression is monotone in each of its edge positions. In particular its derivative in r_{01} has the sign of $a_0b_1 - (r_{11}/r_{00})a_1b_0$, nonnegative by the side condition; the other direct derivatives are nonnegative. The second half gives the remaining derivatives. A common multiplier of the r_{ij} multiplies the whole odds graph by that multiplier. Compactness of the allowed fractions therefore gives a lower response $c(\gamma)$ to a common edge increment and bounded graph slopes. Ordered same-type disks have at least the asserted separation. Their edge orders agree on all crossed edges, since disjoint boundary cuts of the tetrahedron sphere are nested between the two vertex groups. The triangular case follows from the affine separating plane, or the same formula with one group a single vertex.

Different compatible types include a triangular disk. By the face-arc order every crossing vertex of the other disk lies on the specified side of that triangle plane. So does its convex hull, and hence its chosen disk. An edge gap gives the same quantitative separation by the triangle's uniformly

positive separating altitudes. Edges not crossed by the triangle already have vertex clearance. This proves the simultaneous disjointness assertion.

Triangulate a reference disk consistently, and use the standard ordered triangulation of each triangle times an interval. Map its bottom and top vertices to the two placements. A stair tetrahedron contains three mixed-level vertices and one duplicate vertex moving along its crossing edge. Its mixed triangle still has a strictly separating plane. For example the first quadrilateral half has plane coefficients proportional to

$$(-r_{01}, -r_{11}, r_{01}/r_{00}, 1)$$

on (I_0, I_1, J_0, J_1) , with strict separating signs for all allowed mixed placements. Consequently the oriented altitude of the duplicate-edge motion has the correct sign and is at least $c(\gamma)$ times that edge displacement. The other half and triangular types are identical in this respect. All stair tetrahedra have positive orientation and volumes bounded below by that displacement times a fixed normalized area factor.

The common order on neighboring triangle prisms makes the boundary ribbons agree. On a tetrahedron face each ribbon lies between disjoint full chord cuts with monotone endpoint tracks. Thus the prism boundary is embedded and bounds precisely the desired shell. Positive nondegenerate simplex signs give local openness in the interior: an interior preimage is isolated locally, and a nearby generic value in an incident simplex image has positive local degree. The embedded sphere boundary has degree one. Summing the positive local degrees proves that the entire prism map is a homeomorphism. This degree argument also excludes possible branching at internal simplex faces and edges.

On each stair tetrahedron, inversion of its affine matrix and the altitude lower bound give (5.4). One may use two half-prisms with the central disk exactly prescribed. If all placements are close to one template, the horizontal layer tangents are close to that template's main triangle: differentiating at fixed layer parameter compares mixed vertices at that same parameter, so only displacement differences enter the tangential columns. $\square$

5.6 Simultaneous product collars

Normal prisms give product coordinates for each individual wall. Definition 4.2 also requires their intersections to vary as products preserving both scalar parameters. In a candidate collar $\mathcal{E}_j$, write $f_k = u_k \circ \mathcal{E}_j$ for an opposite-colour parameter on its domain. The following criterion obtains the simultaneous products from an intrinsic direction in which f_k increases strictly. The proof of Lemma 5.10 will supply these directions at smooth crossings, main creases, and tetrahedron faces.

Lemma 5.8 (A sufficient pattern-triviality test). *Let Σ be a compact smooth disk or annulus, let J be a compact interval, and let $E : \Sigma \times J \rightarrow C$ be bi-Lipschitz. For finitely many indices k , suppose that a Lipschitz function $f_k(z, \lambda)$ is defined on an open neighborhood of its compact enlarged strip*

$$K_k = \{(z, \lambda) : a_k - \delta_k \leq f_k(z, \lambda) \leq b_k + \delta_k\}, \quad a_k < b_k, \quad \delta_k > 0.$$

The sets K_k are assumed disjoint, and all relevant strip points belong to these compact sets, with a margin inside their domains of definition. Each strip is the full inverse image of its indicated label interval in its isolated neighborhood: apart from $\partial\Sigma$, there is no additional lateral frontier inside the enlarged label interval. At every point of K_k suppose there is a smooth vector field V on a surface neighborhood, tangent to $\partial\Sigma$ at boundary points, and a neighborhood in $\Sigma \times J$ on which

$$D_z f_k(z, \lambda) V(z) \geq c > 0 \quad \text{for almost every } (z, \lambda).$$

Equivalently one may impose the integrated strict increase along the local V -flow for every nearby λ . The constant and direction may depend on the neighborhood.

Then the simultaneous subdivision of C by the strips $a_k \leq f_k \leq b_k$ is jointly bi-Lipschitz trivial over J , preserving f_k on each strip and preserving the manifold boundary. For each fixed λ , every connected strip is also a bi-Lipschitz product over its f_k parameter. Thus, when its level fibres are the prescribed arcs or circles, it has the full product-box form required in Definition 4.2.

Proof. Fix $\lambda_0 \in J$. Compactness and a partition of unity combine the finitely many admissible local directions into a smooth field V_k near the k th enlarged strip at λ_0 . After decreasing the parameter neighborhood, the same field works for every λ in that neighborhood, with a uniform positive lower bound c_k . Indeed differentiation in the direction $\sum_i \psi_i V_i$ is $\sum_i \psi_i D_z f_k V_i$; there is no derivative of the coefficients in this identity. All fields are tangent to the boundary, and the supports belonging to different k can be kept disjoint. Extend them smoothly, using slightly larger neighborhoods, and write Φ_k^t for their flows.

The derivative lower bound implies

$$c_k(t_2 - t_1) \leq f_k(\Phi_k^{t_2} z, \lambda) - f_k(\Phi_k^{t_1} z, \lambda) \leq M_k(t_2 - t_1)$$

whenever the indicated flow segment stays in the enlarged strip, for a finite M_k . To justify this for every flow line, use a smooth flow box. The inequality holds on almost every parallel line for almost every parameter by the Lipschitz chain rule and Fubini, hence on all parallel lines and at every parameter by continuity. It applies to the boundary lines by continuity from the interior.

For z in a smaller enlarged strip and λ close to λ_0 , strict monotonicity gives a unique small number $t_k(z, \lambda)$ satisfying

$$f_k(\Phi_k^{t_k(z, \lambda)} z, \lambda) = f_k(z, \lambda_0).$$

The available strip margin ensures existence in both time directions. The lower slope bound and joint Lipschitz continuity give $|t_k(z, \lambda)| \leq C|\lambda - \lambda_0|$ and joint Lipschitz continuity of t_k in (z, λ) , by subtracting the two defining equations and applying the lower slope bound to the time difference.

Choose a Lipschitz cutoff χ_k , equal to one on $[a_k - \delta_k/2, b_k + \delta_k/2]$ and zero outside a slightly larger interval strictly inside $(a_k - \delta_k, b_k + \delta_k)$, and set

$$P_\lambda(z) = \Phi_k^{\chi_k(f_k(z, \lambda_0))t_k(z, \lambda)} z$$

on its support, extending by the identity elsewhere. The supports for distinct k are disjoint. We verify the required injectivity, since a small displacement by itself would not suffice. In flow-box coordinates (y, s) , write $F_\lambda(y, s) = f_k(z, \lambda)$. The uncut reparametrization h_λ is determined by $F_\lambda(y, h_\lambda(y, s)) = F_{\lambda_0}(y, s)$. For $s_2 > s_1$ it satisfies

$$h_\lambda(y, s_2) - h_\lambda(y, s_1) \geq (c_k/M_k)(s_2 - s_1).$$

The cut reparametrization is $s + \chi_k(F_{\lambda_0}(y, s))(h_\lambda(y, s) - s)$. Its derivative in s , where defined, is at least

$$\min\{1, c_k/M_k\} - \text{Lip}(\chi_k \circ F_{\lambda_0}) \|h_\lambda - s\|_\infty > 0$$

after making $|\lambda - \lambda_0|$ small. It has a finite upper Lipschitz bound as well. Thus it is strictly increasing on every flow line, with a uniform lower slope, and is jointly Lipschitz in the transverse coordinates and in λ . Solving this scalar monotone equation gives a jointly Lipschitz inverse. The maps agree with the identity off their supports. Globally on each flow orbit they are increasing reparametrizations with bounded displacement, or increasing degree-one maps on a periodic orbit, and hence are onto.

Consequently P_λ is a homeomorphism of Σ , and $(z, \lambda) \mapsto (P_\lambda(z), \lambda)$ is bi-Lipschitz on the compact local parameter product. Tangency of the flows preserves $\partial\Sigma$.

On the actual strip the cutoff is one, so $f_k(P_\lambda z, \lambda) = f_k(z, \lambda_0)$. Its image is the entire corresponding strip. Indeed, by smallness of the displacement and the extra half-margin, the inverse of a point in the target strip lies where the cutoff is one; the same identity then identifies its value. Complementary components and boundary components are carried to their counterparts by the resulting ambient surface homeomorphism.

Cover J by finitely many such parameter neighborhoods and subdivide it into successive closed intervals subordinate to this cover. Transport from the first endpoint across one interval at a time, composing the transports already constructed. At a subdivision point the next transport starts with the identity. The resulting transport is continuous, preserves all the specified strip values, and is jointly bi-Lipschitz, together with its inverse, because it has only finitely many bi-Lipschitz pieces in the parameter interval. Composing with E gives the asserted simultaneous trivialization.

Finally fix λ . A field with the preceding strictly positive lower slope can be chosen on the entire compact enlarged strip. Every one of its flow lines starting at $f_k = a_k$ reaches $f_k = b_k$ before it can leave the enlarged strip, and does so in time at most $(b_k - a_k)/c_k$. Every point of the strip reaches $f_k = a_k$ uniquely by the reverse flow. The first hitting time of a specified intermediate value is jointly Lipschitz, by the same monotone-equation estimate. Flow boxes show that $f_k = a_k$ is a compact Lipschitz one-manifold, with its boundary on $\partial\Sigma$. The first hitting maps therefore give the claimed bi-Lipschitz product of each of its arc or circle components with $[a_k, b_k]$. Combining this product with the preceding parameter transport preserves both parameters. $\square$

Remark 5.9. The test is qualitative: its constants may depend on the fixed compact collar and its intersection pattern. A global locally bi-Lipschitz change of coordinates preserving the parameters transfers the conclusion directly. At a crease of a triangulated sheet, a direction along the crease can be used when the required positive one-sided slope holds on both incident sheet pieces; integration in the corresponding intrinsic flow boxes gives precisely the hypothesis used above. The test supplies pattern triviality, not the common ambient changed-PL condition or the later quantitative density bound; those remain separate parts of Definition 4.2.

Lemma 5.10 (Uniform weak templates). *After the separate-colour normalizations, all normal disks can be thickened into pre-stacks satisfying Definition 4.2, with the boundary data of Lemma 5.2. On a physical tetrahedron σ of size h_σ with universally bounded shape, away from special boundary widths, the scalar-slope majorant satisfies*

$$G|_\sigma \leq Ch_\sigma^{-1} \max_{e \subset \sigma} W_e. \quad (5.6)$$

Fixed local chart factors and the fixed boundary terms described below apply in the preselected error regions. No quantitative general-position bound depending on the number of actual sheets is required.

Proof. On each interior edge put each colour's central hits near its own mid-edge reference, with the two reference clusters separated. Preserve the within-colour order and give each hit two-sided fractional widths at least cw_j/W_e , using its prescribed coorientation. The total widths fit for a fixed small c . At boundary edges keep the actual prescribed widths. The warp, obliqueness, and pair-reference avoidance give bounded reference lists there, separated between colours and away from vertices. All actual central positions lie in small windows about these references. Formula (5.1) and (5.4) show that a boundary edge contributes only a fixed local term, of scale $C\ell/(\varepsilon c_*)$ in square coordinates, instead of an inverse mesh or belt scale. Mixed interior and boundary widths are allowed by the maximum in (5.4).

We now arrange opposite-colour transversality uniformly over the finite reference tables. In one tetrahedron move the second system by a smooth diffeomorphism equal to the identity near the boundary. Ordinary relative general position for the finite lists of triangles and main creases makes smooth pieces transverse, makes every crease meet opponent triangles transversely along the crease, and keeps opponent creases disjoint. Near the tetrahedron boundary the face arcs already cross transversely and no pair meets an edge, so the motion can indeed be supported away from it. Approximate this diffeomorphism by a PL homeomorphism whose simplex derivatives are uniformly close to its derivatives, still fixed in a smaller boundary collar. Fine affine interpolation of a smooth diffeomorphism supplies such an approximation; taking the derivative error smaller than its inverse-derivative margin keeps it locally invertible, and its fixed boundary and degree give a homeomorphism. Take the error also smaller than all the finite transversality margins.

There are finitely many combinatorial reference types, while their positions range in compact sets with the fixed endpoint and opposite-colour slack. Each of the preceding choices works on an open neighborhood of its data. A finite subcover therefore gives a finite list of PL motions, a common allowed position and tangent perturbation, and common finite complexity and bi-Lipschitz bounds. No general position within one colour's candidate list is needed: the same ambient motion preserves all its disjointness relations. Make the actual clusters narrow enough for these uniform tolerances. The normal prisms then give the desired broad layers.

Here is the required joint, rather than merely separate, transversality verification. At an interior smooth crossing use an intrinsic direction of one sheet transverse to the other. At a main crease crossing use the crease direction on that sheet. Conversely, in the opponent tangent plane there is a direction crossing both adjacent facets with the same strict sign: the crease-normal-plane normals of the two facets are not opposite, and the opponent plane projects surjectively along the crease. At a tetrahedron-face crossing use the boundary-arc direction, which crosses the opponent arc with the same sign from either tetrahedron. Pair crossings avoid all endpoints of these intrinsic edges. These strict directional cones persist for the actual prism simplices and the fine PL approximation of the smooth motion. Orient them using increasing stack parameter and retain small extension margins. More precisely, extend each edge-displacement formula to a slightly larger parameter interval, with the same orders and shared prism triangulations. Compactness of each wall and the strict separation and directional inequalities permit a positive extension retaining these properties. Main creases and tetrahedron faces are glued internal sides of the enlarged collar. Since the wall is proper, its only remaining sides are its parameter ends and its product side on the manifold boundary. Restriction to a smaller enlarged interval therefore gives the open domain, compact margin, and absence of any additional lateral frontier required in Lemma 5.8.

The sufficient test in Lemma 5.8 therefore supplies the entire pattern trivialization, preserving the opposite parameter. The joint boundary collar from Lemma 5.2 extends it to the true boundary.

Finally apply (5.4) to the proportional widths. The finite template motions multiply the bounds by fixed factors, giving (5.6). Parameters and endpoint arrangements are PL in the working coordinates, followed by the common locally bi-Lipschitz collar changes. Upper local slopes at interfaces may be bounded by the adjacent maxima. This is precisely the regularity required in Definition 4.2; it makes no claim of a sharp derivative constant. $\square$

6 Calibrated islands and sharp parameter estimates

We improve the ordinary pre-stacks of Section 5 on the finitely many aligned source patches used to recover the deficient-rank gradients. The product structures and boundary data of Definition 4.2 are preserved; the two scalar gradient bounds become nearly sharp.

6.1 Calibrated slots in protected lattice islands

Only finitely many protected lattice patches require sharp geometry. Work in their aligned affine units, before the tiny ambient edge-generic jitter. First-derivative chart errors and the jitter fractions will be prescribed as small as needed. A common background step H is available from Lemma 5.1. Keep all sharp operations inside the pure lattice, away from the boundary collar. Reserve nested positive patch margins: the placements become fully clustered while still in the aligned lattice, and all later fixed-number mesh buffers fit inside these margins. On the enlarged islands assume that all eligible nearby labels belong to the specified smooth (D, e) sectors of $q_* = q_0$. Use a common weight $W = w_j$ for every feature relevant there and in the required finite-on-compacts buffer. The ratio W/H is chosen arbitrarily small only after the edges and the preceding geometry have been fixed. The order of these choices is collected in the proof of Proposition 6.10.

The two ideal scalar-covector modes in aligned coordinates are

$$(e_1^T, e_2^T) \quad \text{and} \quad (b_1 e_1^T, b_2 e_1^T), \quad |b_1| + |b_2| = 1. \quad (6.1)$$

Call them independent and parallel, respectively. They are the normal forms of the two scalar derivatives in the nonsingular source frames chosen in Subsection 7.3; here they are inputs to the geometric construction. Signs of the coefficients affect label coorientations but not the following geometric orders. An inter-edge for a family increases its indicated axis by H ; a level edge has constant indicated coordinate. An edge is called *eligible* when it lies in the sharp placement region and its upper counts $N_{e,i}$ for the relevant families satisfy

$$N_{e,i} \leq (B_i + \delta')H/W \quad \text{on inter-edges}, \quad N_{e,i} \leq \delta'H/W \quad \text{on level edges}, \quad (6.2)$$

where $B_i = 1$ in independent mode and $B_i = |b_i|$ in parallel mode. A Kuhn tetrahedron is called *regular* when all its required edges are eligible. The inter-edge excess may use a different small tolerance, as long as it is much smaller than the slot loss below. The level tolerance δ' is free to be made smaller after constants depending on that slot loss have been fixed.

Lemma 6.1 (Slots, slab slopes, and gaps). *Fix a small slot-loss tolerance $\eta > 0$. Choose the tolerances in (6.2) sufficiently small relative to η . Central hits on eligible inter-edges can be placed in consecutive slots of axial spacing at least $(1 - C\eta)W$, starting at a common offset of order ηH from the lower endpoint. Fractions stay in a fixed interval $[\gamma(\eta), 1 - \gamma(\eta)]$.*

On each regular tetrahedron all but $C\delta'H/W$ disks of a family are slabs separating its lower and upper vertex layers. The main triangles of these slabs have slopes $O_\eta(\delta')$ from the indicated coordinate planes. Successive slabs are separated along that axis by at least $(1 - C\eta)W$, with a smaller fixed choice of the slot slack. In parallel mode this also holds between the two families, placed in consecutive separate bands. These statements allow arbitrarily small generic perturbations of the slot positions.

Proof. In independent mode allocate slots separately for the appropriate colour on each inter-edge. In parallel mode use one list, all first-colour slots before all second-colour slots. The sum of the two count bounds is at most $(1 + 2\delta')H/W$. Shortening the slot spacing by a sufficiently large multiple of ηW leaves room for both endpoint margins. Preserve the normal-surface order of each family. An edge failing its eligibility test, or lying outside the sharp placement region, instead uses arbitrary margin-spaced placements with same-colour gap at least $cH/(1 + \#\text{hits})$; in the outer buffer use the clustered placements of Lemma 5.10. A tetrahedron is used for the sharp conclusion only when every required edge uses the slots. Use the same combinatorial quadrilateral diagonal for a fixed cut type, also across the two colours.

A Kuhn tetrahedron has two nonempty vertex layers in each indicated coordinate, at heights separated by H . A normal disk other than the cut between those layers hits a level edge. Hence the total number of non-slabs is at most the sum of the level-edge counts, which is $C\delta'H/W$. Every slab hits every inter-edge once. Slabs of a fixed family are nested, so their order toward the upper layer is the same on every such edge, even when their positive label coorientations differ. Only non-slabs can change their ranks in the complete edge lists. Thus the ranks differ by $O(\delta'H/W)$, and the corresponding fractional positions differ by $O(\delta')$. In parallel mode the length of the preceding first-colour band varies between edges only by the same non-slab count. This proves the same estimate for the second band.

Apply (5.5), or its triangular analogue, to these almost-identical fractional positions. Its graph and the two main triangles are $O_\eta(\delta')$ perturbations of the constant-height template. This gives the claimed slopes. To verify the sharp gap, move all fractions of a lower slab by an increment $\Delta \asymp W/H$ no larger than the least edge-slot gap to the next slab. Its odds multipliers are

$$1 + \frac{\Delta}{t_*(1-t_*)} + O_\eta(\Delta(\delta' + \Delta)),$$

where the old fractions differ from t_* by $O(\delta')$. Monotonicity and the common-multiplier property in Lemma 5.7 give at least $(1 - O_\eta(\delta' + \Delta))\Delta$ response in group height u at fixed probability coordinates (a, b) .

This last comparison must be converted to a fixed physical transverse foot. It suffices to consider points with possible separation $O(W)$. Since u is bounded off 0 and 1, their probability coordinates differ by $O_\eta(W/H)$. The near-horizontal slab graph has (a, b) slopes $O_\eta(\delta')$, so this changes u by only $O_\eta(\delta'W/H)$. The group height component is exactly the indicated coordinate divided by H . This proves the same local gap on a common transverse foot. Choose δ' , W/H , and the further generic displacements small enough compared with the previously reserved slot slack. No lower bound on the number of surviving slabs was used. $\square$

Lemma 6.2 (A neighboring majorant). *On a finite enlarged lattice region containing the equal-weight island and its clustered buffer, give original tetrahedra their star-neighbor graph. Its degree is bounded by a fixed D . Let N_e be the total initial wall-hit upper count of both colours on e ; the counts after separate-colour normalization are bounded by these numbers. Put*

$$m_\sigma = 1 + \frac{W}{H} \max_{e \subset \sigma} N_e, \quad M_\sigma = \sum_{\tau} \theta^{\text{dist}_{\text{star}}(\sigma, \tau)} m_\tau, \quad s_\sigma = \frac{W}{M_\sigma}, \quad (6.3)$$

where $0 < \theta < 1/(2D)$ is fixed. Then $M \geq m$, neighboring values of M differ by at most a factor θ^{-1} , and for $p \geq 1$

$$\sum_{\sigma} H^3 M_\sigma^p \leq C_{p,D,\theta} \sum_{\sigma} H^3 m_\sigma^p. \quad (6.4)$$

If $m_\sigma \leq B + e_\sigma$ with fixed B and a small $\sum H^3 e_\sigma^p$, the portion of M above a sufficiently large fixed threshold, including any fixed number of neighboring layers, has correspondingly small p -cost.

Proof. For neighbors σ, ρ , graph distances to any τ differ by at most one, giving $\theta M_\rho \leq M_\sigma \leq \theta^{-1} M_\rho$. Both row sums and column sums of the symmetric kernel $\theta^{\text{dist}_{\text{star}}}$ are at most $C_\theta = \sum_{k \geq 0} D^k \theta^k$. Jensen's inequality with these row sums, followed by summation in σ , proves (6.4). The same estimate applies separately to the convolution of e . The baseline convolution is at most BC_θ . On $\{M > 2BC_\theta + 1\}$ the error convolution dominates M up to a fixed factor. The asserted small cost follows. Neighbor comparability and bounded degree extend it to a fixed number of graph layers. All statements remain valid with the common lattice-volume weight H^3 . $\square$

Lemma 6.3 (Auxiliary mesh). *On an inset union of the original lattice tetrahedra, with its outer boundary deep in the clustered buffer, there is a conforming auxiliary triangulation $\mathcal{G}$ whose cells in an original tetrahedron have sizes comparable to s_σ . Its shapes and star degrees have fixed bounds. In particular all auxiliary sizes are at most CW .*

Proof. Refine each original tetrahedron dyadically to scale comparable to s_σ . By Lemma 6.2 the dyadic level difference on neighboring original tetrahedra is bounded by a constant depending only on θ . Overlay the nested face subdivisions and cone as in Lemma 5.1. The resulting finite normalized configuration list gives the asserted bounds. Since $M \geq 1$, $s_\sigma \leq W$. $\square$

6.2 A master system and its translated regular portions

We now combine the sharp slot placements with the ordinary clustered collars. The first common construction may have extremely thin strips: its purpose is to establish the simultaneous pattern of both colours. On regular portions we replace these strips by small axial translations of their supporting triangles. We then protect smaller translated portions while changing coordinates elsewhere to control all parameter slopes. The final broadening takes place inside those protected portions, where the axial formulas have remained unchanged.

Lemma 6.4 (Master pre-stacks). *The slot placements and their weak clustered continuations admit compact master pre-stacks satisfying Definition 4.2. Before the uniformly controlled template adjustments in fully clustered tetrahedra, the main central disks are the normal triangles and quadrilaterals. On nonclustered diagrams their widths may be microscopic. In the equal-weight fully clustered belt they may be chosen comparable to s_σ , with all neighboring tetrahedra included in the equal-weight cost region. Each main sheet triangle may be subdivided by one common dyadic factor comparable to H/W , without increasing its baseline tangential error or its inverse-parameter estimate by that factor.*

Proof. First make all nonclustered central diagrams qualitatively generic by arbitrarily small allowed variations of edge positions. Two meeting main planes are transverse generically. Containment of a main crease in an opponent plane imposes an equality between its endpoint fractions and the intersections of that plane with the crossed edges, including the fourth relation if it is part of a quadrilateral. It is therefore avoided generically. Two opponent creases are disjoint generically: an interior point on one crease prescribes, by barycentric projection onto the two skew edges of the other, a possible endpoint pair; these pairs form a one-dimensional null set in the two-dimensional endpoint space. Face crossings and edge separation have the same direct generic description. Only finitely many strict conditions occur locally. In fully clustered tetrahedra use instead the robust finite-list motion from Lemma 5.10, equal to the identity near the tetrahedron faces. No such motion is needed in nonclustered tetrahedra.

Assign two endpoint displacements along each crossed edge, strictly ordered according to the disk's coorientation. In a non-robust diagram make them as small as necessary for all its central transversality margins. This is purely qualitative and may require microscopic widths. In an equal-weight fully clustered belt use instead $c \min_{\sigma \supset e} s_\sigma$ as the aligned displacement scale, with a fixed small c . The neighbor-ratio bound makes this comparable to the local s_σ . Farther out return to the ordinary weak proportional widths. Leave several original tetrahedron layers in the quantitative clustered regime between every possibly microscopic width and the eventual boundary of the snapping patch.

All central edge gaps in this region are at least cs_σ . This follows either from the W -slot gaps, or from $H/(1 + N_e) \geq W/(1 + N_e W/H) \geq s_\sigma$, up to the fixed cluster factor. Edges incident to

a non-robust tetrahedron may retain microscopic widths even if another incident tetrahedron is fully clustered. The robust construction allows such mixed widths. At a mixed face the common boundary-arc direction has the same strict crossing sign from both sides, so sufficiently small baseline tangent errors preserve the joint pattern test. Likewise the last stars on which s_σ widths are used stay inside the equal-weight budget region; their transition to ordinary widths occurs only in the already clustered belt. Nested positive patch margins make all these requirements compatible after H is made small.

Subdivide every main triangle by the same dyadic barycentric factor comparable to H/W . At a new vertex P interpolate the original endpoint displacement vectors affinely on its main triangle, obtaining $d^-(P)$ and $d^+(P)$. Use a globally consistent ordering for the stair prisms on the fine triangles and the two half-intervals. On a stair simplex, choose its duplicate vertex as origin for the tangential columns. At fixed layer parameter each other vertex can be compared with that duplicate at the same layer. The tangential correction is then a difference of the interpolated displacements; their common translation cancels. Since these displacements are affine on a main triangle, an original $O(\epsilon H)$ bound produces a relative $O(\epsilon)$ tangential correction on every fine triangle, independent of the subdivision factor.

The vertical column uses one fine-vertex displacement per half-parameter width. That vector is a convex combination of the consistently oriented crossing-edge displacements. Its component against the main separating normal has the correct sign and is at least a fixed multiple of the same convex combination of their magnitudes. In particular a microscopic width does not lose its relative normal margin. The determinant and inverse parameter estimates of Lemma 5.7 therefore persist. The endpoint sheets and their face ribbons bound embedded shells; the same positive-degree argument fills them. On the subdivided cut-boundary ribbons the endpoint secant velocities and their convex combinations obey the response cone of Lemma 5.2; again tangential columns contain only displacement differences. Thus the fine subdivision is compatible also with the prescribed boundary intervals and the ordinary weak costs elsewhere. Strict opponent directional cones survive, so Lemma 5.8 applies to the entire master system. $\square$

Lemma 6.5 (Thin axial translations). *Fix a low-leakage threshold $M_\sigma \leq C_0$. In regular tetrahedra satisfying this bound, at a sufficiently large fixed W -multiple of clearance from tetrahedron faces, main slab creases, non-slab disks, nonregular tetrahedra, and placement transitions, the master may be changed so that its slab strips are exact axial translations of their supporting main triangles, of half-width $a_0 = c_0 W$, where $c_0 > 0$ is fixed sufficiently small. The modified system remains a valid system of master pre-stacks. Deep in such a portion its layer coordinate is exactly the affine conversion of $l_P/a_0 \in [-1, 1]$ to $[0, W]$, where l_P is signed axial height above the supporting plane.*

Proof. On these tetrahedra $W/C_0 \leq s_\sigma \leq W$, so a displacement of size $a_0 = c_0 W$ is bounded by $c_0 C_0 s_\sigma$. At each fine vertex in the safe portion replace the two endpoint displacements by $\pm a_0 e_i$, with signs determined by the local sheet coorientation. Start the change only with the stated clearance, so every affected triangle and possible opponent has regular near-horizontal slab geometry. Main triangle shapes have fixed bounds from the slot margins; the fine sides are comparable to W . Even an abrupt vertex switch therefore adds at most $C(\eta)c_0$ to the tangential derivative. The old nonclustered displacements here can be taken microscopic in advance. Convex combinations of old and new vertex motions retain their strict normal components, including at unswitched vertices. Take c_0 small relative to the slot margins and the near-axis transversality bounds.

We verify embedding through the switch. On each affected main triangle, projection perpendicular to its indicated axis is a uniformly small Lipschitz perturbation of the central triangle's projection in its abstract sheet coordinates. It is one-to-one and covers the common interior footprints with

margin, because the motion is small and the switch stays away from the triangle boundary. At a fixed projected foot, height increases strictly with the oriented layer parameter. Indeed the tangential compensation needed to fix that foot has only the small central height slope, whereas the vertical column has a definite normal component relative to its magnitude. This remains true even where that magnitude is microscopic. It holds on affine pieces and then everywhere by continuity and integration. All altered images have such interior feet, while outside the change region the original embedded geometry is unchanged.

The axial gaps of Lemma 6.1 keep same-axis slabs disjoint. The full exclusion buffers keep the motion away from non-slabs, main creases, and nonregular diagrams. Opponents are either disjoint in parallel mode or retain independent near-axis normals. The same arguments hold throughout the continuous interpolation of the vertex motions, proving that the pattern trivialities are preserved. When all vertices of a fine prism use the same axial translation, its affine stair maps interpolate that translation exactly at fixed layer parameter. Hence its coordinate is precisely the stated axial-height formula. $\square$

6.3 Finite-type snapping with protected buffers

The change of coordinates used below is directed from the new configuration to the old one. Its forward Lipschitz constant is uniform; its inverse Lipschitz constant need only be finite for the particular configuration. This distinction permits the old affine pieces to have arbitrarily small altitudes.

We use the scales of Lemma 6.2 and the auxiliary mesh of Lemma 6.3. Thus, on an original tetrahedron σ ,

$$s_\sigma = \frac{W}{M_\sigma}, \quad M_\sigma \geq 1, \quad 0 < s_\sigma \leq W,$$

and the scales on incident original tetrahedra are comparable by a fixed factor. Every auxiliary simplex D of positive dimension has diameter h_D comparable to the relevant s_σ , has a uniformly controlled shape after rescaling by h_D , and belongs to a uniformly bounded star. All constants in this subsection may depend on these fixed comparisons.

Lemma 6.6 (Bounded local complexity of the master data). *Consider an equal-weight patch with $W \leq H$. Suppose that the following properties hold in every original tetrahedron relevant to the canonical region.*

1. *The central normal disks of each one of the two families are pairwise separated by at least cs_σ . Each disk consists of a bounded number of main triangles of diameter comparable to H , with uniformly controlled shapes. The constants are those of the fixed endpoint margins and normal templates of Lemma 5.7.*
2. *Each main triangle is subdivided in barycentric coordinates by a common dyadic factor comparable to H/W . Its fine triangles therefore have diameter comparable to W and uniformly controlled shapes. The two half-prisms over a fine triangle are triangulated by the fixed ordered stair rule. All their vertex displacements have magnitude at most As_σ , for a fixed A .*
3. *Any subsequent template adjustment in an original tetrahedron is a PL homeomorphism of that tetrahedron with Lipschitz constant and inverse Lipschitz constant at most L . It is affine on a partition with a uniformly bounded number of convex polyhedral pieces, whose numbers of faces are uniformly bounded. The identity is allowed.*

Then there is a constant N , independent of the number of disks and of the small angles between the two families, with the following property. On each auxiliary simplex, the master collar domains and

all their affine parameter pieces admit a common straight simplicial subdivision having at most N simplices. These subdivisions can be chosen consistently on the whole canonical subcomplex. Each collar domain in that subcomplex is a union of closed simplices, and its parameter is affine on every relevant simplex, with values in $[0, W]$. The simplices of this subdivision are nondegenerate for the fixed configuration, but no uniform shape bound for them is asserted.

Proof. First fix an auxiliary top-dimensional simplex D inside an original tetrahedron σ , and let Ψ be its template adjustment. The set $\Psi^{-1}(D)$ has diameter at most Lh_D , hence at most Cs_σ . If an adjusted prism piece meets D , a point of its unadjusted prism lies in $\Psi^{-1}(D)$. The affine prism construction puts every such point within As_σ of the corresponding central fine triangle: write the point as a convex combination of prism vertices and omit their displacement vectors. Consequently its central disk meets a ball of radius Cs_σ .

Choose one central-disk point in this ball for each relevant disk of one family. Points chosen from different disks have mutual distances at least cs_σ . Disjoint balls of radius $cs_\sigma/3$ about these points fit in a ball of radius $(C + c)s_\sigma$. Comparing their Euclidean volumes bounds the number of relevant disks by a constant depending only on C/c . Apply this argument to both families. There are only a bounded number of original tetrahedra to consider when D is a lower-dimensional auxiliary simplex, since the original lattice stars are bounded and their scales are comparable.

For any one of these disks, a relevant fine triangle meets a ball of radius $Cs_\sigma \leq CW$. Every fine triangle has diameter at most C_1W and area at least c_1W^2 . The interiors of the fine triangles in one main triangle are disjoint. The relevant fine triangles lie in a planar ball of radius $(C + C_1)W$, so area comparison bounds their number uniformly. The number of main triangles is bounded as well. Each fine triangle produces only a fixed number of stair tetrahedra. Cutting one such tetrahedron by the fixed finite partition of Ψ produces a bounded number of affine polyhedral pieces with bounded numbers of faces. We have therefore bounded the entire list of affine collar pieces that can meet D .

For completeness, the asserted central-disk separation is precisely the quantitative consequence of the normal-template hypotheses needed here. For equal disk types, the common-direction graph-order estimate bounds the separation below by a fixed multiple of the minimum crossed-edge gap. For different compatible disk types, one disk is triangular. Every vertex of the other disk lies on the prescribed side of its separating plane. On a shared crossed edge its distance to that plane is at least a fixed multiple of the edge gap; on an uncut edge the endpoint margin gives an even larger bound. Convexity gives the same separation for the whole other disk. The gaps used here are at least a fixed multiple of s_σ : slot gaps are comparable to $W \geq s_\sigma$, and, if an edge has n hits, the alternative margin-spaced placement has gap at least a fixed multiple of $H/(1 + n)$. Since $W \leq H$ and $M_\sigma \geq 1 + nW/H$, one has

$$\frac{H}{1 + n} \geq \frac{W}{1 + nW/H} \geq \frac{W}{M_\sigma} = s_\sigma.$$

The fixed cluster factors only alter the separation constant. Equivalently, after adjustment the separation is retained up to the factor L ; the preceding packing argument used the unadjusted configuration so that no smallness of the adjustment relative to s_σ was needed.

We now construct the common subdivision. In a top-dimensional auxiliary simplex take all supporting planes of its relevant affine polyhedral pieces. A bounded number of planes cuts the simplex into a bounded number of convex polyhedra. Every collar piece is a union of these polyhedra, and its parameter has a single affine formula on each of them. On every auxiliary face, overlay the restrictions of the subdivisions from all incident auxiliary simplices. There are only a bounded number of incident simplices and a bounded number of planes from each, so all these overlays have bounded complexity. On auxiliary edges use all induced cut points, and use the same points from every incident face.

Triangulate each polygonal face consistently, respecting its already subdivided boundary, by joining an interior point to its boundary segments. Apply the same rule to shared interior faces of the convex polyhedra, including any subdivisions forced at their boundaries by the auxiliary-face overlays. Finally, join an interior point of each full-dimensional convex polyhedron to its triangulated boundary. Lower-dimensional pieces are treated first, so all choices on shared pieces agree. Degenerate intersections are treated in their actual dimension; an interior point is chosen in the relative interior of each positive-dimensional piece. Thus all resulting simplices are nondegenerate. The bounded numbers of planes, incidences, boundary segments, and stars give a uniform bound for the number of resulting simplices in each auxiliary simplex. Taking the largest of the finitely many dimension-dependent bounds gives N . $\square$

Lemma 6.7 (Canonical snapping and protected buffers). *Let $\mathcal{G}$ be the finite auxiliary triangulation in the patch. Let $\mathcal{C}$ and $\mathcal{P}$ be disjoint subcomplexes, respectively the canonical and protected subcomplexes. On $\mathcal{C}$ suppose that a common subdivision as in Lemma 6.6 has been fixed, with at most N simplices in each auxiliary simplex. In particular, every master collar domain is a subcomplex of that subdivision and every relevant master parameter u_j is affine there with values in $[0, W]$.*

On each remaining positive-dimensional auxiliary simplex $D \notin \mathcal{C}$, suppose the old master parameters already have upper local slope at most BW/h_D on their closed parameter parts. Here the upper local slope of a function on a closed set A at $x \in A$ means

$$\limsup_{\substack{y \rightarrow x \\ y \in A, y \neq x}} \frac{|u(y) - u(x)|}{|y - x|},$$

with value zero at an isolated point; bounds on different incident cells may be combined by taking their maximum. The master parameters are locally Lipschitz for the fixed configuration.

There is a cell-preserving PL homeomorphism J of the patch, from new coordinates to old coordinates, such that:

1. *J is the identity on $\mathcal{P}$;*
2. *$\text{Lip}(J|_D) \leq C$ on every auxiliary simplex D , with C depending only on N , the auxiliary shape bounds, and the dimension;*
3. *on each positive-dimensional canonical cell D , the transported parameter $u_j \circ J$ has a CW/h_D -Lipschitz extension to all of D ;*
4. *on every other positive-dimensional cell, the transported parameters have upper local slope at most CBW/h_D on their parameter parts.*

Consequently the transported upper slope density is at most $C(1 + B)W/s_\sigma = C(1 + B)M_\sigma$, with incident maxima at interfaces. The constants do not depend on the smallest altitude of an old subdivision simplex or on the smallest angle between opposite master sheets. If the outer boundary of the patch belongs to $\mathcal{P}$, then J extends by the identity outside the patch. The inverse of J is locally Lipschitz for the fixed configuration, without a uniform bound being required.

Proof. We first specify the finite representatives, then give the extension estimate used both in the canonical construction and in the buffers.

Finite marked representatives. For each dimension $d \leq 3$, fix a standard d -simplex Δ_d . The type of a subdivision records its abstract simplicial complex, the ordering of the vertices of its coarse simplex, and, for every subdivision simplex, the smallest coarse face containing it. Thus all coarse corners and faces are marked. A bound on the number of subdivision simplices bounds the number of vertices. There are only finitely many such abstract marked types.

Retain only the types that have a straight nondegenerate realization as a subdivision of a simplex with these markings. For each retained type choose one such realization on Δ_d , once and for all. Existence follows by taking any realization of that type and applying the affine map of its coarse simplex to Δ_d . There is no claim that these representatives are optimally shaped. Each chosen representative has finitely many nondegenerate simplices, and the list of representatives is finite. Hence the inverses of all its simplex edge matrices have a finite common bound. This fixed bound is the only representative-shape bound needed below.

For a canonical auxiliary simplex D , let

$$R_D : D \longrightarrow \Delta_d$$

be the simplicial map carrying its old subdivision to its chosen marked representative. This is a PL homeomorphism and preserves every marked coarse face. For positive-dimensional D , its inverse has the bound

$$\text{Lip}(R_D^{-1}) \leq Ch_D.$$

Indeed, on a representative d -simplex with vertices $a_0, \dots, a_d$, whose corresponding old vertices are $v_0, \dots, v_d$, the derivative of R_D^{-1} , in orthonormal coordinates on the affine spans, is

$$[v_1 - v_0 \ \cdots \ v_d - v_0][a_1 - a_0 \ \cdots \ a_d - a_0]^{-1}.$$

The first matrix has norm at most $C_d h_D$, since its vertices lie in D . The second matrix belongs to the finite representative list. This proves the bound on each affine piece, and hence on the convex simplex Δ_d by subdividing a line segment at the finitely many faces it meets. No inverse edge matrix of an old subdivision simplex appears in this estimate.

An explicit coning estimate. Let D_0, E_0 be convex simplices, with interior centers c, c' , respectively. Suppose

$$B(c, r) \subset D_0 \subset B(c, R), \quad B(c', r') \subset E_0 \subset B(c', R').$$

Let $g : \partial D_0 \rightarrow \partial E_0$ be a PL homeomorphism that is L_0 -Lipschitz for the ambient Euclidean distances. Every $x \neq c$ has a unique expression $x = c + t(z - c)$, with $z \in \partial D_0$ and $0 < t \leq 1$. Define

$$C_g(c) = c', \quad C_g(c + t(z - c)) = c' + t(g(z) - c').$$

This is a homeomorphism, whose inverse is the same construction with g^{-1} , and it is PL after coning a boundary triangulation on which g is affine.

To estimate it, write $x = c + t(z - c)$ and $y = c + s(w - c)$, with $t \geq s$. The radial gauge $p(x - c) = t$ is the maximum of the normalized linear forms defining the facets of D_0 . Their gradients have norm at most $1/r$, so $|t - s| \leq |x - y|/r$. Moreover,

$$t|z - w| \leq |x - y| + (t - s)|w - c| \leq (1 + R/r)|x - y|.$$

It follows that

$$\text{Lip}(C_g) \leq L_0(1 + R/r) + R'/r. \quad (*)$$

The case $y = c$ follows directly from the same estimate or by continuity. If g^{-1} has Lipschitz constant L'_0 , the inverse cone similarly satisfies

$$\text{Lip}(C_g^{-1}) \leq L'_0(1 + R'/r') + R/r'.$$

In particular, the forward estimate in $(*)$ requires no bound on $\text{Lip}(g^{-1})$.

We will also use the fact that compatible Lipschitz bounds on the facets give an ambient Euclidean Lipschitz bound on the whole boundary, with a constant depending only on the coarse simplex shape. Here is a direct verification. Let a be its diameter and h its smallest altitude. If x, y belong to a common facet, use the segment joining them. Otherwise choose facets F_i, F_j containing x, y , with $i \neq j$, and choose $k \neq i, j$. With vertices v_i and barycentric coordinates λ_i , put

$$z_x = x + \lambda_j(x)(v_k - v_j), \quad z_y = y + \lambda_i(y)(v_k - v_i).$$

Both points lie in $F_i \cap F_j$. Since $\lambda_j(y) = \lambda_i(x) = 0$ and each barycentric coordinate has Lipschitz constant at most $1/h$,

$$|x - z_x| + |y - z_y| \leq 2(a/h)|x - y|.$$

The three boundary segments from x to z_x , then to z_y , then to y have total length at most $(1 + 4a/h)|x - y|$. Sum the facetwise Lipschitz estimates along this path. This proves the assertion in dimensions at least two; for a one-dimensional simplex its two boundary points are handled directly. Applying the same argument to the inverse boundary map gives the corresponding two-sided assertion when every facet is mapped to its prescribed facet.

Canonical face corrections. We construct J_D on all canonical simplices by induction on dimension, in the form

$$J_D = R_D^{-1} \circ K_D,$$

where $K_D : D \rightarrow \Delta_d$ is PL and, for $d \geq 1$, satisfies

$$\text{Lip}(K_D) \leq C/h_D, \quad \text{Lip}(K_D^{-1}) \leq Ch_D.$$

These are uniform two-sided estimates in normalized cell units. At vertices the map J_D is the identity. For an edge, prescribe the marked endpoint values and use the affine map K_D to its standard interval.

Suppose now that the maps have been constructed on all proper faces of a canonical simplex D . On a facet F prescribe

$$K_D|_F = R_D|_F \circ J_F = (R_D|_F \circ R_F^{-1}) \circ K_F. \quad (**)$$

The parenthesized map is a simplicial comparison between the chosen representative of the face type and the face subdivision induced by the chosen representative of the type of D . Both are fixed realizations of the same marked old face combinatorics. There are only finitely many possible comparisons, including the finitely many vertex-identification choices, and each comparison is bi-Lipschitz. Their two-sided constants are therefore uniformly bounded. Uniform shape control gives $h_F \asymp h_D$, so the induction hypothesis in $(**)$ gives the required normalized two-sided bounds on every facet.

The facet prescriptions agree on their intersections: on any smaller face they equal $R_D \circ J$ with the already constructed map of that face. They therefore form a PL boundary homeomorphism onto $\partial\Delta_d$, mapping each facet to its marked facet. The preceding boundary estimate supplies uniform two-sided bounds on the whole boundary. Cone from the barycenter of D to the barycenter of Δ_d . The coning estimates give the asserted bounds for K_D ; only the dimension and normalized shape constants enter. Formula $(**)$ also proves $J_D|_F = J_F$, as required for gluing.

The induction has at most three levels. Thus the constants depend only on the fixed representative list and the coarse shape bounds, not on the shapes of any old subdivision simplices. Combining the estimates for R_D^{-1} and K_D gives

$$\text{Lip}(J_D) \leq (Ch_D)(C/h_D) \leq C.$$

Parameter estimate on canonical cells. Fix a positive-dimensional canonical cell D and a parameter u_j on its old closed domain in D . On the representative triangulation its pullback $u_j \circ R_D^{-1}$ is affine on the relevant subcomplex. Give every representative vertex outside that subcomplex the value zero, retain the prescribed values at all vertices in the subcomplex, and extend affinely over every representative simplex. This defines a continuous PL function $\tilde{u}_{j,D}$ on all of Δ_d . It agrees with $u_j \circ R_D^{-1}$ on its domain and all its vertex values lie in $[0, W]$.

The fixed inverse edge-matrix bounds for the representative simplices imply $\text{Lip}(\tilde{u}_{j,D}) \leq CW$ on the convex standard simplex. Consequently

$$\tilde{u}_{j,D} \circ K_D$$

is a CW/h_D -Lipschitz extension to D of the transported parameter $u_j \circ J_D$. This argument applies separately to each label. It does not require the numerical vertex labels to belong to a finite set; only their common range $[0, W]$ is used.

Protected cells and buffer cells. Set J equal to the identity on the protected subcomplex. There is no conflict with the canonical construction because $\mathcal{C} \cap \mathcal{P} = \emptyset$. Extend over all remaining auxiliary vertices by the identity, and then over the remaining positive-dimensional simplices by increasing dimension. On a remaining simplex D , its proper faces already carry compatible cell-preserving PL homeomorphisms with uniformly bounded forward Lipschitz constants. They give a boundary homeomorphism of D , again with a uniform forward constant by the boundary estimate. Cone this map from the barycenter of D to itself. The forward bound in $(*)$ gives $\text{Lip}(J_D) \leq C$, regardless of the inverse Lipschitz constants on its boundary. Since the dimension is bounded, iterating this step preserves a uniform forward bound. No regular subdivision of a protected trace is needed for this step: its map is the identity, and only the forward bound of the prescribed boundary map enters the cone estimate.

Let A_j be an old parameter domain and put $A_j^{\text{new}} = J^{-1}(A_j)$. The map is cell-preserving, so on a buffer cell D it maps $A_j^{\text{new}} \cap D$ into $A_j \cap D$. The elementary composition inequality for upper local slopes, together with $\text{Lip}(J_D) \leq C$, yields

$$\text{lip}_{A_j^{\text{new}} \cap D}(u_j \circ J)(x) \leq C \text{lip}_{A_j \cap D} u_j(J(x)) \leq CBW/h_D.$$

On a protected cell the same statement holds with J the identity. At an interface, every sequence approaching a point lies, after passing to a subsequence, in one of its finitely many incident closed auxiliary cells. Taking the maximum of their bounds gives the asserted upper local slope on the whole parameter domain. Scale comparability changes W/h_D to at most CW/s_σ .

All the cell maps agree on common faces and are homeomorphisms preserving each cell. They therefore glue to a PL homeomorphism of the finite patch. Each affine piece is nondegenerate for the fixed input, so the inverse is locally Lipschitz for that input; its constant may deteriorate with old sliver degeneracy. If the outer boundary is protected, the identity extension agrees there and gives the asserted ambient change of coordinates. Pulling the collar system back by J preserves its intersections, labels, and parameter trivializations exactly. A previously given locally bi-Lipschitz trivialization remains such after composition with J^{-1} , since only a finite, input-dependent inverse bound is needed for that qualitative statement. $\square$

Figure 4 summarizes the directions of the maps in the preceding proof and distinguishes their uniform forward bounds from the input-dependent inverse bound.

The disjoint subcomplexes required in Lemma 6.7 are selected using the already quantitative belts of the master construction. Put in the canonical complex every closed auxiliary cell on which a microscopic master width can cause an uncontrolled parameter slope, together with all its faces.

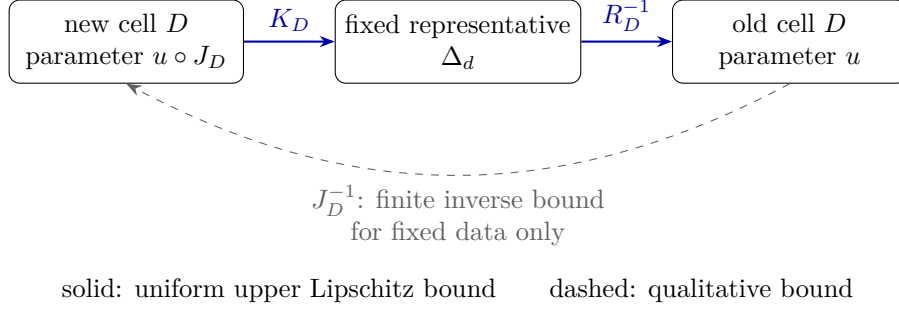


Figure 4: The direction of canonical snapping, in normalized cell units. Both solid arrows have uniform upper Lipschitz bounds; K_D also has a uniform inverse bound. No uniform bound is required for R_D or J_D^{-1} . The new parameter is the old parameter composed with $J_D = R_D^{-1}K_D$. The diagram records map directions and bounds, not the shapes of the triangulations.

Select the protected cells strictly inside the fully translated regular region, and at the outer edge of the snap patch strictly inside the quantitative clustered belt. Leave several auxiliary cell layers of the same quantitative regime between these protected cells and the uncontrolled cells. Equivalently, trim the initial protected regions by a fixed number of closed auxiliary stars before selecting their closed cell subcomplexes. Since every auxiliary cell has diameter at most CW , this only enlarges the prescribed geometric buffers by a fixed multiple of W . The two closed subcomplexes are then disjoint, and every remaining buffer cell has the master slope bound used in the Lemma. At the outer transition the canonical complex is kept inside the region where displacements are at most Cs_σ ; any return to larger ordinary weak widths takes place in the quantitative cells outside it. Thus the bounded-complexity hypothesis is used exactly where it is available.

The separation from protected cells is essential to the stated proof. On canonical faces, the uniformly controlled correction is obtained by comparing two finite representatives as in (**). The identity is prescribed only on the disjoint protected subcomplex; the intervening cells use the forward coning estimate and the old quantitative parameter bound. No uniform inverse estimate is imposed on those buffer maps.

Figure 5 shows how the quantitative buffer separates the canonical construction from an unchanged translated region. Broadening will take place farther inside that protected region.

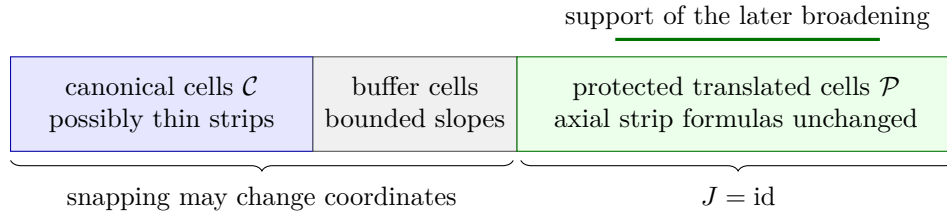


Figure 5: A schematic local passage from canonical cells to an interior protected region. The buffer already has quantitative parameter bounds, so only the forward Lipschitz bound for its snapping map is needed. Snapping fixes the protected region; broadening is supported strictly inside it. The bands depict adjacency, not metric widths or the global topology of the subcomplexes. Protected cells at the outer patch boundary instead lie in the quantitative clustered belt.

6.4 Broadening the protected strips

Lemma 6.8 (Simultaneous broadening by an ambient flow). *On the deep protected translation regions of Lemma 6.5, after snapping, the half-width can be increased from $a_0 = c_0 W$ to*

$$a_1 = (1 - C\eta)W/2, \quad (6.5)$$

with fixed slack below half the regular slab gap. It equals a_1 outside larger excluded-set buffers and equals a_0 near the edge of the protected region. The change is induced by a common ambient locally bi-Lipschitz homeomorphism preserving every layer parameter. The resulting pre-slope density is bounded by CM_σ throughout the sharp construction. On a region of constant width a_1 ,

$$\nabla u_j = \epsilon_j e_i + O(\eta), \quad |\nabla u_j| \leq 1 + C\eta, \quad \epsilon_j \in \{-1, 1\}, \quad (6.6)$$

where the error is in aligned coordinates and may be made smaller than any prescribed multiple of η except for the fixed slot-loss contribution in the principal component.

Proof. Choose a common smooth cutoff supported well inside the unchanged translation subcomplexes, equal to one beyond larger excluded-set buffers. Mollified distance cutoffs give derivative $O(1/(NW))$ when the buffer width is NW . First take the fixed integer N sufficiently large, depending on the slot margins and the previously fixed constants. Let $a(x)$ vary between a_0 and a_1 using this cutoff, so $|\nabla a| \leq C/N$.

On a supporting main triangle P , let $l_P(x)$ be signed axial height above its plane, with sign chosen according to increasing layer parameter. Use the strip $|l_P| \leq a(x)$ and the layer parameter given by affine conversion of l_P/a :

$$u_P(x) = \frac{W}{2} \left(1 + \frac{l_P(x)}{a(x)} \right). \quad (6.7)$$

The support of the change is far enough from the main triangle boundary, from all unswitched fine triangles, and from all non-slab, crease, face, and nonregular exclusions that every supporting-plane formula required within several multiples of W is valid. Same-family gaps from Lemma 6.1, including the opposite parallel band, stay larger than the two half-widths with fixed slack. The slice slopes are small because the supporting planes are near horizontal and $|\nabla a|$ is small. In independent mode the only possible concurrent strips have near-independent coordinate normals. These facts remain true on a small open enlargement of each interval $l_P/a \in [-1, 1]$.

We verify that this simultaneous change preserves the parameter topology. Interpolate smoothly in time from a_0 to $a(x)$, writing $a_\tau(x)$, $0 \leq \tau \leq 1$, with the same spatial cutoff. On each relevant plane neighborhood impose

$$D_x(l_P/a_\tau)X_\tau + \partial_\tau(l_P/a_\tau) = 0. \quad (6.8)$$

Locally the list of possible constraints contains one member or an independent pair. Their gradients have the same independence as the near-axis normals, since

$$D_x(l_P/a_\tau) = a_\tau^{-1} (\nabla l_P - (l_P/a_\tau) \nabla a_\tau).$$

Thus the linear equations have a smooth local solution. For example use the right inverse of the one-row or two-row gradient matrix. The right sides are linear in $\partial_\tau a_\tau$, and this choice has size $O(W)$ on the necessary open strip neighborhoods. Take a finite space-time cover such that every neighborhood's list includes all constraints potentially present there. A smooth partition of unity preserves the linear equations on their required sets. Include the complementary region with zero

field, and arrange zero wherever $\partial_\tau a_\tau = 0$; the local choices are linear in that quantity. The field is compactly supported in the finite patch.

Its flow exists on the whole time interval. Increase the buffer factor if necessary so its $O(W)$ motion stays within all the supporting-plane and endpoint extension margins. Equation (6.8) makes l_P/a_τ constant along the flow. The backward flow gives the converse, so all strip boundaries and all layer parameters are transported exactly. In particular this is a common ambient change of both systems. It leaves the nonflat seams unchanged and preserves the joint product trivializations and the PL-in-changed-coordinates property. No uniform derivative bound for the flow or its inverse is needed.

The quantitative bound is obtained directly from (6.7), not by differentiating that flow:

$$\nabla u_P = \frac{W}{2a} (\nabla l_P - (l_P/a) \nabla a).$$

Since $a \geq c_0 W$, $|l_P| \leq a$, and the plane slopes and $|\nabla a|$ are bounded, this is bounded by a fixed constant. Broadening occurs only on low-leakage tetrahedra, where s_σ is comparable to W . Elsewhere the snapped bound is CM_σ . Thus this bound holds everywhere. When $a = a_1$, $\nabla a = 0$, $\nabla l_P = \epsilon_j e_i + O_\eta(\delta')$, and $W/(2a_1) = 1 + O(\eta)$. Taking the remaining tolerances as in Lemma 6.1 proves (6.6). $\square$

Proposition 6.9 (Sharp-island geometric contract). *Fix η , the compact chart and basis bounds, and the finite template choices above. The calibrated islands admit pre-stacks with all the qualitative properties of Definition 4.2 and with the following quantitative properties.*

1. *Throughout an enlarged equal-weight island the pre-slope density is at most CM_σ in aligned units. This constant is independent of microscopic master widths, central crossing angles, final guard gaps, H , W , and the final level-count tolerance δ' .*
2. *In deep regular low-leakage portions, the scalar gradients obey (6.6). At concurrent independent strips the two-coordinate squared Frobenius cost is at most $2 + C\eta$. At an occupied diagonal transition after routing, the active coordinate has gradient $\pm \nabla u_j \pm \nabla u_k$, whose squared norm has the same bound. In parallel mode the opponent bands are disjoint in these portions and the squared norm is at most $1 + C\eta$. Occupied-stair speeds multiply these bounds by at most $(1 + C\zeta)^2$.*
3. *In a regular tetrahedron the excluded CNW neighborhoods of its faces, all its main slab creases, and its non-slab disks occupy relative volume at most*

$$C_N \left(\frac{W}{H} + \delta' \right). \tag{6.9}$$

The canonical protection and broadening margins only change the fixed factor C_N . Nonregular tetrahedra, high-leakage tetrahedra, their required fixed-neighbor layers, and patch boundary margins are charged by their full M^P -cost.

The norm statements concern the smooth compatible label sectors after the combined label barrier, away from central radial activation cutoffs. They are geometric scalar-pair estimates; their physical conversion and the saturation argument are made in Lemma 7.6.

Proof. The master construction, the translation switch, the canonical snapping, and the common broadening flow preserve the entire two-colour pattern and its parameters, as proved above. In the canonical region all displacements are at most Cs_σ . In particular the larger translation displacement

a_0 is used only when s_σ is comparable to W . Thus Lemma 6.6 applies there, and Lemma 6.7 gives the first assertion in the formerly uncontrolled region. Its buffer cells use only a uniform upper Lipschitz bound for the map from new to old coordinates and an already quantitative master bound. Ordinary weak widths may be larger outside the canonical region, but their transition is fully clustered and has the same CM_σ bound in the equal-weight belt, or the ordinary edge bound outside it. Finally Lemma 6.8 supplies the remaining transition and deep-strip estimates. The finite representatives, slot margins, and chart factors were all fixed before the small count errors, so no new dependence on those errors has entered.

In independent mode the two indicated axes are orthogonal. Equation (6.6) gives squared Frobenius norm at most $2 + C\eta$ when both coordinates contribute. A routed diagonal has parameter $d = w_j - u_j + u_k$, with possible coorientation changes. Hence its gradient is the asserted signed sum of two near-orthogonal covectors, with the same bound. Only one output coordinate is active on that transition. In parallel mode the two bands are separated by the axial gap, so they have no such overlap in the protected region. The occupied-interval stair conversion has speed at most $1 + C\zeta$, giving the stated factor. These statements use the combined label pinning to identify the intended smooth sector; they assert no estimate across an unrelated sector or an active central cutoff.

For the volume estimate, there are $O(H/W)$ slab disks, each with at most one main crease of length $O(H)$. Their CNW -tubes cost at most $C_N H^2 W$. There are at most $C\delta' H/W$ non-slabs, each of area $O(H^2)$; thickening their finitely many main triangles by CNW costs at most $C_N \delta' H^3$ (the smaller edge and vertex terms are absorbed when W/H is small). The tetrahedron-face buffers cost $C_N H^2 W$. Divide by volume comparable to H^3 to obtain (6.9). Auxiliary cells have size at most CW , so trimming protected subcomplexes and reserving the flow neighborhoods only enlarge this fixed buffer factor. For nonregular and high-leakage sets use Lemma 6.2; the appropriate analytic error tests in Lemma 7.2 make their full costs small. No small volume assertion for those uncontrolled tetrahedra is being used in place of that M^p payment. $\square$

6.5 The pre-stack contract and the order of constants

Proposition 6.10 (Geometric realization of pre-stacks). *Suppose the central starting systems have the boundary data of Lemma 5.2 and the wall-by-wall edge upper counts used in the separate-colour normalization of Proposition 5.6. Suppose also that the weighted edge tests have been selected as in Lemma 5.3. The constructions of Sections 5 and 6 give the pre-stacks required in Definition 4.2: same-colour compact disjoint collars; opposite arc or circle products with the full parameter rectangles; individual pattern trivializations preserving the opposite parameter; the exact boundary correspondence; and locally bi-Lipschitz changed coordinates in which the relevant endpoint arrangements and parameters are PL.*

Their pre-parameter slopes admit a density G satisfying the ordinary bound (5.6), the fixed boundary and chart bounds of Lemma 5.2, and the island bound $G \leq CM_\sigma$ of Proposition 6.9. All constants in these inequalities may be fixed before choosing final guard gaps, mesh resolutions, and sample weights. Consequently any crude compact-local integral capacities deduced from these inequalities and the conditional edge-variation bounds are independent of those later choices. The actual simultaneous choice of the capacities, meshes, and samples is supplied by Lemma 7.3 and Proposition 7.4.

Proof. Normalize one colour at a time, keeping its specified boundary traces and without increasing any individual wall-edge count. For this conclusion only the resulting normal data are needed; an ambient normalization isotopy need not fix the other colour. The weak templates give the claimed collar and pattern properties away from the calibrated islands. On the islands the master

construction has the same properties and the subsequent common ambient changes preserve them exactly. All attachments to the outer collar use its joint trivialization. In particular no step merely chooses independent individual product charts and assumes they preserve the other coordinate. Compactness of each individual sampled wall and local finiteness of the family give the required qualitative local bi-Lipschitz statements for each fixed construction.

We make the constant order explicit. First fix coarse working charts, compact basis bounds, feature lengths and aspect bounds, the slot loss η , and the weak boundary-jitter slack. Fix the resulting finite weak template list, its angle tolerances, and the boundary compression ε . These fix the normal-template shapes, the auxiliary neighbor-ratio bounds, the cardinality of the canonical arrangements, all finite representatives in Lemma 6.7, and the translation and broadening buffer factors. On a given compact their constants are finite, though they need not be small. Next choose the level-edge tolerance and generic perturbation fractions small against those fixed bounds; in calibrated patches fix also the chart and ambient-jitter error fractions. Their possibly large density constants are then known. The analytic mean errors are chosen smaller afterward. None of these choices involves the eventual guard gaps.

Fix the local edge-variation and density bounds, including any chart weights, before the guard nets are set. They give finite candidate capacities for $\int G^p$ by (5.6), (6.4), and the fixed boundary terms. Having chosen the guards and their forbidden label neighborhoods, take d_0 , the mesh, and every displacement fine enough for the associated spatial buffers. Boundary warp motion has size $O(\delta)$ with $\delta \ll d_0$; its inverse label constants do not deteriorate under this thinning. The interior meshes keep their shape lists under refinement and their fixed jitter density factors. Sample last, taking all feature slots sufficiently fine and, in the finitely many equal-weight islands, W/H as small as necessary. Arbitrarily small allowed feature components do not obstruct this final refinement. Whole-interval and occupied-interval stair errors charge absolute gap lengths and rounding errors, so they may also honor the already prescribed guard and label accuracies.

Microscopic master widths introduce no additional capacity constant. They are removed precisely on the canonical cells by a map with uniformly bounded upper Lipschitz constant, while the direct representative-label estimate is CW/s_σ . Subdivision uses displacement differences and introduces no H/W loss. The final broadening estimate comes directly from (6.7). The qualitative inverse bounds of the common coordinate changes, the pattern trivializations, and the eventual product or ball extensions in Theorem 4.11 are never substituted into these estimates.

For path estimates take upper slopes relative to the closed parameter domains, as defined in Lemma 6.7, with adjacent-cell maxima. The fixed-input parameters are locally Lipschitz on their compact collars. On exceptional lower-dimensional interface strata one may enlarge G further to any necessary actual local collar Lipschitz bound, with locally finite prescriptions. Such strata have zero ambient volume. This gives the pointwise control needed for one-dimensional coarea on restricted path portions without asserting a uniform two-point Lipschitz bound across arbitrary gaps between parameter pieces. It changes none of the stated integral capacities. $\square$

7 Assembly for exponents greater than two

We now choose the data in the carrier, quantizer, wall, and mesh constructions simultaneously. Throughout this section, put

$$\Lambda = \Omega', \quad E = \int_{\Omega} |Df|^p, \quad p > 2.$$

Matrix norms without a subscript are Frobenius norms. All local finiteness statements refer to the open source or target, rather than to their closures.

Our objective is to prove Theorem 1.1 by constructing an approximant with any prescribed positive error in Equation (1.1). The construction first gives a locally bi-Lipschitz homeomorphism. The last use of Theorem A.1 gives the smooth map. We keep separate the constants allowed to multiply an initial energy tail and those allowed only in later absolute-error prescriptions. This distinction is essential: the constants of an individual topological normalization or an individual chart need not be bounded in terms of p .

Two kinds of estimates enter the proof. Ordinary meshes give the *weak estimate*: an energy bound in terms of the carrier energy. Around the retained rank-one and rank-two data, calibrated meshes give the *sharp estimate*: an energy bound close to that of the affine comparison. Label pinning controls the mean gradient there, so a quantitative uniform-convexity argument yields strong gradient accuracy; this is the classical energy-to-strong-convergence principle associated with Clarkson [14]. The precise pointwise estimate used here is proved in Lemma 7.6. The complete order of choices is collected in Remark 7.5.

The estimates have four spatial roles. The sharp estimate recovers the derivative on small source patches containing almost all the retained rank-one and rank-two energy. Reserved full-rank cubes will instead receive their affine comparison maps at the end. When $2 < p < 3$, marked interiors are also set aside, for a later source compression. On the remaining region, and on the shells and collars needed to make these replacements, the weak estimate must give small energy. These buffered regions can overlap; their separate bounds will be imposed simultaneously. We first reserve the full-rank cubes and the carrier budgets, then choose the sharp patches and realize the wall parameters that recover their gradients.

7.1 Initial parameters and carrier budgets

Choose a small aspect $\gamma > 0$ for the protected rank-one rectangles; use squares for rank two. Choose a small slot-loss parameter $\eta > 0$, and then $\zeta, c_* > 0$ sufficiently small compared with η . Here ζ controls occupied root length and sampling density, and c_* is the ratio of a true target interval length to its root weight. The ordinary weak constant, denoted C_w , may depend on

$$p, \gamma, \eta, \zeta, c_*,$$

but it is independent of all later jet truncations, chart bounds, mesh resolutions, and guard spacings. The physical bulk meshes and clustered templates of Proposition 6.10 supply precisely this kind of constant. Protected rectangle aspects are included in C_w . We will prove a leading sharp error bound

$$C_p(\eta + \gamma)(1 + E) + \text{freely small errors}, \quad (7.1)$$

where C_p is independent of γ, η and of the finite jet bounds. Thus γ and η can be fixed at the outset to make the first term as small as required.

In what follows, a freely small error is prescribed only after every factor that will multiply it has been fixed. On noncompact parts we use a locally finite compact cover and positive prescriptions with sum as small as required. A finite number of additional local weights can be included in these prescriptions by dividing the allowed error by their maximum. Equivalently, for a countable collection of local costs, prescribe the j th weighted cost to be less than $\delta 2^{-j}$, after its weights are known. Positive tolerance minorants and simultaneous locally summable choices are supplied by Lemma 1.2. This convention never replaces the initial choice of tails against C_w .

By Lemma 2.1, we may choose compact sets of regular rank-one and rank-two points covering those ranks except for an arbitrarily small tail in the measure

$$d\mu = (1 + |Df|^p) dx.$$

First make this tail small compared with the already fixed weak constant. On the selected sets the positive singular values can be bounded above and bounded away from zero. Separately choose a compact set of full-rank points with finite jet and inverse-jet bounds, leaving an equally affordable tail in $\int |Df|^p$ on that rank. The full-rank points and the deficient compact data are separated.

Use Proposition 2.16 to reserve finitely many full-rank patches. Write B_i for their inner cubes and b_i for the positive affine comparison jets. The enlarged support and comparison cubes are disjoint across distinct patches. In normalized cube coordinates, let $t_i = 0$ on ∂B_i and let $s > 0$ be the cut-band parameter. The finite jet factors are fixed before s is made small. For any fixed large C' , all bands

$$|t_i| \leq C's \quad (7.2)$$

can consequently have arbitrarily small total $|Df|^p$ integral, even with the jet factors and C_w included. The constant C' includes the finitely many buffer enlargements used below. The power-mean and normalized value tests at the full-rank points are then imposed on sufficiently small outer cubes. Vitali selection and finite truncation give coverage by the inner cubes, and even by $\{t_i < -3s\}$, up to the required full-rank tail. We retain

$$\sum_i \int_{B_i} |Df - Db_i|^p \text{ as small as required,} \quad |b_i^{-1}f - \text{Id}| \ll cs \quad \text{in normalized cube units.} \quad (7.3)$$

The cube sizes also make the eventual value cost of these corrections arbitrarily small. Subsequent value approximations will preserve the second test with room.

Before fixing the flattening and endpoint data, choose a positive continuous source tolerance a small enough for the desired value error and the finite normalized full-rank tests, reserving room for all later changes. In Proposition 2.6 use a target tolerance $a_\Lambda(y) \leq a(f^{-1}(y))/4$. Then $|F_\kappa - f| < a/4$ for every $0 \leq \kappa \leq 1$. After this share has been reserved, the two carrier rounds use the remaining allowance in a , as in Proposition 2.13. When $2 < p < 3$, fix the marked-collar enlargement factor large enough for the fixed two-sided buffers used below, before assigning the carrier's facet budgets; only the collar widths are chosen after both rounds.

Apply the flattening and two-round carrier construction, using Proposition 2.6 and Proposition 2.13, and then the quantizer calibration of Lemma 3.4. Choose the line directions and the flattening tolerance λ so accurately that, on the ultimately retained rank-one data, the range direction of Dq_0 differs from the long rectangle axis by $o(\gamma^2)$. The derivative of the flattening has relative error $C\lambda$ there. The central fraction of the quantizer uses a common scalar times orthogonal projection onto the protected plane; its scalar $1/a_*$ can be taken arbitrarily close to one. Trimming to these fractions loses arbitrarily little μ -mass. Kernel inclusion follows from Lipschitz composition on the retained data, and the positive singular-value bounds preserve the rank when the errors are small enough.

Denote by K_c the common compact data surviving the carrier rounds, before the later phase losses. In both rounds the implant supports *and their regular comparison boxes* avoid K_c with room. Indeed the regular assignments are first restricted to compact sets separated from the central data; only finitely many operations meet a neighborhood of K_c . The two rounds can therefore use common positive separations. In particular,

$$F_b = F_\kappa \quad \text{near } K_c, \quad q_0 = TJF_b, \quad |Dq_0| \leq C|DF_b| \quad \text{a.e.} \quad (7.4)$$

All supports and comparisons are chosen sufficiently fine relative to the reserved full-rank patches and their bands. The regional charging conclusion of Proposition 2.13 has only the two preassigned buffer enlargements, so this is a requirement on those comparisons before they are fixed.

Here are the resulting regional budgets, stated explicitly for their later uses. Put

$$Z = \Omega \setminus \bigcup_i \{t_i \leq -s\}.$$

Outside K_c and the marked cubes, the $|DF_b|^p$ integral on Z is small against C_w . Each regular implant contributing there charges a disjoint comparison away from K_c . Exterior comparisons in the second round also avoid the first-round marked cubes. Both charging enlargements stay in $\{t_i \geq -3s\}$ for every i . The unchanged derivative is bounded by $C|Df|$, while exceptional implant costs are confined to marked interiors. The rank-zero set contributes no $|Df|^p$ cost. The same assertion holds on the enlarged full-rank shells outside marked interiors, with the finite correction jet factors included, because their charging enlargements lie in the larger bands (7.2).

If $2 < p < 3$, use the marked cubes U of Proposition 2.13. Their enlarged union has arbitrarily small $\int(1 + |Df|^p)$. After both carrier rounds choose collar widths $d_U > 0$, with fixed-factor two-sided buffers and mutual separation, so that

$$\sum_U \frac{\ell_U}{d_U} \int_{\mathcal{C}_U} |DF_b|^p \quad \text{is as small as required.} \quad (7.5)$$

Here ℓ_U is the side scale, and $\mathcal{C}_U$ is an enlarged collar of width comparable to d_U about ∂U . The prescribed smallness includes C_w and the full-rank jet factors wherever those tests overlap. The unchanged-base slice estimate in Proposition 2.13 allows the widths to be chosen after the rounds, however small the unchanged neighborhoods have become. The marked cubes, their enlargements and collars avoid the deficient protected data. Their sizes are also fine enough to make the oscillation of f on each enlarged cube meet all required value tests. There are no marked cubes for $p \geq 3$. The weights ℓ_U/d_U are now fixed for all later absolute-error prescriptions.

7.2 Polar localization and the active derivative estimate

The central-tube modification will change q_0 to a map q_* with only finite local Sobolev bounds in its low-radius regions. We arrange that the final nonstrict map is constant there. Thus the weak derivative estimate will use $q_* = q_0$ wherever an output parameter is active; the larger low-region bounds will enter only the crude capacities needed to pin the labels.

We next choose the protected arrays, phases and subcells from Lemma 3.4 and the polar setup of Lemma 3.7. Arrays can be arbitrarily fine for the value accuracy and for the quantizer's maximum-activity and clearance tests. Phase selection retains the protected μ -weight in rectangular interiors, up to arbitrarily small loss, and avoids atoms at the centers and the rays to corners. Additional subedge divisions along straight protected sides can also avoid ray levels of positive μ -mass: before overlaying the polygon meshes, take strict interior cuts in the individual clipped diagrams and slide these finitely localized cuts generically in their available intervals.

All full-cell and lower-face geometries, and their compact-local aspect bounds, are now fixed. For unprotected polygons, use the original-vertex localization in Lemma 3.6 before fixing the fine subcells. In detail, the worst relevant aspect factor on each true polygon is finite at this stage. Moreover $Dq_0 = 0$ a.e. on each vertex level set. Thus the integral of $|Dq_0|^p$, multiplied by that fixed aspect factor and any already assigned local weights, tends to zero on the preimages of shrinking vertex neighborhoods. Properness makes these tests locally finite. Choose the neighborhoods with summable costs, and then fit the fine subcells. Everywhere else the aspect factors in the ordinary weak estimate are absolute or are the already fixed protected rectangle factor.

Choose first the conical boundary-layer widths in the full output cells, as in (4.16). Their weighted excess costs can be made summably small before choosing the central activation radii. To

see the order, in one boundary polygon the two-factor interpolation has a Lipschitz bound depending on its fixed aspect and c_* : its radial factor satisfies $\rho/r \leq C$ and $|\rho'| \leq C/c_*$, while the inverse angular derivative contributes an aspect factor times $1/r$. The r cancels. Once the layer thickness is fixed, its normal derivative is controlled by requiring the factor motions to be sufficiently small. The parametrizations of the full input blocks are already fixed exact locally bi-Lipschitz data, so these are ordinary absolute-continuity choices on known maps.

Now take the activation radii a_D sufficiently small for those motion tests, and retain compact data of both positive deficient ranks where $r > 5a_D$, strictly inside smooth sectors. Center and ray avoidance makes the additional loss arbitrarily small; no absolute continuity of the projected protected measure is being assumed. Choose $r_{u,D} \ll a_D$ and use Lemma 3.9 to obtain M_y, h_*, M_x, F_*, q_* . The exterior changes from q_0 are confined to deep low regions within individual subcells, which we may require to satisfy

$$r(q_*) < a_D/100. \quad (7.6)$$

The relative radius is set equal to one on subedges, giving a continuous function on the two-complex. All openings, central changes and labels retain the preliminary fine value tests.

Write $K^\sharp \subset K_c$ for the final protected compact set. It lies in the interior of M_x ; the slab heights, tube clearances and strict sector insets give room to require $q_* = q_0$ on its neighborhood. Lemma 3.4 gives, a.e. there,

$$|Dq_0 - Df| \leq C(\lambda + 1 - a_*)|Df|. \quad (7.7)$$

All losses from K_c made so far are small against C_w .

The wall barriers will require arbitrarily fine local closeness of $q_h = Th$ to q_* . Prescribe that closeness so that the relevant radii are comparable, including on all mesh-neighborhood enlargements. The radii are positive and locally bounded away from zero on M_x . Whenever q_h lies on an edge, or in the possible active face range $r(q_h) \geq a_D/2$, the comparison neighborhood is contained in an open high-budget region where $q_* = q_0$. Such neighborhoods exist by (7.6). Prescribe sufficiently small stair shifts, also relative to L_D in the logarithmic coordinate, that the nonstrict map is constant below this active range.

Let G be the upper physical pre-parameter slope supplied by the mesh and wall constructions, allowing a fixed factor for sums of two slopes. On the active region, Lemma 4.1, (4.15), and Proposition 4.12 give

$$|D(P_0q_h)| \leq Cr(q_h)G \quad \text{a.e.} \quad (7.8)$$

Indeed a nonzero stair derivative occurs in an occupied stack. Its root parameter has one pre-parameter derivative or a sum/difference of two derivatives on a switched diagonal. Every intervening OUT-to-IN identification has absolute slope one, so there is no factor depending on itinerary length. The occupied stair has root speed at most $1 + O(\zeta)$, and the forward polar map costs $O(r)$, including the activation ramp. On edge interiors this is the perimeter estimate with $r = 1$; on vertex levels the derivative vanishes. The formulas suffice a.e. at knots and piece interfaces: h is locally bi-Lipschitz, the face and edge product projections are absolutely continuous for their respective dimensional measures, and derivatives vanish on constant level sets. In full blocks, away from the already paid conical layers, $P_0q_h = q_0$. On the remaining central tube interiors outside M_x the nonstrict map is constant by the exact correspondence. Thus only the prescribed high-budget regions require the weak estimate (7.8).

7.3 Calibrated source patches

Use the fixed cuboidal and coarse chart structures on M_x . Their walls and nonsmooth piece loci are locally finite ambient-null sets. Trim $K^\sharp$ away from them with room, losing arbitrarily little μ -mass.

At a retained Lebesgue gradient point z , within a single smooth (D, e) sector, put

$$Y_0 = (t(q_0), s(q_0)), \quad B = D_x Y_0(z). \quad (7.9)$$

We choose a nonsingular spatial basis V as follows. If B has rank two, let k be a unit vector in its kernel and take

$$V = [B^\dagger, k/\|B\|_{\text{op}}].$$

The columns of $B^\dagger$ are orthogonal to k , and the singular values of V show that

$$BV = [I_2, 0], \quad \|V^{-1}\|_{\text{op}} = \|B\|_{\text{op}}. \quad (7.10)$$

If B has rank one, write $B = av^T$, where $|v| = 1$, put $\alpha = |a_1| + |a_2|$, and choose an orthogonal Q with $Qe_1 = v$. Then $V = Q/\alpha$ gives

$$BV = \begin{pmatrix} b_1 & 0 & 0 \\ b_2 & 0 & 0 \end{pmatrix}, \quad |b_1| + |b_2| = 1. \quad (7.11)$$

For the rank-one rectangle,

$$\min(|b_1|, |b_2|) \leq C\gamma. \quad (7.12)$$

On a long-side sector the long-axis direction has only perimeter speed, of order $1/r$. On an end sector it has logarithmic speed of order $1/r$ and perimeter speed at most $C\gamma/r$. The $o(\gamma^2)$ direction error, combined with inverse sector cost $O(1/(\gamma r))$, preserves these estimates. For rank two the rectangles are squares. In both cases Lemma 3.7 and (7.7) give the absolute bound

$$r(z)\|V^{-1}\|_{\text{op}} \leq C|Df(z)|, \quad r(z) = r(q_0(z)). \quad (7.13)$$

This is the cancellation needed when the normalized sharp estimate is returned to physical coordinates: the target reconstruction costs $O(r)$, while the source change of coordinates costs $\|V^{-1}\|_{\text{op}}$. Their product is controlled by the original gradient, without a factor from the finite jet bounds. On the retained compact set, V and V^{-1} have finite bounds: use the positive rank, the singular-value truncations, the fixed sector insets, and the finite chart bounds.

In a smooth coarse chart $x = g(\xi)$ use aligned coordinates

$$\xi = \xi_z + Ay, \quad A = Dg(\xi_z)^{-1}V.$$

Choose small outer y -cubes centered at $y = 0$, corresponding to z , with nested inner cubes. On their enlargements require: a single smooth sector with small label oscillation; chart derivatives close to their center values; and arbitrarily small power means of

$$DY_0 - B, \quad Df - Df(z).$$

All required eccentricities are bounded on the retained compact, so these are a Vitali differentiation family. Select disjoint outer cubes and then a finite subfamily. Choose the nested ratios close enough to one that the inner cubes still capture the required weight, including a further inset for residual tests. We call them the *saturation cubes*. Their physical images are separated from the boundary of M_x , the marked cubes and the full-rank correction supports.

There is no circular dependence between their small tests and their geometric constants. First fix the finite chart, sector, basis, slot-shape and template bounds on the prospective compact data. The finite triangulation lists in Lemma 6.7 depend on bounded cardinalities and fixed slot margins, not on microscopic noncluster angles or master widths. The transition mesh bounds in Lemma 5.1

depend on the fixed bases, not on the relative thickness of patch buffers. Next fix the level-edge count tolerance, then smaller chart-error and jitter fractions, and then still smaller power-mean data errors, allowing the inverse jitter-density factors. Finally choose patch radii realizing these tests. The mesh step H and then W/H can be arbitrarily smaller.

Thin patch margins, including equal-weight, clustering and Delaunay exit belts, are chosen after the fixed local derivative and density factors, so their volume and needed Sobolev integrals are small. They can be resolved without deterioration of the shape factors. Keep every equal-weight convolution cost inside enlarged pure lattice regions where the affine tests remain valid. Microscopic master widths occur only inside the snapping construction; leave several layers of widths comparable to s_σ in the clustered equal-weight belt before returning to ordinary widths. Sharp and mixed-position tetrahedra use the noncluster generic convention; there is no robust-template motion of an opponent system in the sharp bulk. The wholly clustered convention resumes past the buffer. Outside-chart and unequal-orientation transitions use the fixed coarse-shape weak bound on their paid regions.

7.4 Ambient edge tests and weak costs

Outside saturation interiors use the physical-mode overrides of Lemma 5.1 on smooth coarse chart interiors. Reserve small neighborhoods of coarse facets, chart-piece seams, patch and Delaunay transitions, and nonphysical mesh regions. Their integrals of $1 + |Dq_*|^p$, with every assigned compact-local factor and residual weight, can be made summably small. Indeed these factors are fixed before the neighborhood widths, and finitely many inset physical patches cover a coarse compact except for arbitrarily small weighted error. Enlarge the error sets slightly to include neighboring mesh stars. The actual cut-boundary adjustment and exact-region jitter taper can be made equally thin, later if necessary.

The last assertion uses the width-independent bound in Lemma 5.2. The horizontal warp and chord inverse response have fixed bounds, and motions depending on d/d_0 have size comparable to the boundary mesh scale, which is much smaller than d_0 . The slopes of d_0 are controlled. Thus the label slope, including its conversion to a root parameter, has no inverse power of d_0 ; a fixed factor $1/c_*$ is allowed. Equal subdivision of the boundary chords changes only displacement differences in tangential derivatives and adds no subdivision factor.

The exterior residual test lies in Z , outside slightly shrunk saturation interiors, and enters a marked interior only through its enlarged collar. The other two tests are the buffered full-rank correction shells and the marked collars with weight ℓ_U/d_U . Mesh stars and jitter displacements will fit inside the enlargements already reserved for these tests.

Lemma 7.1 (Ambient edge tests and weak costs). *With the preceding geometry and local weights fixed, the ordinary weak costs are controlled by the carrier budgets with constant C_w , plus freely small absolute errors. The control is uniform over meshes sufficiently fine for prescribed buffer and guard requirements. It applies simultaneously to the exterior residual, the full-rank correction shells, and the marked collars weighted by ℓ_U/d_U .*

Proof. For a mesh edge or edge portion l , let W_l be the sum of the weights of its initial central hits. On whole edges these sums are upper bounds for the normalized systems: cleaning, opening and normalization do not increase any individual wall-edge count. Edge portions are used only to estimate the initial counts by coarea; their contributions are added before normalization. On an edge portion outside the modified boundary collar, conditional on a suitable ACL mesh, Lemma 4.1 gives

$$W_l \leq (1 + C(\zeta + c_*)) \text{Var}_l(q_*) + \varepsilon_l. \quad (7.14)$$

Here the variation is the sum of star variation for the radial family and graph arclength variation for the perimeter family; the two may also be tested separately. The additive errors ε_l may be as small as required relative to the actual edge and its weights. Each feature has only finitely many relevant edge tests on a compact localization, by properness.

The coefficient close to one is the weighted sampling density of Lemma 4.1, not the width of I_j . Its stratified sampling argument, applied to the integrable crossing-count tests in Lemma 5.3, gives the inequality with arbitrarily high probability after the edges have been chosen. It also permits a common weight on finitely many features and all almost-sure null avoidances.

In a high-budget region, the two graph speeds satisfy

$$|D(\text{graph labels})| \leq \frac{C_{\text{asp}}}{r} |Dq_0| \quad (7.15)$$

along the relevant ACL edge paths. The bound is a stratumwise length bound; an inverse ambient dihedral angle is unnecessary. On a radial spoke the perimeter-vertex coordinate is constant, and $q = d_D + r(\xi - d_D)$ gives $|dt| \leq (L_D/|\xi - d_D|)|dq|/r$. On a subedge the perimeter projection is the identity and the star coordinate is constant. At vertices or graph nodes the corresponding coordinate speed vanishes a.e. on its level set. Restrictions at density and differentiability times have the ambient ACL derivatives, so these bounds combine along the path. Local Lipschitz graph coordinates, possibly with larger fixed seam constants, ensure the required absolute continuity. The estimate itself uses just the sector aspect factors.

On modified cut-collar portions replace (7.15) by the fixed local label-slope bound of Lemma 5.2. The moving graphs and chords have inverse-response bounds on the allowed bins, also across the triangle pieces. Restricted one-dimensional coarea on the disjoint allowed parameter strips gives the corresponding weighted-count bound in terms of these prepared labels, with the stated fixed factors. The central prepared trace family is fixed before sampling; its later interval-ribbon adaptation keeps the sampled central chords unchanged. We split an edge into its unchanged and modified collar portions when applying these bounds. The depth-stretched portions use the adjoining exact-region bounds.

For an ordinary all-clustered tetrahedron σ of physical scale h_σ , Proposition 6.10 and (7.8) give the active slope majorant

$$Cr_\sigma h_\sigma^{-1} \max_{l \subset \sigma} W_l. \quad (7.16)$$

The radii on the required stars are comparable to r_σ . There is no fixed baseline in (7.16): a contributing pre-parameter has width proportional to its root weight, and zero crossings require no disk. Apply (7.14), (7.15) and Hölder's inequality on an edge $x_l(u)$, $0 \leq u \leq 1$, parametrized at speed $O(h_\sigma)$. The p th power of (7.16), times $|\sigma|$, is at most an arbitrarily small additive error plus

$$Ch_\sigma^3 \sum_{l \subset \sigma} \int_0^1 C_{\text{asp}}(x_l(u))^p |Dq_0(x_l(u))|^p du. \quad (7.17)$$

Only tetrahedra in deterministic high-budget buffers require this estimate. Costs which become inactive after pinning the labels are not included in the weak objective.

Lemma 5.3 supplies the ambient expectation estimate: on an unconstrained jittered edge, the point at fixed normalized edge time has density at most Ch_σ^{-3} in a neighboring star, and the normalized edge speed is at most Ch_σ . The lemma also supplies ACL and the chain rule. Thus for any nonnegative ambient integrable weight F ,

$$\mathbb{E} \left[h_\sigma^3 \int_0^1 F(x_l(u)) du \right] \leq C \int_{\text{star}^+(\sigma)} F(x) dx. \quad (7.18)$$

Here the enlarged stars have bounded overlap. Jitter radii are chosen to stay within those stars. The taper into nonrandom data is wholly in exact regions, where fixed local Lipschitz bounds replace the expectation estimate. In the physical bulk, shapes, displacement fractions and overlap are universal. Keep any additional mesh layers needed to reach that bulk inside the paid transition regions. Summing (7.18) therefore charges (7.17) to the prescribed ambient budgets with only C_w in the ordinary bulk. In the exceptional zones it charges $1 + |Dq_*|^p$ with the already fixed local factors. Near the original polygon vertices it charges the affordable weighted $|Dq_0|^p$ selected before subdivision.

We now charge the deterministic envelopes specified above. The part of the exterior-residual envelope in K_c is small in $|Df|^p$ by weighted capture and (7.4); on its complement use the regional DF_b budget and (7.5). The full-rank shells and marked-collar envelopes lie in their preassigned enlargements and avoid the sharp construction. Mesh stars and jitter displacements are finer than all these buffers. The sharp transition and equal-weight belts use the margin and leakage estimate proved below. Add the already small full-cell conical-layer costs; elsewhere in full blocks the map is q_0 . The objectives are a bounded number of aggregate nonnegative costs, and their local weighted summands can be combined into one objective. Their expectations have the asserted smallness. Lemma 7.3 selects the mesh and sample data simultaneously with the crude guard capacities; Proposition 7.4 realizes the actual wall systems with these same bounds. $\square$

7.5 Sharp counts, convolution and discarded regions

On an enlarged calibrated patch let H be the common aligned Kuhn step. The ambient volume jitter is included in its chart g' from y to x . The small first-chart errors and jitter fraction ensure that Dg' is as close to V as prescribed, in relative matrix norm. All relevant features use a common root weight W , with W/H arbitrarily small.

Lemma 7.2 (Sharp counts and discarded costs). *The meshes and the subsequent stratified samples can be chosen so that the sharp-template count conditions hold outside tetrahedra of arbitrarily small aggregate pre-derivative p -cost. More precisely, for the two family counts on an edge l ,*

$$\frac{N_{l,i}W}{H} \leq (1 + C(\zeta + c_*)) \left| (BV)_i \frac{\Delta y_l}{H} \right| + e_l, \quad e_l \geq 0, \quad (7.19)$$

and the aggregate $\sum_l H^3 e_l^p$ can be arbitrarily small after all fixed template, transition and physical conversion factors. The dirty portions left by Proposition 6.9 have arbitrarily small aggregate evaluated pre-derivative p -cost in aligned coordinates.

Proof. Use the separate coordinate variation tests in the single smooth sector. Along an edge, subtract the affine comparison BVy . The resulting variation is bounded by the integral of $|DY_0 - B|$ times the bounded chart speed, plus the chart/jitter derivative error times $\|B\|$. Normalize by H and use Hölder's inequality. The expectation estimate (7.18) bounds the sum of the resulting p -errors by the power-mean data errors on the enlarged patch. The inverse jitter-density factor may be large, but was fixed before those tests. The small chart/jitter terms are deterministic. Conditional sampling gives the multiplicative factor in (7.19) and arbitrarily small additive errors, which are included in e_l . There are only finitely many sharp patches, and all matrix bounds are finite. We may therefore impose an aggregate error with any prescribed finite patchwise weights; there is no requirement of simultaneous probabilistic control of a relative mean on every individual small cube.

Fix first a count threshold much smaller than η , and also a level-edge threshold $\delta' > 0$, which may be arbitrarily small after the η -dependent template constants. Discard tetrahedra where a

relevant e_l exceeds these thresholds. On an inter-edge, (7.19) bounds each independent count by $(1+o(\eta))H/W$. In parallel mode the sum of the two ideal counts is H/W , by (7.11), so the combined excess is also $o(\eta)H/W$. On a level edge the ideal variation vanishes: its count is at most $\delta'H/W$, with no multiplicative bias. These distinct tolerances have distinct uses. Inter-edge excess controls fitting the lists into slots of spacing $(1-C\eta)W$; level-edge errors control non-slabs and discrepancies in slab ranks. Start the lists at common offsets of order ηH from the lower axis end and perturb only within the unused slack. Across a regular tetrahedron the rank offsets of corresponding slabs differ by at most $O(\delta'H/W)$, including the first band before the second band in parallel mode. These are the hypotheses of Proposition 6.9; no lower bound for the number of surviving slabs is needed.

We give the cost calculation for discarded tetrahedra. In the finite enlarged equal-weight region, put

$$m_\sigma = 1 + \max_{l \subset \sigma} N_l W/H, \quad M_\sigma = \sum_{\tau} \theta^{d(\sigma, \tau)} m_\tau,$$

where $N_l = N_{l,1} + N_{l,2}$ is the total two-colour upper count, the star graph has bounded degree, and θ is smaller than a fixed reciprocal of that degree. The initial weighted counts remain upper bounds for all normalized counts, so they can be used in these majorants. Equation (7.19) implies $m_\sigma \leq m_0 + C e_\sigma$, where m_0 is an absolute baseline and e_σ is the largest of its finitely many edge errors. The kernel has uniformly bounded row and column sums. Thus its convolution is bounded on every ℓ^p , and

$$M_\sigma \leq M_0 + L_\sigma, \quad \sum_{\sigma} H^3 L_\sigma^p \leq C \sum_{\sigma} H^3 e_\sigma^p. \quad (7.20)$$

Adjacent M values also have bounded ratios.

For a fixed threshold $\tau > 0$, the number-weighted volume of $\{e_\sigma > \tau\}$ is at most $\tau^{-p} \sum H^3 e_\sigma^p$. Its M^p cost is small as well: pay the baseline M_0^p by that volume and the remainder by (7.20). On $\{M_\sigma > 2M_0\}$, the full M^p cost is bounded by a constant times the L^p cost. Fixed neighbor layers of these sets have the same conclusions, using bounded degree and adjacent ratios. In thin patch margins within the convolution region, pay the baseline by volume and the leakage by (7.20). Mixed-width tetrahedra in the last quantitative belt can use both this majorant and the ordinary edge bound; they still lie in the paid margin.

On low-leakage regular tetrahedra, fix the buffer multiple N sufficiently large after the slot margins. It accommodates the flat translations, the auxiliary-cell trimming for snapping, and the slow broadening to strip half-width $a_1 = (1 - O(\eta))W/2$. Proposition 6.9 gives relative dirty volume at most

$$C_N(W/H + \delta'). \quad (7.21)$$

The pre-slope there is at most CM_σ , and M_σ is bounded on these tetrahedra. First make δ' sufficiently small, then W/H sufficiently small. The remaining bad-tetrahedron and high-leakage costs are small by (7.20), and the patch-boundary buffers are paid margins. Altogether the integral of the pre-slope to the p th power on the dirty portions is arbitrarily small. The opposite-band gap and slope errors away from these portions are $o(\eta)$ by the same level tolerance, negligible generic motions, and W/H . All constants used here were fixed before the final aggregate error was prescribed. $\square$

7.6 Capacities before guards and simultaneous choices

Here a capacity means a finite upper bound for a local integral $\int G^p$, not a Sobolev capacity of a set. The preceding estimates are uniform over sufficiently fine mesh resolutions. They therefore allow the crude G capacities to be fixed before the guard nets, even though the final meshes must be finer than the guard buffers.

There are two tasks. We first select edges and samples for which the count-based slope bounds fit the chosen capacities and energy budgets. We then construct the actual initial walls and pre-stacks, retaining those very counts. The second task is carried out in Proposition 7.4; until then, the selected bounds are upper budgets for the eventual slope density G .

Lemma 7.3 (Prior capacities and simultaneous choice). *For each member of a locally finite family of buffered compact barrier charts one can fix a finite capacity A_j , independently of guard spacing and final mesh resolution. After fixing these capacities, one can choose the guards, mesh and samples so that substitution of the selected edge counts into the slope bounds of Proposition 6.10 gives local integral upper budgets at most A_j . The same data meet the weak aggregate bounds of Lemma 7.1 and the sharp count and discarded-cost bounds of Lemma 7.2.*

Proof. Uniform preliminary budgets. For the crude bound on a compact chart, include the upper bound for every pre-slope, including ones that will later be inactive. In nonsharp tetrahedra, Proposition 6.10 bounds the slope by the weighted edge variations with the fixed local chart, aspect and inverse-radius factors, together with the allowed fixed width terms and the cut-collar bounds. Equations (7.14) and (7.18) give a finite expectation majorant using $1 + |Dq_*|^p$ on an enlarged compact. The local factors are bounded there. In equal-weight regions the aligned pre-slope is at most CM_σ , and the bounded convolution estimate charges the whole enlarged patch to its m budget. Physical conversion has only fixed compact factors. Sharp overrides occur in a finite interior compact. Thus, for the j th barrier chart, there is a finite number B_j bounding the expected crude upper budget, uniformly over all sufficiently fine resolutions and all permitted guard gaps.

This expected cost is an upper-budget construction. First require the weighted sample counts to be bounded by their variations plus the prescribed errors, conditional on the edges. Take the p th power of those variation upper bounds and then the ambient jitter expectation. No high moment of the unweighted crossing count is being assumed. The additive count errors can be chosen to obey both the small aggregate objectives and the crude capacities. The boundary-collar constants are independent of the excluded belt widths and of d_0 , and the mesh constants are independent of guard spacing. A guard-induced allowed component may be very short; only the eventual sample weight, not these constants, must then be smaller.

Choose capacities, for example,

$$A_j = 2^{j+6}(1 + B_j).$$

The probability that the corresponding upper budget exceeds A_j is at most 2^{-j-6} . The small objectives are finitely many aggregate nonnegative random variables. Their expected values can be made smaller than their desired bounds by any prescribed factor, using the initial affordable tails and the subsequent absolute errors already described. Fix these with enough room that their Markov failure probabilities sum to less than $1/8$. The sharp affine-error objective is treated in the same way. The sum of the crude-capacity failure probabilities is also less than $1/8$. Thus a positive-probability set of edge jitters satisfies all these upper-budget inequalities.

Guards, meshes, and samples. Now use these fixed capacities in Proposition 4.9 to choose the guard spacings and their buffers. The desired accuracies include those used in Subsection 7.2, all local value tests, and the following sharper requirement on every saturation cube:

$$Y = (S_D(t(q_h)), S_e(s(q_h))). \quad (7.22)$$

Require this pair to be uniformly as close to Y_0 as prescribed relative to the aligned cube side length. Both labels stay in the same smooth sector, and the activation function H_{a_D} is the identity at their resulting radii. These are physical label requirements after the finite patches have been selected. They can be imposed by a sufficiently fine combined label accuracy and sufficiently close stairs.

Properness bounds the number of guard-label gap exclusions relevant to a compact feature. Choose the preliminary guard realization accuracies relative to these prescribed gap budgets, which may be smaller than the relative tolerance ζ to ensure fine stair motion. The dyadic sampling belts in Lemma 5.2 can meet these prescriptions by using summably small fractions across scales and requiring every scale actually used on a compact feature to lie sufficiently far down the tail. Only finitely many such scales are used there once a mesh has been selected.

Choose d_0 and the meshes small enough for the guard, high-label and regional buffers. The size functions have constant cores and a common sharp-interior step H , as in Lemma 5.1. The push into the cut boundary preserves the already fixed local bi-Lipschitz constants. Apply the allowed boundary and volume jitter. The uniform upper-budget bounds just proved apply to this mesh, so choose a jitter satisfying them and all almost-sure ACL conditions. Finally choose the central sampling weights after these edges. In the finite equal-weight regions take W/H as small as required. Split allowed components sufficiently finely and place intervals I_j of width comparable to $c_* w_j$, with small sampling margins. Lemma 4.1 makes all required weighted count inequalities hold with simultaneous positive conditional probability. Countably many local sampling tests use summable conditional failure probabilities. Hence at least one joint choice of edges and samples satisfies everything. The displacement of whole-interval and occupied-interval stairs is charged only to absolute gaps, rounding and slot size; keeping $c_* > 0$ fixed introduces no residual motion error.

In the same conditional selection impose the almost-sure tests needed to open the carrier: avoid labels of nonsingleton fibre values, labels attained on nonexact seams along edges, and mesh vertices; require finite original-carrier hit sets, including in the exact collar. These tests use only coarea and null avoidance, and add no numerical energy constraint. Their applicability is verified in the next proof. Intersecting them with the positive-probability set just obtained completes the selection. $\square$

7.7 From sampled counts to the actual wall systems

The sampled estimates refer to the original carrier away from the boundary adjustment and to the prepared trace family inside its exact collar. We now obtain actual walls with these counts. A single opening is used for both colours; its equality with the carrier in the exact collar then allows the boundary preparation to realize the second set of tests.

Proposition 7.4 (Initial walls with the selected budgets). *The guards, mesh, samples and count bounds selected in Lemma 7.3 admit pre-stacks satisfying Definition 4.2 and avoiding the selected guards. Their actual slope majorant G satisfies*

$$\int_{U_j} G^p \leq A_j$$

on every buffered barrier chart U_j , together with the weak and sharp estimates of Lemmas 7.1 and 7.2. The mesh and samples are unchanged. Consequently the hypotheses of Theorem 4.11 hold.

Proof. A common opening and the initial walls. We now pass from the sampled carrier levels to the initial systems required by Proposition 6.10; this step precedes routing. Let $\mathcal{W}$ be the entire two-colour family of sampled standard walls in M_y . Each wall is compact and properly embedded, and there are finitely many samples per feature. The standard feature family is locally finite. Hence $\mathcal{W}$ is locally finite and its union is relatively closed in M_y . The nonsingleton values of the central carrier in Lemma 3.9 are countable. Each such value forbids at most finitely many radial or intermediate meridian levels, and a graph vertex is not an intermediate meridian level. These countably many labels are among the almost-sure sample avoidances in Lemma 7.3. Thus the whole wall union, not merely its edge-hit values, avoids every nonsingleton value.

For the opening use the *original* carrier F_* and the actual final mesh edges, before the boundary preparation has changed the walls. Away from the exact collar, the already chosen ACL and coarea tests give finite hit sets on compact localizations. The nonexact seam set S of Lemma 3.9 has edge-parameter measure zero by volume jitter. The relevant label paths are absolutely continuous, so the images of these parameter sets have one-dimensional measure zero. These scalar label sets and the mesh-vertex labels were also excluded in the same sample selection. This gives no hit on S outside the exact regions. In the exact collar the unmodified carrier labels along the final edges are locally Lipschitz. Their additional coarea tests therefore give finite hits almost surely; these tests require no numerical energy budget. The finite feature lists permit all of them in the selection of Lemma 7.3. A compact localization meets finitely many edges and sampled walls, so its total actual hit set is finite.

Lemma 3.10 therefore supplies *one common* opening homeomorphism $f_o : \Omega \rightarrow \Lambda$ for the entire two-colour wall union, with exactly its original carrier hits on every mesh edge. Choose it sufficiently finely close to F_* for the prescribed guard-gap buffers. It equals $F_* = h_*$ on an exact neighborhood containing all supports of the boundary preparation, as well as on the other prescribed exact data, and maps M_x onto M_y with the fixed boundary correspondence. Pull back both standard families by this same f_o . They are compact properly embedded locally flat systems of the required individual types, with noncontractible annular cores.

Now apply the common joint boundary adjustment of Lemma 5.2: the warp and chord preparation, its mirrored continuation inside the cut, and the depth stretch. This operation is supported entirely in the exact neighborhood where $f_o = F_* = h_*$. Therefore the resulting collar walls and their actual edge counts are precisely the prepared data whose weighted coarea bounds were used above. Outside that neighborhood their edge counts are the original carrier counts, preserved by the opening. The interval-ribbon adaptation keeps the central sample chords fixed. Consequently all previously selected weighted and directional upper counts apply to these actual initial systems. No sampling estimate on a sample-dependent pullback of an edge is being used.

Apply Lemma 5.4 separately to each colour, retaining the smaller prescribed PL boundary product collars. Its hypotheses hold by the finite isolated edge hits, vertex avoidance and the exact prepared boundary data. Choose its fine position tolerances within the guard buffers. It gives the required PL systems and flat transverse interior crossing germs, without increasing any individual wall-edge count. Apply Proposition 5.6 separately to these two systems, retaining their boundary traces, topologies and coorientations, and again without increasing an individual edge count. Neither application requires a relative isotopy fixing the other colour. Proposition 6.10 now supplies the pre-stacks. No individual edge count has increased. The same count-based slope upper budgets selected in Lemma 7.3 therefore bound the actual integrals $\int_{U_j} G^p$ by A_j and give the stated weak and sharp estimates. Their guard avoidance follows from the reserved label gaps and the chosen finer mesh: each final normal disk lies in a tetrahedron incident to an original possible hit, and placements, thickening and snapping stay in its prescribed mesh buffers. The preliminary opening and PL changes were chosen within the same positive buffers. Thus this execution establishes the pre-stack hypotheses of Theorem 4.11 without changing the previously fixed capacities or the guard order.

Slopes on lower-dimensional interfaces. For the rectifiable-path use of the barrier, define G on mesh interfaces and exceptional collar-boundary strata to include the larger adjacent actual local collar slopes. These strata have ambient volume zero and locally finite prescriptions, so this changes none of the volume budgets. The pre-parameters are Lipschitz on their compact product collars; restricted path coarea therefore uses these pointwise upper slopes. On open pieces the volume bounds are exactly those already proved. In particular, no estimate for an inverse distance across a gap at a manipulated boundary is needed in the integral capacity. $\square$

Remark 7.5. The order established above can be recorded without a forward parameter dependency. Fix p and the desired power error; choose γ, η , then ζ, c_* and the ordinary weak constant. Choose affordable rank tails and finite jet truncations, then the full-rank shell tests. Construct the carrier and fix its regional and weighted marked-collar budgets. Fix the polar data, aspect localizations, full-cell layer widths, activation radii and exact central carrier. Fix compact sharp-template and basis bounds, then level tolerances, chart/jitter fractions and smaller power-mean tests; select the finite sharp patches. Choose the paid transition regions and the crude capacities. Only then choose guard accuracies and spacings. Finally take the boundary widths and meshes fine enough for all buffers, select a permissible jitter, and sample with sufficiently small weights. The expectation bounds are uniform in these last resolutions, which is why the capacities may precede the guards. Strict parameters, marked-cube collapses and the final smoothing are chosen afterward.

7.8 Saturation and physical gradient accuracy

Apply the wall construction to the pre-stacks of Proposition 7.4, with the capacities and samples selected in Lemma 7.3. Theorem 4.11 gives the locally bi-Lipschitz h , the actual occupied intervals and the label pinning. We estimate P_0Th , not the derivative of the qualitative product and ball extension used to construct h .

Lemma 7.6 (Saturation and physical gradient accuracy). *On the union of the saturation interiors,*

$$\int |D(P_0Th) - Df|^p \leq C_p(\eta + \gamma)(1 + E) + \text{freely small errors.} \quad (7.23)$$

The constant in the displayed leading term is independent of the compact jet and chart bounds.

Proof. In a saturation cube Q in aligned coordinates put

$$\tilde{Y} = Y \circ g', \quad D_0 = BV.$$

Proposition 6.9 and Proposition 4.12 give, off the dirty portions,

$$|D_y \tilde{Y}| \leq \begin{cases} \sqrt{2} + C\eta, & \text{rank } D_0 = 2, \\ 1 + C\eta, & \text{rank } D_0 = 1. \end{cases} \quad (7.24)$$

At an essential both-active crossing, the two scalar gradients are separately near-unit orthogonal axis gradients. On a switched diagonal only one output scalar is active and its derivative is their sum or difference, with the same squared-norm bound. In parallel mode the opponent strips in this deep flat region are disjoint. Strip width a_1 , flat slopes and occupied-root speeds contribute $O(\eta)$ after the subsequent smaller choices. Thus arbitrary partial switches do not increase the upper bound in (7.24). By Lemma 7.2, the evaluated p -integrals on dirty portions are arbitrarily small in aggregate, with any fixed physical conversion weights.

The mean gradient has the required affine value:

$$\left| \frac{1}{|Q|} \int_Q D_y \tilde{Y} - D_0 \right| \text{ is arbitrarily small.} \quad (7.25)$$

For $Y_0 \circ g'$ this follows from the enlarged power-mean tests, the bounded Jacobian factors and the chart/jitter error. For the difference, integration on the boundary of the cube gives

$$\left| \frac{1}{|Q|} \int_Q D_y ((Y - Y_0) \circ g') \right| \leq \frac{C}{\text{side } Q} \|(Y - Y_0) \circ g'\|_{L^\infty(Q)}.$$

This is valid for the continuous Sobolev functions at issue by the trace formula; the right side is small by (7.22). The cube remains compactly in one smooth sector.

We use the elementary uniform-convexity inequality

$$c_p |Z - D_0|^p \leq |Z|^p - |D_0|^p - p|D_0|^{p-2} D_0 : (Z - D_0), \quad p > 2. \quad (7.26)$$

One proof integrates the Hessian of $|X|^p$ along the segment $D_0 + t(Z - D_0)$. Its quadratic form is at least $p|X|^{p-2}|Z - D_0|^2$. After scaling $|Z - D_0|$ to one, an interval of fixed positive length in $0 \leq t \leq 3/4$ is a fixed positive distance from the possible zero of the projection of that segment onto $Z - D_0$. The integrated Hessian is therefore bounded below by a positive constant depending only on p . This proves (7.26) uniformly in both matrices.

In rank two, $|D_0| = \sqrt{2}$. In rank one, (7.11) and (7.12) imply $1 - C\gamma \leq |D_0| \leq 1$. Integrate (7.26) with $Z = D_y \tilde{Y}$. Equation (7.24) bounds its energy on the clean set; the dirty energy is already small. The linear term is controlled by (7.25), since $|D_0| \leq \sqrt{2}$. We obtain

$$\int_Q |D_y \tilde{Y} - D_0|^p \leq C_p(\eta + \gamma)|Q| + \text{mean and dirty errors}. \quad (7.27)$$

It remains to verify that the physical conversion does not multiply the leading error by a large jet condition number. On this cube $P_0 q_h = \mathcal{R}(Y)$, with the smooth sector reconstruction

$$\mathcal{R}(t, s) = d_D + \exp(t/L_D)(\xi_e(s) - d_D).$$

There is no central activation transition. Since s is arclength and L_D controls the polygon diameter, its two derivative columns are bounded by Cr ; this bound is independent of the thin rectangle aspect. Require the chart error to satisfy $\|(Dg' - V)V^{-1}\|_{\text{op}} \ll 1$. Then

$$\|D\mathcal{R}(\tilde{Y})\|_{\text{op}} \|(Dg')^{-1}\|_{\text{op}} \leq Cr(z)\|V^{-1}\|_{\text{op}} \leq C|Df(z)|, \quad (7.28)$$

and $|\det Dg'| = (1 + o(1))|\det V|$ relatively. All these requirements are possible after the finite basis bounds.

Split the physical derivative into the term containing $D_y \tilde{Y} - D_0$ and the comparison term

$$D\mathcal{R}(\tilde{Y})D_0(Dg')^{-1}.$$

The latter differs as little as prescribed from $Dq_0(z)$, by the label, patch and chart tests. At the central data the identity $D\mathcal{R}(Y_0(z))B = Dq_0(z)$ holds. Now use (7.7) and the affine-gradient tests on Df to compare further with $Df(x)$. The mean, dirty and comparison errors may use every finite conversion factor because they were chosen after those factors. For the leading term, however, (7.28) and (7.27) give exactly

$$C_p(\eta + \gamma)|Df(z)|^p |\det V| |Q|.$$

The selected outer patches are disjoint, their affine-gradient tests are accurate, and their chart Jacobians are relatively close to $|\det V|$. Hence

$$\sum_Q |Df(z_Q)|^p |\det V_Q| |Q| \leq C(E + 1).$$

Summing proves (7.23). The flattening angle and central-fraction losses in (7.7) were freely chosen small relative to this same finite energy. $\square$

7.9 Strict maps, marked interiors and completion

Proof of Theorem 1.1. Carry out the preceding choices with sufficient room in each target error. Let $u = P_0Th$ be the nonstrict map. On saturation interiors it satisfies Lemma 7.6. On the exterior residual, full-rank shells and weighted marked collars, the derivative costs are arbitrarily small by Lemma 7.1 and the simultaneous choice and transfer in Lemma 7.3 and Proposition 7.4. The $|Df|^p$ integral on the residual is also small: the positive deficient ranks are captured by the saturation interiors up to the affordable losses, the full-rank part is captured by its inner cubes up to the chosen tail, and the rank-zero derivative vanishes. The corresponding $|Df|^p$ costs on the correction shells and enlarged marked cubes were fixed small at the initial stage.

Use Proposition 4.13 and Lemma 3.5 to replace u by

$$k = P^\epsilon T_\delta h.$$

The symbols ϵ and δ here denote the strict target parameters, not the slot loss. Choose them locally with summably small derivative and value errors on every prescribed test, including the finite full-rank tests and the marked-collar weights. The stated tangential compatibility on faces and edges ensures this derivative convergence; at vertex levels the nonstrict derivative is zero. The map k is a locally bi-Lipschitz homeomorphism of Ω onto Λ . We impose no global energy bound on its still exempt marked interiors at this stage.

If $2 < p < 3$, apply Lemma 2.17, using (7.5) and its transferred weighted volume tests for k . Choose the rescaled boundary in a slightly outer collar so that the entire exempt cube is enclosed. We recall why this step can make its energy small despite an arbitrarily large fixed interior Lipschitz constant. In cube polar coordinates, the source self-homeomorphism sends $[0, \alpha]$ linearly to $[0, 1 - \rho]$ and $[\alpha, 1]$ to $[1 - \rho, 1]$, fixing the boundary. The inner energy is bounded by a fixed compact constant times α^{3-p} and can be made arbitrarily small after k has been chosen. For fixed α , take $\rho \downarrow 0$ at the chosen radial Lebesgue boundary. The outer energy is bounded by a constant times

$$\ell_U \int_{\text{chosen facets}} |Dk|^p \int_\alpha^1 t^{2-p} dt, \quad \int_\alpha^1 t^{2-p} dt \leq \frac{1}{3-p}.$$

Slice selection bounds the first factor by the weighted collar volume test. The factor depending on $p < 3$ is fixed before those budgets are prescribed. Use summably small choices for the marked cubes, also including any finite full-rank shell factors. The changes avoid all saturation patches. Their value error is small because f has the prescribed small oscillation on each enlarged cube and k already meets the fine value test there. Denote the resulting locally bi-Lipschitz homeomorphism by k_1 . If $p \geq 3$, put $k_1 = k$.

The map k_1 now belongs to global $W^{1,p}$. The residual, saturation, shell and marked-interior derivative costs have summable bounds. The only possibly unpaid full-rank interiors lie in a finite compact subset of Ω , where local bi-Lipschitzness gives finite p -energy. Finally its values lie in the bounded Λ , and Ω has finite measure. This proves actual membership, rather than merely local membership, before invoking the smoothing Theorem.

Apply Theorem A.1 to k_1 , keeping room in the energy and value tests, in particular the normalized full-rank sup tests. Perform the finitely many smooth cubical jet corrections of Proposition 2.16. The map equals b_i on the exact inner cubes, where the error is (7.3). Its changed shells have arbitrarily small cost by the buffered budgets including the fixed jet factors. Their supports avoid the saturation patches. On the complement, the sharp estimate remains in force and the residual small-energy estimates for both maps control the gradient difference. Smooth once more by Theorem A.1, with an arbitrarily small additional $W^{1,p}$ error.

All operations have the required fixed target. The strict target maps are self-homeomorphisms of Λ ; the wall realization maps onto Λ ; the marked-cube operations are source self-homeomorphisms fixed at their boundaries; the cubical corrections preserve their previous images; and the smoothing Theorem preserves the source and target. Local finiteness and the fine properness tests in the preceding constructions apply on the open domains throughout. Thus surjectivity holds for every individual approximant, rather than only for a limiting image. Its everywhere invertible smooth differential supplies local smooth inverses, which bijectivity assembles into a smooth inverse on Λ .

We may also make the value power error arbitrarily small. One explicit way to pass from the fine local tests to this assertion is to choose a compact subset of Ω whose complement has arbitrarily small measure. On the compact impose arbitrarily small uniform value error through the preliminary constructions, corrections and smoothing. On the complement use the fixed bound on the diameter of Λ . Full-rank correction supports and marked cubes were chosen with the additional small oscillation requirements needed for their localized motions. Thus none of the last operations obstructs this value estimate.

Combining the estimates gives a final derivative error bounded by

$$C_p(\eta + \gamma)(1 + E) + o(1),$$

where $o(1)$ denotes the later freely prescribed errors, after all of their multipliers are fixed. First choose η, γ so that the displayed leading term is smaller than the desired error share; then choose the tails against C_w and the successive absolute errors in Remark 7.5. The value error and the two smoothing errors receive the remaining shares. This gives the prescribed bound ε . Apply the construction with $\varepsilon = 2^{-j}$ and undo the initial reflection if needed. The resulting sequence satisfies Equation (1.1), with each approximant in global $W^{1,p}$ and onto the same fixed target. $\square$

A Qualitative topology and strong smoothing

All PL structures in this Section are the ordinary structures on open subsets of $\mathbb{R}^3$. Triangulations are locally finite in the open domain. In particular, neither a triangulation nor the local constants used below need have uniform behavior at the boundary.

The qualitative PL tools belong to the three-dimensional triangulation and approximation theory of Moise and Bing [8, 31, 32]; Hamilton's relative formulation [21] is the precise interface used below. Edwards–Kirby's deformation theorem supplies the supported ambient extensions [18]. We separate these topological existence statements from the derivative estimates: finite local model libraries and subsequently chosen small exceptional sets provide the latter.

Theorem A.1 (Strong smoothing of locally bi-Lipschitz maps). *Let $\Omega, \Omega' \subset \mathbb{R}^3$ be bounded domains, let $1 \leq p < \infty$, and let $H : \Omega \rightarrow \Omega'$ be a locally bi-Lipschitz homeomorphism in $W^{1,p}(\Omega; \mathbb{R}^3)$. For every $\varepsilon > 0$ and every continuous function $\xi : \Omega \rightarrow (0, \infty)$, there is a C^∞ diffeomorphism $g : \Omega \rightarrow \Omega'$ such that*

$$\|g - H\|_{W^{1,p}(\Omega)} < \varepsilon, \quad |g(x) - H(x)| < \xi(x) \quad (x \in \Omega). \quad (\text{A.1})$$

In particular, g belongs to $W^{1,p}(\Omega; \mathbb{R}^3)$, and the approximations have exactly the target Ω' .

The proof has two parts. First we smooth a locally finite PL homeomorphism with summable local derivative errors. We then approximate H by such a PL map. In the second part a finite collection of local models gives derivative bounds before the measure of the exceptional mesh cubes is made small. The qualitative topology used to choose those models supplies no derivative estimate.

A.1 Relative qualitative tools

Lemma A.2 (Fine relative PL approximation). *Let M, N be PL three-manifolds without boundary, with N metrized, and let $f : M \rightarrow N$ be a homeomorphism. Suppose that $K \subset M$ is closed and that f is PL on an open neighborhood of K . Given a continuous function $\eta : M \rightarrow (0, \infty)$, there is a PL homeomorphism $f_1 : M \rightarrow N$ which agrees with f on a neighborhood of K and satisfies*

$$d_N(f_1(x), f(x)) < \eta(x) \quad (x \in M).$$

Noncompact manifolds are allowed. For manifolds with boundary the same statement holds if K contains ∂M and f is PL near K .

Proof. Choose a closed neighborhood K^+ of K contained in the open set on which f is PL. Transport the PL structure of N to M by f . The identity from the original structure to the transported structure is PL near K^+ . Hamilton's relative approximation Theorem [21, Section 3, Theorem 2.2, pp. 68–69] applies to these two structures. Use the compatible metric $d_f(x, y) = d_N(f(x), f(y))$ and the fine tolerance η . It gives a homeomorphism a that is PL between the two structures, is the identity on K^+ , and satisfies $d_f(a(x), x) < \eta(x)$. Then $f_1 = f \circ a$ has all the asserted properties. The boundary condition in Hamilton's Theorem is precisely the additional condition stated here; for an open Euclidean domain its manifold boundary is empty. $\square$

Lemma A.3 (Small local ambient extension). *Fix a closed Euclidean cube $B \subset \mathbb{R}^3$, compact sets $C, D \subset B$ with $C \subset \text{int } B$, and an open neighborhood O of $C \cup D \cup \partial B$. For every $\lambda > 0$ there is a $\delta > 0$, depending only on these source sets and on λ , with the following property. If $e : O \rightarrow \mathbb{R}^3$ is an embedding,*

$$\sup_{x \in O} |e(x) - x| < \delta, \quad e = \text{id} \text{ on } D \cup \partial B,$$

then there is a homeomorphism $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that

$$A = e \text{ on } C, \quad A = \text{id} \text{ on } D \cup (\mathbb{R}^3 \setminus \text{int } B), \quad \sup_{\mathbb{R}^3} |A - \text{id}| < \lambda. \quad (\text{A.2})$$

The number δ is uniform over a finite list of fixed normalized configurations (B, C, D, O) . No derivative bound on e is required.

Proof. Put $E = C \cup D \cup \partial B$. The deformation Theorem of Edwards–Kirby [18, Theorem 5.1], applied at the inclusion of O in $\mathbb{R}^3$, deforms every sufficiently close embedding e through embeddings e_t , starting at $e_0 = e$, so that $e_1 = \text{id}$ on E . The deformation is unchanged outside a fixed compact neighborhood L of E in O , and fixes the inclusion. Its continuity at the inclusion permits us to require

$$|e_t(x) - x| < \lambda/2 \quad (x \in L, 0 \leq t \leq 1), \quad |e(x) - x| < \lambda/2 \quad (x \in L).$$

A sufficiently small uniform δ on O meets the finitely many compact-open conditions that specify this neighborhood of the inclusion. In the terminology of that Theorem a proper embedding respects manifold boundaries; this condition is automatic for the ambient manifold $\mathbb{R}^3$.

All the open images $e_t(O)$ coincide. To see this, enclose L in a finite union of relatively compact subdomains of O whose boundary collars are outside the region where the deformation changes the map. The embeddings agree on that collar. Invariance of domain [22, Theorem 2B.3] and degree relative to the common boundary show that the images of the enclosed subdomain are the same. Outside it the maps already agree. On this common open image define

$$A_0 = e \circ e_1^{-1},$$

and define A_0 to be the identity elsewhere. The map is already the identity off the compact set $e_1(L)$ in the common image, so this is an ambient homeomorphism. It agrees with e on E and has displacement less than λ . Since it fixes ∂B pointwise, it preserves the bounded complementary component $\text{int } B$. Restrict A_0 to B and extend it by the identity outside B to obtain A .

The construction uses only the displayed source configuration and continuity at the inclusion. For finitely many normalized configurations take the minimum of the resulting positive tolerances. $\square$

We will use Lemma A.3 when the input embedding is defined by a small map η near a new core and by the identity near older protected sets. The following observation specifies the required gluing condition. Choose open sets $\overline{O_0} \subset U_0$, where η is defined on U_0 , and an open set V containing the protected sets, such that $\eta = \text{id}$ on $U_0 \cap V$. If η is sufficiently close to the identity that $\eta(O_0) \subset U_0$, the map

$$e = \eta \text{ on } O_0, \quad e = \text{id} \text{ on } V \quad (\text{A.3})$$

is an embedding. Indeed, a collision $\eta(x) = y$ with $x \in O_0$ and $y \in V$ has $y \in U_0 \cap V$, whence $\eta(y) = y$ and injectivity of η gives $x = y$. The open embedding property then follows from invariance of domain. All the sets in this observation can be fixed in advance when the source configurations range over a finite list.

Lemma A.4 (Fine control, properness, and the target). *Let $h : \Omega \rightarrow \Omega'$ be an orientation-preserving homeomorphism between bounded domains, and let $g : \Omega \rightarrow \mathbb{R}^3$ be a smooth map with $\det Dg > 0$. If*

$$|g(x) - h(x)| < \frac{1}{2} \text{dist}(h(x), \mathbb{R}^3 \setminus \Omega') \quad (x \in \Omega), \quad (\text{A.4})$$

then g is a diffeomorphism of Ω onto Ω' .

Proof. Write $d(y) = \text{dist}(y, \mathbb{R}^3 \setminus \Omega')$ and $g_t = (1-t)h + tg$. Condition (A.4) puts the entire segment $g_t(x)$ in Ω' . Since d is 1-Lipschitz,

$$d(g_t(x)) \leq \frac{3}{2}d(h(x)).$$

If $K \subseteq \Omega'$ and $m = \min_K d > 0$, the inverse image of K under the homotopy lies in

$$h^{-1}(\{y \in \Omega' : d(y) \geq 2m/3\}) \times [0, 1].$$

The set in braces is compact because Ω' is bounded. Thus g_t is a proper homotopy. Its endpoint g has the same degree as h , namely one. A proper local diffeomorphism has finitely many preimages of each point, and here each preimage has positive local degree. The degree-one identity therefore says that every point of Ω' has exactly one preimage. The local smooth inverses patch to a smooth inverse on Ω' . $\square$

A.2 Smoothing a locally finite PL homeomorphism

Campbell–D’Onofrio–Vítek proved diffeomorphic approximation of locally finite piecewise affine homeomorphisms in dimensions three and four, with uniform value control and derivative-error control for both the map and its inverse [13, Theorem A]. We give the local proof needed here, including arbitrary positive continuous value tolerances.

Proposition A.5 (Strong PL smoothing). *The conclusion of Theorem A.1 holds if H is replaced by a locally finite PL homeomorphism $h : \Omega \rightarrow \Omega'$ in $W^{1,p}(\Omega; \mathbb{R}^3)$.*

Proof. An orientation-reversing Euclidean isometry in the target reduces the proof to the orientation-preserving case. Fix a locally finite affine triangulation for h . We prescribe summable error budgets for its edges, vertices, and a locally finite family of compact sets covering the remainder of the domain. All modifications below can also obey an arbitrary positive continuous value tolerance.

The open edges. Around each open edge choose a tube of radius $d(t)$, where $0 < t < l$ is its axial coordinate. The tubes of distinct open edges are disjoint, their radii are bounded by a small multiple of $\min(t, l - t)$, and d is exactly linear near both endpoints. These choices follow from the finite geometry of each simplex star; local finiteness makes them compatible throughout Ω . We may make $\|d'\|_\infty$ small by decreasing the width.

In positively oriented orthogonal frames, and after translations, the original map on this tube is

$$(t, r, \theta) \mapsto (at + rb(\theta), rs(\theta)e_{\phi(\theta)}), \quad e_\alpha = (\cos \alpha, \sin \alpha), \quad (\text{A.5})$$

where $a > 0$, $s > 0$, and the coefficients are smooth on the closed angular sectors. The transverse homogeneous map is injective: otherwise two equal transverse images, taken at arbitrarily small radius, would have their axial difference canceled by changing t , contradicting injectivity of h in the edge star. Its angular map is consequently an orientation-preserving circle homeomorphism. We choose a lift ϕ with $\phi(\theta + 2\pi) = \phi(\theta) + 2\pi$. Positive determinants on the finitely many sectors give upper and positive lower bounds for s and for the one-sided derivatives ϕ' .

Choose constants $c > 0$ and c' , and for $0 \leq \tau \leq 1$ put

$$b_\tau = (1 - \tau)b, \quad s_\tau = (1 - \tau)s + \tau c, \quad \phi_\tau = (1 - \tau)\phi + \tau(\theta + c').$$

For fixed τ , the determinant of the corresponding map in the frame $(\partial_t, \partial_r, r^{-1}\partial_\theta)$ is

$$as_\tau^2\phi'_\tau > 0. \quad (\text{A.6})$$

On all the closed sectors and all $\tau \in [0, 1]$ this has a positive minimum. Let χ be a smooth cutoff equal to 1 on $(-\infty, -1]$ and to 0 on $[-1/2, \infty)$, and substitute

$$\tau(t, r) = \chi\left(\frac{\log(r/d(t))}{N}\right)$$

in the preceding formula. The additional derivative columns have size at most

$$\frac{C}{N}(1 + \|d'\|_\infty). \quad (\text{A.7})$$

Indeed, $|r\partial_r\tau| \leq C/N$ and $|r\partial_t\tau| \leq C(r/d)|d'|/N$, while $r/d \leq 1$ on the support of the modification. The remaining factors come from the fixed angular data. Taking N large preserves positive determinants on every sector, including one-sided limits. Near the axis the result is the nonsingular affine map $(t, r, \theta) \mapsto (at, cre_{\theta+c'})$. Near the tube boundary it agrees with h .

All first derivatives are bounded by an edge-dependent constant independent of a further common decrease of d . The volume of the tube tends to zero under that decrease. Both its derivative error and its value error therefore have arbitrarily small $W^{1,p}$ cost. Choose these costs summably over the edges and call the result h_1 . It is continuous and locally Lipschitz, fixes all vertices, and has the required small total error. Because the tube widths are exactly linear near endpoints, $h_1 - h_1(v)$ is homogeneous of degree one on a sufficiently small ball about every vertex v .

The faces and the punctured vertex balls. Away from the vertices, the closed convex hull of the nearby almost-everywhere gradients of h_1 is contained in $\text{GL}^+(3)$ on a sufficiently small neighborhood of each point. At a smooth point this follows by continuity. At a nonsmooth point the only remaining interface is one face. If A_- and A_+ are its two gradient traces, continuity gives agreement on the tangent plane, so $A_+ - A_-$ has rank at most one with normal covector to that face. Consequently

$$\det((1 - t)A_- + tA_+) = (1 - t)\det A_- + t\det A_+ > 0 \quad (0 \leq t \leq 1). \quad (\text{A.8})$$

The traces vary continuously on the two sides. Their nearby convex hull lies in a sufficiently small neighborhood of this compact positive-determinant segment, proving the assertion. Homogeneity makes this property uniform at relative scale on a punctured ball about a vertex: cover the unit sphere by finitely many of the corresponding neighborhoods and rescale.

Let $\rho \geq 0$ be a smooth probability kernel supported in the unit ball. Away from vertices define

$$h_2(x) = \int_{\mathbb{R}^3} h_1(x + w(x)z) \rho(z) dz, \quad (\text{A.9})$$

where w is smooth and positive there and the integration balls lie in Ω . Its derivative is

$$Dh_2(x) = \int Dh_1(x + w(x)z) \rho(z) dz + \int (Dh_1(x + w(x)z) \otimes Dw(x) \rho(z) dz. \quad (\text{A.10})$$

The first integral is in the positive-determinant convex hull just described. Taking w and $|Dw|$ sufficiently small locally makes the second integral a small perturbation, and hence gives $\det Dh_2 > 0$. Formula (A.9) is smooth where $w > 0$, as is also seen by writing its smooth variable kernel in the integration variable $y = x + w(x)z$.

On a small punctured ball at v choose $w(x) = c_v|x - v|$, with $c_v > 0$ sufficiently small. This is consistent with the relative-scale convex-hull condition and makes $h_2 - h_1(v)$ homogeneous on a smaller ball. Set $h_2(v) = h_1(v)$ there. The required radius function exists by a locally finite partition of unity with sufficiently small positive coefficients. Near the vertices blend this function with $c_v|x - v|$ on disjoint annuli. The annular cutoff terms are bounded by a constant times c_v if the competing radii are chosen at that same small scale, so both the radius and gradient restrictions survive. This is also an instance of the local minorant construction in Lemma 1.2.

Here is the global error choice. First take disjoint vertex balls so small that their volumes, multiplied by the fixed local gradient bounds to the power p , meet a summable error budget. On these balls the gradients in Equation (A.10) are bounded independently of a further decrease of their radii. On a compact set outside the balls, the local gradients are bounded and converge at every piecewise smooth point as $w, |Dw| \rightarrow 0$. Dominated convergence therefore gives an arbitrarily small derivative error on that compact set. A locally finite compact covering with summable budgets makes the total $W^{1,p}$ error as small as prescribed. The value error follows from local Lipschitz continuity and the small radius. These arguments include $p = 1$.

The vertices. Center a homogeneous vertex ball at the origin and subtract $h_1(v)$. Write the resulting map as F . Euler's identity is $DF(x)x = F(x)$. Since $DF(x)$ is invertible for $x \neq 0$, $F(x)$ never vanishes there. We can thus write

$$F(r\theta) = rR(\theta)\Phi(\theta), \quad R > 0, \quad \theta \in S^2. \quad (\text{A.11})$$

Its positive determinant implies that $\Phi : S^2 \rightarrow S^2$ is a smooth orientation-preserving local diffeomorphism. It is a covering by compactness, and it is one-sheeted because S^2 is simply connected.

Smale's Theorem gives a smooth isotopy Φ_τ from Φ to a rotation, smooth jointly in (τ, θ) [40, Theorem 6]; for smooth dependence on the sphere diffeomorphism see also [30, Theorem 1.5]. Interpolate R through positive functions R_τ to a positive constant. For fixed τ the homogeneous map $F_\tau(r\theta) = rR_\tau(\theta)\Phi_\tau(\theta)$ has determinant

$$R_\tau(\theta)^3 J_{S^2} \Phi_\tau(\theta) > 0.$$

The family has bounded first derivatives and a positive determinant minimum, by compactness. On an arbitrarily small ball replace F by $F_{\tau(r)}$, where τ is 0 near the outer boundary, 1 near the center,

and $\sup_r |r\tau'(r)|$ is sufficiently small. Such a transition is obtained by spreading a fixed cutoff over a sufficiently long interval of $\log r$. Its extra derivative term is bounded by $C|r\tau'(r)|$, so positive determinant persists. At the center the map is a dilation followed by a rotation. It is therefore smooth and nonsingular there.

The derivative bound for this last modification depends on the fixed spherical isotopy but not on the support radius. Shrinking that radius gives arbitrarily small summable $W^{1,p}$ costs over all vertices. The resulting map g is smooth and has positive determinant throughout Ω .

Finally, in each of the three operations impose local value tolerances whose sum is less than

$$\min\left\{\xi(x), \frac{1}{2} \text{dist}(h(x), \mathbb{R}^3 \setminus \Omega')\right\}.$$

The preceding constructions permit these fine tolerances and a total Sobolev error less than ε . Lemma A.4 makes g a diffeomorphism onto the original target. This proves the Proposition. $\square$

It remains to approximate the locally bi-Lipschitz map by a PL homeomorphism with strong derivative control. The next two subsections establish this reduction.

A.3 Fine meshes and controlled patch insertion

Fix a coarse, locally finite Whitney decomposition $\mathcal{P}$ of Ω into closed dyadic cubes with disjoint interiors. Choose relatively compact enlarged neighborhoods U_P that are still locally finite. Enlarge them a fixed finite number of times when necessary. Since H is locally bi-Lipschitz, there are constants $K_P \geq 1$ such that

$$K_P^{-1}|x - y| \leq |H(x) - H(y)| \leq K_P|x - y| \quad (x, y \in U_P). \quad (\text{A.12})$$

Enlarge K_P also to bound $|DH|$ almost everywhere on U_P in the chosen matrix norm. The constants may incorporate the data from finitely many neighboring coarse cubes. All references below to data local to P allow this finite enlargement, but never an enlargement depending on the final fine-mesh resolution.

Lemma A.6 (Adapted dyadic meshes). *There is an absolute integer n_0 with the following property. Given arbitrary positive local upper bounds on the fine scale, there is a locally finite dyadic cube decomposition $\mathcal{Q}$ of Ω with centers c_Q and side lengths l_Q for which, on writing*

$$B_Q(s) = c_Q + [-sl_Q, sl_Q]^3, \quad (\text{A.13})$$

the following hold:

- (i) $B_Q(100) \subseteq \Omega$, and cubes meeting $B_Q(50)$ have side lengths comparable to l_Q by absolute constants.
- (ii) The normalized cube configurations in $B_Q(6)$ belong to a finite list. There is a conforming affine triangulation of $\mathcal{Q}$ with finitely many normalized nondegenerate shapes on these configurations.
- (iii) The cubes have n_0 colors so that the closed boxes $B_Q(6)$ of a single color are pairwise disjoint.
- (iv) Attach to Q a coarse index $P(Q)$ containing c_Q , using a fixed convention on coarse boundaries. The initial scale bounds can be reduced so that all interactions through any fixed number of neighboring patch layers are confined to a fixed locally finite graph of coarse indices, independent of further mesh refinement. The corresponding patches lie in the neighborhoods on which the required bounds in Equation (A.12) hold.

The upper bounds on the fine scale remain arbitrarily prescribable.

Proof. Choose a positive size function s on Ω with sufficiently small absolute Lipschitz constant, below all the prescribed local bounds and below $10^{-3} \text{dist}(x, \mathbb{R}^3 \setminus \Omega)$. Lemma 1.2 supplies such a minorant. Take the maximal dyadic cubes for which

$$l_Q \leq \inf_Q s.$$

Failure of the condition for the dyadic parent gives $s(x) \leq (2 + C \text{Lip}(s))l_Q$ on Q . The small Lipschitz constant then gives local comparability on $B_Q(50)$. For example, choosing it sufficiently small makes every dyadic ratio in this neighborhood belong to $\{1/2, 1, 2\}$. The distance bound puts $B_Q(100)$ inside Ω . Maximality gives a decomposition, and the positive lower bound for s near every compact subset gives local finiteness.

The cube vertices have dyadic coordinates. Bounded local scale ratios and bounded normalized distance therefore give only finitely many normalized configurations. To triangulate, decompose each cube face into its square contact facets with adjacent cubes. Put every contact vertex and hanging subdivision point on the perimeter of each such square. Cone these perimeter segments from the square center, and then cone the resulting triangulation of each cube boundary from the cube center. The constructions on shared facets agree. They give nondegenerate simplices, and their normalized shapes range over a finite list.

Local comparability bounds the degree of the graph joining cubes whose $B_Q(6)$ boxes meet. A coloring with one more than that degree proves (iii). Finally choose the initial local upper bounds so small relative to the coarse Whitney sizes that the finitely many required patch layers stay in fixed coarse neighborhoods. Since the coarse enlarged cover is locally finite, its interaction graph is locally finite. Taking a still smaller minorant s does not change any of these conclusions. $\square$

The finite lists record actual normalized geometric configurations, including the positions of faces and vertices. The same dyadic-position argument applies in $B_Q(20)$, using the scale comparability on $B_Q(50)$. This includes the older protected boxes needed below. Indeed, if a box $B_R(t)$ with $t \leq 2$ meets $B_Q(5)$, applying scale comparability at the larger cube gives $l_R/l_Q \in \{1/2, 1, 2\}$. Thus $|c_R - c_Q|_\infty \leq 5l_Q + 2l_R \leq 9l_Q$, and $B_R(2) \subset B_Q(13)$. Shrinking these boxes by one of finitely many fixed amounts therefore leaves only finitely many possible intersections and protection margins.

Lemma A.7 (Insertion with protected cores). *Consider a patch Q and a finite collection of older protected boxes with the relative sizes and positions provided by Lemma A.6. For every older box prescribe a larger half-shrunk box and a smaller fully-shrunk box, separated by a fixed positive normalized margin. Let $G : \Omega \rightarrow \Omega'$ be a homeomorphism, and let h_Q be a PL embedding on a neighborhood of $B_Q(3)$. Assume that $h_Q = G$ on the overlaps with the half-shrunk boxes.*

For every $\lambda > 0$ there is an $a > 0$, depending only on λ , the normalized source configuration, and a local constant K_P in Equation (A.12), such that

$$\sup_{B_Q(5)} |G - H| < al_Q, \quad \sup_{B_Q(3)} |h_Q - H| < al_Q \quad (\text{A.14})$$

implies the following conclusion. There is a domain homeomorphism A supported in $B_Q(5)$, with

$$\sup |A - \text{id}| < \lambda l_Q,$$

such that $G \circ A = h_Q$ on a neighborhood of $B_Q(2)$ and $G \circ A = G$ on the fully-shrunk older boxes. The tolerances are uniform over a finite list of source configurations. They do not depend on the derivatives of G or of h_Q .

Proof. Scale the patch to $l_Q = 1$. First reduce a using K_P so that $h_Q(B_Q(3)) \subset G(\text{int } B_Q(4))$. Indeed, for $x \in B_Q(3)$ and $z \in \partial B_Q(4)$, the distance $|H(z) - H(x)|$ is bounded below by K_P^{-1} . The small errors in Equation (A.14) preserve this separation. Degree on $\text{int } B_Q(4)$, comparing G with H on the boundary, shows that $h_Q(x)$ is in its image. The same statement holds with a small margin around $B_Q(3)$.

The embedding

$$\eta = G^{-1} \circ h_Q$$

is consequently defined there. If $z = \eta(x)$, then

$$|z - x| \leq K_P |H(z) - H(x)| \leq K_P (|H(z) - G(z)| + |h_Q(x) - H(x)|) < 2K_P a. \quad (\text{A.15})$$

Thus its closeness to the inclusion uses no inverse-Lipschitz bound for G .

Take a fixed closed neighborhood C of $B_Q(2)$ inside $B_Q(3)$, and let D be the union of the fully-shrunk older boxes intersected with $B_Q(5)$. The half/full shrink margins give neighborhoods on which the map η near C agrees with the identity near D . Include also an identity neighborhood of $\partial B_Q(5)$, separated from C . Choose these neighborhoods once for each normalized source configuration. By Equation (A.15), sufficiently small a makes the union of η and these identity maps an embedding, as in Equation (A.3). Apply Lemma A.3 with $B = B_Q(5)$ and displacement λ . Precomposition by the resulting A installs h_Q on C and preserves the fully-shrunk boxes. Rescaling proves the Lemma. $\square$

We record explicitly how tolerances in repeated insertions are chosen. For a patch insertion supported in $B_Q(5)$,

$$|G(A(x)) - H(x)| \leq |G(A(x)) - H(A(x))| + K_P |A(x) - x|. \quad (\text{A.16})$$

After division by the local cube size, interacting patches introduce only the bounded ratios from Lemma A.6. Hence, for a fixed number of stages, all tolerances can be assigned by backward recursion: start with the desired final value bounds; choose a small allowed displacement at the last stage; use Lemma A.7 to obtain its input tolerance; require the preceding value error to be a small fraction of that tolerance; and continue backward. At a coarse index take the minimum over its finitely many interacting indices at each step. Only finitely many steps occur. Every tolerance remains positive, and none depends on a PL derivative bound selected later.

A.4 Finite PL libraries before resolution selection

The use of finite normalized PL families with exact agreement on earlier overlaps has a precedent in Väisälä's construction [42, Sections 2.7–2.11].

Proposition A.8 (Strong PL approximation of a locally bi-Lipschitz map). *Under the hypotheses of Theorem A.1, there is a locally finite PL homeomorphism $G : \Omega \rightarrow \Omega'$ satisfying*

$$\|G - H\|_{W^{1,p}(\Omega)} < \varepsilon, \quad |G(x) - H(x)| < \xi(x) \quad (x \in \Omega).$$

Proof. Fix the coarse neighborhoods, local bounds K_P , and finite mesh configurations of Lemma A.6. We use two rounds of n_0 insertion stages. The first installs a single piecewise affine interpolant on the cubes where H is nearly affine; its gradients there are close to DH . The second installs PL models on every cube while preserving protected neighborhoods of the first-round cubes. It gives local derivative bounds everywhere, fixed before the remaining cubes are made small in measure. For the next steps, let $\mathcal{Q}$ be any admissible mesh. We choose the tolerances and model libraries

uniformly over all such meshes; the mesh used for the final map will be selected only after the libraries and their derivative bounds have been fixed.

Write $S = 2n_0$ for the number of stages. A newly installed patch has protected radius 2. At each subsequent stage decrease the radius of every older protected patch by

$$d_* = (4n_0 + 4)^{-1} \quad (\text{A.17})$$

in its own B_Q coordinates. Its final radius is greater than $3/2$, and in particular its closed original cube Q remains in the interior of its protected core. At an insertion we use half of the current prescribed decrease for agreement of the new model and the old map, and the full decrease for the sets fixed by the ambient change. These are precisely the margins in Lemma A.7.

Choose all insertion and displacement tolerances by the backward procedure following Equation (A.16). They can be chosen to give an arbitrarily small final normalized value error on every patch. If $a_{s,P}$ is the accuracy required of a new model at stage s , require the current $G - H$ bound before that stage to be less than $a_{s,P}/8$, also on its interacting neighborhoods. This additional fraction is imposed during the same backward recursion. All these numbers are now fixed.

Quantized affine interpolation and good patches. Choose positive thresholds δ_P so small that errors $C\delta_P$ suffice at every first-round insertion involving that coarse neighborhood, and so that

$$\sum_{P \in \mathcal{P}} (C\delta_P)^p |U_P| < \varepsilon^p/8. \quad (\text{A.18})$$

Here and below a local tolerance is reduced using finitely many neighboring indices when necessary. We also require $C\delta_P < (2K_P)^{-1}$, where C is the interpolation constant for the finite mesh shapes.

We quantize the interpolation data so that their exact values on protected overlaps, after normalization, have only finitely many possibilities. At each vertex v of the fine triangulation, round $H(v)$ to a target dyadic grid whose spacing is a small dyadic multiple of the smallest incident cube size. The dyadic factor is fixed from the coarse indices of those cubes, small enough that all the corresponding normalized rounding errors are at most δ_P . Interpolate the rounded values affinely on the conforming triangulation to obtain a continuous piecewise affine map L . We do not assert that L is globally injective.

Call a cube Q , with $P = P(Q)$, good if there is an affine map A_Q of bi-Lipschitz constant at most $2K_P$ such that

$$\sup_{B_Q(20)} |H - A_Q| \leq \delta_P l_Q, \quad \left(\frac{1}{|Q|} \int_Q |DH - DA_Q|^p \right)^{1/p} \leq \delta_P. \quad (\text{A.19})$$

Changing the fixed factor 20 by a larger absolute factor would have the same effect. The rounding and the finite shape list imply

$$\begin{aligned} |DL - DA_Q| &\leq C\delta_P \quad \text{near } B_Q(3), \\ \sup_{B_Q(3)} |L - H| &\leq C\delta_P l_Q. \end{aligned} \quad (\text{A.20})$$

For completeness, subtract A_Q at the vertices of any such tetrahedron. The vertex errors are at most $C\delta_P l_Q$; the inverse matrix of its three edge vectors has norm at most C/l_Q , by finite normalized shapes. Multiplication gives the first estimate. The value estimate follows by affine interpolation and Equation (A.19).

The first estimate in Equation (A.20) makes L an embedding on a neighborhood of $B_Q(3)$. Indeed, $L - A_Q$ is Lipschitz on a slightly larger convex box with constant at most $C\delta_P$; integrate its piecewise constant derivatives on segments. Thus

$$|L(x) - L(y)| \geq ((2K_P)^{-1} - C\delta_P)|x - y| > 0.$$

Invariance of domain gives the open embedding assertion.

The first round. Start with $G_0 = H$. For stages $1, \dots, n_0$ process the colors in order, inserting a patch only if it is good, and taking its model to be $h_Q = L$. On every overlap with an older protected core this agrees with the already installed map, because both equal the single global map L . Equation (A.20) and the chosen thresholds supply the input accuracy for Lemma A.7. Supports $B_Q(5)$ of one color are disjoint, so these insertions are simultaneous. Their local finiteness makes the union a domain homeomorphism. The value invariants follow from Equation (A.16). At the end of this round the map agrees with L on every good cube and on a protected neighborhood of it.

Normalized data and their finiteness. For each Q choose $m_Q \in \mathbb{Z}^3$ nearest to $H(c_Q)/l_Q$ and use normalized coordinates

$$z = \frac{x - c_Q}{l_Q}, \quad \hat{H}_Q(z) = \frac{H(c_Q + l_Q z) - l_Q m_Q}{l_Q}. \quad (\text{A.21})$$

The value at $z = 0$ is bounded by an absolute constant, and Equation (A.12) gives a uniform local Lipschitz bound. The normalized vertex values of L on $B_Q(4)$ lie in a bounded set on finitely many dyadic grids. They therefore take only finitely many values for each coarse index. Here it matters that the translation $l_Q m_Q$ is included in the data: without it one would not have finiteness of affine maps, as opposed to finiteness of their gradients.

If R interacts with Q , then

$$\frac{l_R m_R - l_Q m_Q}{l_Q} \quad (\text{A.22})$$

is a bounded dyadic vector with a denominator from a finite list. Boundedness follows by comparing $H(c_R)$ and $H(c_Q)$ using Equation (A.12); the denominators are fixed by the bounded dyadic scale ratios. Consequently the first-round exact PL data, as viewed in any interacting normalized patch, have only finitely many possibilities.

Choosing the second-round libraries. For stages $n_0 + 1, \dots, 2n_0$ we insert at every patch of the current color. The models are chosen as follows, before the final fine scale is selected. For each coarse index, the normalized maps $\hat{H}_Q$ on a fixed larger patch form a uniformly bounded equicontinuous family. They admit finitely many sup-norm bins of any prescribed positive diameter. At stage s use diameter less than $a_{s,P}/8$.

Inductively assume that the already installed normalized PL models have finite lists. Let $\hat{h}_R$ be an older model in its own R -normalization. In the normalized coordinates of an interacting patch Q , this same map is

$$z \mapsto \frac{l_R \hat{h}_R \left(\frac{c_Q - c_R}{l_R} + \frac{l_Q}{l_R} z \right) + l_R m_R - l_Q m_Q}{l_Q}. \quad (\text{A.23})$$

The source similarities and protected-core positions have finitely many possibilities by Lemma A.6 and Equation (A.17); the target translations do as well by Equation (A.22). Only boundedly many older cores meet $B_Q(5)$, and their coarse indices lie in a fixed finite neighborhood. Record each

older model's library label, its normalized source placement and target translation, and its current protected radius. These records determine the model on the whole protected core and have only finitely many possibilities. In particular, their restrictions to the half-shrunk overlaps give a finite list of exact PL data. No quantization of the later models' affine coefficients is needed. There are consequently only finitely many combinations of

$$\text{source configuration, normalized } H\text{-bin, and exact protected-model records.} \quad (\text{A.24})$$

For each combination which can occur, select one representative normalized embedding G^* , close to a map H^* in that bin by the prescribed current value tolerance, and realizing the specified older data. Take the representative on the fixed open patch $(-4, 4)^3$. If $\widehat{K}_i$ are the normalized half-shrunk older boxes, the set

$$K = [-3, 3]^3 \cap \bigcup_i \widehat{K}_i$$

is compact in that open patch. It lies strictly inside the currently protected cores, so G^* is PL on a neighborhood of K . Apply Lemma A.2 to G^* on the open patch, keeping K fixed. Obtain a PL embedding h^* there with uniform error less than $a_{s,P}/8$ on the required compact patch.

The full protected-model records also ensure agreement on a neighborhood of K with every actual tuple having the same combination: both its current map and G^* equal the recorded models on the protected cores, which contain the half-shrunk boxes in their interiors. Relative approximation makes $h^* = G^*$ on a neighborhood of K . Since there are only finitely many closed half-shrunk boxes, we can choose a neighborhood N of $[-3, 3]^3$ inside $(-4, 4)^3$ whose intersection with each such box lies in this common agreement neighborhood. Store $h^*|_N$ as the library entry indexed by the coarse region, stage, and combination in Equation (A.24). Its entire overlap with each half-shrunk box now has the equality required by Lemma A.7.

If a different actual tuple has the same combination in Equation (A.24), this single model agrees with all of its prescribed PL overlap data. Moreover,

$$\|h^* - \widehat{H}_Q\|_\infty \leq \|h^* - G^*\|_\infty + \|G^* - H^*\|_\infty + \|H^* - \widehat{H}_Q\|_\infty < \frac{3}{8}a_{s,P}. \quad (\text{A.25})$$

The three norms are on the fixed patch needed for insertion. The actual current map is already within $a_{s,P}/8$ of its H , so Lemma A.7 applies after undoing the normalization. This installs h_Q while preserving the fully-shrunk older cores.

This selection does not require compactness of the family of possible intermediate maps G . Only the H -family is binned; the intermediate map is used as an existence witness for an embedding realizing the exact finite overlap data. All possible normalized meshes and inputs obeying the fixed coarse bounds may be included when deciding which combinations occur. At any stage the choices for a coarse index depend only on libraries from strictly earlier stages at its finitely many interacting indices, so they can be made simultaneously over all coarse indices. Hence the choices are independent of the eventual mesh resolution.

Each stored map is PL on a neighborhood of a fixed compact patch and thus has finitely many affine pieces there. It has a finite upper derivative bound, unchanged by the common source and target rescaling. There are finitely many stored maps for each local coarse combination, and only finitely many stages. After the second round every cube retains its own installed model on a protected neighborhood, because its protected radius remains greater than $3/2$. Thus the final derivative on that cube comes from a stored PL model, not from an ambient transition. This proves by induction the finiteness of the data needed at the next stage and gives constants $M_P < \infty$ such that the final map satisfies

$$|DG| \leq M_P \quad \text{almost everywhere on } Q, \quad P = P(Q). \quad (\text{A.26})$$

The constants M_P are fixed before reducing the fine scale. They may be arbitrarily large; no efficient dependence on K_P is asserted.

The final map is locally PL. If a single triangulation is desired, take common refinements of the finitely many affine subdivisions meeting each compact set and triangulate the resulting polyhedral cells. Local finiteness gives a locally finite affine triangulation on Ω . The map is still onto Ω' because every stage was a precomposition by a domain homeomorphism. On every good cube the map still equals L , since its first-round protected neighborhood was preserved throughout.

Selecting the resolution and paying for bad cubes. Only now choose the final local upper size bounds of Lemma A.6. On a fixed coarse compact region, almost every x is both a differentiability point of H and a Lebesgue point of DH . The derivative at such a point is invertible, with the upper and lower bounds inherited from local bi-Lipschitzness. The affine map

$$A_x(y) = H(x) + DH(x)(y - x)$$

then passes both tests in Equation (A.19) on every sufficiently small cube containing x . Indeed its enlargement is contained in a ball of radius Cl_Q about x , which gives the sup-norm test by differentiability. The volume of that ball is at most a fixed multiple of $|Q|$, which gives the derivative mean test by the Lebesgue-point property. Only finitely many coarse thresholds occur near a fixed compact set.

It follows that the measure of the union of bad cubes with a fixed coarse index P tends to zero as their local upper side length tends to zero. This statement is uniform over the admissible meshes. One can see the uniformity by taking the union of all bad dyadic cubes with that index and side at most r . These measurable unions decrease as $r \downarrow 0$, and their intersection is null by the preceding pointwise argument. They lie in the fixed finite-volume neighborhood U_P .

Choose positive numbers b_P with $\sum_P b_P < \varepsilon^p/8$, and reduce the local mesh bounds until

$$(M_P + K_P)^p \left| \bigcup_{\substack{Q \text{ bad} \\ P(Q)=P}} Q \right| < b_P. \quad (\text{A.27})$$

All bounds can be imposed simultaneously using Lemma A.6. The M_P in this inequality were fixed in Equation (A.26); they are unaffected by this refinement.

The stage tolerances also bound $|G - H|/l_Q$ by fixed local constants on each cube. Further reduce the local size bounds so that the actual value error is below ξ and has p th integral less than the remaining part of ε^p . For example require it to be below a global constant $\varepsilon/[4(1 + |\Omega|)^{1/p}]$ and below the positive minimum of $\xi/2$ on the relevant compact neighborhood. These reductions only help the bad-cube estimate. Choose a mesh satisfying all these bounds and carry out the two rounds with the already selected libraries, obtaining the final map G .

On good cubes, Equation (A.19) and Equation (A.20) give

$$\int_Q |DG - DH|^p \leq (C\delta_P)^p |Q|.$$

On bad cubes use Equation (A.26) and Equation (A.12). Summing and using Equation (A.18) and Equation (A.27) makes the derivative error less than the allocated part of ε^p .

The map is locally Lipschitz, its derivative is globally p -integrable by the estimates just proved, and its values lie in the bounded target. Thus it belongs to $W^{1,p}(\Omega; \mathbb{R}^3)$ and has the required strong error. $\square$

Proof of Theorem A.1. Apply Proposition A.8 with Sobolev tolerance $\varepsilon/2$ and value tolerance $\xi/2$, obtaining a PL homeomorphism $h : \Omega \rightarrow \Omega'$. Apply Proposition A.5 to h with the same tolerances. The triangle inequality proves Equation (A.1). Both approximations have exactly the target Ω' . $\square$

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